E-Chatco: An Integrated Smart-Hailing and Fleet Management Platform for the Modernized Philippine Jeepney

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Presented to the Faculty of the
Information and Communications Technology Program
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In Partial Fulfilment
of the Requirements for the Degree
Bachelor of Science in Information Technology

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ENDORSEMENT FORM FOR PROPOSAL DEFENSE

TITLE OF RESEARCH: E-Chatco: An Integrated Smart-Hailing and Fleet Management Platform for the Modernized Philippine Jeepney

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In Partial Fulfilment of the Requirements
for the degree Bachelor of Science in Information Technology
has been examined and is recommended for Proposal Defense.

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This capstone project proposal titled **E-Chatco: An Integrated Smart-Hailing and Fleet Management Platform for the Modernized Philippine Jeepney**, prepared and submitted by **Marinel T. Carbonell, Mark Arone M. Dela Cruz, Rod Erick T. Dulalia, and Mhaku Jose M. Manalili,** in partial fulfillment of the requirements for the degree of Bachelor of Science in Information Technology, has been examined and is recommended for acceptance and approval.

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ABSTRACT

Title of research: **E-Chatco: An Integrated Smart-Hailing and Fleet Management Platform for the Modernized Philippine Jeepney**

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The Philippine jeepney industry is undergoing modernization under the Public Utility Vehicle Modernization Program (PUVMP), yet operators continue to face challenges in fare collection transparency, boundary remittance tracking, fleet monitoring, and commuter visibility. This study developed E-Chatco, an integrated smart-hailing and fleet management platform for the modernized Philippine jeepney, in partnership with CHATCO Cooperative along the Calumpit to Meycauayan route. The review of related literature and systems examined existing transport management platforms and digital payment integrations, identifying gaps in integrated boundary remittance visualization and role-based access for conductor task management.

The system was developed using the Agile Scrum methodology across seventeen sprints, with Laravel (PHP) as the backend, Next.js 16 / React 19 as the frontend, Hostinger

MySQL for production and local development, Laravel Echo with Pusher for real-time communication, and React Native with Expo for the mobile applications.

The platform features real-time GPS tracking with proximity-based smart hailing within a 1-kilometer radius, point-area fare computation, dual cash and GCash payment tracking with Paymongo integration, automated end-of-day remittance with separate cash and GCash recording, fleet visualization, digital lost and found management, and safety features including Share My Ride.

The completed system was evaluated using the ISO/IEC 25010:2023 software quality criteria across eight characteristics: Functional Suitability, Reliability, Performance Efficiency, Usability, Compatibility, Security, Maintainability, and Portability. The platform achieved an overall mean rating of 4.71, interpreted as Excellent, with Functional Suitability obtaining the highest mean at 4.86. A total of 346 test cases were executed across the commuter, conductor, and administrator applications, with all cases passing after the second round of testing following the resolution of defects identified in the initial round. These results affirm E-Chatco's readiness for deployment as a comprehensive digital solution for modernized jeepney operations, directly addressing the commuter visibility, fare transparency, and boundary remittance concerns identified by the CHATCO Cooperative.

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Introduction

Project Context

The Philippine public transportation system, particularly the iconic jeepney, is currently undergoing a critical phase of modernization. The government's Public Utility Vehicle Modernization Program (PUVMP), launched in 2017, aims to phase out traditional jeepneys and replace them with modern, eco-friendly alternatives (Buizing et al., 2025). However, recent research indicates that while modernization offers environmental benefits and improved safety features, significant challenges persist regarding affordability, accessibility, and livelihood sustainability for drivers and operators (Dimalanta & Atienza, 2025).

Another reason why this change is urgently needed is the growing mismatch between the traditional jeepney system and the demands of a rapidly urbanizing society. The old fleet, while culturally iconic, is increasingly unable to cope with worsening traffic congestion, rising commuter volumes, and stricter environmental standards. Modernization is not just about replacing vehicles—it is about creating a more efficient, reliable, and sustainable transport network that can support economic growth and improve the daily lives of millions of Filipinos. Without this shift, the country risks locking itself into outdated infrastructure that hampers mobility, contributes to pollution, and undermines long-term development goals.

The informal nature of jeepney operations has historically relied on manual, cash-based transactions, creating inefficiencies in fare collection and limited operational oversight. However, a more pressing challenge identified in recent studies is the boundary remittance system—the unique financial arrangement where drivers pay a fixed daily

amount to operators and keep excess earnings. Research by Mettke et al. (2019) and recent policy developments under PUVMP reveal that tracking these daily remittances, ensuring compliance, and reconciling cash declarations with actual earnings remains a significant operational burden for operators (Respicio & Co., 2025). The integration of digital payment systems such as GCash alongside traditional cash payments is intended to provide transparent transaction records and reduce discrepancies in boundary remittances, while still accommodating commuters who prefer cash payments. The system records cash and GCash transactions separately for tracking and validation, ensuring both payment methods are accounted for during the end-of-day remittance process.

While modernization aims to solve these structural issues, the daily reality for the commuter remains problematic. To quantify the specific pain points of the target demographic, the proponents conducted a preliminary survey of 71 commuters along the CHATCO Modernized Jeepney route. The data revealed a critical lack of trip visibility: 69.0% of respondents wait 10 minutes or more for a unit, and 60.6% frequently wait without any indication of when a jeepney will arrive. Furthermore, 80.3% reported experiencing long waiting times "often" or "always," with environmental factors like extreme heat (cited by over 50% of respondents) and severe overcrowding (57.7%) compounding the inconvenience. Most notably, an overwhelming 91.5% of respondents stated that the ability to see a jeepney's location before leaving their location is "Important" or "Very Important," yet this capability is entirely absent in the current local transit setup.

Recent studies on paratransit and informal transport in developing countries demonstrate that technology adoption can improve service delivery and address these exact visibility issues. Research on ride-hailing app adoption by paratransit operators in Cambodia indicates that flexible regulatory approaches to technology integration can increase trip

numbers and driver revenue (Phun et al., 2018).In the Philippine context, recent developments include the Cebu People's Multipurpose Cooperative (CPMC) partnership with Maya in 2023, which accepts cashless payments through RFID/tap technology, and academic proposals such as PAYSADA (2020) and the JeePoint (2017). However, these systems remain either hardware-dependent or lack integrated payment and operational management features.

The project emerges from this context as a comprehensive web-based digital solution designed specifically for the Philippine jeepney ecosystem, developed in partnership with CHATCO Cooperative. Unlike generic ride-hailing applications or hardware-dependent payment systems, the proposed system addresses the unique operational structure of jeepneys—fixed routes with flexible stopping points, conductor-mediated fare collection using predefined operational point areas, the boundary remittance system with separate cash and GCash recording, and high-frequency short-distance trips—through an accessible web platform that requires a dedicated mobile app installations for commuters, while conductors can utilize their mobile phones paired with bluetooth thermal printers for transaction processing. By directly addressing the local survey findings—specifically the demand for real-time visibility to eliminate uncertain waiting times—E-Chatco aims to bridge the gap between traditional manual operations and modern digital expectations, offering both cash and GCash cashless payment options to accommodate the varying technological readiness of commuters.

Purpose and Description

E-Chatco is a modern jeepney management and e-payment solution that aims to make managing transport operations easier and better. The project will incorporate the latest technology with conventional transit operations to give a more integrated and engaging

experience for both the commuter and the transport cooperative. By merging fare collection, safety tracking, and operational management functionalities, E-Chatco provides a well-structured method for jeepney modernization.

This project aims to improve efficiency in jeepney operations by providing a digital platform where commuters can track rides within a 1-kilometer radius, calculate fares based on predefined CHATCO operational point areas, and pay directly through GCash integration or with cash with ease. The fare calculator uses the cooperative's official fare matrix and point-area references, ensuring that fare computation is always aligned with actual CHATCO operational procedures. This platform will also provide a way for cooperative staff (conductors and administrators) to coordinate by having role-based accounts and tasks. Conductors may use mobile phones paired with bluetooth thermal printers for ticket generation during fare transactions. Through updates and efficient processes, commuters and cooperatives can have more organized transactions. The system minimizes the risk of miscommunication and human error by automating important functions such as boundary remittance tracking with separate cash and GCash recording.

What sets E-Chatco apart is the inclusion of an interactive tool for operational visualization and safety. Instead of relying on manual cash declarations or physical panic buttons, conductors can verify GCash payments via their operational devices with thermal printers. Commuters can share their real-time location with family members and view all active CHATCO units system-wide on the map. The hailing function (“Pick Me Up”) is only enabled when a CHATCO unit is within a 1-kilometer radius of the commuter’s current location, with vibration feedback provided while the hail signal is active and a sound notification alerting the commuter when a nearby unit enters the radius. This interactive feature enables users to make a judgment as to whether their

transit experience is safe and efficient. The ability to see all units on the map while restricting hailing to nearby units minimizes unnecessary hailing signals and maximizes commuter satisfaction, while the point-area-based fare computation ensures that all fare calculations align with actual CHATCO cooperative operational zones.

E-Chatco is a solution that digitizes jeepney services by adopting technology. It not only simplifies transit organization but also delivers an interactive experience for its users. The project is particularly applicable to today's fast-moving industry, in which convenience, precision, and customer interaction are key factors to delivering successful public transport solutions.

The study benefited several stakeholders by providing an integrated smart-hailing and fleet management platform that digitizes jeepney operations for the modernized Philippine transport system.

Commuters. Commuters benefited from the platform by gaining the ability to view all active CHATCO units on a real-time GPS map, allowing them to plan their commutes with greater certainty and reduce uncertain waiting times. The proximity-based hailing feature, which activates only when a unit is within a 1-kilometer radius, ensures that pickup requests are actionable and relevant. Sound notifications and vibration feedback alert commuters when a nearby unit approaches or when their hail signal is active, eliminating the need to constantly monitor the screen. The direct GCash payment integration provides a seamless cashless payment option without the burden of maintaining a stored-value wallet, while the point-area-based fare computation ensures transparent and accurate fare charges aligned with the approved CHATCO fare matrix.

The “Share My Ride” feature enhances personal safety by allowing commuters to share their live location with family members during transit.

Conductors. Conductors benefited from the system through the use of a dedicated operational device similar to a mobile phone with a built-in thermal printer, which streamlined their fare transaction workflow. The device enables them to select predefined operational point areas for fare computation; verify GCash payments through direct integration; record cash payments digitally; and generate thermal-printed tickets for commuters—all within a single interface. This eliminates the reliance on the existing fare computation gadget, which has a known flaw where incorrect entries cannot be reversed or deleted. The separate tracking of cash and GCash payments throughout the shift provides conductors with a clear and organized record of their collections, simplifying the end-of-day remittance declaration process and reducing the risk of discrepancies.

CHATCO Cooperative Administrators. Administrators benefited from the platform by gaining a centralized dashboard for monitoring fleet operations, boundary remittance compliance, and financial records with separate cash and GCash breakdowns. The real-time fleet visualization provides oversight of all 39 active units, including speed monitoring and overspeeding flags for passive safety enforcement. The remittance tracking module, with its separate cash and GCash declarations, enables administrators to validate GCash transactions against reference IDs and cross-check cash declarations for accuracy.

CHATCO Cooperative as an Organization. The cooperative as a whole benefited from the platform’s ability to digitize and organize daily operations that were previously handled through manual, cash-based processes and paper-based records. The system

provides a transparent and auditable record of all fare transactions, boundary remittances, and fleet activities, reducing the risk of data loss, boundary remittance disputes, and payment tracking difficulties. The integration of GCash as the cashless payment method addresses the cooperative's previous unsuccessful attempt to implement tap card payments, providing a proven and widely adopted digital payment infrastructure without the need for specialized hardware. The digital lost and found matching system replaces the manual Facebook-based process, improving the efficiency of item recovery for commuters.

Local Government and Transport Regulators. The platform supports the objectives of the Public Utility Vehicle Modernization Program (PUVMP) by demonstrating a viable digital solution for modernized jeepney operations. The system's data on ridership patterns, fare collections, fleet utilization, and route coverage provides valuable information that can support transport planning and regulatory compliance monitoring. The cashless payment integration and structured boundary remittance tracking contribute to the government's push for financial transparency in the public transport sector.

During account registration, commuters are required to verify their identity through a valid ID submission to determine their discount eligibility status (Student, Senior Citizen, or PWD). This ensures that fare discounts are accurately applied and prevents misuse of discount privileges. Conductors and administrators are assigned role-based accounts with access permissions tailored to their respective functions, reinforcing accountability and preventing unauthorized access to sensitive operational and financial data.

Objectives

The general objective of this study is to develop E-Chatco, an integrated smart-hailing and fleet management platform for the modernized Philippine jeepney. This study aims to address the challenges specifically faced by the CHATCO Cooperative.

The specific objectives are as follows:

1. To develop an Integrated Smart-Hailing and Fleet Management Platform that features significant functionalities and modules:

- Landing Page Module for All System Users
- Login and Registration Module for All System Users (with email verification, valid ID submission, and administrator approval of commuter accounts)
- Dashboard Module for All System Users
- User Account Monitoring and Management Module for the Administrator
- Real-time GPS Tracking and Hailing Module for Commuters and Conductors (with 1-kilometer radius optimization)
- GCash E-Payment Module for Commuters (QR-based payment through the PayMongo gateway, with group payment, voucher application, and cash receipt claiming)
- Fare Calculator Module Using Predefined Operational Point Areas (with automatic Student, Senior Citizen, and PWD discounts)
- Boundary Remittance Module for Conductors and Administrators (with Separate Cash/GCash Recording and End-of-Day Close-out)
- Conductor Shift and Fare Recording Module (with unit verification, live location broadcasting, capacity and break status, thermal receipt printing, and offline cash recording)
- Fleet Management and Traffic Visualization Module for Administrators (with overspeeding detection and commuter demand zones)

- Lost and Found Management Module for Commuters and Administrators
- Safety and Loyalty Page Module for Commuters (with Share My Ride, free-ride rewards, and vouchers)
- Discount Application and Student Renewal Module for Commuters and Administrators
- Ride Feedback and Rating Module
- Announcements Module
- Receipts, Transactions, and Analytics Module for Administrators
- System Settings and Configuration Module for Administrators (fare matrix, route, financial rules, operations rules, voucher generator, and receipt settings)

2. To implement role-based access and user permissions for administrators, conductors, and commuters relevant to the process workflow of CHATCO Cooperative.

3. To evaluate the system's acceptability using the ISO/IEC 25010:2023 software quality criteria as perceived by the respondents on the following:

- Functional Suitability;
- Reliability;
- Performance Efficiency;
- Usability;
- Compatibility;
- Security;
- Maintainability, and;
- Portability.

By addressing these objectives, the study aims to enhance the overall efficiency, internal coordination, and commuter satisfaction for CHATCO Cooperative while adopting modern technological innovations like web-based GCash payment integration to revolutionize the local jeepney industry.

Scope and Limitations

The general purpose of this study is to design and develop an integrated smart-hailing and fleet management platform tailored for the modernized Philippine jeepney. The scope of this study covers the core functionalities of the CHATCO smart-hailing system, including proximity-based smart hailing with 1-kilometer radius detection; a dual cash and GCash payment system via QR code scanning processed through the PayMongo gateway; point-area-based fare computation with automatic discounts for verified Student, Senior Citizen, and PWD commuters; admin-reviewed discount application and student renewal processing; automated fleet management and real-time monitoring; ridership, revenue, and commuter demand analytics; a digital lost and found system; safety features including Share My Ride; a loyalty program with free-ride rewards and promotional vouchers; and separate cash and GCash transaction tracking for remittance purposes.

The target users and beneficiaries are the management, conductors, and commuters of CHATCO Cooperative. The research will cover the end-to-end digital transaction process—from the moment a commuter hails a ride and pays through either cash or GCash to the conductor's end-of-day boundary remittance logging with separate cash and GCash declarations, up to the administrator's fleet and financial monitoring.

The study will be conducted over a period of nine (9) months, starting from February 2026, and will consider the 39 modern jeepney units currently operating under the cooperative's single, fixed route (Calumpit to Meycauayan).

While the system proposes a comprehensive digital solution within this scope, several operational and technical limitations have been established to ensure the project remains feasible and aligned with the cooperative's actual environment:

First, automated turn-by-turn directional mapping is excluded. Given that the cooperative operates on a single, fixed route and the fast-paced nature of the trip, a GPS navigation system is deemed redundant and will not be developed.

Second, live tracking is strictly an online feature. Commuters can view all on-duty CHATCO units system-wide on the real-time GPS map; units on break are hidden from the map. However, the hailing function ("Pick Me Up") is only enabled when a CHATCO unit is within a 1-kilometer radius of the commuter's current location. A hail remains active for three minutes and is automatically cancelled if the commuter moves out of range or the conductor does not respond. While the hail is active, the commuter's device provides continuous vibration feedback to indicate that the hail signal is being broadcast to nearby units. A sound notification also alerts the commuter when a CHATCO unit enters the 1-kilometer radius. This proximity-based hailing restriction optimizes the system by reducing unnecessary hailing signal processing and network broadcasts, lowering bandwidth usage for hail signals, decreasing system load from processing out-of-range hails, and minimizing synchronization issues. If a commuter loses internet connection or runs out of data load, the live map will pause until connectivity is restored. On the conductor side, cash fares can still be recorded while offline and are synchronized automatically once connectivity is restored; however, GCash payments and voucher validation require a continuous internet connection, and all offline cash must be synchronized before the end-of-day remittance can be submitted.

Third, regarding device deployment, conductors will use an operational device similar to a mobile phone equipped with a built-in thermal printer for ticket generation during fare transactions. This device is dedicated to E-Chatco fare processing, displaying GCash payment QR codes and confirmations, and receipt or ticket issuance. The use of this operational device replaces the previous arrangement that relied on conductors' personal smartphones. The built-in thermal printer enables conductors to provide physical tickets to

commuters during each fare transaction, whether the payment is made in cash or through GCash. Each printed receipt contains a QR code that commuters can scan within six hours to link a cash-paid ride to their account, by scanning each code per ride they can accumulate 10 rides in order for them to have a free ride voucher.

Fourth, in terms of financial features, the system supports both cash and cashless (GCash) payment methods. Conductors process cash payments manually and record them in the system, while GCash payments are initiated by the conductor through a payment QR code that the commuter scans, and are confirmed automatically through the PayMongo payment gateway. The same fare rates apply regardless of payment method. Cash and GCash payments are recorded separately during operations for tracking and validation purposes, and both are totaled during the end-of-day remittance process. Direct bank account linking is not supported; GCash is the sole cashless payment method integrated into the system.

Fifth, fare computation is based on predefined operational point areas as established in the CHATCO fare matrix, not on unrestricted GPS point-to-point calculations. A minimum fare covers the first set of barangays, and a succeeding fare is added for each additional barangay, with both rates configured by the administrator. When a commuter travels from one point area toward another and the actual drop-off location lies between two official point areas, the conductor selects the nearest official operational point area for fare computation. The conductor manually selects the operational point area during each transaction to ensure accuracy and alignment with CHATCO's approved fare structure. Discounted fares are applied automatically only to commuters whose Student, Senior Citizen, or PWD status has been verified and approved by the administrator.

Sixth, the loyalty system grants a free-ride voucher after a configurable number of qualifying paid rides (by default, every 10 rides), with each voucher valid for three days. In addition,

administrators may generate promotional voucher batches that the system distributes to randomly selected eligible commuters. The system does not offer tiered or personalized loyalty programs.

Seventh, regarding safety monitoring, the overspeeding detection feature is strictly for passive monitoring only. The system will visualize speed data and flag overspeeding instances on the administrative dashboard for recording and disciplinary purposes, but it will not physically intervene in the jeepney's operation.

Finally, the mobile applications (Conductor App and Commuter App) are developed for the Android platform only and are not available on iOS. The applications are distributed through direct APK sideloading rather than the Google Play Store, since they are intended for the cooperative's internal use and the operational devices issued to conductors. Minimum Android version and storage requirements apply, and installation on unsupported devices may result in degraded performance or missing features. A stable internet connection is required for real-time features such as live tracking, GCash payment verification through PayMongo, although offline cash fare recording is supported on the Conductor App and syncs automatically once connectivity is restored.

Review of Related Literature/Studies/Systems

For years, the jeepney transit industry has contributed to the daily commute of Filipinos by providing transport services. However, most jeepney operators rely on manual, cash-based booking systems, verbal arrangements, and paper-based records that may increase the risk of data loss, boundary remittance disputes, and difficulties in payment tracking and fleet management.

In response to these challenges, digital/web-based transport management systems have emerged, offering booking solutions, payment tracking, and staff coordination. Systems like these incorporate QR customization tools and GPS tracking features to make operations efficient and easier to manage. However, despite these features, visualizing fleet operations and boundary remittances that are offered by some transport businesses still remains a challenge, relying on manual logbooks that may not accurately represent the conductor's actual earnings.

This literature review discusses existing transport management systems and digital payment integrations in public transit. By studying and analyzing past research and available technologies, this study aims to establish supporting references for the development of E-Chatco, a smart-hailing and fleet management platform for a more efficient jeepney operation, improved planning, and an innovative transit experience.

Related Literature/Studies

Foreign-Related Literature/Studies

Assessment of Informal Transport Payment Systems and Operational Challenges in Developing Countries

Research by Mettke et al. (2019) discusses the effect of non-digital fare collection in paratransit operations, highlighting its advantages and disadvantages. With the evolution of technology, transport sectors are adopting digital solutions like mobile payments and intelligent inventory management to enhance efficiency. Nevertheless, these developments also have disadvantages, including data security threats and employees' adoption requirements. The research highlights that although digitalization has the potential to greatly enhance efficiency in operations, overcoming privacy, employee training, and data protection challenges is necessary in order to maximize its benefits.

This study, therefore, supports the digitalization of industries related to public transport, thus supporting E-Chatco. However, the study also advises safe and effective digitalization of systems by watching for data privacy, conductor training, and data protection to be truly considered effective.

Cashless Payment Adoption Challenges in Lagos Public Transport

Mogaji and Nguyen (2024) present a study examining commuter preferences for contactless payment systems in Lagos, Nigeria. Their research employs quantitative methods to map current digital interventions. A key insight is the resistance to cashless systems due to technological literacy disparities and consumer trust, which enables planners to trial configurations virtually, reducing setup errors and improving stakeholder communication. However, their descriptive study stops short of proposing an integrated platform, leaving out modules for conductor-mediated GCash payment verification and multi-role collaboration.

Meanwhile, E-Chatco: An Integrated Smart-Hailing and Fleet Management Platform for the Modernized Philippine Jeepney embeds direct GCash payment integration within a full-service transit management system. E-Chatco allows direct GCash payments that are verified in real time by the conductor's operational device. Beyond payment processing, E-Chatco offers end-to-end operations—hailing, boundary remittance tracking with separate cash and GCash recording, point-area fare computation, and role-based dashboards for admins and conductors. By transforming standalone cashless payments into one cohesive platform that also supports traditional cash payments, E-Chatco delivers both the payment benefits identified by Mogaji and Nguyen and the operational efficiencies required for real-world jeepney execution.

Local Related Literature/Studies

Challenges and Opportunities in the Modernization of Public Utility Vehicles in the Philippines

Salazar and Berba (2022) conducted a study examining existing practices and systemic challenges in the Philippine PUV modernization process and its impact on operational success. Their study delved into several areas: specifically, financial sustainability, logistical coordination among cooperatives, driver livelihood, and the communication gap between transport regulators and operators. Their study and proposed framework recommend a couple of objectives that can help improve and develop a more adaptive and responsive modernization process. These recommendations include clarifying financial records, enhancing cooperative capacity, strengthening coordination, and investing in modern digital management tools.

E-Chatco will consider the study's recommendations for the development of the system. Through assigning roles to staff members (e.g., Admin vs. Conductor), E-Chatco can clarify operational responsibilities, as users can check which tasks and remittances are assigned to them. Coordination may also be strengthened, according to the study, which could be realized in E-Chatco through role-based access, where conductors can update statuses, strengthening coordination between field staff and management.

Jeepney Operator Experiences in Dasmariñas City, Cavite

Antonio et al. (2025) conducted a study examining existing practices and challenges in jeepney boundary remittance processes and their impact on operational success. Their study delved into four areas: specifically, revenue tracking, expense logging, communication, and manual declarations. Their study and proposed framework recommend a couple of objectives that can help improve and develop a more adaptive and responsive remittance process. These recommendations include clarifying financial records, enhancing conductor capacity, strengthening coordination, and investing in modern communication tools.

E-Chatco will consider the study's recommendations for the development of the system. Through assigning roles to staff members, E-Chatco can clarify role assignments, as the users can also check which expenses they are assigned to. Coordination may also be strengthened, according to the study, which could be realized in E-Chatco through role-based access, where conductors can perform their respective roles and update statuses, which may strengthen coordination between the members in organizing boundary remittances.

Related Systems

Foreign-Related Systems

Web-Based Transport Management System

Mama and Akinsanya (2021) developed a system called a ‘Web-based Transport Management System,' a web-based and automated application designed to increase efficiency in the transport process. It connects commuters and administrators through a centralized database for the purpose of easy communication and operational efficiency. Commuters can view routes, make payments, and receive confirmations through notifications. For administrators, the system features a dashboard to manage transit logs and update statuses. By automating tasks, the system helps save time and reduce errors.

Similar to the system, E-Chatco features centralized management and web-based accessibility. However, in contrast to Mama and Akinsanya's systems, E-Chatco also offers a localized fleet management and boundary remittance tracking feature. Additionally, in contrast to their system, which provides a standard digital customer interaction model, E-Chatco utilizes a conductor-mediated QR scanning approach tailored to the unique "boundary" culture of Philippine jeepneys.

Comprehensive Study on QR Code-Based Ticketing System for Public Transport

The study by Gupta and Mittal (2022) introduces an interactive system that assists users in navigating transit networks by leveraging QR code ticketing. The system provides a robust framework for generating and validating QR codes, ensuring functional and secure fare transactions. This approach simplifies the transit process for users, allowing them to generate tickets efficiently. However, the system is primarily focused on formal, fixed-route ticketing and does not encompass broader informal transit management functionalities such as conductor boundary tracking, dynamic hailing, or real-time collaboration among transport staff.

In contrast, E-Chatco offers a solution that integrates QR code payments specifically for informal paratransit (jeepneys). Beyond simple ticketing, it includes features for boundary remittance logging, expense tracking, and coordination between administrators and conductors. While Gupta and Mittal's system enhances the efficiency of formal QR ticketing, it supports the development of E-Chatco by highlighting the technical feasibility and security benefits of user-friendly QR tools in public transport contexts.

Local Related Systems

PAYSADA: An Architectural Design for an Automated Fare Collection System

The study by Andalecio et al. (2020) entitled PAYSADA explores the transition from manual to digital fare collection in the Philippine jeepney industry. Their system manages automated fare collection, reduces manual counting errors, and improves transaction security by allowing clients to pay via RFID. Like E-Chatco, this study addresses inefficiencies in traditional jeepney operations, such as manual cash handling. However, while E-Chatco aims to integrate a web-based QR visualization and boundary tracking tool, PAYSADA focuses only on digitizing bookings via hardware-dependent RFID technology.

Both systems emphasize operational efficiency—E-Chatco through its planned web-based QR planning and staff coordination features and PAYSADA through automated RFID reservations. The key difference lies in their technological scope: PAYSADA’s system is an RFID-based platform requiring specialized hardware. E-CHATCO seeks to combine management tools with accessible web-based QR visualization, eliminating the need for expensive POS hardware while enhancing client-staff collaboration.

JeePoint: A Secure Mobile and Web-Based Platform for Modern Jeepney Transport

The study of Quijano et al. (2017) on JeePoint examines the implementation of a digital platform to streamline both operational management and passenger bookings in the jeepney industry. Their system enhances business efficiency by enabling online reservations, package customization, and secure encrypted transactions while allowing owners to monitor utilities and manpower allocation. Similar to E-Chatco, this research addresses traditional critical challenges like manual record-keeping and passenger-staff miscommunication. However, while E-Chatco proposes a boundary remittance tool tailored for cooperative operations, JeePoint concentrates exclusively on digitizing backend security and frontend bookings.

Both prioritize operational efficiency—E-Chatco through its planned boundary-tracking tools and JeePoint through comprehensive web-based security automation. The primary distinction lies in their technological approach: JeePoint serves as an all-in-one secure booking tool, whereas E-Chatco seeks to augment traditional cooperative processes with specific boundary remittance and smart-hailing features for jeepney setups.

Synthesis and Research Gap

The reviewed literature shows valuable insights into how transport management systems and digital payments are currently advancing in society. Previous studies highlight the benefits of digital platforms, particularly in transport businesses, in improving efficiency in operations, while in today's technological era, QR tools continue to make their advancements in several industries.

Despite the advancements made within the digitalization of transport operations, existing transport management systems still lack integrated boundary remittance visualization functionality as well as role-based access for conductor task management. Additionally, current platforms, as observed in the previous studies and developments of transport management systems, still focus on online transactions, such as customer booking and payment, focusing less on internal operations like boundary execution. These gaps in the studies highlight the need for a more integrated and adaptive solution like E-Chatco.

E-Chatco builds and takes into consideration these previous works by combining a transport management system and web-based QR payment planning. Through this integration, E-Chatco can offer more functionalities beneficial to the jeepney business, such as a transport operations platform for better fleet management, as well as offer a way to visualize what a remittance status will look like. By addressing the limitations shown by previous studies and systems, the system aims to provide a more comprehensive and organized experience for jeepney businesses, their conductors, and their commuters. The reviewed literature shows valuable insights into how transport management systems and digital payments are currently advancing in society. Previous studies highlight the benefits of digital platforms, particularly in transport businesses, in improving efficiency in operations, while in today's technological era, QR tools and mobile payment integrations continue to make their advancements in several industries.

Despite the advancements made within the digitalization of transport operations, existing transport management systems still lack integrated boundary remittance visualization functionality as well as role-based access for conductor task management. Additionally, current platforms, as observed in the previous studies and developments of transport management systems, still focus on online transactions, such as customer booking and payment, focusing less on internal operations like boundary execution. Furthermore, no

existing system addresses the need for point-area-based fare computation aligned with cooperative-specific fare matrices, nor do they provide proximity-optimized real-time tracking that reduces bandwidth and system load. These gaps in the studies highlight the need for a more integrated and adaptive solution like E-Chatco.

E-Chatco builds and takes into consideration these previous works by combining a transport management system with direct GCash payment integration and point-area-based fare computation. Through this integration, E-Chatco can offer more functionalities beneficial to the jeepney business, such as a transport operations platform for better fleet management, system-wide unit visibility with proximity-based hailing restricted to a 1-kilometer radius for optimized performance, separate cash and GCash transaction tracking for transparent remittance processing, and point-area fare selection that aligns with actual CHATCO cooperative operations. By addressing the limitations shown by previous studies and systems, the system aims to provide a more comprehensive and organized experience for jeepney businesses, their conductors, and their commuters.

Methodology

Technical Background

The current jeepney operations at CHATCO Cooperative rely on boundary remittances and manual hailing coordination. This traditional approach lacks the technological infrastructure needed for efficient fleet management, conductor coordination, and accurate payment tracking, resulting in frequent scheduling conflicts and recording errors. The proposed system will use modern programming languages and development environments to turn these manual processes into an integrated digital solution:

1. Core Development Technologies

- **Laravel 12 (PHP 8.2):** Server-side framework for backend logic, database interaction, scheduled tasks, and the REST API used by the web platform and the Android applications.
- **Laravel Sanctum:** Authentication package for token-based log-in sessions on the web and Android applications.
- **Next.js 16 (React 19):** Frontend framework for the structure, routing, and interactivity of the commuter, conductor, and administrator web interfaces.
- **React Native:** Framework used to develop the Commuter and Conductor Android applications, which provide the same features as their web counterparts.
- **Pusher with Laravel Echo:** Real-time broadcasting service for live GPS tracking, hailing notifications, payment confirmations.
- **MySQL:** Primary relational database management system for secure data storage.

2. Integrated Development Environments (IDEs)

- **Visual Studio Code:** A lightweight code editor used for web, mobile, and backend development.

3. Real-Time and Mapping Components

- **Leaflet.js with React-Leaflet:** JavaScript mapping library used for interactive maps, live unit tracking, route display, and location picking.
- **Haversine Formula:** Distance computation method used to enforce the 1-kilometer hailing radius between commuters and on-duty units.
- **qrcode.react:** Library used for generating payment, receipt, and unit QR codes.
- **html5-qrcode:** Library used for scanning QR codes through the device camera.

4. Supporting Tools

- **Git/GitHub:** Version control system for collaborative development.
- **Tailwind CSS:** Utility-first CSS framework used for styling the user interface.
- **Framer Motion:** Animation library used for interface transitions.
- **Lucide React:** Icon library used throughout the user interface.
- **jsPDF and SheetJS (xlsx):** Libraries used to export reports and records as PDF and Excel files.
- **Postman:** Tool used for API testing.

5. Other Tools and Services

- **PayMongo:** Payment gateway used to process GCash fare payments.
- **Cloudflare R2:** Cloud object storage for uploaded files such as valid IDs, discount documents, profile photos, and lost-and-found photos.
- **Brevo:** Email service used to send verification codes, password reset codes, and account and discount notifications.
- **Hostinger:** Hosting service for the Laravel backend and MySQL database.
- **Vercel:** Hosting service for the Next.js frontend.

Software Development Life Cycle (SDLC) Approach

Agile software development is a flexible and collaborative approach to creating software systems. It focuses on building working software in small, manageable parts called sprints, which usually last between one and four weeks. Instead of completing the whole system at once, agile encourages development teams to release functional features early and continuously improve them based on feedback from users or stakeholders. This method allows for faster delivery, higher quality, and more effective collaboration between developers and clients.

The core principles of Agile were established in 2001 through the Agile Manifesto, which emphasizes four key values: individuals and interactions over processes and tools, working software over comprehensive documentation, customer collaboration over contract negotiation, and responding to change over following a strict plan (Agile Alliance, 2024). These values guide Agile teams to prioritize human communication and adaptability, important traits when developing software in rapidly changing environments.

Agile is often implemented through frameworks like Scrum, where tasks are organized into product backlogs and distributed across sprints. At the end of each sprint, teams hold a review to demonstrate the completed work and a retrospective to reflect on what went well and what can be improved. This cycle ensures that the team is always learning, adjusting, and growing. It also encourages transparency, allowing clients to be involved throughout the process, not just at the beginning or the end.

According to Atlassian (n.d.), one of the biggest advantages of Agile is its ability to deliver value quickly and consistently. By releasing smaller parts of the system early, users can interact with the product and give feedback sooner. This prevents costly errors that might only be discovered

at the end of traditional development cycles. Agile also supports continuous testing and integration, which helps maintain quality throughout the project.

Figure 2.1 represents the specific Agile development method to be used by the proponents for the system.

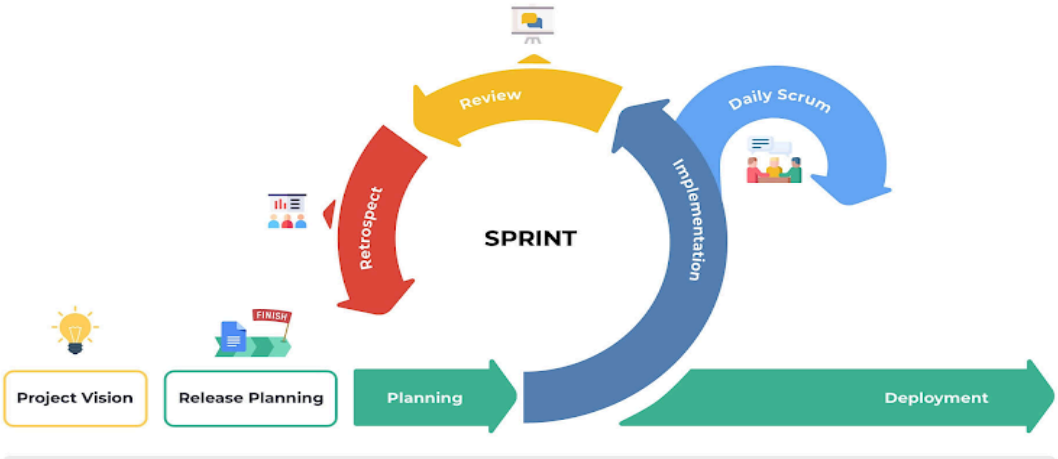


Figure 2.1: Agile Scrum Development Model Overview

Scrum is a framework used in agile project management to help teams deliver products efficiently through short, focused cycles called 'sprints.' It consists of five distinct phases that guide the team from project initiation to final release. The process starts with the team defining the project's goal during the initiation phase. The project's prioritized goals, or epics, are compiled into a product backlog, and roles are assigned.

Next is the Planning and Estimates Phase, where the team breaks these epics down into smaller user stories. These are then estimated and selected for the upcoming sprint based on priority and effort. During the implementation phase, the team carries out the sprint. Daily stand-up meetings are held to track progress, discuss any obstacles, and stay aligned. The Review and Retrospective

Phase begins at the conclusion of the sprint. The team demonstrates what was completed, gathers feedback, and reflects on how the sprint went to identify improvements for the next cycle. The finished product is made available to users during the deployment phase. A final retrospective can also be held to assess the project's overall success.

In Agile, different ceremonies are conducted in order to enable a smooth workflow for the success of each sprint. Table 2.1 lists the types of Agile ceremonies and their purpose.

CEREMONY	SEGMENT	DESCRIPTION
Sprint Planning	Initiation	These sessions are what initiate each sprint. The team decides what work from the product backlog can be completed in the time period, who will complete what tasks, and how to be successful.
Daily Scrum	Routine	These daily check-ups help teams stay on track and mark progress. Each morning, team members discuss what they worked on yesterday, what they're doing today, and what's blocking them from moving forward.
Product backlog	Continuous process	While not always listed as an 'official' ceremony, product backlog refinement is a continuous process of cleaning up, adjusting, and updating tasks and the feature list.
Sprint Review	End	At the end of each sprint cycle, teams meet to demo what they've shipped and get early feedback from stakeholders.
Sprint Retrospective	End	At the end of each sprint cycle, the team also meets to discuss what's working (or not) on a process level, including blockers, communication issues, or a lack of resources.

Table 2.1: Agile Scrum Ceremonies

- ***Calendar of Activities***

From the second week of February to the second week of May, the proponents have conducted various activities for the planning and development of the study. The activities were done collaboratively and were done through various collaborative software applications, such as Google Docs, and through face-to-face meetings.

The proponents consist of four (4) people, namely Marinel T. Carbonell, Mark Arone M. Dela Cruz, Rod Erick T. Dulalia, and Mhaku Jose M. Manalili. Each contributed to the study according to the tasks assigned to them. Guiding them is their capstone adviser, Mr. Jose Crisanto S. Austria, who approved and gave his insights on the research being conducted by the proponents through online communication, such as Messenger and Google Meet, and through face-to-face meetings.

After determining the main objectives and the software development life cycle to be used on the project, the proponents formulated a Gantt chart based on the Agile Scrum method, which will help determine the other phases of the project development as well as determine the subsequent steps and activities for each phase

Figure 2.3 shows the Gantt chart of the project, representing the perceived timeline of the project development as well as its key tasks along the different project phases.

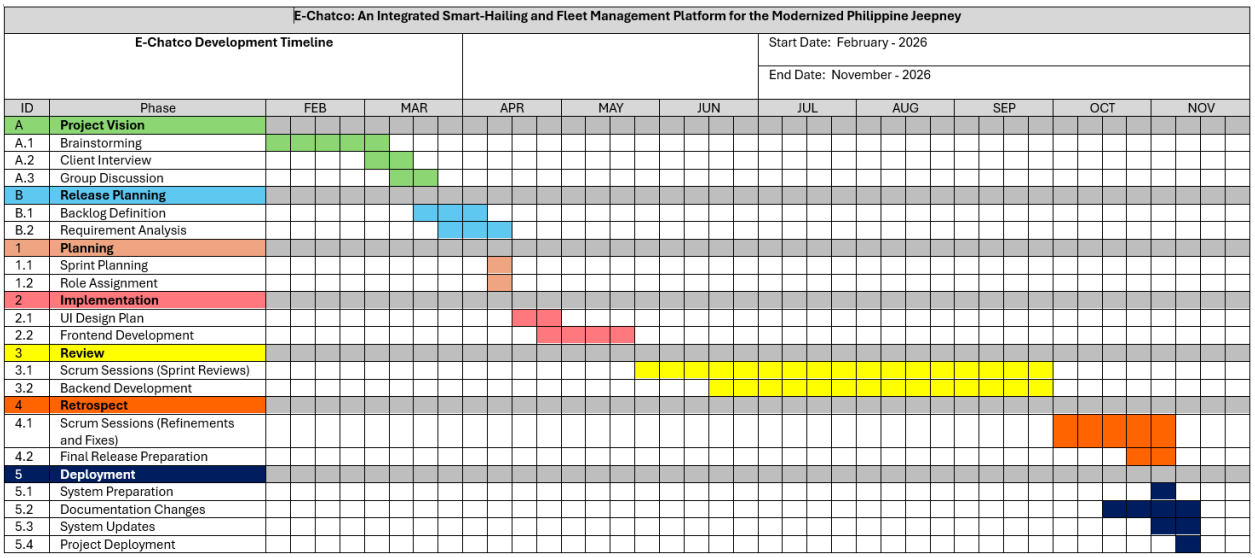


Figure 2.3: E-Chatco Project Gantt Chart

- Resources

Hardware Resources

HARDWARE	MINIMUM	RECOMMENDED
CPU	Intel/AMD 1.6 GHz (dual-core)	Intel/AMD 2.5 GHz (quad-core) or higher
RAM	4GB	8GB
Storage	20GB Free Space	50GB Free Space
Graphics	Integrated GPU (any)	Integrated GPU (any)
Monitor	HD 720p	HD 1080p
Internet	Stable 3 Mbps	10 Mbps or higher
Mobile Device	Xiaomi Redmi 9/10	Realme 10/11

Table 2.2: Hardware Requirements

Software Requirements

SOFTWARE	MINIMUM	RECOMMENDED
Operating System	Windows 10 SL 64-bit	Windows 10 SL 64-bit r
Web Browser	Chrome & Edge (Version 110)	Latest Stable (Version 123 or higher)
Code Editor	VS Code v1.75	VS Code (latest stable version)
Mapping Library	Leaflet.js 1.9 with React-Leaflet 5	Leaflet.js 1.9 with React-Leaflet 5
Backend Framework	Laravel 12 / PHP 8.2	PHP 8.2+, Laravel 12.x (Laravel Sanctum for API authentication)
Runtime Environment	Node v16.x	Node v20.x LTS or later
Frontend Framework	Next 16.2.3 / React 19	Next.js 16 (React 19, Typescript 5), Tailwind CSS 4

Table 2.3: Software Requirements

Requirements Analysis

To ensure that the proposed system effectively addresses the challenges faced by CHATCO Cooperative, a business requirements analysis was conducted by the proponents through interviews and the development of computing solutions. This analysis focuses on identifying the current practices and limitations in the key aspects of the business process, as well as proposing computing solutions to improve the client's current business process.

Requirements Gathering Methods

Interview

The interviewers conducted interviews to gather information regarding the business process of CHATCO Cooperative. According to Zaremba and Liaskos (2021), interviews are driven by several questions for the purpose of receiving responses that will help understand the domain and the needs of the users. Through this method, the proponents of the system were able to interview

the manager of CHATCO Cooperative, Mrs. Alicia Tiongco, who brings her extensive knowledge of the business processes to the system**.**

Interview Notes from Formal Interview (April 6, 2026) **Approved by: Alicia Tiongco**
Person Interviewed: Alicia Tiongco, General Manager of CHATCO Cooperative
Interviewer: Marinel T. Carbonell
Purpose of the Interview: To gather comprehensive information regarding the business process, operational challenges, boundary remittances, and safety procedures to refine system requirements.
Summary of the Interview:
● The cooperative operates 39 modern jeepney units running from Calumpit to Meycauayan (6 roundtrips daily). ● Drivers and conductors may use any available vehicle unit. ● The cooperative handles fuel purchasing directly; expenses are not deducted from the driver's boundary; therefore, expense management is outside the scope of the system. ● Fare calculation relies on a gadget, but a major flaw is that incorrect entries cannot be reversed or deleted. ● Cashless payments (tap cards) are currently unavailable due to implementation troubles with the hardware. ● Drivers are strictly prohibited from using phones during operations, while conductors are permitted to use them when necessary. ● Lost items are reported and claimed via their Facebook page with no digital matching system. ● Emergencies or breakdowns are reported via manual phone calls or SMS ● Drivers frequently miss picking up commuters due to overcrowding or poor visibility.
Needs expressed by the business: ● Boundary remittance visibility and tracking to ensure quota compliance. ● A "Pick Me Up" or hailing mechanism to address missed commuters due to overcrowding. ● A digital lost and found matching system to replace the manual Facebook page process.
Detailed Notes: See Appendix B for the Transcript

Table 2.4: Formal Interview Report

- **5W1H Framework Analysis**

With the needs and current processes of the business determined, Table 2.6 illustrates the computing solutions the system will provide to meet the needs of the client in terms of their current business process.

ASPECT	CURRENT PROCESS	COMPUTING SOLUTION
Who	Manual coordination between the administrator and conductors.	Role-based user accounts with task-specific dashboards.
What	Manual boundary declaration, hailing, and task assignments.	Centralized remittance tracking with separate cash/GCash records, smart-hailing, and task tools
Where	On-site jeepney units; paper-based logs.	Mobile-friendly, web-based mounted devices held by the conductors
When	Payment deadlines are tracked manually at the end of the day.	Automated real-time GCash transaction verification and cash declaration tracking
Why	Existing methods are familiar but inefficient as transit demand grows.	Modern system to improve efficiency, accuracy, and safety.
How	Paper logs, defective gadgets, hand signals, and manual phone calls.	Digital records, structured communication through QR-based GCash payment verification, GPS hailing, and point-area fare computation.

Table 2.6: 5W1H Framework

Requirements Documentation

User Requirements

The Integrated Smart-Hailing and Fleet Management Platform is primarily intended for internal use by the staff of CHATCO Cooperative. Commuters may browse publicly available information about the project. However, full access to tracking of E-jeeps' e-payment features—including GCash transaction history and ride records—requires commuter login.

Meanwhile, staff members such as the administrator and conductor are assigned system accounts with access permissions tailored to their respective roles. The roles and corresponding access privileges established from the client interview are summarized in Table 2.7.

ID	ROLE	CAPABILITIES
1	Administrator	Log in to the system; view the dashboard with daily rides, revenue, active vehicles, pending remittances, and a live fleet map; manage user accounts, including approving or rejecting commuter registrations, creating conductor accounts, registering commuters on-site, and suspending or disabling accounts; review fare discount applications and student renewals; manage vehicles and drivers, and assign a driver and conductor to each unit; monitor units on shift in real time and receive overspeeding; review conductor remittances with separate cash and GCash totals; view fare receipts; manage the fare matrix and the discount rates for students, senior citizens, and PWDs; record found items and review claims in lost and found; post announcements; generate vouchers and set the loyalty ride threshold; view ride and revenue analytics; view and filter activity logs; and configure the operations rules, safety notifications, receipt settings, maintenance mode, and FAQs.
2	Conductor	Log in through the web portal or the Conductor Android App; select the assigned unit and driver to start a shift; share the unit's live location while on shift; set the unit's capacity status to Available, Standing, or Full, and start or end a break; receive and accept or reject Pick Me Up requests from commuters within 1 kilometer; compute fares from boarding and drop-off points using the fare matrix; record cash payments, generate a GCash payment QR code for the commuter to scan, and apply vouchers; record a single payment for a group of passengers; print receipts through a paired Bluetooth thermal printer (58mm or 80mm); keep recording cash fares offline and sync them once the connection returns; report accidents or breakdowns; view ratings and feedback; move an active shift to another device; and

		submit the end-of-day remittance with separate cash and GCash totals.
3	Commuter	View the system website; register with a valid ID through the web or the Commuter Android App, and log in once approved by the administrator; track active jeepneys on a live map; receive a sound and vibration alert when a CHATCO unit is within 1 kilometer; use the Pick Me Up feature (available only within 1 kilometer of a unit, with a repeating sound and vibration while active); calculate fares between pickup and drop-off points; pay through GCash by scanning the conductor's payment QR code; view the GCash payment history; earn and redeem reward vouchers, including claiming paper cash receipts toward rewards; apply for a student, senior citizen, or PWD discount and renew the student discount each school year; share their live ride location with family or friends; search, claim, and watchlist found items in lost and found; rate the driver and conductor; and read announcements.

Table 2.7: User Requirements

User Stories

The system proponents also established user stories based on the interview with the manager. Each module has specific user stories with several roles. Table 2.8 lists the user stories that will help shape the system.

MODULE	**USER STORIES**
User Authentication	● As an administrator, conductor, or commuter, I want to log in to the system so that I can do my tasks as intended. ● As a commuter, I want to create an account with a valid ID so that the administrator can verify my identity before I use the system. ● As a commuter, I want to reset my account password so that I can regain access if I forget my credentials. ● As an administrator, I want to review commuter registrations so that only verified commuters can use the system. ● As a conductor, I want to continue my active shift on another device so that

	a lost or dead phone does not stop my shift.
User Dashboard	● As an administrator, I want to view the dashboard with a concise layout so that I can see the day's rides, revenue, active vehicles, and pending remittances at a glance. ● As a conductor, I want to view my assigned unit, driver, and the cash and GCash collections of my shift so that I am aware of my progress. ● As a commuter, I want to view nearby units, my GCash payment history, and my reward progress from the home page so that I can reach them quickly.
Fleet Management	● As an administrator, I want to view, add, and update vehicles, drivers, and conductors so that the fleet records match the units in operation. ● As an administrator, I want to assign a driver and a conductor to a unit so that each shift runs with the right crew. ● As a conductor, I want to select the unit and driver I will work with before starting my shift so that my fares and location are recorded under the right vehicle. ● As a conductor, I want to set my unit's capacity status and start or end a break so that commuters and administrators know if the unit can take passengers.
Hailing & Tracking	● As a commuter, I want to view all active CHATCO units on the map so that I can see the fleet status and plan my commute. ● As a commuter, I want to receive a sound and vibration alert when a CHATCO unit enters my 1-kilometer radius so that I know a unit is approaching. ● As a commuter, I want to send a "Pick Me Up" signal when I am within 1 kilometer of a CHATCO unit so that the conductor knows I am waiting. ● As a commuter, I want to receive a repeating sound and vibration while my hail signal is active so that I know my request is still being broadcast. ● As a conductor, I want to receive Pick Me Up requests from commuters within range and accept or reject them so that I can pick up waiting commuters. ● As a conductor, I want my unit's location to be shared while I am on shift so that commuters and administrators can track the jeepney. ● As an administrator, I want to monitor units on shift on a live map so that I can track fleet operations in real time. ● As an administrator, I want to be alerted when a unit is overspeeding so that I can act on unsafe driving.
Safety	● As a conductor, I want to report an accident or a vehicle breakdown so that the operator can respond to problems on the road. ● As a commuter, I want to share my live location through a link so that my

	family or friends can follow my ride.
GCash Payment Integration	● As a commuter, I want to pay my fare through GCash by scanning the conductor's payment QR code so that I can complete cashless transactions. ● As a commuter, I want the freedom to pay with cash or GCash so that I can choose the most convenient payment method. ● As a conductor, I want to generate a GCash payment QR code for the commuter to scan so that the payment is linked to the fare I recorded. ● As a conductor, I want to record both cash and GCash fares, including one payment for a group of passengers, so that every payment is recorded digitally. ● As a conductor, I want to keep recording cash fares when the connection drops so that no payment is lost. ● As a conductor, I want to print a receipt through a paired Bluetooth thermal printer so that commuters receive proof of payment. ● As a commuter, I want to view my GCash payment history so that I can keep track of my fares. ● As an administrator, I want to view the receipts of fare transactions with their payment method and status so that I have a record of every payment in the system.
Point-Area Fare Calculator	● As an administrator, I want to manage the fare matrix so that fares follow the approved rates. ● As a conductor, I want to select the boarding and drop-off points from the CHATCO fare matrix so that the fare computation is accurate and aligned with cooperative operations. ● As a commuter, I want to select my pickup and drop-off points from the approved CHATCO stops so that I know my fare before I ride.
Discount Management	● As an administrator, I want to set the discount rates for students, senior citizens, and PWDs so that discounted fares are computed correctly. ● As a commuter, I want to apply for a student, senior citizen, or PWD discount with a supporting document so that I can pay the discounted fare. ● As a student, I want to renew my discount every school year so that I can keep paying the student fare. ● As an administrator, I want to review discount applications and student renewals so that only eligible commuters receive discounted fares.

Remittance	● As a conductor, I want to view the cash and GCash totals of my shift and submit my remittance so that I can end my shift with both payment categories recorded. ● As an administrator, I want to view the cash and GCash remittances of conductors separately so that I can track and validate them.
Lost and Found	● As an administrator, I want to record a found item with its description, unit, and photo so that its owner can find and claim it. ● As a commuter, I want to search the found items and add them to a watchlist so that I can check if my lost item has been found. ● As a commuter, I want to submit a claim for an item and track its status so that I can get it back. ● As an administrator, I want to review claims and mark items as released so that items go only to their rightful owners.
Rewards	● As an administrator, I want to generate vouchers and set the number of rides needed to earn a reward so that the loyalty program can be adjusted. ● As a commuter, I want to earn and use reward vouchers for riding regularly so that I can get discounts or free rides. ● As a commuter, I want to claim my paper cash receipts toward rewards so that I am not left out when paying in cash.
Feedback	● As a commuter, I want to rate the driver and the conductor by scanning the unit's QR code so that the operator knows the quality of their service. ● As a conductor, I want to view my ratings and feedback so that I know how commuters rate my service.
Announcements	● As an administrator, I want to post announcements so that commuters are informed of service updates. ● As a commuter, I want to read announcements and see which ones I have not read yet so that I stay informed.
System Activities	● As an administrator, I want to view and filter the activity logs so that I know the history of changes within the system. ● As an administrator, I want to view ride and revenue analytics so that I can see trends and make operational decisions. ● As an administrator, I want to configure the operations and safety rules, maintenance mode, and FAQs so that the system follows the operator's policies.

Table 2.8: User Stories

Interface Requirements

Any software system's success depends on its interface design. To ensure that users can easily access and utilize the system's features relevant to their jobs, each interface, as shown in Table 2.9, was carefully built to fit various user roles.

ID	INTERFACE	SPECIFICATION
Administrator		
1.1	Dashboard Page	To view the day's total rides and revenue, active vehicles, pending remittances, a live map of units on shift, and recent users, vehicles, announcements, and lost and found items.
1.2	User Management Page	To view commuter, conductor, and administrator accounts; review commuter registrations, discount applications, and student renewals; create conductor accounts; register commuters on-site; and suspend or disable accounts.
1.3	Fleet Management Page	To view, add, and update vehicles, drivers, and conductors; assign a driver and a conductor to each unit; and view each unit's details, QR code, and shift history.
1.4	Monitoring Page	To view the live location, status, and capacity of each unit on shift, flag units that stopped sending updates, and respond to overspeeding.
1.5	Remittances Page	To view conductor remittances with separate cash and GCash totals, monitor pending remittances, and view the fare transactions behind each remittance.
1.6	Receipts Page	To view all fare transactions with their payment method and status, filtered by

		date.
1.7	Analytics Page	To view ride and revenue totals for a selected period compared with the previous period, peak hours, a demand map of pickup points, and voucher usage.
1.8	Lost & Found Page	To record found items with a description, unit, and photo, review commuter claims, and mark items as released.
1.9	Announcements Page	To create, update, and delete announcements for commuters.
1.10	Activity Logs Page	To view and filter the history of actions performed by each user in the system.
1.11	Settings Page	To manage the fare matrix, discount rates, operations and safety rules, receipt settings, vouchers and the loyalty ride threshold, maintenance mode, FAQs, and the administrator's own profile and password.
Conductor		
2.1	Unit Verification Page	To select the assigned unit and driver and start the shift.
2.2	Dashboard Page	To view the assigned unit and route on a live map, set the unit's capacity status, start or end a break, receive and accept or reject Pick Me Up requests.
2.3	Payment Page	To compute the fare from the boarding and drop-off points using the CHATCO fare matrix, record cash payments, generate a GCash payment QR code, apply vouchers, record one payment for a group of passengers, and print receipts.
2.4	End-of-Day Page	To view the shift's cash and GCash totals, submit the remittance, end the shift, and view the remittance history.

2.5	Performance Metrics Page	To view the conductor's average rating, rating distribution, and recent feedback.
2.6	Settings Page	To manage the conductor's account, the thermal printer and auto-print, and voice announcements of upcoming stops.
Commuter		
3.1	Landing Page	To learn about CHATCO's features, read answers to common questions through the FAQ chat, and sign up or log in.
3.2	Dashboard Page	To view active jeepneys on a live map, receive a sound and vibration alert when a CHATCO unit is within 1 kilometer, send Pick Me Up signals to units within 1 kilometer (with a repeating sound and vibration while active).
3.3	Fare Calculator Page	To select a pickup point and a destination from the approved CHATCO stops and view the fare.
3.4	GCash PaymentPage	To scan the conductor's payment QR code, apply a voucher, pay through GCash, and view the payment confirmation.
3.5	Payment History Page	To view past fare payments with their date, amount, payment method, and status.
3.6	Share My Ride Page	To generate a live-location link that family or friends can open without an account.
3.7	Rewards Page	To view progress toward a free ride, view available, used, and expired vouchers, and claim paper cash receipts toward rewards.
3.8	Lost and Found Page	To search found items, add them to a watchlist, submit claims, and track claim status.
3.9	Discount Application Page	To apply for a student, senior citizen, or PWD discount, renew the student discount

		each school year, and view the application status.
3.10	Feedback Page	To rate the driver and the conductor by scanning the unit's QR code.
3.11	Announcement Page	To read announcements and see which ones are unread.
3.12	Profile Page	To view and update personal details and change the password.

Table 2.9: Interface Requirements

Design of Software, System, Product, and/or Processes

Effective planning and communication during the design phase of software, systems, products, and processes depend on the creation of well-structured models. The tools and diagrams for visualizing requirements, processes, and system structures include the Data Flow Diagram Level 0, Entity-Relationship Diagram (ERD), and flowcharts.

Data Flow Diagram Level 0 (CFD)

Figure 2.6 shows the DFD Level 0 of E-CHATCO. Each user has an input and output to and from the system. This serves as a visual representation of how data generally flows between the system and the user.

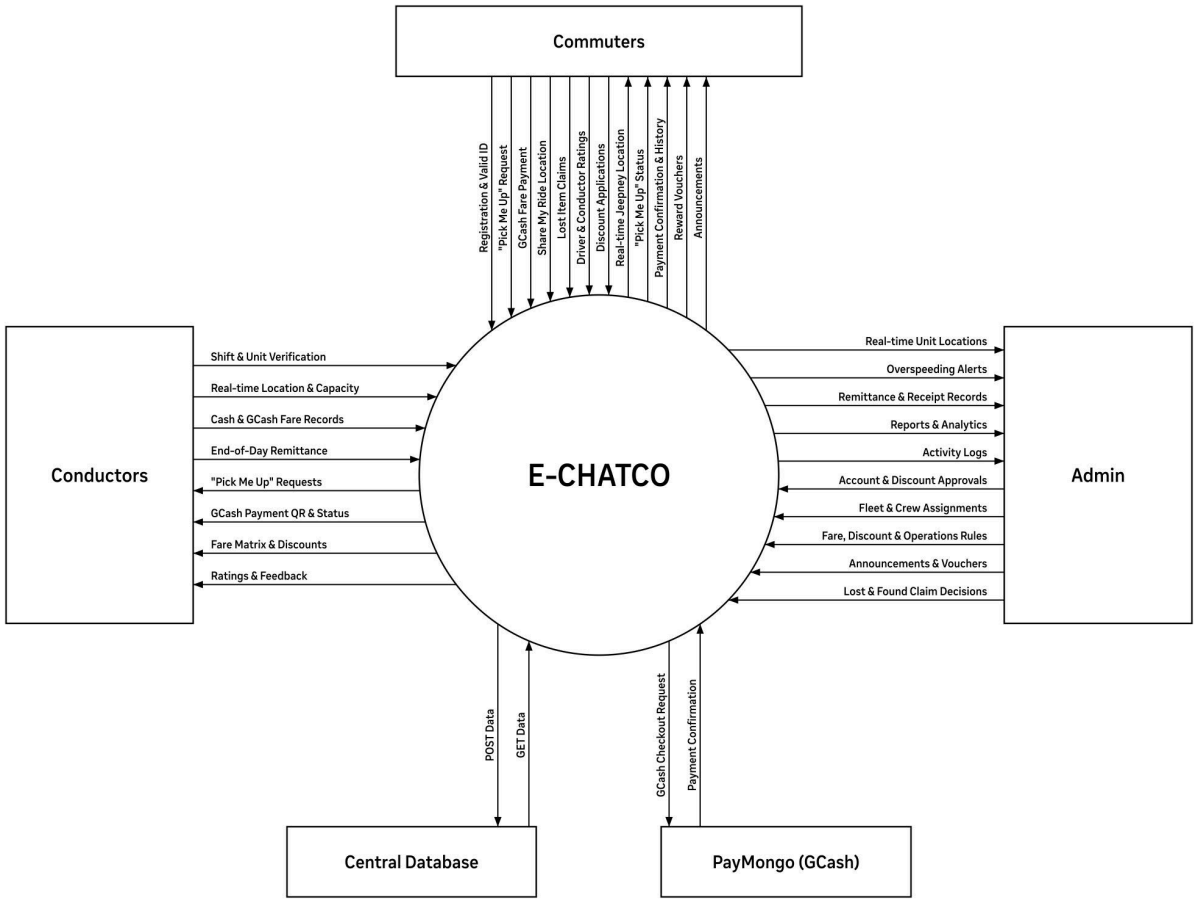


Figure 2.6: Data Flow Diagram Level 0 (Context Flow Diagram)

Entity Relationship Diagram (ERD)

Figure 2.7 illustrates the ERD of the system. Utilizing this helps in understanding the data flow, ensuring efficient database design and sensible relationships between entities.

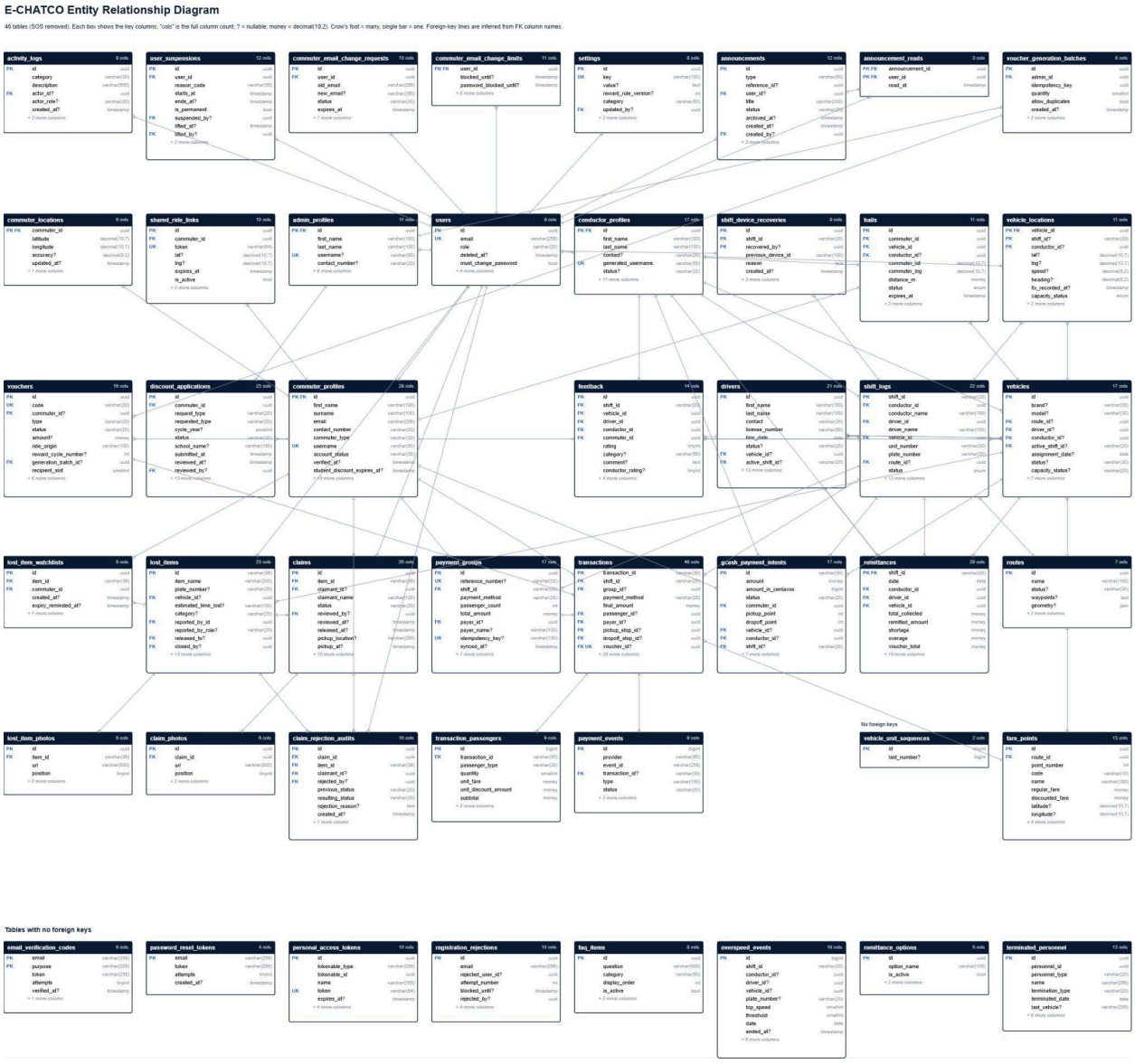


Figure 2.7: Entity Relationship Diagram

Flowchart

Figure 2.8 outlines the process for navigating a login/sign-up page, beginning with the display of the login form and including checks for credential validation. This is essential for checking user roles so the system can show their role-based tasks inside the system.

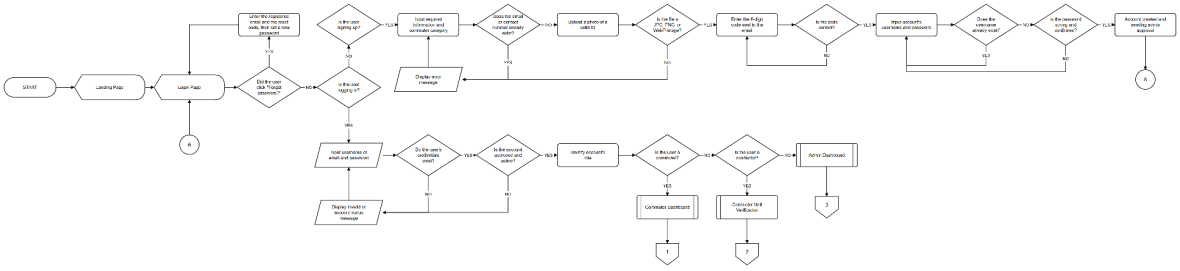


Figure 2.8: Login/Signup Page

Figure 2.9-2.14 displays the navigation capabilities and options available to a conductor within the application, guiding them through various functionalities based on user interactions.

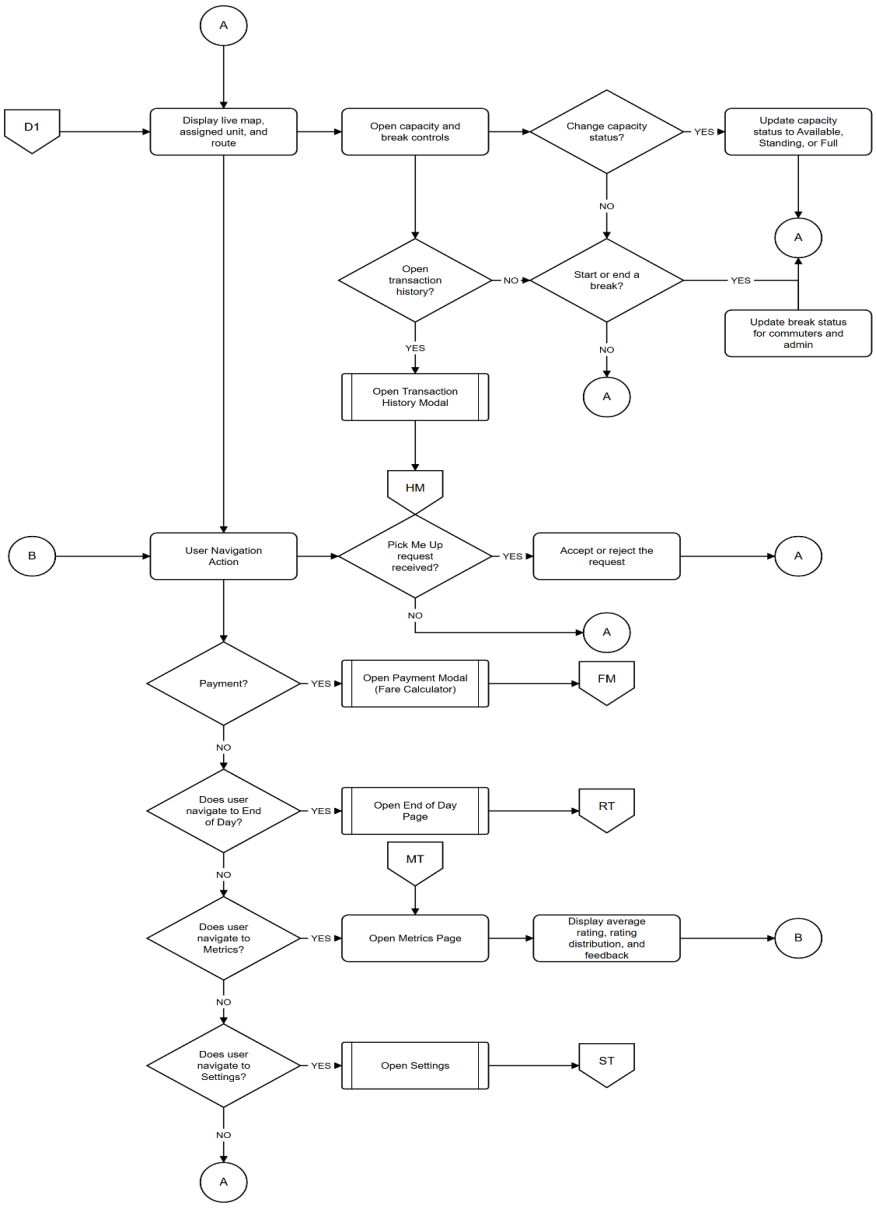
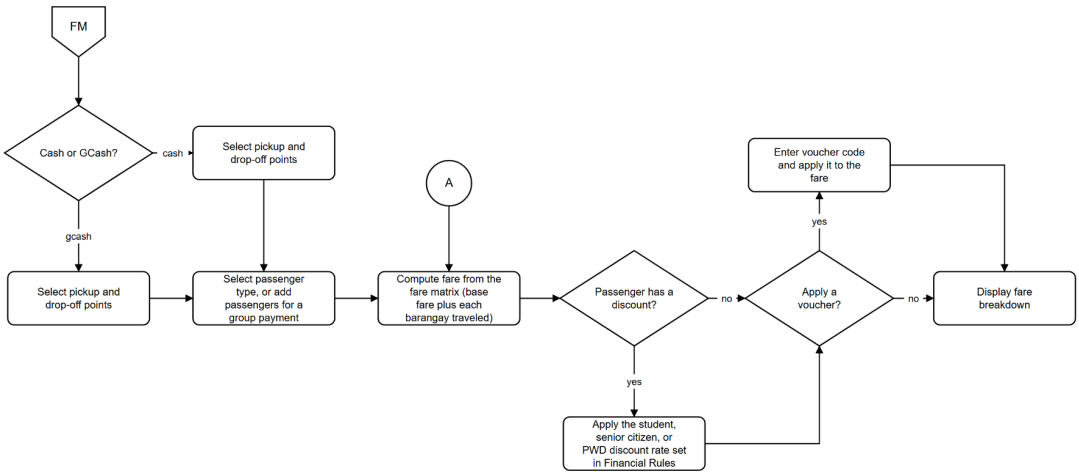


Figure 2.9: Dashboard

Figure 2.10: Fare Calculator Flow 1



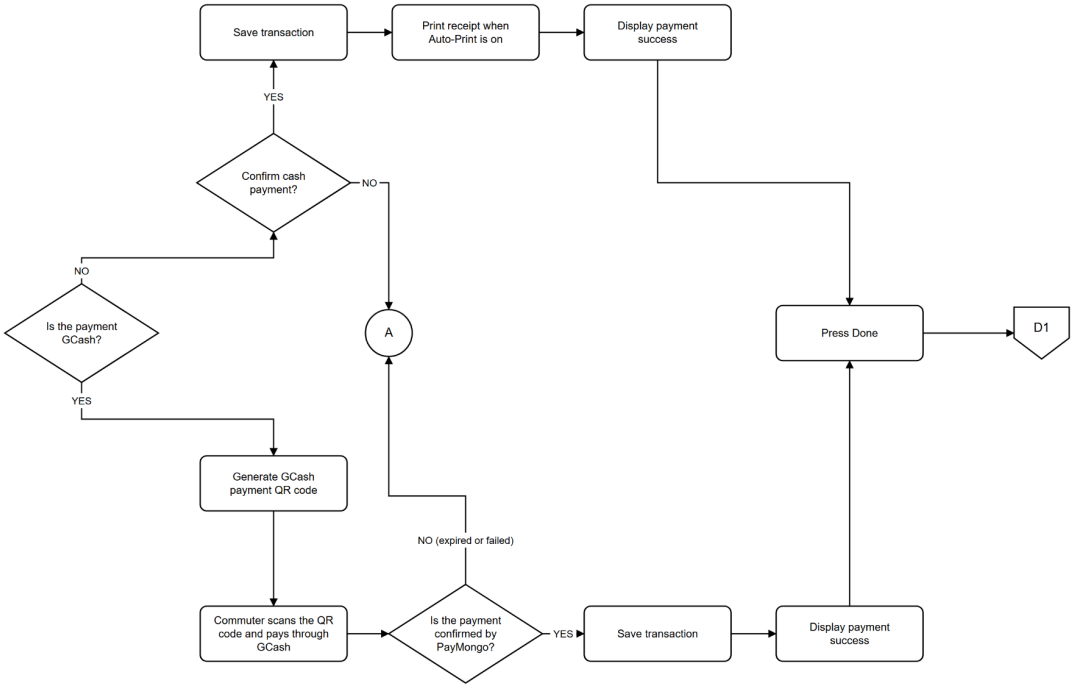


Figure 2.11: Fare Calculator Flow 2

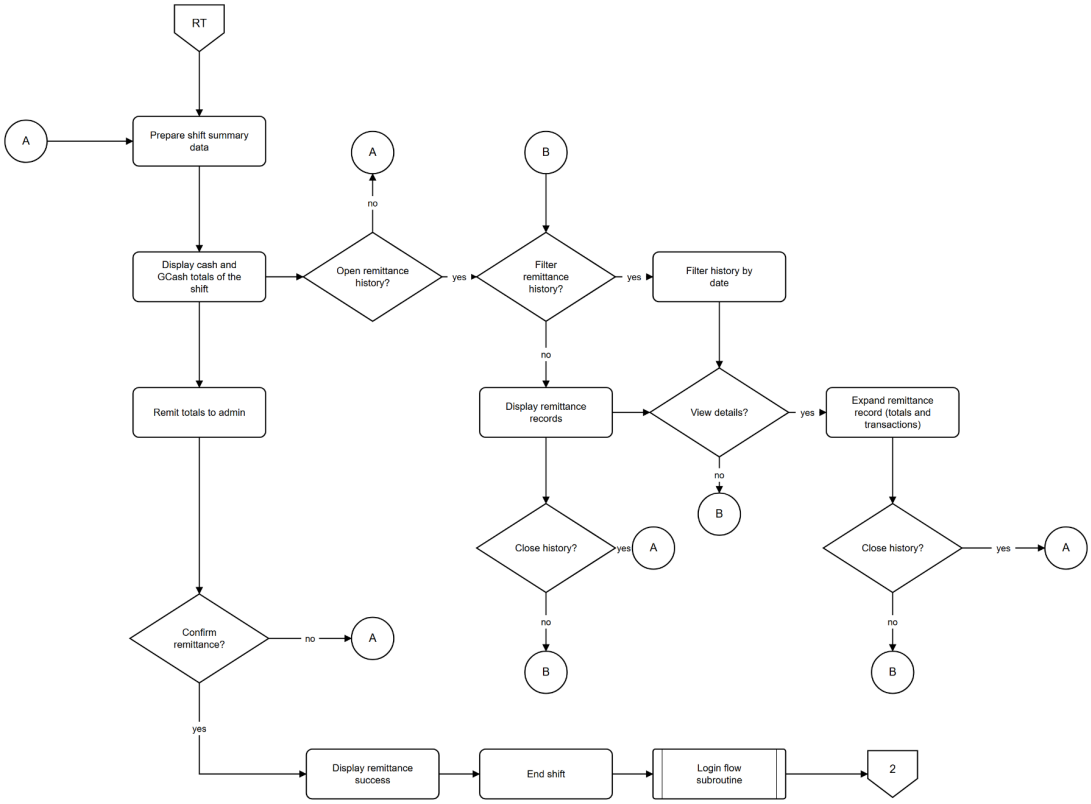


Figure 2.12: Reports Module (End of Day)

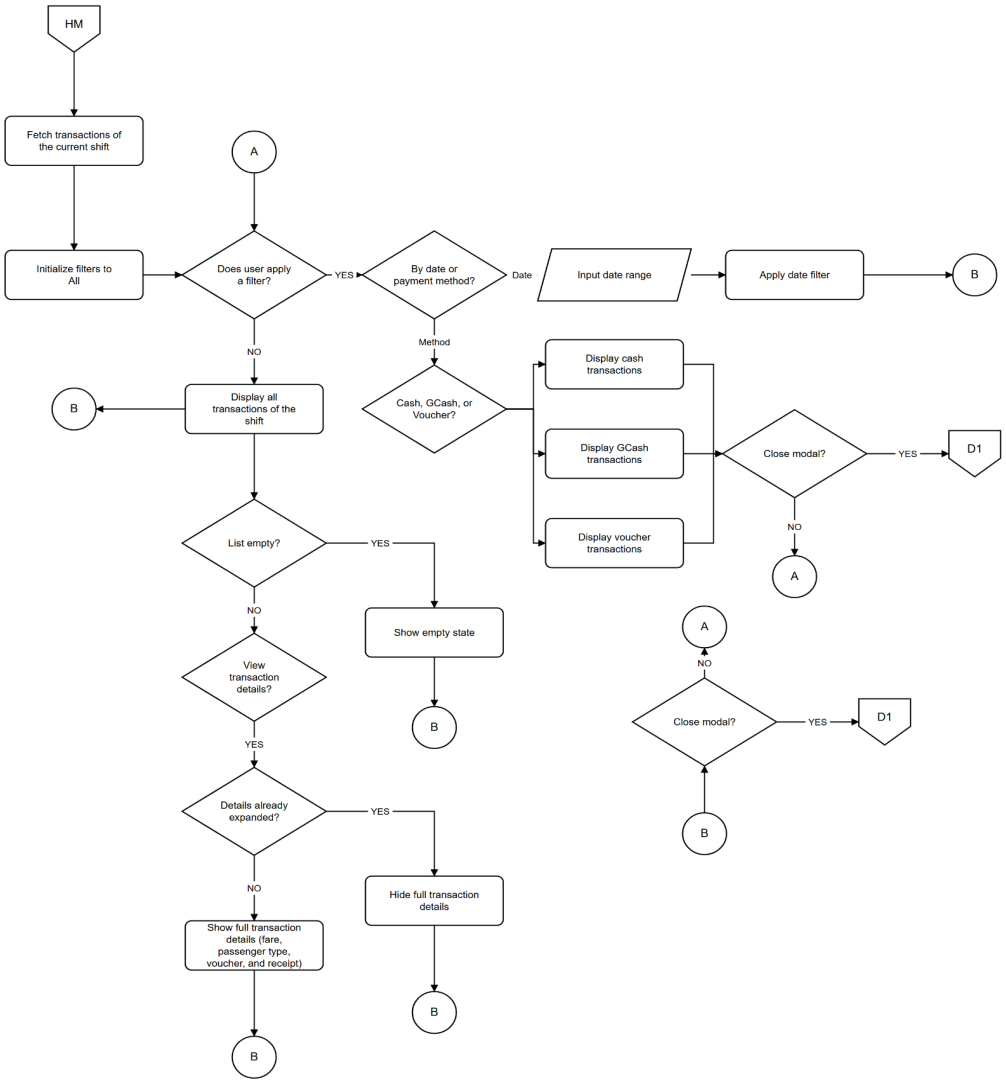


Figure 2.13: Transaction History

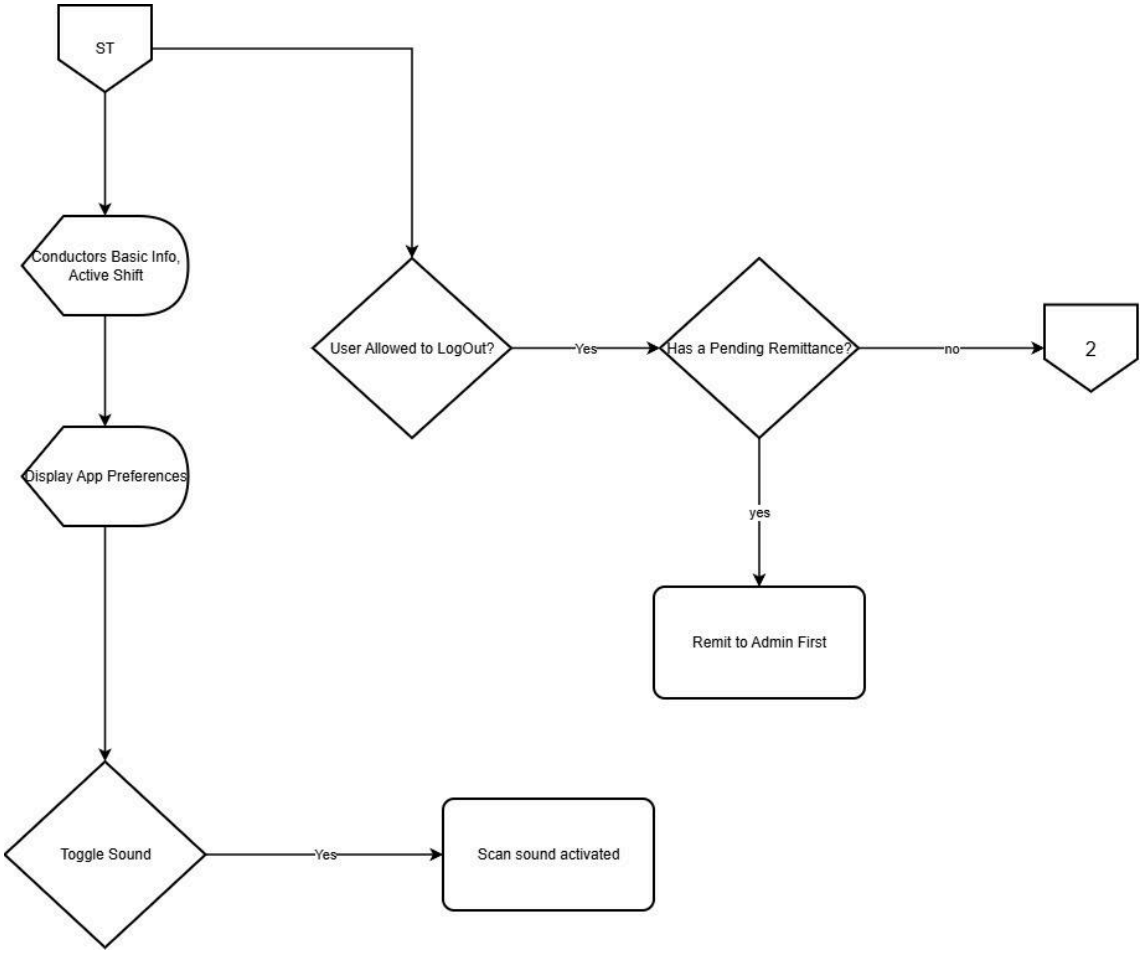


Figure 2.14: Settings (Conductor Side)

Figure 2.15 shows the process flow for the Commuter. It illustrates the decision points from accessing the landing page, tracking jeepney locations, sending a "Pick Me Up" signal, and generating a QR code for payment to utilizing safety features like "Share My Ride."

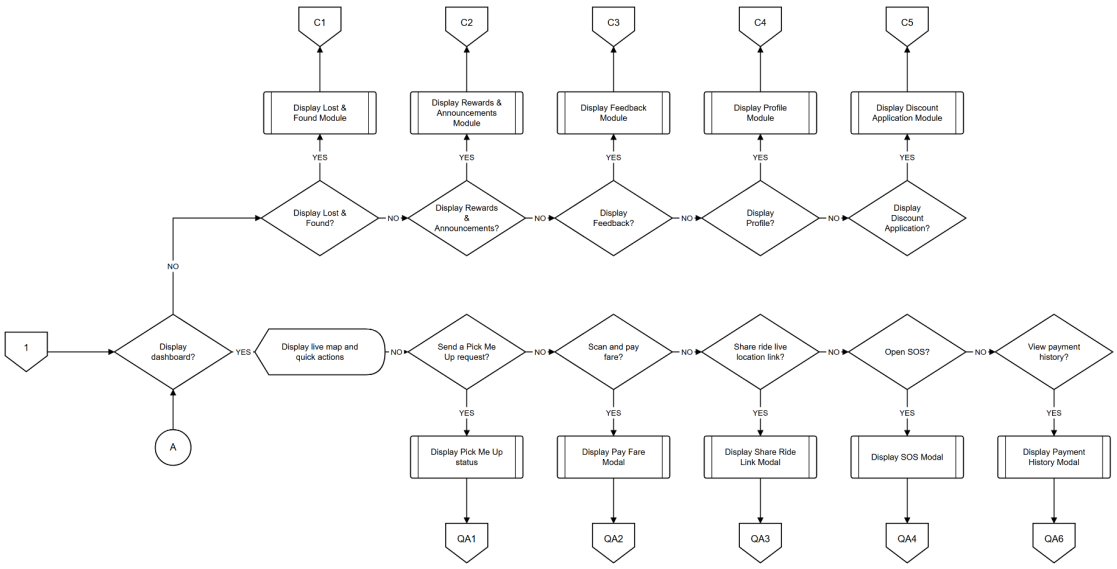


Figure 2.15: Commuter Side 1 - Display Flow Module

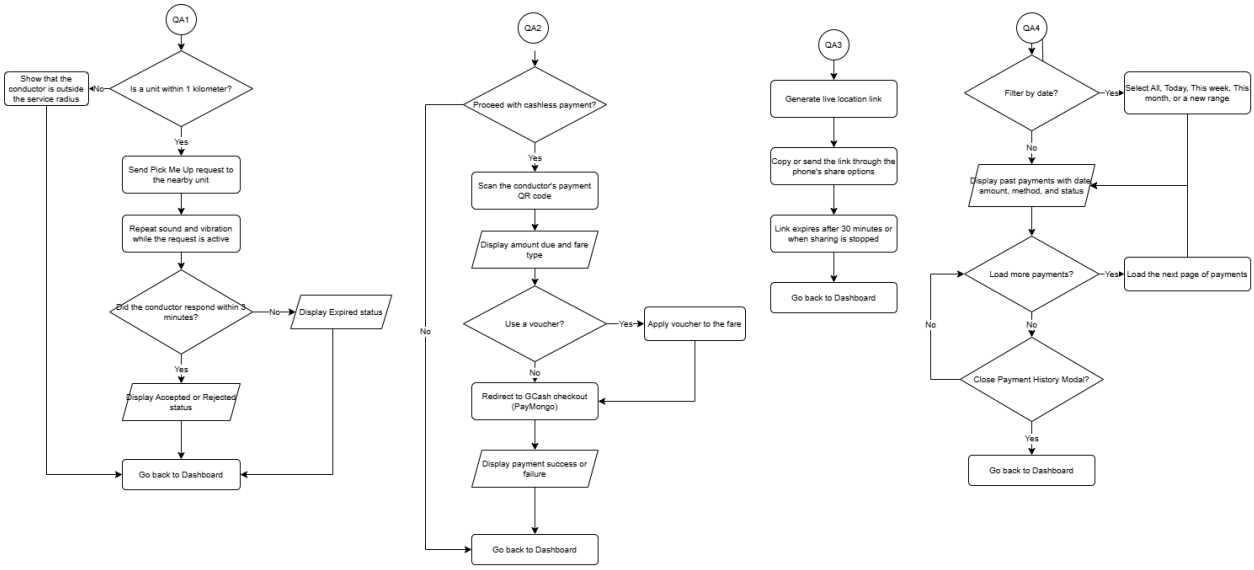


Figure 2.16: Commuter Side 2 - QR, Pay Fare, Share Ride, Top-up, Transaction

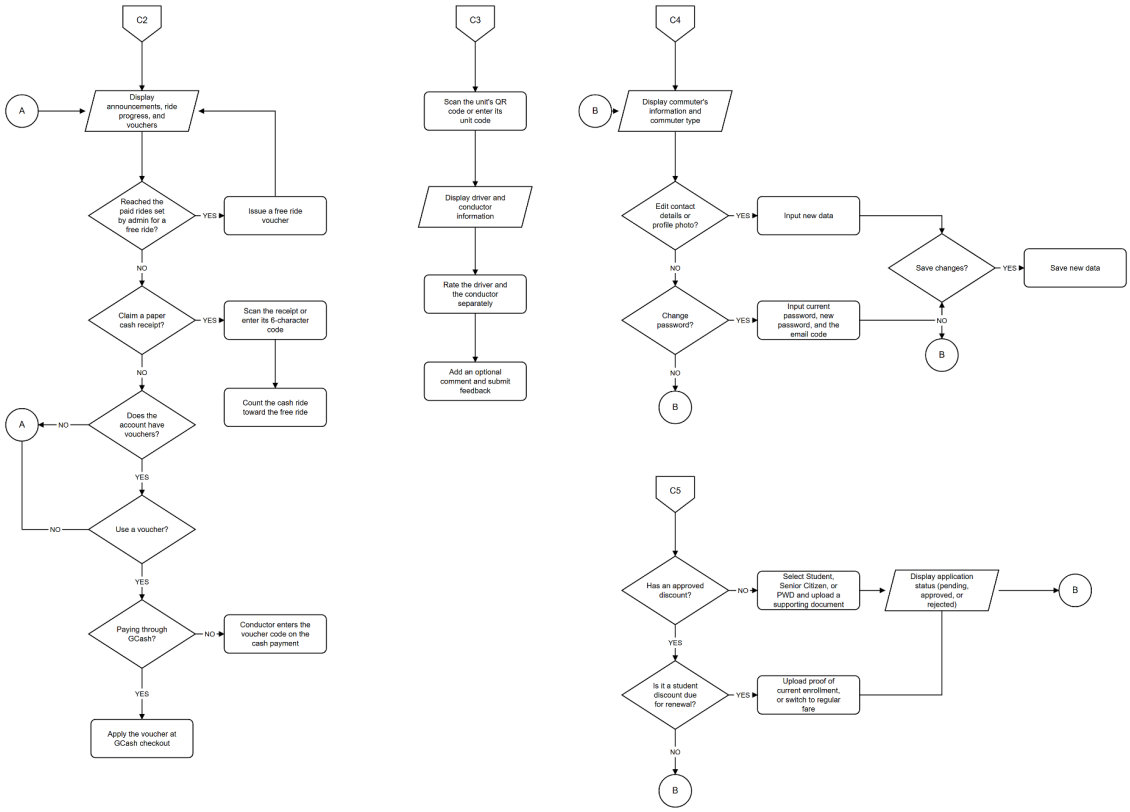


Figure 2.17: Rewards, Feedback,Profile, and Discount Application Module (Commuter)

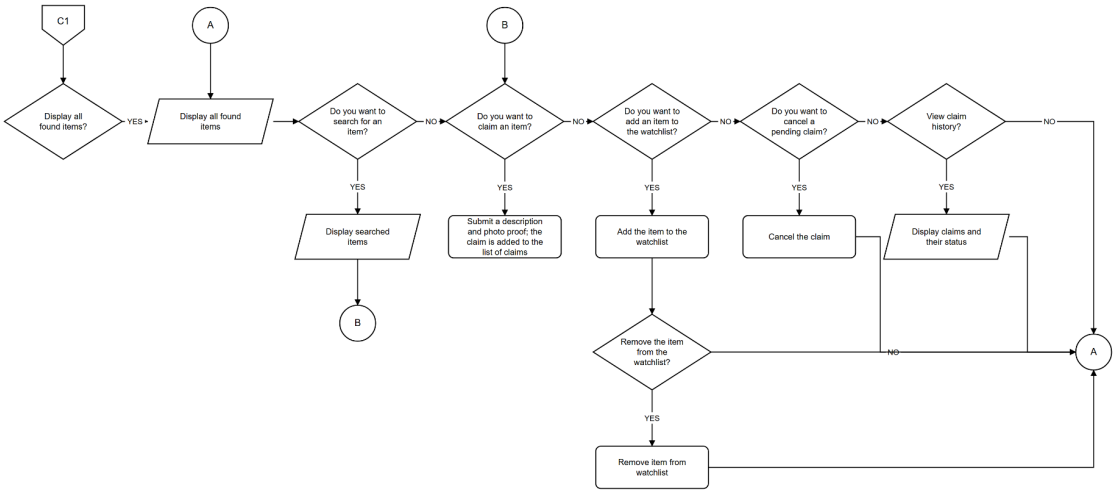


Figure 2.18: Lost and Found Module (Commuter)

Figure 3 provides a structured navigation path for the system administrator. This diagram demonstrates the flow for accessing high-level operational data, monitoring real-time fleet status, reviewing financial dashboards for boundary compliance, and managing the lost and found matching system.

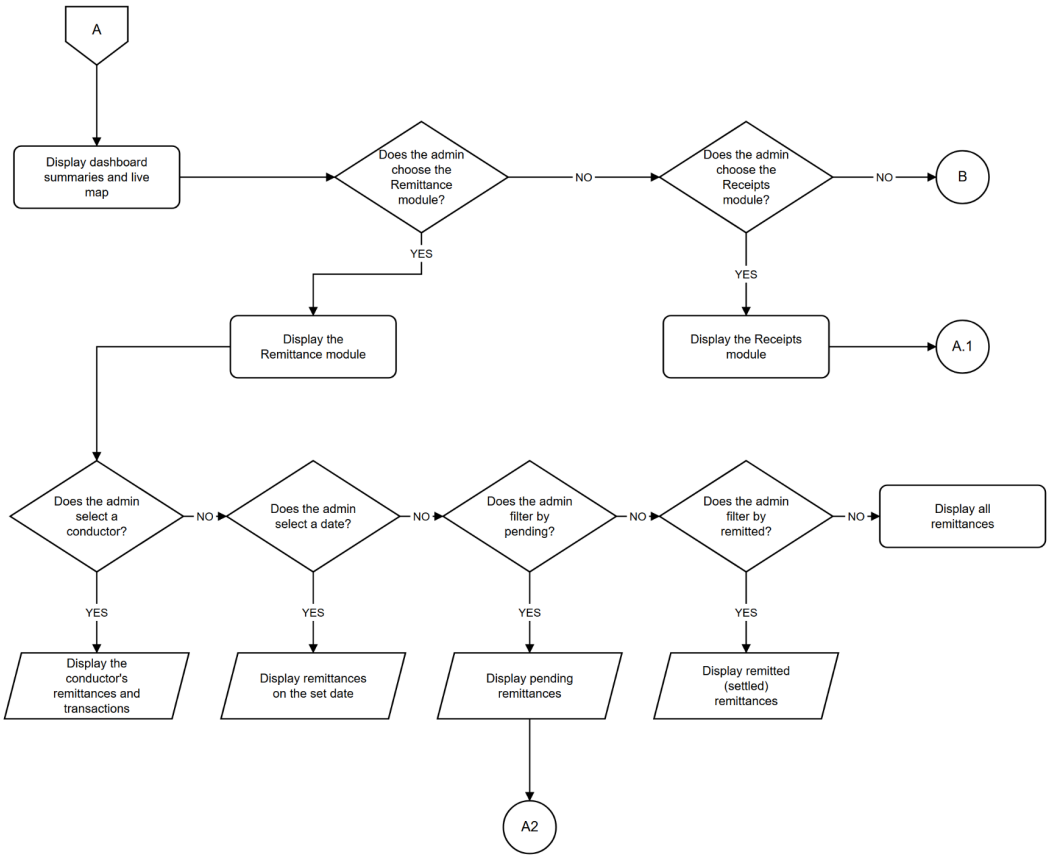


Figure 3: Administrator Flow 1

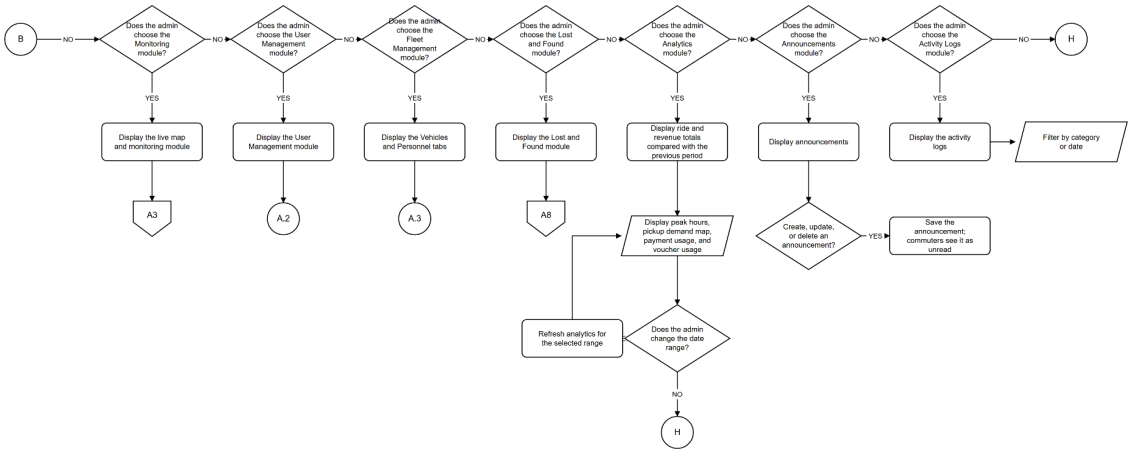


Figure 3.1: Administrator Flow 2

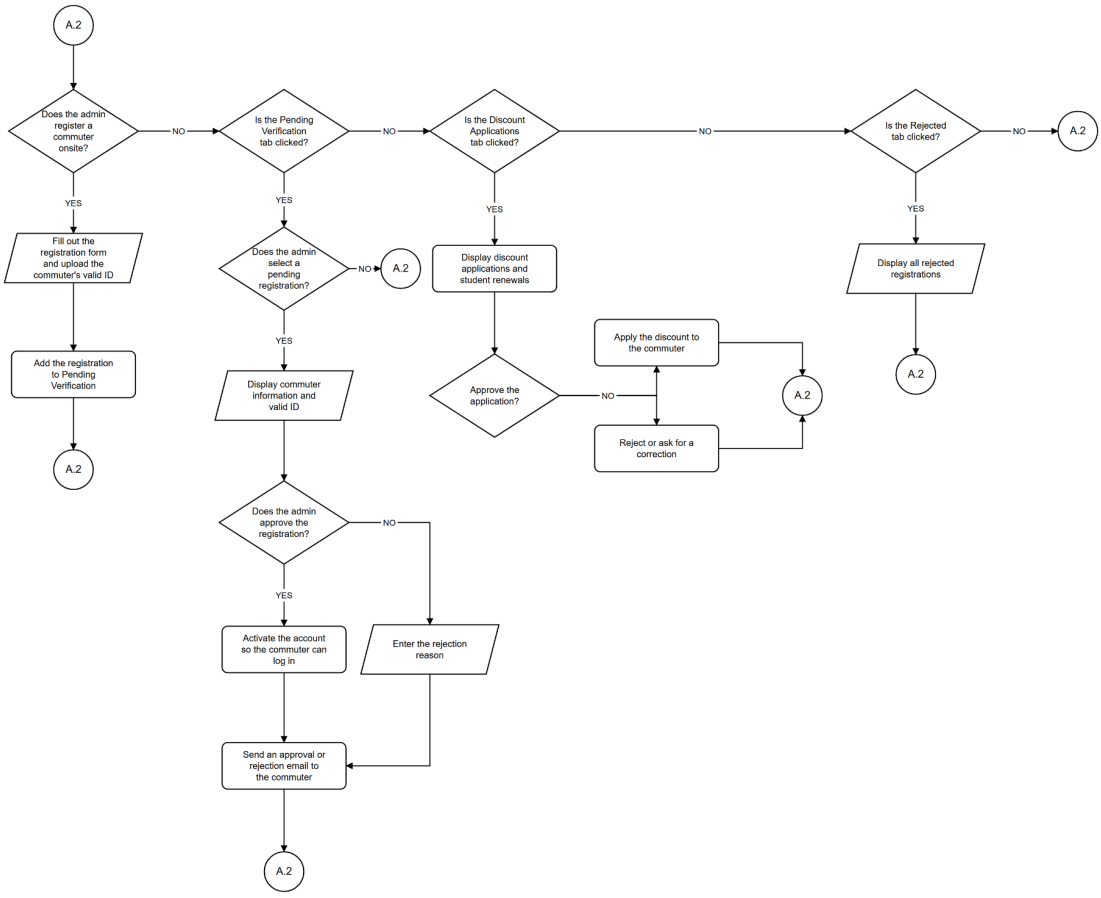


Figure 3.2: Admin Side - User Management

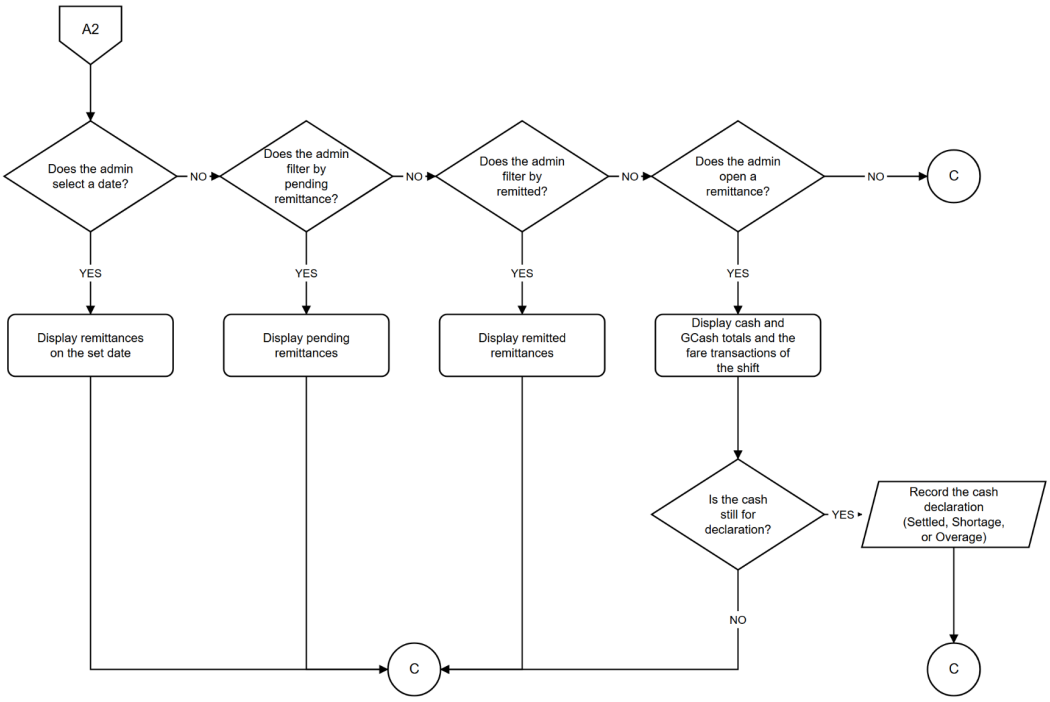


Figure 3.3 - Remittance Flow Module

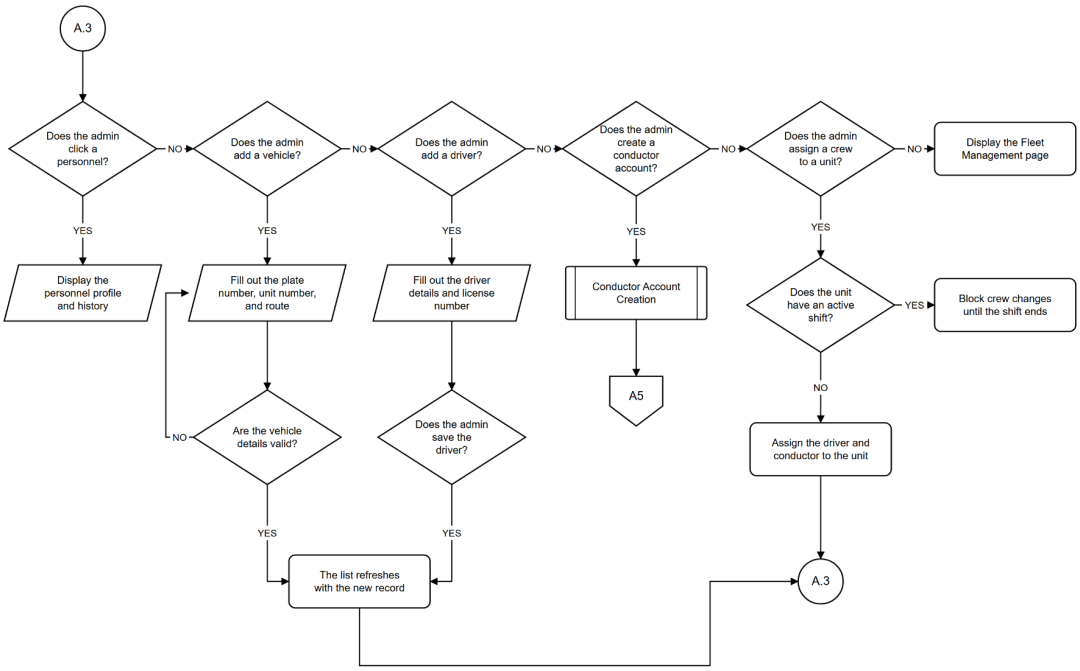


Figure 3.4: Fleet Management Flow

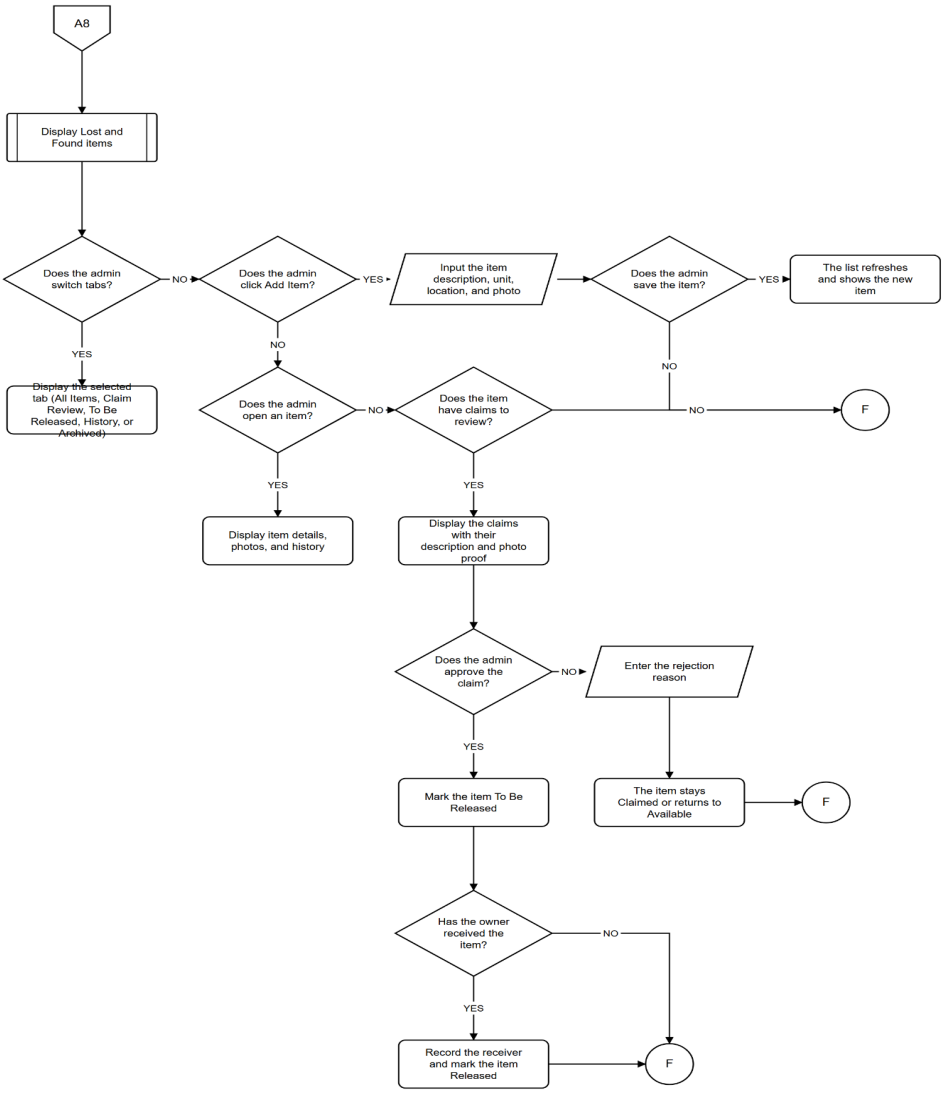


Figure 3.5: Lost and Found Module

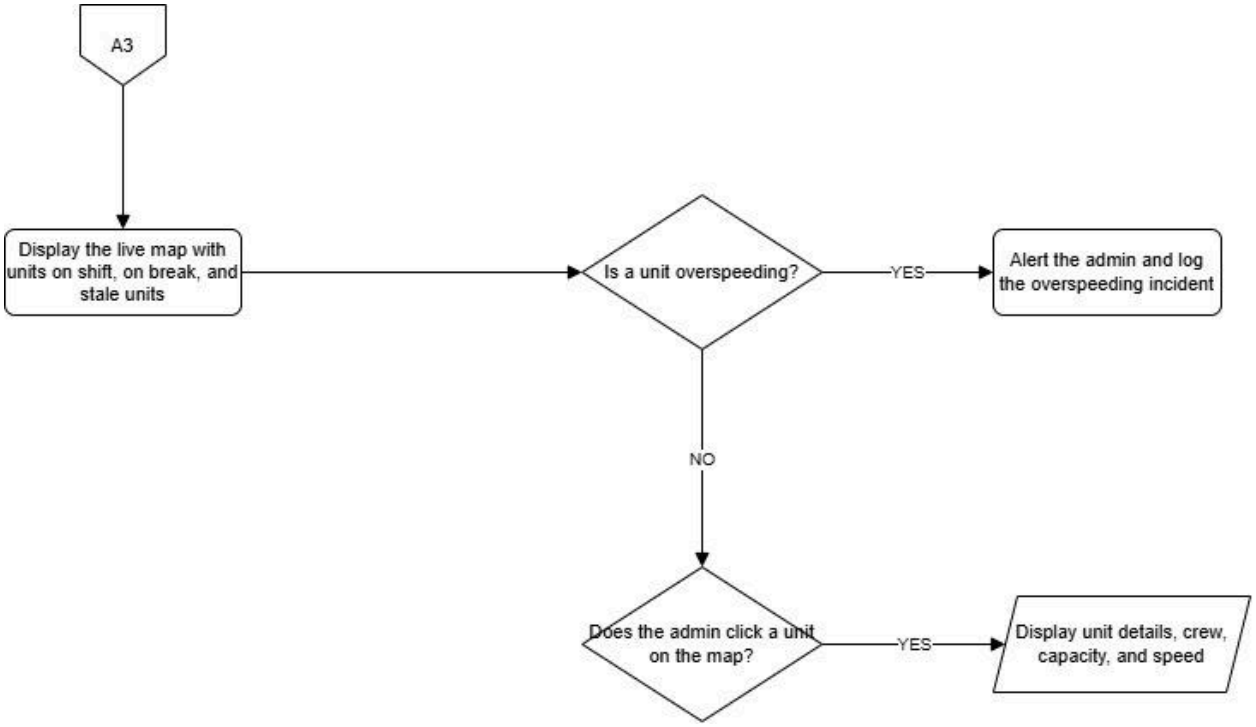


Figure 3.6: Monitoring Module

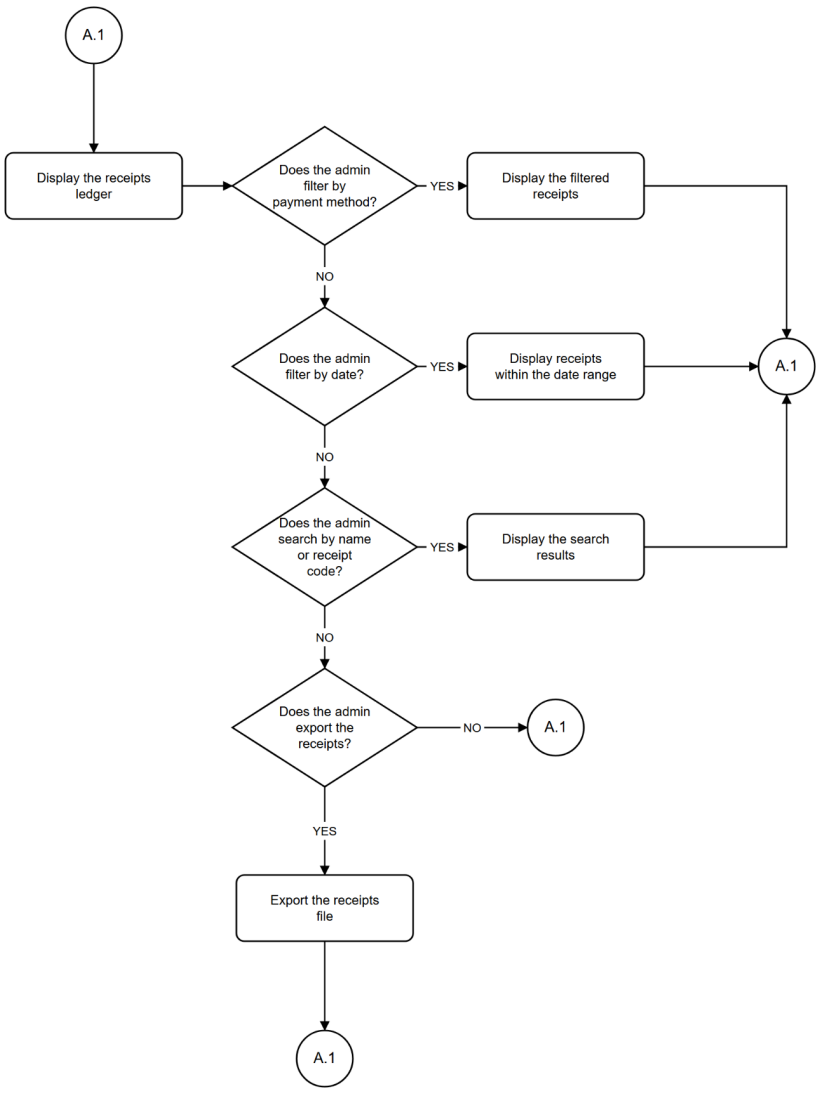


Figure 3.7: Receipts Module Flow

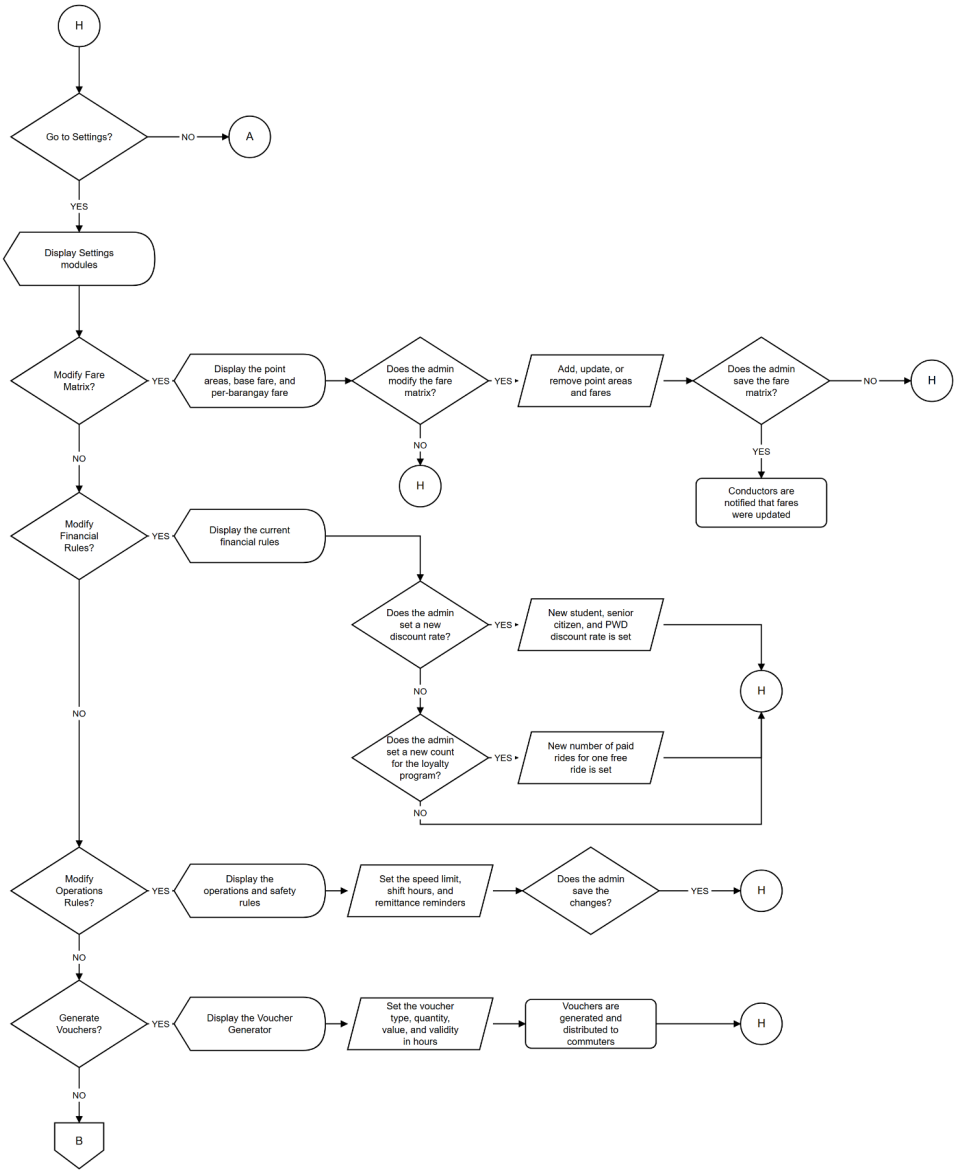


Figure 3.8: Settings Module Flow 1

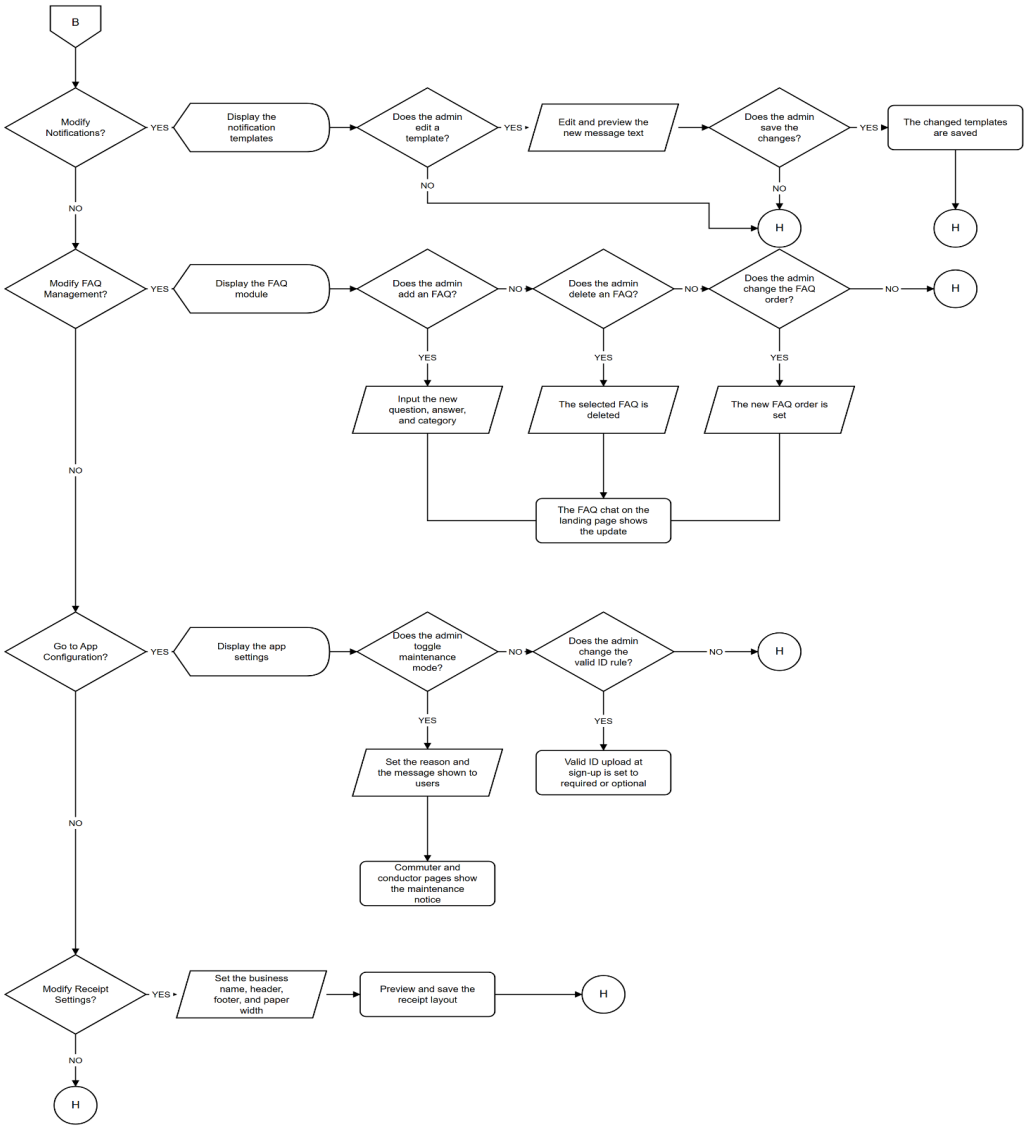


Figure 3.9: Settings Module Flow 2

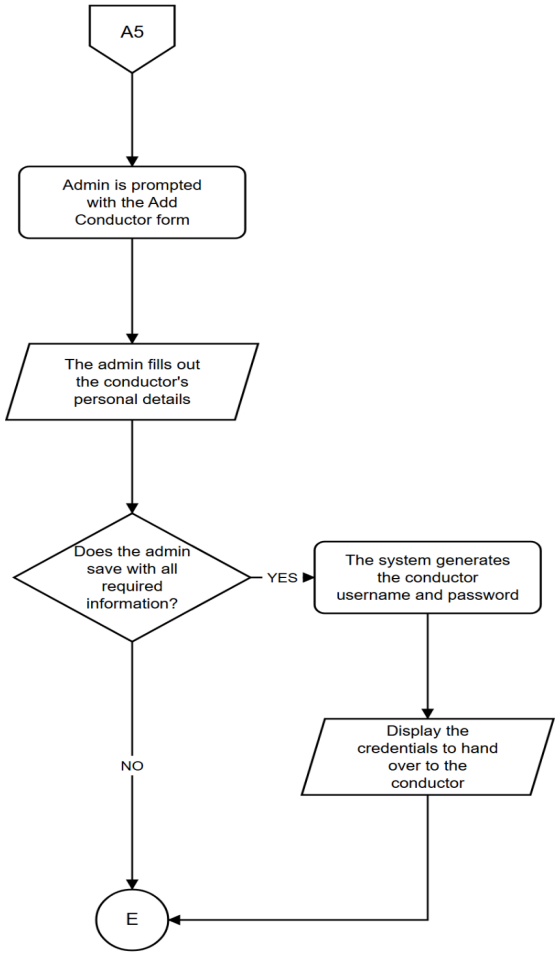


Figure 3.10: Conductor Account Creation

Development

The system was developed using Agile methodology, specifically the Scrum framework. Development was carried out through structured sprint cycles using project management tools to organize and monitor tasks. To ensure consistency and ongoing progress, regular Scrum ceremonies were carried out, including sprint planning and daily stand-ups. This approach was expected to make it easier to collaborate efficiently, be adaptable, and successfully integrate system features.

Phases of Agile Scrum Development Cycle (ASDLC)

Preliminaries A: Project Vision

Setting the vision for the development of the system, this project aims to assist the CHATCO Cooperative in modernizing their jeepney operations and enhancing their transport management services. Aligned with the goal of improving efficiency, communication, and service delivery, this project focuses on addressing boundary remittance disputes, commuter visibility issues, and coordination gaps in jeepney operations.

Table 3.1 represents the Project Vision Board that illustrates the client information, project objectives, and desired outcomes.

E-Chatco: An Integrated Smart-Hailing and Fleet Management Platform For Modernized Philippine Jeepneys		
Project Objectives	1	Development of an integrated smart-hailing and fleet management platform.
	2	Implementation of real-time GPS tracking, GCash e-payment, and boundary remittance tracking features.
	3	Ensuring system accessibility through web-based applications for conductors, commuters, and administrators.

<table>
<tr><th>Target Client</th><th>Client Story</th><th>Product</th><th>Expected Outcomes</th></tr>
<tr><td>Client</td><td>Neccessity</td><td rowspan="4">E-Chatco: Integrated Smart-Hailing and Fleet Management Platform for the Modernized Philippine Jeepney</td><td rowspan="4">1. Faster and more transparent boundary remittance tracking with separate cash and GCash recording.
2. Reduced commuter waiting times through real-time GPS visibility and proximity-based hailing.
3. Improved coordination between cooperative management and conductors.
4. Digital record-keeping for all fare transactions and fleet activities .</td></tr>
<tr><td>CHATCO Cooperative</td><td>Need for a modernized transport management system to address boundary remittance disputes, lack of real-time commuter visibility, manual cash-based fare collection, and inefficient coordination between the cooperative management and jeepney conductors along the Calumpit to Meycauayan route.</td></tr>
<tr><td>Roles</td><td>Specification</td></tr>
<tr><td>Administrator

Conductor

Commuter</td><td>-Monitoring of fleet operations, boundary remittance validation, and management of system users and records.
-Processing of fare transactions, QR-based GCash payment scanning, and submission of end-of-day cash and GCash remittance declarations.
-Real-time GPS tracking of jeepney units, proximity-based hailing, and cashless fare payment through GCash integration, discount application</td></tr>
</table>

Table 3.1 Project Vision Board

Preliminaries B: Release Planning

The system's initial release planning phase defined the project's overall scope and chosen components that are critical for fulfilling CHATCO Cooperative transport management requirements. Project backlogs including user stories, tasks, and requirements that guide the

system development were established at this stage. Early system requirements classification facilitated the development of understanding between stakeholders and proponents during the release planning phase.

Table 3.2 displays a selection of the project backlog items, each addressing particular user requirements and expectations appropriate for CHATCO Cooperative.

ID	Client	Initiative		Type	Priority
	As a/an…	I want to be able to…	So that I can…		
1	Commuter	View a live map of active jeepney units	See any available units near my area and plan my commute	Dashboard	**Must**
2	Commuter	Register with a valid ID and log in once my account is approved	Use the system securely	User Authentication	**Must**
3	Commuter	Send a"Pick Me Up" signal when within a 1-km radius of a CHATCO unit	Hail a jeepney conveniently	Hailing and Tracking	**Must**
4	Commuter	Pay my fare through GCash by scanning the conductor's payment QR code	Complete cashless transactions	GCash E-payment	**Must**
5	Commuter	Share my live location with family members during transit	Enhance personal safety during my ride	Safety Features	**Must**
7	Commuter	Apply for a student, senior citizen, or PWD discount	Pay the discounted fare	Discount Management	**Must**
8	Conductor	Select my assigned unit and driver before starting a shift	Record my fares and location under the right vehicle	Fleet Management	**Must**
9	Conductor	View my assigned unit, route, and the cash and GCash collections of my shift	Monitor my shift progress	Dashboard	**Must**

10	Conductor	Receive Pick Me Up requests from nearby commuters	Pick up waiting commuters	Hailing & Tracking	**Must**
11	Conductor	Select boarding and drop-off points for fare computation	Compute fares accurately	Fare Calculator	**Must**
12	Conductor	Generate a GCash payment QR code for the commuter to scan	Process cashless fare transactions	GCash E-Payment	**Must**
13	Conductor	Submit my end-of-day remittance with separate cash and GCash totals	Record my remittance accurately	Remittance	**Must**
14	Admin	View the dashboard with fleet overview, financial summaries, and alerts	Monitor operations at a glance	Dashboard	**Must**
15	Admin	Approve or reject commuter registrations and create conductor accounts	Control who can access the system	User Authentication	**Must**
16	Admin	Manage vehicles and drivers and assign a crew to each unit	Keep fleet records accurate	Fleet Management	**Must**
17	Admin	Monitor all active units on the map with overspeeding	Ensure fleet safety and compliance	Fleet Management	**Must**
18	Admin	View remittance records with separate cash and GCash breakdowns	Keep fares consistent with the approved rates	Remittance	**Must**
19	Admin	Manage the fare matrix and discount rates	Benefit from promotional campaigns	Rewards	**Must**
20	Admin	Review discount applications and student renewals	Give discounted fares only to eligible commuters	Discount Management	**Should**
21	Conductor	Print receipts through a paired Bluetooth thermal printer	Provide commuters proof of payment	GCash E-Payment	**Should**
22	Commuter	Calculate the fare between my pickup and drop-off points	Know how much to pay before I ride	Fare Calculator	**Should**

23	Commuter	View my GCash payment history	Review my cashless fare records	GCash E-Payment	**Should**
24	Admin	Record found items and review claims	Release items only to their rightful owners	Lost and Found	**Should**
25	Commuter	Search, claim, and watchlist found items	Recover my belongings	Lost and Found	**Should**
26	Commuter	Rate the driver and the conductor	Help the operator monitor service quality	Feedback	**Should**
27	Admin	Post announcements	Inform commuters of service updates	Announcement	**Should**
28	Admin	View where commuters usually get picked up and the peak hours	Make data-driven deployment decisions	System Activities	**Should**
29	Admin	View activity logs of system changes and user activities	Track system changes and user activities	System Activities	**Should**
30	Commuter	Earn and redeem reward vouchers, including from cash rides	Benefit from the Loyalty Program	Rewards	**Could**

Table 3.2: Project Backlogs

Each backlog item is categorized using the MoSCoW Rules, a popular method to ensure predictability in projects with incremental deliveries (Miranda, 2021). This method consists of four categories of features: Must Have, Should Have, Could Have, and Won't Have. The "Must Have" items are the most important functionalities required for the system to operate. The "Should Have" and the "Could Have" items are functionalities to enhance user experience and engagement but are not critical to have.

Table 3.3 lists the types of features in the project backlog along with a summary of each type's intended usage for a specific user.

Type	Description	Role/s
User Authentication	Features for account registration with a valid ID, email verification, admin approval, log-in, and password reset	Commuter, Admin, Conductor
Dashboard	Role-specific overviews of fleet activity, shift collections, and nearby units	Admin, Conductor, Commuter
Fleet Management	Functionalities to manage vehicles, drivers, and crew assignments, verify the unit at shift start, and monitor units and their speed	Admin, Conductor
Hailing and Tracking	Features related to GPS tracking, 1-kilometer proximity alerts, and Pick Me Up requests	Commuter, Conductor, Admin
GCash E-payment	Functionalities for QR-based GCash fare payments through PayMongo, payment history, and receipts	Commuter, Conductor, Admin
Fare Calculator	Features for point-area-based fare computation using the CHATCO fare matrix and discount rates	Conductor, Commuter, Admin
Discount Management	Features for student, senior citizen, and PWD discount applications and student renewals	Commuter, Admin
Remittance	Features for end-of-day remittance with separate cash and GCash totals and cash declaration	Conductor, Admin
Lost and Found	Functionalities for recording found items, submitting and reviewing claims, watchlists, and item release	Commuter, Admin
Safety Features	Features for Share My Ride, overspeeding detection, and incident reports	Commuter, Conductor, Admin
Feedback	Features for rating the driver and the conductor	Commuter, Conductor, Admin

Announcements	Features for posting and reading service updates	Commuter, Admin
Rewards	Features for reward vouchers, the loyalty program, and cash receipt claims	Commuter, Admin
System Activities	Features for activity logs, analytics, and system configuration	Admin

Table 3.3: Type of Features

Phase 1: Sprint Planning

Sprint Planning is the core component of the Agile Scrum methodology which contributes to a successful software development phase. A sprint is a predetermined amount of time where a set of work is completed, with clear goals defined beforehand. The steps that were completed during each sprint are listed in the tables below, with particular attention given to the user stories, acceptance criteria, and completion status.

Table 2.1 lists the activities to be performed and developed in the first sprint, circling around the administrator's access to the system. This aims to establish a secure, role-based log-in and a dashboard that gives the administrator an overview of the fleet and fare activities of the day.

Sprint 1: Administrator Landing				June 15 - 21
ID	Initiative			Status
1	I want to be able to log in to the system so that I can do my tasks as intended.			Completed
Feature Type	User Authentication	*Priority*	**Must**	
Requisite	N/A	*Module*	Login Page	

Acceptance Criteria (Functional Requirements):		
1. The user shall be able to access the login page.		✓
2. The user shall be able to log in using Administrator credentials registered within the system.		✓
3. The user shall be able to log in and proceed with the dashboard as the administrator.		✓
4. The system shall deny access to administrator pages for accounts that do not have the administrator role.		✓
2	I want to be able to view the dashboard with a concise and comprehensive layout so that I can have an overview of the system's operations.	Completed
Feature Type	User Dashboard *Priority*	**Must**
Requisite	ID #1 *Module*	Dashboard
Acceptance Criteria (Functional Requirements):		
1. The user shall be able to access the dashboard.		✓
2. The user shall be able to access and view the navigation menu, grouped into Operations, Management, and System, upon logging in.		✓
3	I want to be able to view summaries of the day's rides, revenue, active vehicles, and pending remittances in the dashboard so that I can monitor fleet and fare activities.	Completed
Feature Type	User Dashboard *Priority*	**Must**
Requisite	ID #1, #2 *Module*	Dashboard
Acceptance Criteria (Functional Requirements):		

1. The administrator dashboard shall display the total rides and total revenue for the current day.	✓
2. The administrator dashboard shall display the number of active vehicles and pending remittances.	✓

3. The administrator dashboard shall display recent users, vehicles, announcements, and lost and found items for quick access.		✓
4	I want to be able to view the live location of units on the dashboard so that I can see the fleet status straight away without navigating anywhere else in the system.	Completed
Feature Type	User Dashboard *Priority* **Should**	
Requisite	ID #1, #2 *Module* Dashboard	
Acceptance Criteria (Functional Requirements):		
1. The dashboard shall display a map showing the current location of units on shift.		✓
2. The map shall update as conductors send new location data.		✓
5	I want to be able to log out of the system so that no one else can use my account on the same device.	Completed
Feature Type	User Authentication *Priority* **Should**	
Requisite	ID #1 *Module* Dashboard	
Acceptance Criteria (Functional Requirements):		
1. The administrator shall be able to log out from the navigation menu.		✓
2. The system shall end the administrator's session upon logging out.		✓
3. The system shall redirect the administrator to the login page after logging out.		✓

Table 2.1. *Sprint 1: Administrator Landing*

Table 2.2 lists the activities to be performed and developed in the second sprint, circling around the administrator's ability to manage the accounts of commuters and conductors. This aims to make sure that only verified commuters and authorized conductors can access the system.

Sprint 2: Administrator User and Registration Management				June 22 - 28
ID	Initiative			Status
6	I want to be able to view the list of accounts registered in the system so that I can manage commuters, conductors, and administrators in one place.			Completed
Feature Type	User Management	*Priority*	**Must**	
Requisite	ID #1	*Module*	User Management	
Acceptance Criteria (Functional Requirements):				
1. The user management page shall be accessible to the administrator.				✓
2. The page shall load and display the existing accounts upon navigation to the page.				✓
3. The page shall separate the accounts into tabs by role and account status.				✓
4. The administrator shall be able to search for accounts.				✓
7	I want to be able to review commuter registrations so that only verified commuters can use the system.			Completed
Feature Type	User Management	*Priority*	**Must**	
Requisite	ID #1, #6	*Module*	User Management	
Acceptance Criteria (Functional Requirements):				
1. The system shall list the commuter registrations that are pending approval.				✓
2. The administrator shall be able to view the valid ID submitted by each applicant.				✓

3. The administrator shall be able to approve a registration, allowing the commuter to log in.				✓
4. The administrator shall be able to reject a registration with a reason.				✓
5. The system shall notify the commuter through email once the registration is approved or rejected.				✓
8	I want to be able to create accounts for conductors so that they can access the system and perform their shift duties.			Completed
Feature Type	User Management	*Priority*	**Must**	
Requisite	ID #1, #6	*Module*	User Management, Add Conductor Form	
Acceptance Criteria (Functional Requirements):				
1. The system shall allow the administrator to create conductor accounts.				✓
2. The system shall generate the log-in credentials of each new conductor account.				✓
3. Conductor accounts shall only have access to the features relevant to the conductor role.				✓
9	I want to be able to register commuters on-site so that commuters who register at the terminal can still be verified.			Completed
Feature Type	User Management	*Priority*	**Should**	
Requisite	ID #1, #7	*Module*	Onsite Registration Form	
Acceptance Criteria (Functional Requirements):				
1. The administrator shall be able to fill out the registration form and upload the valid ID on the commuter's behalf.				✓
2. The system shall add the on-site registration to the pending registrations for approval.				✓
3. The commuter shall be able to set their own password through password recovery once the account is approved.				✓

10	I want to be able to update, suspend, or disable accounts so that account details stay accurate and problematic accounts cannot access the system.		Completed
Feature Type	User Management	*Priority*	**Must**
Requisite	ID #1, #6	*Module*	User Management
Acceptance Criteria (Functional Requirements):			
1. The administrator shall be able to update the profile information of commuters and conductors.			✓
2. The administrator shall be able to suspend a commuter account, blocking it from logging in.			✓
3. The administrator shall be able to disable a conductor account and reset its credentials.			✓
4. The system shall sign out a disabled account from its active sessions.			✓

Table 2.2. *Sprint 2: Administrator User and Registration Management*

Table 2.3 lists the activities to be performed and developed in the third sprint, circling around the administrator's ability to manage the jeepney units and the personnel assigned to them. This aims to keep the records of vehicles, drivers, and conductors accurate for every shift.

Sprint 3: Administrator Fleet Management			June 29 - July 5
ID	Initiative		Status
11	I want to be able to view the list of units and personnel so that I know which vehicles, drivers, and conductors are in the fleet.		Completed
Feature Type	Fleet Management	*Priority*	**Must**
Requisite	ID #1	*Module*	Fleet Management
Acceptance Criteria (Functional Requirements):			

1. The fleet management page shall be accessible to the administrator.			✓
2. The page shall load and display the existing vehicles, drivers, and conductors upon navigation to the page.			✓
3. The page shall display the status and the assigned crew of each unit.			✓
12	I want to be able to add and update vehicles so that the fleet records match the units in operation.		Completed
Feature Type	Fleet Management	*Priority*	**Must**
Requisite	ID #1, #11	*Module*	Add Vehicle Form
Acceptance Criteria (Functional Requirements):			
1. The system shall allow the administrator to add a vehicle with its plate number, unit number, and route.			✓
2. The system shall allow the administrator to update the details of a vehicle.			✓
3. The system shall validate the vehicle details and display an error for invalid entries.			✓

13	I want to be able to add and update driver records so that each unit has a recorded, licensed driver.	Completed
Feature Type	Fleet Management *Priority*	**Must**
Requisite	ID #1, #11 *Module*	Add Driver Form

Acceptance Criteria (Functional Requirements):	
1. The system shall allow the administrator to add drivers with their license number.	✓
2. The system shall allow the administrator to update the information of a driver.	✓
3. The administrator shall be able to view the profile of each driver and conductor.	✓
14 I want to be able to assign a driver and a conductor to a unit so that each shift runs with the right crew.	Completed
Feature Type Fleet Management *Priority*	**Must**
Requisite ID #8, #12, #13 *Module*	Fleet Management
Acceptance Criteria (Functional Requirements):	
1. The administrator shall be able to assign a driver and a conductor to a unit.	✓
2. The system shall prevent crew changes while the unit has an active shift.	✓
3. The fleet management page shall display the crew currently on duty for each unit.	✓
15 I want to be able to view the details and history of each unit so that I can trace its past shifts and personnel.	Completed
Feature Type Fleet Management *Priority*	**Should**
Requisite ID #11 *Module*	Vehicle Details
Acceptance Criteria (Functional Requirements):	

1. The administrator shall be able to open the details of a vehicle, including its permanent unit QR code.	✓
2. The system shall display the shift history of each unit.	✓
3. The system shall keep a record of terminated personnel.	✓

Table 2.3. *Sprint 3: Administrator Fleet Management*

Table 2.4 lists the activities to be performed and developed in the fourth sprint, circling around the administrator's ability to monitor units in real time and respond to safety concerns. This aims to give the operator immediate visibility of the fleet and of commuters who need help.

Sprint 4: Administrator Monitoring and Safety				July 6 - 12
ID		Initiative		Status
	16	I want to be able to monitor the live location of units on shift so that I can track fleet operations in real time.		Completed
Feature Type	Live Monitoring	*Priority*	**Must**	
Requisite	ID #1	*Module*	Monitoring	
Acceptance Criteria (Functional Requirements):				
1. The monitoring page shall be accessible to the administrator.				✓
2. The page shall display all units currently on shift on a live map.				✓
3. The page shall flag units that have stopped sending location updates.				✓

4. The administrator shall be able to select a unit to focus on its location.	✓
17 I want to be notified when a unit is overspeeding so that I can act on unsafe driving.	Completed
Feature Type Safety Management *Priority* **Must**	
Requisite ID #16 *Module* Monitoring	
Acceptance Criteria (Functional Requirements):	
1. The system shall detect when a unit exceeds the speed limit set in the operations rules.	✓
2. The system shall alert the administrator of overspeeding units.	✓
3. The system shall keep a history of overspeeding incidents for review.	✓

Table 2.4. *Sprint 4: Administrator Monitoring and Safety*

Table 2.5 lists the activities to be performed and developed in the fifth sprint, circling around the administrator's ability to track fare collections and manage fare rates. This aims to reconcile the cash and GCash collected by conductors and to keep fares consistent across the system.

Sprint 5: Administrator Financial Management	July 13 - 18
ID Initiative	Status
18 I want to be able to view conductor remittances so that I can track the cash and GCash collected in each shift.	Completed
Feature Type Financial Management *Priority* **Must**	

Requisite	ID #1	*Module*	Remittance	
Acceptance Criteria (Functional Requirements):				
1. The remittance page shall be accessible to the administrator.				✓
2. The page shall display each remittance with its cash and GCash totals.				✓
3. The page shall display pending remittances separately from completed ones.				✓
4. The administrator shall be able to filter the remittance history.				✓
	19	I want to be able to view the details of each remittance so that I can verify the fares recorded during the shift.		Completed
Feature Type	Financial Management	*Priority*	**Must**	
Requisite	ID #19	*Module*	Remittance Details	
Acceptance Criteria (Functional Requirements):				
1. The administrator shall be able to open a remittance to view its details.				✓
2. The details shall list the paid fare transactions recorded during that shift.				✓
3. The system shall compute the remittance totals from the recorded transactions.				✓

	20	I want to be able to view fare receipts so that I have a record of every payment made in the system.		Completed
Feature Type	Financial Management	*Priority*	**Must**	
Requisite	ID #1	*Module*	Receipts	
Acceptance Criteria (Functional Requirements):				
1. The receipts page shall be accessible to the administrator.				✓

2. The page shall display fare transactions with their payment method and status.		✓
3. The administrator shall be able to filter the receipts by date.		✓
21	I want to be able to manage the fare matrix so that fares charged to commuters follow the approved rates.	Completed
Feature Type	Financial Management *Priority*	**Must**
Requisite	ID #1 *Module* Fare Matrix	
Acceptance Criteria (Functional Requirements):		
1. The fare matrix page shall be accessible to the administrator.		✓
2. The administrator shall be able to add, update, and remove fare points.		✓
3. The system shall use the saved fare matrix when computing fares for commuters and conductors.		✓
22	I want to be able to set the commuter discount rates so that student, senior citizen, and PWD fares are computed correctly.	Completed
Feature Type	Financial Management *Priority*	**Should**
Requisite	ID #22 *Module* Financial Rules	
Acceptance Criteria (Functional Requirements):		
1. The administrator shall be able to set the discount rates for students, senior citizens, and PWDs.		✓
2. The system shall apply the saved discount rates when computing fares.		✓

Table 2.5. *Sprint 5: Administrator Financial Management*

Table 2.6 lists the activities to be performed and developed in the sixth sprint, circling around the administrator's ability to handle lost items, post announcements, and manage

rewards. This aims to support commuters beyond the ride itself.

Sprint 6: Administrator Lost and Found, Announcements, and Rewards	July 20 - 26
ID — Initiative	Status
23 — I want to be able to record items left in units so that their owners can find and claim them.	Completed
Feature Type: Lost and Found — *Priority*: **Must**	
Requisite: ID #1 — *Module*: Lost and Found	
Acceptance Criteria (Functional Requirements):	
1. The lost and found page shall be accessible to the administrator.	✓
2. The administrator shall be able to record a found item with its description, unit, and photo.	✓
3. The administrator shall be able to update and close item records.	✓
24 — I want to be able to review claims for found items so that items are released only to their rightful owners.	Completed
Feature Type: Lost and Found — *Priority*: **Must**	
Requisite: ID #24 — *Module*: Lost and Found	
Acceptance Criteria (Functional Requirements):	
1. The system shall list the claims submitted for each item.	✓
2. The administrator shall be able to approve or reject a claim.	✓
3. The administrator shall be able to mark an item as released once it is claimed.	✓
25 — I want to be able to post announcements so that commuters are informed of service updates.	Completed

Feature Type	Announcements	*Priority*	**Must**
Requisite	ID #1	*Module*	Announcements
Acceptance Criteria (Functional Requirements):			
1. The announcements page shall be accessible to the administrator.			✓
2. The administrator shall be able to create, update, and delete announcements.			✓

3. The system shall show commuters which announcements they have not read yet.			✓
26	I want to be able to generate vouchers so that I can give discounts and free rides to commuters.		Completed
Feature Type	Rewards Management	*Priority*	**Should**
Requisite	ID #1	*Module*	Voucher Generator
Acceptance Criteria (Functional Requirements):			
1. The administrator shall be able to generate vouchers by type, such as discount and free ride.			✓
2. The system shall display the status of each voucher as active, used, or expired.			✓
27	I want to be able to set the number of rides needed to earn a reward so that the loyalty program can be adjusted.		Completed
Feature Type	Rewards Management	*Priority*	**Should**
Requisite	ID #27	*Module*	Loyalty Program
Acceptance Criteria (Functional Requirements):			

1. The administrator shall be able to set the ride threshold for earning a reward voucher.	✓
2. The system shall apply a new threshold only to rides taken after the change.	✓

Table 2.6. *Sprint 6: Administrator Lost and Found, Announcements, and Rewards*

Table 2.7 lists the activities to be performed and developed in the seventh sprint, circling around the system activities and configurations that are accessible only by the administrator. This includes the activity log to track user digital footprints within the system.

Sprint 7: System Activities			July 27 - August 2
ID	Initiative		Status
28	I want to view the activity logs so that I can know the history of each user and the changes that happened within the system.		Completed
Feature Type	System Activities	*Priority*	**Must**
Requisite	ID #1	*Module*	Activity Logs
Acceptance Criteria (Functional Requirements):			
1. The activity logs page shall be accessible to the administrator.			✓
2. The system shall display the history of actions performed by each user within the system.			✓
3. The system shall show the details of changes made, including what was changed, when, and by whom.			✓
29	I want to filter the activity logs so that I can find specific actions quickly.		Completed
Feature Type	System Activities	*Priority*	**Could**

Requisite	ID #29	*Module*	Activity Logs
Acceptance Criteria (Functional Requirements):			
1. The system shall allow the administrator to filter the activity logs by category.			✓
2. The system shall allow the administrator to filter the activity logs by date.			✓
30	I want to view the analytics of rides and revenue so that I can see trends and make operational decisions.		Completed
Feature Type	Analytics	*Priority*	**Should**
Requisite	ID #1	*Module*	Analytics
Acceptance Criteria (Functional Requirements):			
1. The analytics page shall display ride and revenue totals for a selected date range.			✓
2. The page shall compare the selected period with the previous period.			✓
3. The page shall display the peak hours and a demand map of pickup points.			✓
4. The page shall display how vouchers were used.			✓
31	I want to configure the operations and safety rules so that shifts and alerts follow the operator's policies.		Completed
Feature Type	System Configuration	*Priority*	**Must**
Requisite	ID #1	*Module*	Operations Rules, Safety Notifications
Acceptance Criteria (Functional Requirements):			
1. The administrator shall be able to set the overspeeding limit.			✓

2. The administrator shall be able to set the maximum shift duration and the shift closing time.			✓
3. The administrator shall be able to set the remittance reminder schedule.			✓
32	I want to put the system under maintenance so that users are informed when the service is temporarily unavailable.		Completed
Feature Type	System Configuration	*Priority*	**Should**
Requisite	ID #1	*Module*	App Configuration
Acceptance Criteria (Functional Requirements):			
1. The administrator shall be able to turn maintenance mode on or off with a reason.			✓
2. The system shall block commuter and conductor pages and display a notice while maintenance mode is on.			✓
33	I want to manage the frequently asked questions so that the public site answers common questions accurately.		Completed
Feature Type	System Configuration	*Priority*	**Could**
Requisite	ID #1	*Module*	FAQ Management
Acceptance Criteria (Functional Requirements):			
1. The administrator shall be able to add, update, and remove FAQs under categories.			✓
2. The FAQ chat on the landing page shall display the saved FAQs.			✓

Table 2.7. *Sprint 7: System Activities*

Table 2.8 lists the activities to be performed and developed in the eighth sprint, circling around all the tasks and activities that the conductor performs during a shift. This aims to

connect the unit on the road to the system through live location, Pick Me Up requests, and fare recording.

Sprint 8: Conductor Shift and Fare Activities			August 3 - 9
ID	Initiative		Status
34	I want to be able to log in to the system so that I can do my tasks as intended.		Completed
Feature Type	User Authentication	*Priority*	**Must**
Requisite	ID #8	*Module*	Login Page
Acceptance Criteria (Functional Requirements):			
1. The user shall be able to access the login page.			✓
2. The user shall be able to log in using their conductor credentials registered within the system.			✓
3. The user shall be able to log in and proceed with the unit verification as the conductor.			✓
35	I want to access only the features related to my tasks so that I can focus on my responsibilities.		Completed
Feature Type	User Authentication	*Priority*	**Must**
Requisite	ID #35	*Module*	Conductor Dashboard
Acceptance Criteria (Functional Requirements):			
1. The conductor shall only be able to access the conductor pages.			✓
2. The system shall redirect the conductor away from administrator and commuter pages.			✓
36	I want to select my unit and driver before starting my shift so that my fares and location are recorded under the right		Completed

	vehicle.			
Feature Type	Shift Management	*Priority*	**Must**	
Requisite	ID #35	*Module*	Unit Verification	
Acceptance Criteria (Functional Requirements):				
1. The system shall display the units and drivers available for the shift.				✓
2. The conductor shall be able to select a unit and a driver to start a shift.				✓

3. The system shall display a message when no units or drivers are available.				✓
	37	I want my unit's location to be shared while I am on shift so that commuters and administrators can track the jeepney.		Completed
Feature Type	Ride Tracking	*Priority*	**Must**	
Requisite	ID #37	*Module*	Conductor Dashboard	
Acceptance Criteria (Functional Requirements):				
1. The system shall send the GPS location of the unit while the shift is active.				✓
2. Commuters and administrators shall see the location of the unit update on their maps in real time.				✓
3. The system shall stop sending the location once the shift ends.				✓
	38	I want to receive Pick Me Up requests from nearby commuters so that I can pick them up along the route.		Completed
Feature Type	Pick Me Up	*Priority*	**Must**	
Requisite	ID #38	*Module*	Conductor Dashboard	
Acceptance Criteria (Functional Requirements):				

1. The system shall notify the conductor of Pick Me Up requests from commuters within 1 kilometer of the unit.	✓
2. The conductor shall be able to accept or reject a request.	✓
3. The system shall expire requests that are not answered within 3 minutes.	✓
39 I want to record each passenger's fare in cash or GCash so that every payment is accounted for.	Completed
Feature Type Fare Collection *Priority* **Must**	
Requisite ID #37 *Module* Fare Calculator, Payment	
Acceptance Criteria (Functional Requirements):	
1. The system shall compute the fare from the passenger's boarding and drop-off points.	✓
2. The conductor shall be able to record a cash payment.	✓
3. The conductor shall be able to generate a GCash payment QR code for the passenger to scan.	✓
4. The system shall save each recorded fare under the current shift.	✓
40 I want to end my shift and remit my collections so that the operator receives an accurate report of the day's earnings.	Completed

Feature Type Remittance *Priority* **Must**	
Requisite ID #40 *Module* End of Day	
Acceptance Criteria (Functional Requirements):	
1. The system shall display the cash and GCash totals of the shift.	✓
2. The conductor shall be able to end the shift and submit the remittance.	✓

3. The administrator shall be able to see the submitted remittance in the remittance page.	✓

Table 2.8. *Sprint 8: Conductor Shift and Fare Activities*

Table 2.9 lists the activities to be performed and developed in the ninth sprint, circling around the conductor's day-to-day operations beyond basic fare recording. This aims to make the shift workable under real road conditions, such as group passengers, breaks, and unstable connections.

Sprint 9: Conductor Operations Activities			August 10 - 16
ID	Initiative		Status
41	I want to record one payment for a group of passengers so that I can serve groups quickly.		Completed
Feature Type	Fare Collection	*Priority*	**Must**
Requisite	ID #40	*Module*	Payment
Acceptance Criteria (Functional Requirements):			
1. The conductor shall be able to add multiple passengers to one transaction.			✓
2. The system shall apply the discount of each discounted passenger in the group.			✓
3. The system shall support both cash and GCash for group payments.			✓
42	I want to print fare receipts so that passengers have proof of payment.		Completed
Feature Type	Fare Collection	*Priority*	**Should**

Requisite	ID #40	*Module*	Payment, Receipt	
Acceptance Criteria (Functional Requirements):				
1. The system shall print a receipt automatically when a fare is completed.				✓

2. The conductor shall be able to print a receipt manually.				✓
3. The receipt content shall follow the receipt settings set by the administrator.				✓
	43	I want to set my unit on break so that the administrator knows the unit is temporarily not in service.		Completed
Feature Type	Shift Management	*Priority*	**Should**	
Requisite	ID #37	*Module*	Conductor Dashboard	
Acceptance Criteria (Functional Requirements):				
1. The conductor shall be able to start and end a break during a shift.				✓
2. The system shall show the break status on the administrator's dashboard and map.				✓
	44	I want to keep recording cash fares when the connection drops so that no payment is lost.		Completed
Feature Type	Fare Collection	*Priority*	**Must**	
Requisite	ID #40	*Module*	Conductor Dashboard	
Acceptance Criteria (Functional Requirements):				
1. The system shall save cash fares on the device while offline.				✓
2. The system shall send the saved fares to the server once the connection returns.				✓

3. The system shall prevent duplicate fares when syncing.		✓
45	I want to continue my active shift on another device so that a lost or dead phone does not stop my shift.	Completed
Feature Type	Shift Management *Priority*	**Should**
Requisite	ID #35, #37 *Module*	Login Page
Acceptance Criteria (Functional Requirements):		
1. The system shall detect an active shift when the conductor logs in on a new device.		✓
2. The conductor shall be able to move the active shift to the new device.		✓
3. The system shall end the shift session on the previous device.		✓

Table 2.9. *Sprint 9: Conductor Operations Activities*

Table 2.10 lists the activities to be performed and developed in the tenth sprint, circling around the commuter's account and the core riding experience. This aims to let commuters register, verify their identity, and locate and hail jeepneys in real time.

Sprint 10: Commuter Account and Ride Activities		August 17 - 23
ID	Initiative	Status
46	I want to be able to create an account so that I can use the features of the system.	Completed
Feature Type	User Registration *Priority*	**Must**
Requisite	N/A *Module*	Sign Up Page
Acceptance Criteria (Functional Requirements):		

1. The commuter shall be able to access the sign-up page.	✓
2. The commuter shall be able to fill out their personal details and upload a photo of a valid ID.	✓
3. The system shall verify the commuter's email address with a code.	✓
4. The system shall require a strong password.	✓
5. The account shall remain pending until it is approved by the administrator.	✓
47 I want to be able to log in to the system so that I can use the features of the system.	Completed
Feature Type User Authentication *Priority* **Must**	
Requisite ID #7, #47 *Module* Login Page	
Acceptance Criteria (Functional Requirements):	
1. The user shall be able to access the login page.	✓
2. The user shall be able to log in using their username or email and password.	✓
3. The system shall prevent pending, rejected, and suspended accounts from logging in.	✓
4. The user shall be able to log in and proceed with the dashboard as the commuter.	✓
48 I want to be able to reset my account password so that I can regain access if I forget my account credentials.	Completed
Feature Type User Authentication *Priority* **Should**	

Requisite ID #48 *Module* Login Page
Acceptance Criteria (Functional Requirements):

1. The commuter shall be able to click on the "Forgot Password" option on the login page.	✓
2. The commuter shall enter their registered email address to receive a reset code.	✓
3. The commuter shall enter the code to set a new password.	✓
4. The system shall limit the number of attempts to enter the code.	✓
5. Once the password is reset, the commuter shall be able to log in using the new password.	✓
49 I want to see the live location of jeepneys on the map so that I know when a unit is coming.	Completed
Feature Type Ride Tracking *Priority* **Must**	
Requisite ID #48 *Module* Commuter Dashboard	
Acceptance Criteria (Functional Requirements):	
1. The dashboard shall display the current location of the commuter on the map.	✓
2. The map shall display the units on shift in real time.	✓
3. The map shall update the position of a unit as new location data arrives.	✓
50 I want to signal a nearby jeepney to pick me up so that the conductor knows I am waiting.	Completed
Feature Type Pick Me Up *Priority* **Must**	
Requisite ID #50 *Module* Commuter Dashboard	
Acceptance Criteria (Functional Requirements):	
1. The commuter shall be able to send a Pick Me Up request to units within 1 kilometer.	✓
2. The system shall prevent requests to units farther than 1 kilometer.	✓

3. The system shall show the commuter whether the request was accepted, rejected, or expired.	✓
4. The request shall expire after 3 minutes without a response.	✓
51 I want to update my profile and change my password so that my account details stay accurate and secure.	Completed
Feature Type Profile Management *Priority* **Should**	

Requisite ID #48 *Module* Profile	
Acceptance Criteria (Functional Requirements):	
1. The commuter shall be able to view and update their contact details.	✓
2. The commuter shall be able to change their password by entering the current password.	✓

Table 2.10. *Sprint 10: Commuter Account and Ride Activities*

Table 2.11 lists the activities to be performed and developed in the eleventh sprint, circling around the commuter's cashless fare payment and the rewards earned from riding. This aims to make GCash the default way to pay while rewarding regular commuters.

Sprint 11: Commuter Payment and Rewards Activities	August 24 - 30
ID Initiative	Status
52 I want to pay my fare through GCash so that I do not need to carry cash.	Completed
Feature Type Fare Payment *Priority* **Must**	

Requisite	ID #40, #48	*Module*	Payment
Acceptance Criteria (Functional Requirements):			
1. The commuter shall be able to scan the GCash payment QR code generated by the conductor.			✓
2. The system shall redirect the commuter to the GCash checkout to complete the payment.			✓
3. The system shall confirm the payment and update its status for both the commuter and the conductor.			✓
4. The commuter shall be able to cancel a pending payment.			✓
53	I want to view my payment history so that I can keep track of my fares.		Completed
Feature Type	Fare Payment	*Priority*	**Must**
Requisite	ID #53	*Module*	Payment History
Acceptance Criteria (Functional Requirements):			
1. The commuter shall be able to view a list of past fare payments.			✓
2. Each payment shall show its date, amount, payment method, and status.			✓

54	I want to earn rewards for riding regularly so that I can get discounts or free rides.		Completed
Feature Type	Rewards	*Priority*	**Should**
Requisite	ID #28, #53	*Module*	Rewards
Acceptance Criteria (Functional Requirements):			
1. The system shall track the commuter's progress toward the ride threshold.			✓

2. The system shall issue a reward voucher once the threshold is reached.			✓
3. The commuter shall be able to view their available, used, and expired vouchers.			✓
55	I want my cash rides to count toward rewards so that I am not left out when paying in cash.		Completed
Feature Type	Rewards	*Priority*	**Could**
Requisite	ID #55	*Module*	Rewards
Acceptance Criteria (Functional Requirements):			
1. The commuter shall be able to claim a paper cash receipt toward the reward cycle.			✓
2. The system shall prevent a receipt from being claimed more than once.			✓

Table 2.11. *Sprint 11: Commuter Payment and Rewards Activities*

Table 2.12 lists the activities to be performed and developed in the twelfth sprint, circling around the safety and support features available to the commuter. This aims to keep help one tap away during a ride and to support commuters after it.

Sprint 12: Commuter Safety and Support Activities			August 31 - September 6
ID	Initiative		Status
56	I want to share my live location so that my family or friends can follow my ride.		Completed
Feature Type	Safety	*Priority*	**Must**
Requisite	ID #48	*Module*	Share Ride

Acceptance Criteria (Functional Requirements):	
1. The commuter shall be able to generate a link to share their live location.	✓
2. Anyone with the link shall be able to view the commuter's location on a live map without an account.	✓
3. The shared location shall update while sharing is active.	✓
57 I want to search for and claim items I lost in a jeepney so that I can get them back.	Completed
Feature Type Lost and Found *Priority* **Must**	
Requisite ID #24, #48 *Module* Lost and Found	
Acceptance Criteria (Functional Requirements):	
1. The commuter shall be able to view and search the found items.	✓
2. The commuter shall be able to submit a claim for an item.	✓
3. The commuter shall be able to track the status of their claims.	✓

4. The commuter shall be able to add items to a watchlist.	✓
58 I want to rate the driver and the conductor so that the operator knows the quality of their service.	Completed
Feature Type Feedback *Priority* **Should**	
Requisite ID #15, #48 *Module* Feedback	
Acceptance Criteria (Functional Requirements):	
1. The commuter shall be able to scan the QR code of a unit to give feedback.	✓
2. The commuter shall be able to rate the driver and the conductor separately.	✓

3. The commuter shall be able to give feedback from their ride history.			✓
59	I want to read announcements so that I am informed of service updates.		Completed
Feature Type	Announcements	*Priority*	**Should**
Requisite	ID #26, #48	*Module*	Updates
Acceptance Criteria (Functional Requirements):			
1. The commuter shall be able to view the announcements posted by the administrator.			✓
2. The system shall mark announcements that the commuter has not read yet.			✓

Table 2.12. *Sprint 12: Commuter Safety and Support Activities*

Table 2.13 lists the activities to be performed and developed in the thirteenth sprint, circling around the fare discounts given to students, senior citizens, and PWDs. This aims to let commuters apply for and keep their discount, with every application verified by the administrator.

Sprint 13: Fare Discount Applications			September 7 - 13
ID	Initiative		Status
60	I want to apply for a fare discount so that I can pay the discounted fare as a student, senior citizen, or PWD.		Completed
Feature Type	Discount Management	*Priority*	**Must**
Requisite	ID #23, #48	*Module*	Discount Application
Acceptance Criteria (Functional Requirements):			

1. The commuter shall be able to select a discount type and upload a supporting document.	✓
2. The commuter shall be able to view the status of their application.	✓
3. The commuter shall be able to cancel a pending application.	✓
4. The system shall apply the discount to the commuter's fares once the application is approved.	✓
61 I want to renew my student discount every school year so that I can keep paying the student fare.	Completed
Feature Type Discount Management *Priority* **Should**	
Requisite ID #62 *Module* Student Renewal	
Acceptance Criteria (Functional Requirements):	
1. The system shall remind students to renew their discount before the renewal cutoff.	✓
2. The student shall be able to upload proof of their current enrollment.	✓
3. The student shall be able to switch to a regular fare instead of renewing.	✓
62 I want to review discount applications so that only eligible commuters receive discounted fares.	Completed
Feature Type Discount Management *Priority* **Must**	
Requisite ID #1, #62 *Module* User Management	

Acceptance Criteria (Functional Requirements):	
1. The system shall list the discount applications that are pending review.	✓
2. The administrator shall be able to view the supporting document of each application.	✓

3. The administrator shall be able to approve or reject an application.	✓
4. The system shall notify the commuter of the result of their application.	✓
63 I want to review student renewals so that student discounts stay limited to currently enrolled students.	Completed
Feature Type Discount Management *Priority* **Must**	
Requisite ID #1, #63 *Module* User Management	
Acceptance Criteria (Functional Requirements):	
1. The system shall list the student renewals that are pending review.	✓
2. The administrator shall be able to view the proof of enrollment submitted by the student.	✓
3. The administrator shall be able to approve or reject a renewal.	✓

Table 2.13. *Sprint 13: Fare Discount Applications*

Table 2.14 lists the activities to be performed and developed in the fourteenth sprint, circling around the Conductor Android App, which lets the conductor run a shift from a phone. The app uses the same accounts, shifts, and fare records as the web system, so the screens developed for the conductor web portal in the previous sprints serve as its reference.

Sprint 14: Conductor Mobile Shift and Fare Activities	September 14 - 20
ID Initiative	Status
64 I want to be able to log in to the mobile app so that I can do my tasks on my phone.	Completed
Feature Type User Authentication *Priority* **Must**	

Requisite	ID #8	*Module*	Login Screen
Acceptance Criteria (Functional Requirements):			
1. The conductor shall be able to access the login screen of the Conductor Android App.			✓
2. The conductor shall be able to log in using their username or email and password registered within the system.			✓
3. The app shall only allow conductor accounts to sign in.			✓
4. The app shall keep the conductor signed in when the app is reopened until they sign out.			✓
65	I want to verify my assigned unit and driver in the app so that my shift is recorded under the right vehicle.		Completed
Feature Type	Shift Management	*Priority*	**Must**
Requisite	ID #14, #66	*Module*	Unit Verification Screen
Acceptance Criteria (Functional Requirements):			
1. The app shall display the assigned vehicle unit, driver, and route.			✓
2. The conductor shall be able to select their assigned vehicle and driver to start the shift.			✓
3. The app shall show a shift setup summary before the shift starts.			✓
66	I want the app to share my unit's location while I am on shift so that commuters and administrators can track the jeepney.		Completed
Feature Type	Ride Tracking	*Priority*	**Must**
Requisite	ID #67	*Module*	Dashboard
Acceptance Criteria (Functional Requirements):			

1. The app shall send the GPS location of the unit while the shift is active.	✓
2. The dashboard shall display the position of the unit on the route map.	✓
3. The conductor shall be able to set the capacity status of the unit to Available, Standing, or Full.	✓
67 I want to receive Pick Me Up requests in the app so that I can pick up waiting commuters.	Completed
Feature Type Pick Me Up *Priority* **Must**	
Requisite ID #68 *Module* Dashboard	
Acceptance Criteria (Functional Requirements):	
1. The app shall display Pick Me Up requests from nearby commuters.	✓
2. The conductor shall be able to accept or reject a request.	✓
68 I want to collect fares in cash, GCash, or voucher from the app so that I can serve passengers without the web portal.	Completed

Feature Type Fare Collection *Priority* **Must**	
Requisite ID #22, #67 *Module* Payment	
Acceptance Criteria (Functional Requirements):	
1. The app shall compute the fare from the selected stops and the passenger type using the fare matrix.	✓
2. The conductor shall be able to record a cash payment.	✓
3. The conductor shall be able to generate a GCash payment QR code and the app shall detect when the payment succeeds.	✓
4. The conductor shall be able to apply a voucher code to a fare.	✓

5. The conductor shall be able to view the transaction history of the shift.		✓
69	I want to view my collections and remit them at the end of my shift so that the administrator receives an accurate report.	Completed
Feature Type	Remittance *Priority*	**Must**
Requisite	ID #70 *Module*	End of Day Report
Acceptance Criteria (Functional Requirements):		
1. The app shall display the cash, GCash, and grand total of the shift.		✓
2. The conductor shall be able to remit to the administrator and end the shift.		✓
3. The conductor shall be able to view their remittance history.		✓
70	I want to view my ratings in the app so that I know how commuters rate my service.	Completed
Feature Type	Performance Metrics *Priority*	**Could**
Requisite	ID #68 *Module*	Performance Metrics
Acceptance Criteria (Functional Requirements):		
1. The app shall display the average rating, total ratings, and rating distribution of the current shift.		✓
2. The app shall display recent feedback for the conductor and the driver.		✓
71	I want to keep recording cash fares in the app when the connection drops so that no payment is lost.	Completed
Feature Type	Fare Collection *Priority*	**Must**
Requisite	ID #70 *Module*	Payment, Dashboard
Acceptance Criteria (Functional Requirements):		

1. The app shall save cash fares on the device when there is no connection.	✓
2. The app shall display the number of cash fares waiting to be synced.	✓
3. The app shall sync the saved fares automatically once the connection returns, without creating duplicates.	✓
72 I want to move my active shift between the web portal and the app so that a change of device does not stop my shift.	Completed
Feature Type Shift Management *Priority* **Should**	
Requisite ID #67 *Module* Unit Verification Screen, Dashboard	
Acceptance Criteria (Functional Requirements):	
1. The app shall show when the conductor's shift is active on another device.	✓
2. The conductor shall be able to claim or release the shift to move it between the web portal and the app.	✓
3. The app shall sign out a device that was replaced or released by the administrator.	✓

Table 2.14. *Sprint 14: Conductor Mobile Shift and Fare Activities*

Table 2.15 lists the activities to be performed and developed in the fifteenth sprint, circling around the conductor's on-road operations in the Conductor Android App. This aims to support group passengers, receipt printing, and safety reporting, and to help the conductor stay on the franchise route.

Sprint 15: Conductor Mobile Operations Activities	September 21 - 27

ID	Initiative	Status
73	I want to record one payment for a group of passengers in the app so that I can serve groups quickly.	Completed
Feature Type	Fare Collection *Priority*	**Must**
Requisite	ID #70 *Module* Payment	
Acceptance Criteria (Functional Requirements):		
1. The conductor shall be able to add several passengers of different types to one payment.		✓
2. The app shall apply the discount of each discounted passenger in the group.		✓

74	I want to set my unit on break in the app so that commuters and administrators know the unit is temporarily not in service.	Completed
Feature Type	Shift Management *Priority*	**Should**
Requisite	ID #67 *Module* Dashboard	
Acceptance Criteria (Functional Requirements):		
1. The conductor shall be able to start and end a break after confirming the action.		✓
2. Commuters shall see that the vehicle is on break.		✓
75	I want to print fare receipts from the app so that passengers have proof of payment.	Completed
Feature Type	Fare Collection *Priority*	**Should**
Requisite	ID #70 *Module* Settings, Payment	
Acceptance Criteria (Functional Requirements):		

1. The conductor shall be able to pair a Bluetooth thermal printer from the settings.	✓
2. The app shall print a receipt automatically when a ticket is completed while Auto-Print is turned on.	✓
3. The app shall support common 58mm and 80mm thermal printers.	✓
76 I want the app to alert me about speed, route, and upcoming stops so that I can keep my eyes on the road.	Completed
Feature Type Ride Tracking *Priority* **Could**	
Requisite ID #68 *Module* Dashboard, Settings	

Acceptance Criteria (Functional Requirements):	
1. The app shall warn the conductor when the unit goes above 50 km/h.	✓
2. The app shall alert the conductor when the unit leaves the franchise route by more than 300 meters.	✓
3. The app shall announce upcoming stops by voice, and the conductor shall be able to turn this off in the settings.	✓
77 I want to track each trip between Calumpit and Meycauayan so that I can see how each trip performed.	Completed
Feature Type Ride Tracking *Priority* **Could**	
Requisite ID #68 *Module* Trip Cycle Logbook	
Acceptance Criteria (Functional Requirements):	
1. The app shall log each trip cycle and terminal turnaround of the shift.	✓
2. The app shall display the passengers and revenue of each trip, with their averages.	✓

Table 2.15. *Sprint 15: Conductor Mobile Operations Activities*

Table 2.16 lists the activities to be performed and developed in the sixteenth sprint, circling around the Commuter Android App, which brings the commuter web features to a phone. This sprint covers the commuter's account and riding activities, using the same accounts and approval process as the web system.

Sprint 16: Commuter Mobile Account and Ride Activities			September 28 - October 4
ID	Initiative		Status
78	I want to be able to create an account in the mobile app so that I can use the system on my phone.		Completed
Feature Type	User Registration	*Priority*	**Must**
Requisite	ID #7	*Module*	Onboarding, Sign Up Screen
Acceptance Criteria (Functional Requirements):			
1. The app shall show a walkthrough of its features on first launch.			✓
2. The commuter shall be able to fill out their personal details, select a commuter category, and upload a valid ID.			✓
3. The app shall verify the commuter's email address with a code.			✓
4. The account shall remain pending until it is approved by the administrator.			✓
79	I want to be able to log in to the mobile app so that I can use the features of the system.		Completed
Feature Type	User Authentication	*Priority*	**Must**
Requisite	ID #81	*Module*	Login Screen

Acceptance Criteria (Functional Requirements):	
1. The commuter shall be able to log in using their phone number or email and password.	✓
2. The app shall prevent pending, rejected, and suspended accounts from logging in.	✓
3. The app shall keep the commuter signed in until they log out.	✓
80 I want to be able to reset my password in the mobile app so that I can regain access if I forget it.	Completed
Feature Type User Authentication *Priority* **Should**	
Requisite ID #82 *Module* Forgot Password Screen	
Acceptance Criteria (Functional Requirements):	
1. The commuter shall be able to click on the "Forgot password?" option on the login screen.	✓
2. The commuter shall enter their registered email address to receive a reset code.	✓
3. The commuter shall enter the code to set a new password.	✓
81 I want to manage my profile in the mobile app so that my account details stay accurate and secure.	Completed
Feature Type Profile Management *Priority* **Should**	
Requisite ID #82 *Module* Profile Screen	
Acceptance Criteria (Functional Requirements):	
1. The commuter shall be able to view their personal information and commuter type.	✓
2. The commuter shall be able to update their contact number.	✓

3. The commuter shall be able to upload a JPG, PNG, or WebP profile photo of up to 3 MB.	✓

4. The commuter shall be able to change their password with their current password and a 6-digit code sent to their email.			✓
83	I want to see the live location of jeepneys in the mobile app so that I know when a unit is coming.		Completed
Feature Type	Ride Tracking	*Priority*	**Must**
Requisite	ID #82	*Module*	Dashboard
Acceptance Criteria (Functional Requirements):			
1. The app shall ask for location access and show a message when the location is unavailable.			✓
2. The map shall display the units on shift in real time.			✓
3. The app shall show when a unit is full or on break.			✓
84	I want to send a Pick Me Up request from the mobile app so that the conductor knows I am waiting.		Completed
Feature Type	Pick Me Up	*Priority*	**Must**
Requisite	ID #85	*Module*	Dashboard
Acceptance Criteria (Functional Requirements):			
1. The commuter shall be able to send a Pick Me Up request to a nearby unit.			✓
2. The app shall inform the commuter when they are outside the service radius.			✓
3. The app shall show updates on the status of the request.			✓
85	I want to calculate my fare in the mobile app so that I know how much to pay before I ride.		Completed

Feature Type	Fare Payment	*Priority*	**Could**	
Requisite	ID #22, #82	*Module*	Fare Calculator	
Acceptance Criteria (Functional Requirements):				
1. The commuter shall be able to select a pickup point and a destination to view the fare.				✓
2. The commuter shall be able to swap the pickup point and destination.				✓

Table 2.16. *Sprint 16: Commuter Mobile Account and Ride Activities*

Table 2.17 lists the activities to be performed and developed in the seventeenth sprint, circling around the payment, rewards, safety, and support features of the Commuter Android App. This aims to give commuters on mobile the same cashless payment and safety tools available on the web.

Sprint 17: Commuter Mobile Payment, Safety, and Support Activities				October 5 - 11
ID		Initiative		Status
86		I want to pay my fare through GCash in the mobile app so that I do not need to carry cash.		Completed
Feature Type	Fare Payment	*Priority*	**Must**	
Requisite	ID #70, #82	*Module*	QR Scanner	
Acceptance Criteria (Functional Requirements):				
1. The commuter shall be able to scan the conductor's payment QR code with the phone camera.				✓
2. The app shall display the amount due and the fare type before payment.				✓

3. The app shall hand the payment over to GCash and verify the transaction afterward.		✓
4. The commuter shall be able to use a voucher for the payment.		✓
87	I want to view my payment history in the mobile app so that I can keep track of my fares.	Completed
Feature Type	Fare Payment *Priority*	**Must**
Requisite	ID #88 *Module*	Payment History
Acceptance Criteria (Functional Requirements):		
1. The commuter shall be able to view a list of past fare payments.		✓
2. Each payment shall show its details, including the drop-off point and the conductor.		✓
3. The commuter shall be able to load more payments from the history.		✓
88	I want to view my rewards and updates in the mobile app so that I can use my vouchers and stay informed.	Completed
Feature Type	Rewards *Priority*	**Should**
Requisite	ID #28, #88 *Module*	Rewards and Updates

Acceptance Criteria (Functional Requirements):	
1. The app shall display the commuter's total rides and progress toward a free ride.	✓
2. The app shall display the commuter's available, redeemed, and expired vouchers.	✓
3. The commuter shall be able to claim a paper cash receipt by scanning it or entering its 6-character code.	✓

4. The app shall display the latest announcements and mark those that are unread.		✓
89	I want to share my live location from the mobile app so that my family or friends can follow my ride.	Completed
Feature Type	Safety *Priority*	**Must**
Requisite	ID #82 *Module*	Share Ride
Acceptance Criteria (Functional Requirements):		
1. The commuter shall be able to create a link to share their live location.		✓
2. The commuter shall be able to send the link through the phone's share options.		✓
3. The link shall expire after 30 minutes, and the commuter shall be able to stop sharing at any time.		✓
90	I want to search for and claim lost items in the mobile app so that I can get them back.	Completed
Feature Type	Lost and Found *Priority*	**Must**
Requisite	ID #24, #82 *Module*	Lost and Found

Acceptance Criteria (Functional Requirements):	
1. The commuter shall be able to view and search the found items.	✓
2. The commuter shall be able to submit and cancel a claim for an item.	✓
3. The commuter shall be able to view their claim history and its status.	✓
4. The commuter shall be able to add items to a watchlist.	✓
91 I want to rate the driver and the conductor in the mobile app so that the operator knows the quality of their service.	Completed

Feature Type	Feedback	*Priority*	**Should**	
Requisite	ID #15, #82	*Module*	Feedback	
Acceptance Criteria (Functional Requirements):				
1. The commuter shall be able to scan the QR code of a unit or enter its unit code to give feedback.				✓
2. The commuter shall be able to rate the driver and the conductor separately.				✓
92	I want to apply for a fare discount in the mobile app so that I can pay the discounted fare.			Completed
Feature Type	Discount Management	*Priority*	**Must**	
Requisite	ID #62, #82	*Module*	Discount Application	
Acceptance Criteria (Functional Requirements):				
1. The commuter shall be able to submit a discount application with a supporting document.				✓
2. The commuter shall be able to view and cancel their application.				✓
93	I want to renew my student discount in the mobile app so that I can keep paying the student fare.			Completed
Feature Type	Discount Management	*Priority*	**Should**	
Requisite	ID #63, #82	*Module*	Student Renewal	
Acceptance Criteria (Functional Requirements):				
1. The app shall remind students when their renewal is due.				✓
2. The student shall be able to submit proof of their current enrollment.				✓
3. The student shall be able to switch to a regular fare instead of renewing.				✓

Table 2.17. *Sprint 17: Commuter Mobile Payment, Safety, and Support Activities*

Phase 1.2: Role Assignment

Scrum is an Agile framework used in project management that focuses on teamwork and continuous progress. It includes the Product Owner, who manages the product goals and backlog; the Scrum Master, who guides the team in following Scrum practices; and the Development Team, which is responsible for building and testing the system. Scrum promotes collaboration, accountability, and flexibility, helping the team adjust to changing needs during development. For the E-Chatco project, these roles help organize tasks and support the development of the smart-hailing and fleet management platform.

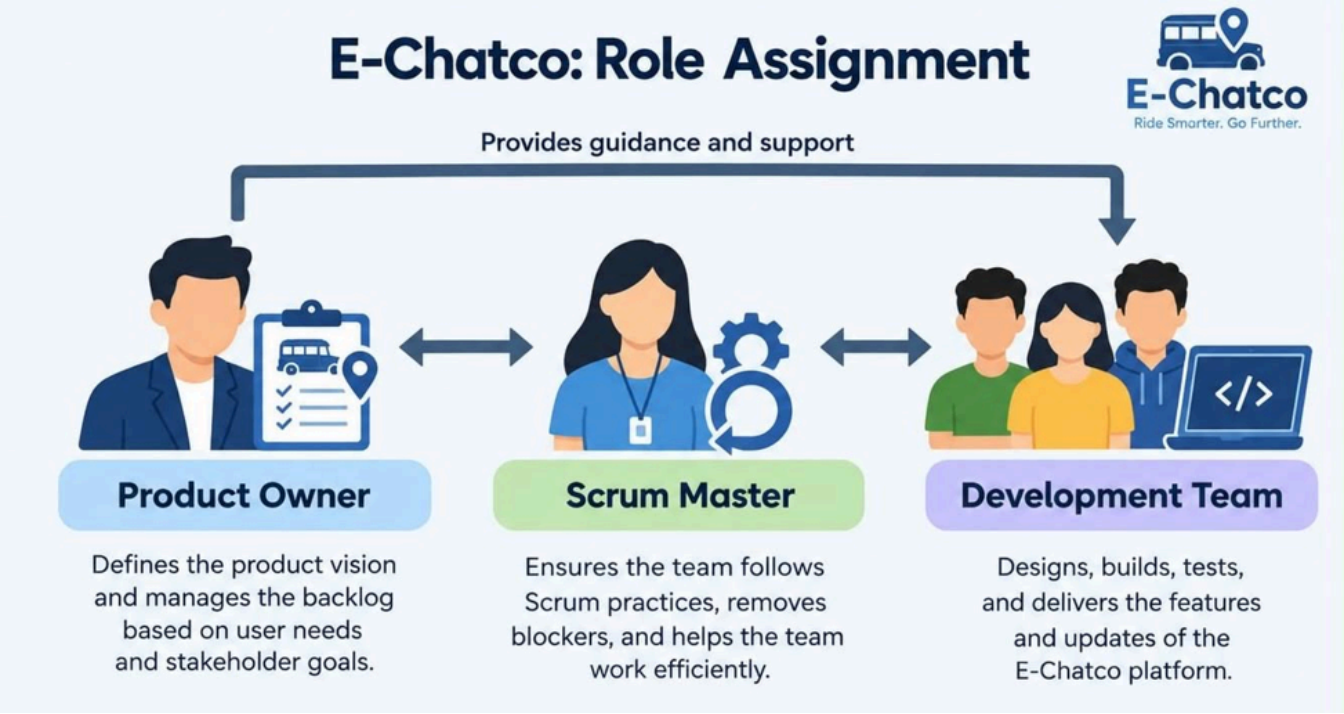


Figure 2.32: *Scrum Roles*

In relation to this, the proponents of the E-Chatco system assigned roles according to Scrum practices. These roles are listed in Table 2.22, including the Product Owner, Scrum Master, and Development Team Members. Each member has a specific responsibility in the development process. This role assignment helps organize tasks, improve teamwork, and ensure that each member understands their responsibilities throughout the development cycle.

Role	Individuals Aligned
Product Owner	• CHATCO
Scrum Master	• Mark Arone M. Dela Cruz
Development Team Member	• Mark Arone M. Dela Cruz • Mhaku Jose M. Manalili • Rod Erick T. Dulalia • Marinel T. Carbonell

Table 2.22: *Scrum Roles Breakdown*

Phase 2.1. UI Design Plan

The proponents selected shades of blue as the primary colors of the CHATCO system to establish a clean, modern, and professional appearance. Deep Navy and CHATCO Blue serve as the main colors, while Signal Orange is used to highlight important actions such as the "Pick Me Up" feature. Neutral colors are also incorporated to maintain readability and consistency throughout the system.

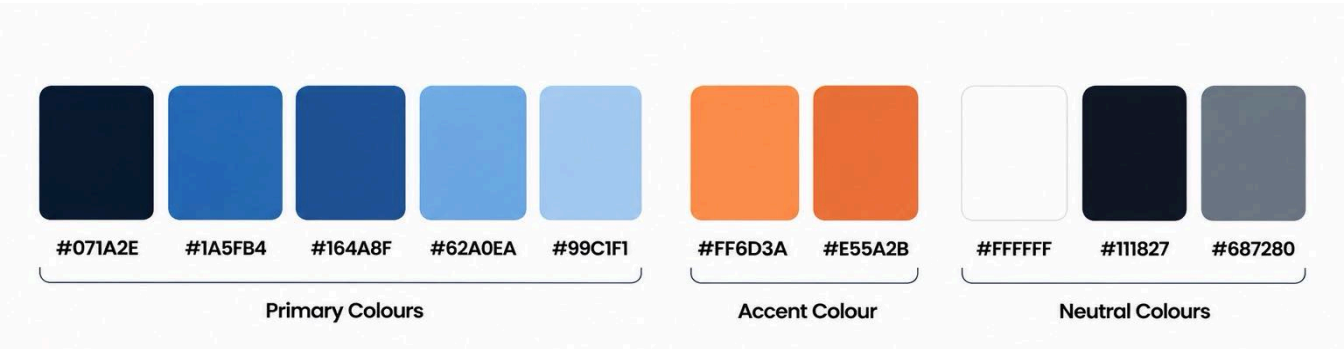


Figure 2.33: CHATCO Color Palette

The team decided on using one typeface for the whole system. This font was chosen due to its clean and modern appearance, and with its geometric structure displaying a sense of professionalism, simplicity, and readability that fits well with the CHATCO system's branding. Its clear letterforms also help users easily read information and navigate through the different interfaces of the system. Figure 2.34 shows the font the proponents will use for the system. It was designed by Indian Type Foundry and is available through Google Fonts. The typeface uses different font weights, including Regular, Medium, SemiBold, and Bold, to establish a consistent visual hierarchy throughout the landing page, commuter, conductor, and admin applications.

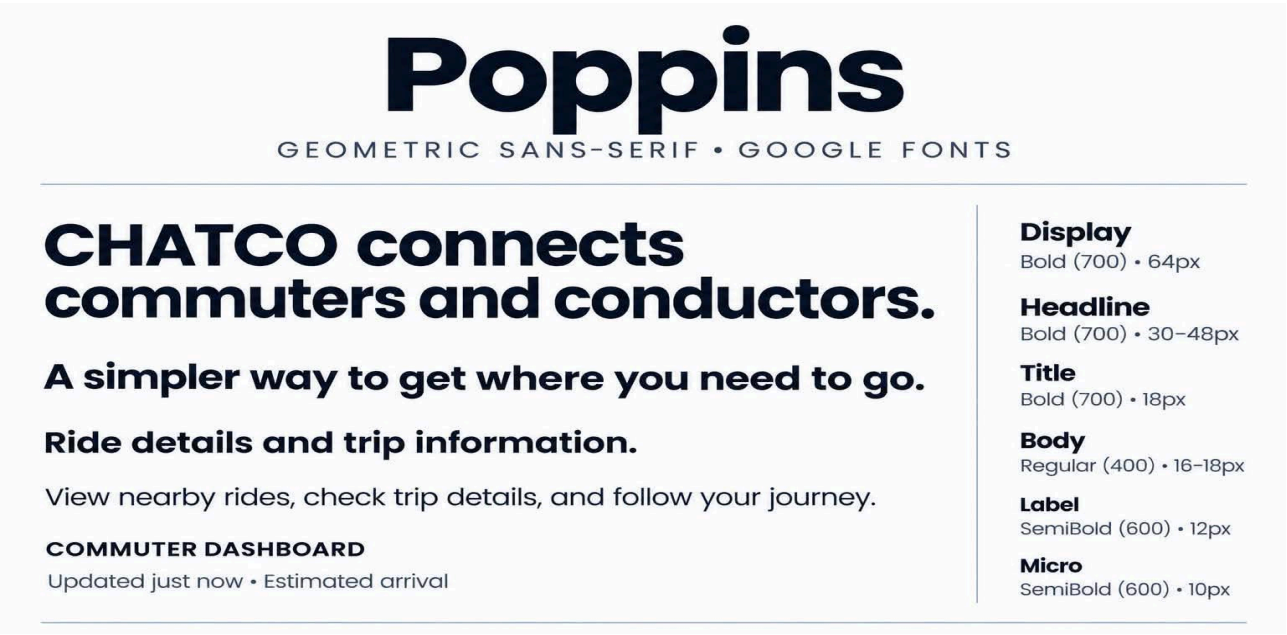


Figure 2.34: CHATCO Typography

Phase 2.2. System Designing

The graphical system design of E-CHATCO: An Integrated Smart-Hailing and Fleet Management Platform for the Modernized Philippine Jeepney is presented in this section. The system is organized around the primary functions and modules assigned to each type of user, focusing on the transportation needs of commuters and the operational responsibilities of conductors and administrators. It incorporates essential features such as real-time jeepney tracking, smart hailing, cashless fare transactions, fleet monitoring, and safety management. The system design illustrates how the different interfaces and their corresponding functionalities are arranged to provide an accessible, organized, and efficient platform for managing daily transportation activities along the Calumpit-Meycauayan route.

During the system design process, the team organized the interfaces according to the roles, responsibilities, and access privileges of the intended users. The system is divided into five main interface categories, namely the Landing Interface, Login and Account Interface, Commuter Interface, Conductor Interface, and Administrator Interface. Each interface contains specific modules that support the activities and requirements of its respective users. The Commuter Interface focuses on passenger services, including jeepney tracking, ride hailing, fare payments, rewards, and safety features. The Conductor Interface provides tools for shift management, fare collection, transaction recording, and daily remittance. Meanwhile, the Administrator Interface contains management and monitoring functions for users, vehicles, financial transactions, routes, and system operations. Shared functionalities, such as account management, notifications, and payment-related processes, are discussed according to their primary purpose to minimize repetition while maintaining a clear presentation of the system's overall functionality. This organization provides a structured representation of the E-CHATCO system and demonstrates how its different interfaces work together to support an integrated and coordinated public transportation service.

Landing Interface

Figure 2.35 displays the Landing Interface serves as the public-facing section of the E-CHATCO system. It provides visitors with information about the platform, its available features, safety services, and the Calumpit-Meycauayan route. It also provides access to important information and services that users can access before signing in.

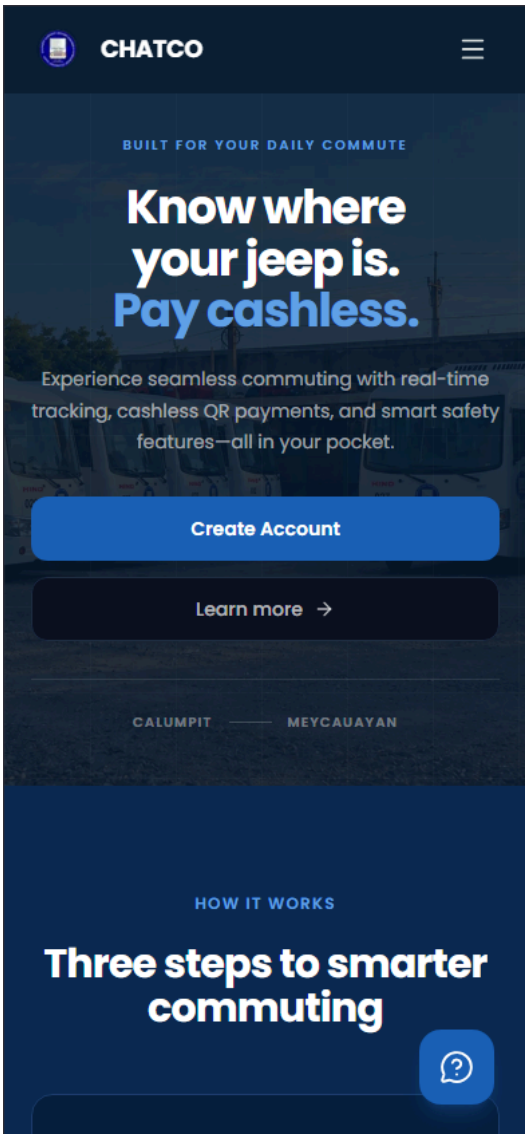


Figure 2.35: *Landing Page*

The landing page introduces the E-CHATCO system and provides an overview of its main features and services. It allows visitors to understand the purpose of the platform, particularly its jeepney tracking, smart-hailing, cashless payment, and safety functionalities. It also presents information about the Calumpit-Meycauayan route to help users become familiar with the transportation service.

Figure 2.36 displays the FAQ chat bubble provides visitors with a question-and-answer feature that helps them find information about the system. It displays predefined questions and corresponding answers regarding the available services, account procedures, and other common concerns. The questions and answers can be maintained and updated by the administrators.

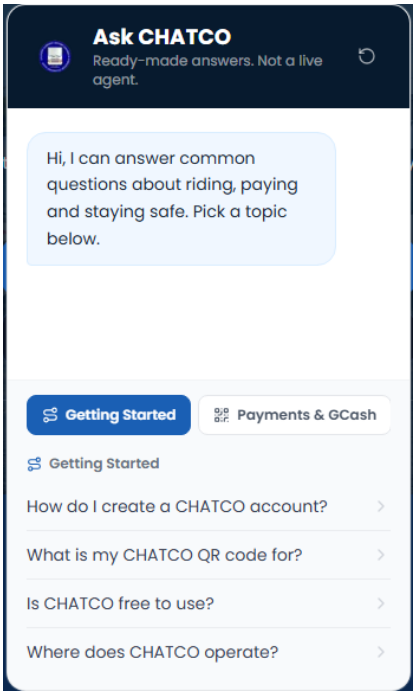


Figure 2.36: FAQ Chat Bubble

Figure 2.37 shows the Login and Account Interface, It manages the authentication and registration processes of the E-CHATCO system. It provides the necessary functions for commuters, conductors, and administrators to access their respective applications. It also includes account verification and password recovery features to support secure account access.

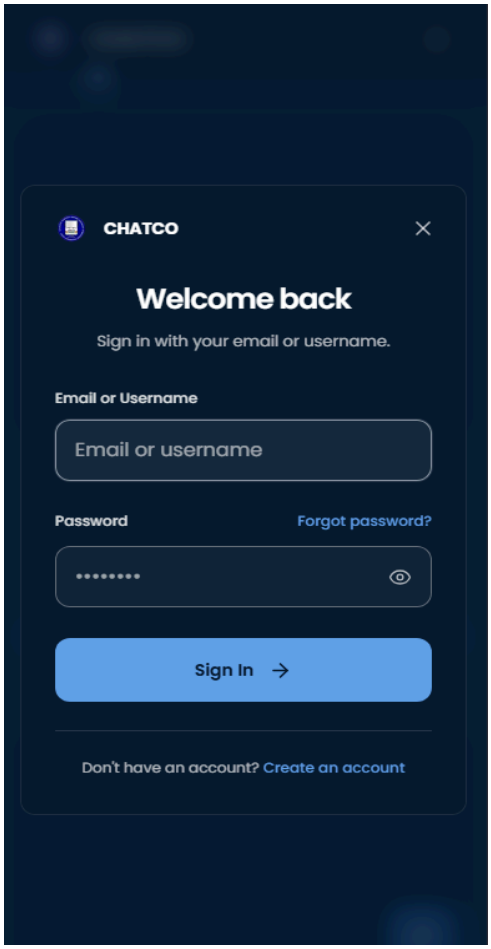


Figure 2.37: Login and Account Interface

Figure 2.38 shows the commuter sign-up page allows new commuters to create an account by providing the required registration information. During registration, users must verify their email address through a verification code, with an option to upload a valid identification document when required. Submitted registrations are forwarded to the administrator for review and approval or rejection before the account can be used.

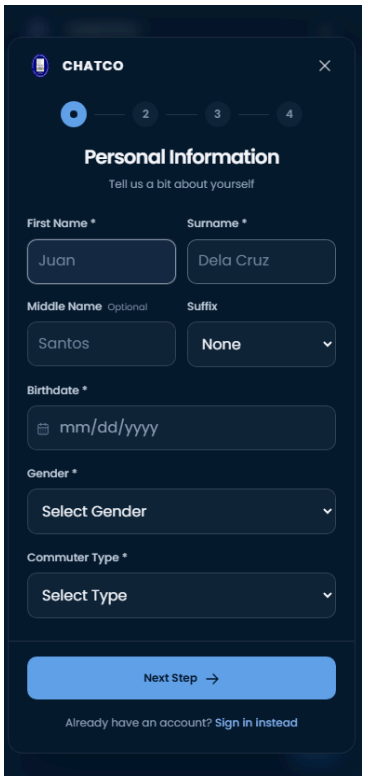


Figure 2.38: Commuter Sign-up

Figure 2.39 displays the forgot and reset password page that allows users who cannot remember their account passwords to recover access to their accounts. Users can request a password reset through their registered email address and verify their identity using an emailed verification code. Once verified, they can proceed with creating a new password.

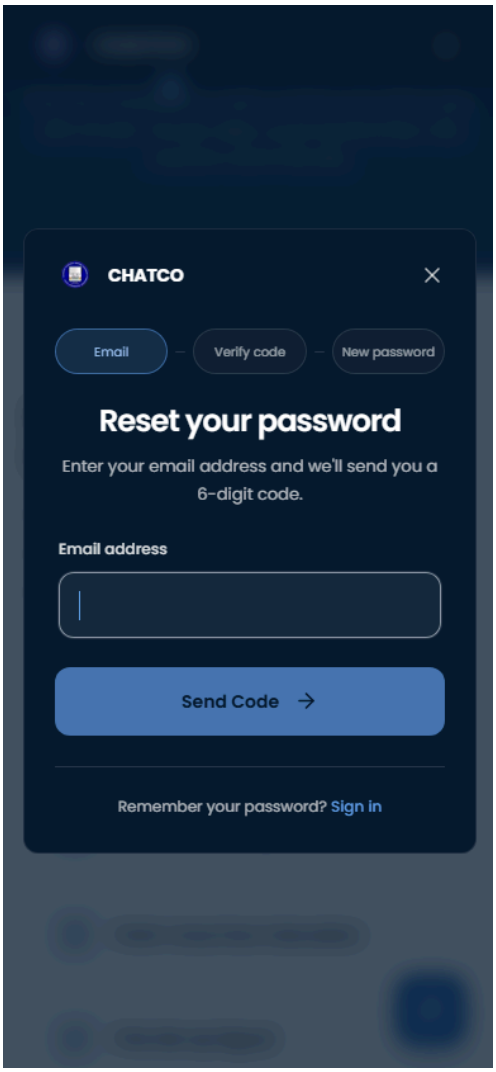


Figure 2.39: Forgot / Reset Password

Commuter Interface

Figure 2.40 displays the Commuter Interface, It is designed for passengers who regularly use jeepneys operating along the Calumpit-Meycauayan route. It provides access to transportation services, including real-time vehicle tracking, smart hailing, fare information, cashless payments, rewards, and safety features. The interface also allows commuters to manage their personal information, monitor transactions, and submit feedback regarding their rides.

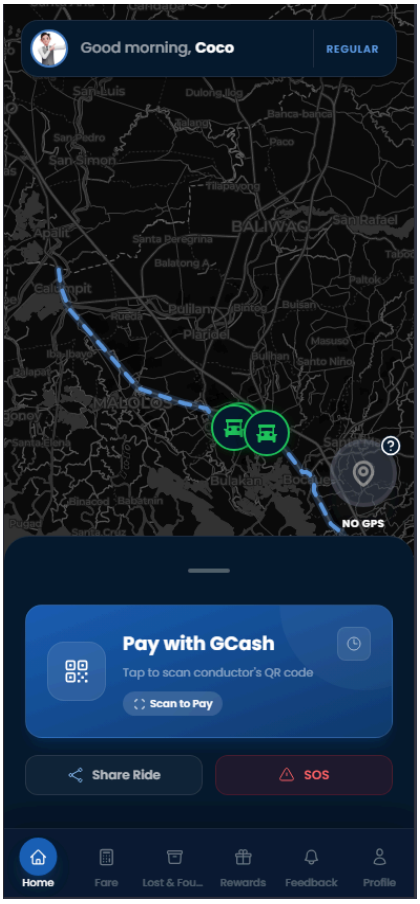


Figure 2.40 Ride and Tracking

Figure 2.41 shows the live map it displays the real-time locations of jeepneys operating along the designated route. It allows commuters to monitor available vehicles and view relevant unit information, including the plate number, assigned crew, and passenger capacity. This feature helps passengers identify nearby transportation options and make informed decisions when planning their trips.

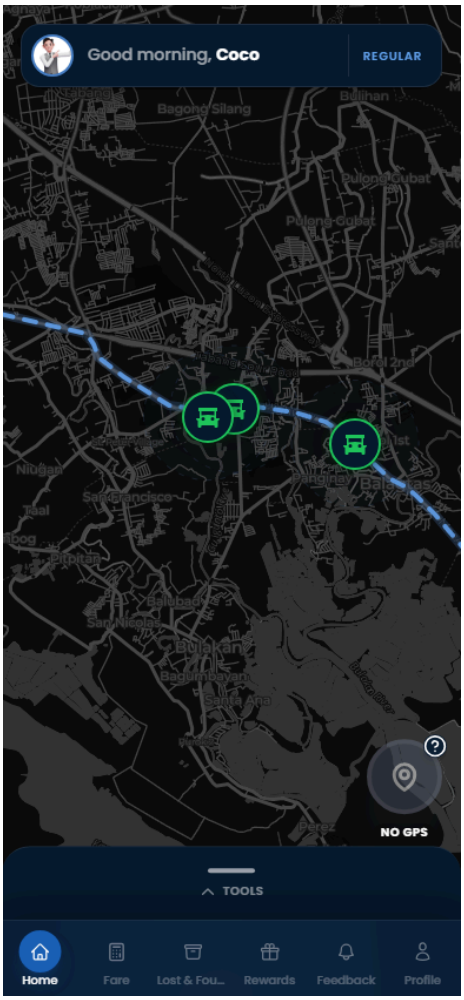


Figure 2.41: Live Map

Figure 2.42 displays the Pick Me Up feature, It allows commuters to send a hailing request to a nearby jeepney when they are waiting for transportation. The request is transmitted to the assigned conductor, who can either accept or reject it depending on the vehicle's availability and operating conditions. This feature provides a more convenient way for commuters to request a ride through the E-CHATCO system.

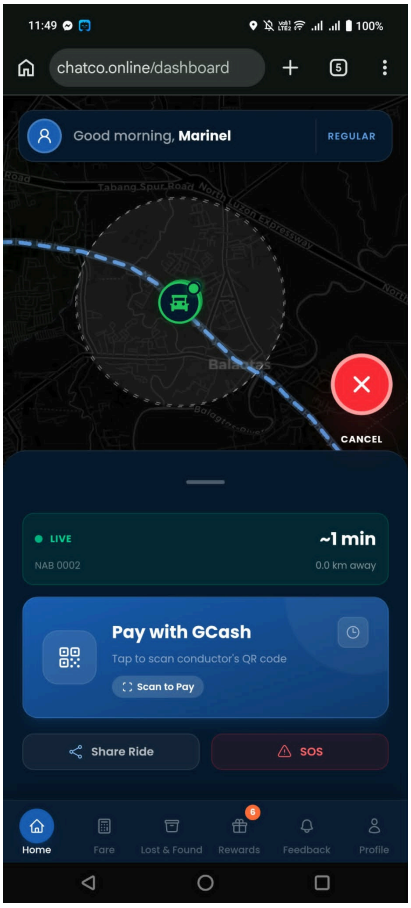


Figure 2.42: Pick Me Up (Hail)

Figure 2.43 displays the fare guide, It provides commuters with fare information between two selected stops along the Calumpit-Meycauayan route. It displays the corresponding fare based on the selected starting point and destination, including applicable discounted rates for qualified passengers. This feature helps commuters determine their expected transportation expenses before boarding a jeepney.

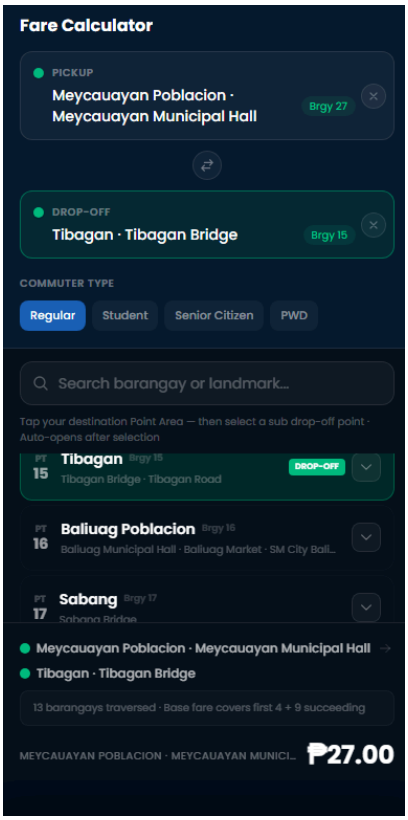


Figure 2.43: Fare Guide

Payment

Figure 2.44 shows the GCash payment feature allows commuters to settle their jeepney fares through the GCash payment service by scanning the QR code presented by the conductor. After scanning, Users are directed through the payment checkout process to complete their transactions before returning to the E-CHATCO application. the system processes the payment based on the recorded fare and applicable discounts. This feature provides an alternative to cash payments and supports convenient digital fare transactions.

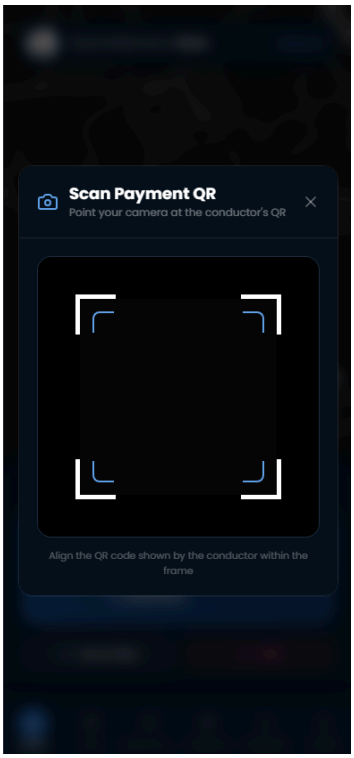


Figure 2.44: Gcash/QR Fare Payment

Figure 2.45 displays the cash receipt claim feature that allows commuters to associate a cash-paid ride with their E-CHATCO account by scanning the QR code printed on their receipt. The scanned code contains the necessary transaction reference to identify and link the payment to the commuter's account. This allows cash-paying passengers to maintain a record of their rides and participate in the system's available reward features.

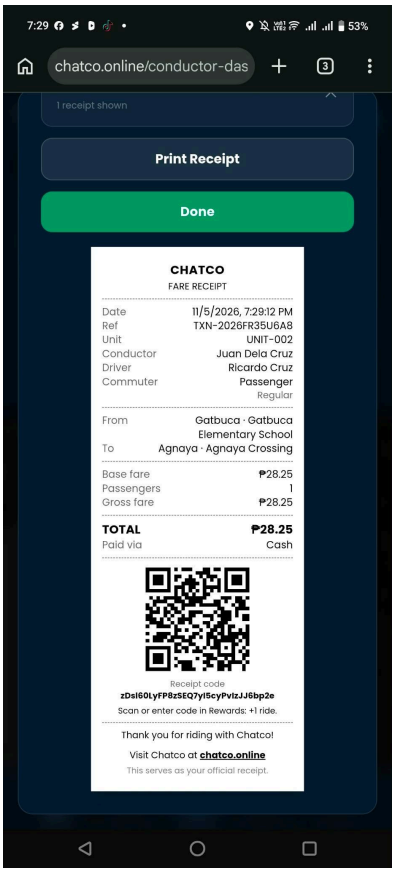


Figure 2.45: Cash Receipt Claim

Figure 2.46 displays the group payment feature that allows one commuter to pay the transportation fares of multiple passengers in a single transaction. The payer can cover the applicable fares for the group, while each participating commuter can view the corresponding shared payment receipt. This feature simplifies fare collection for families, friends, and other groups traveling together.

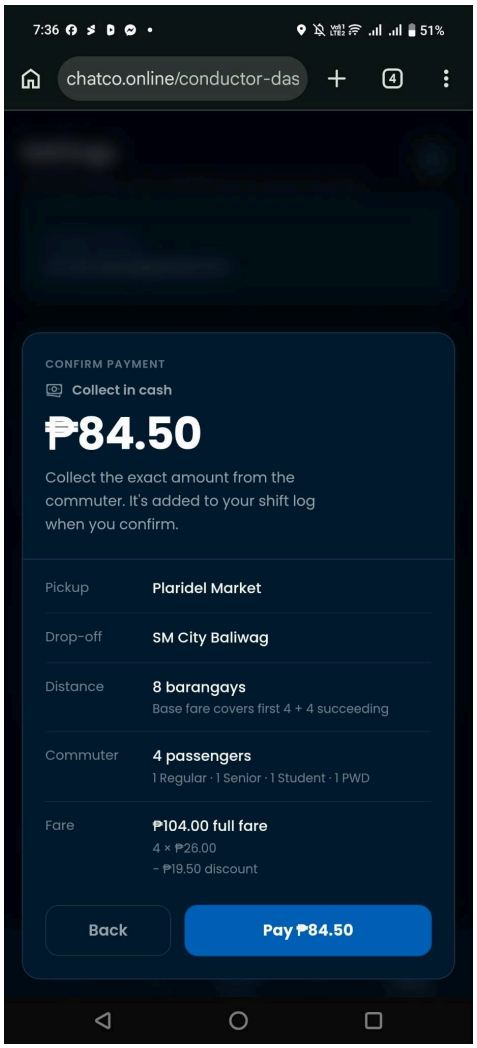


Figure 2.46: Group Payment

Figure 2.47 shows the payment history feature that displays a record of the commuter's previous fare transactions made through the E-CHATCO system. It provides transaction totals and filtering options based on daily, weekly, or monthly periods, as well as a specific date or date range. This allows commuters to monitor their transportation expenses and review their previous payments conveniently.

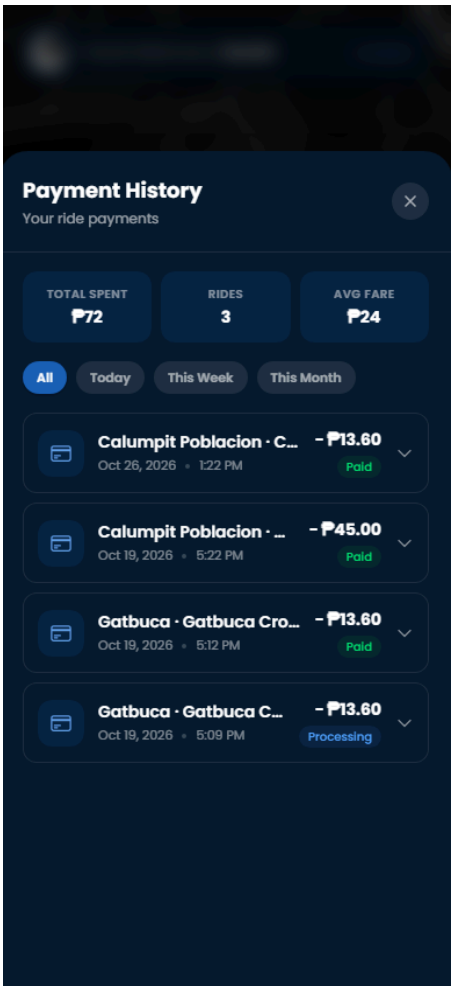


Figure 2.47: Payment History

Rewards

The loyalty rewards feature allows commuters to earn free-ride vouchers after completing a predetermined number of qualifying paid rides. The system monitors eligible transactions and provides a reward once the required number of rides has been reached. This feature encourages continued use of the E-CHATCO system by providing incentives to regular commuters.

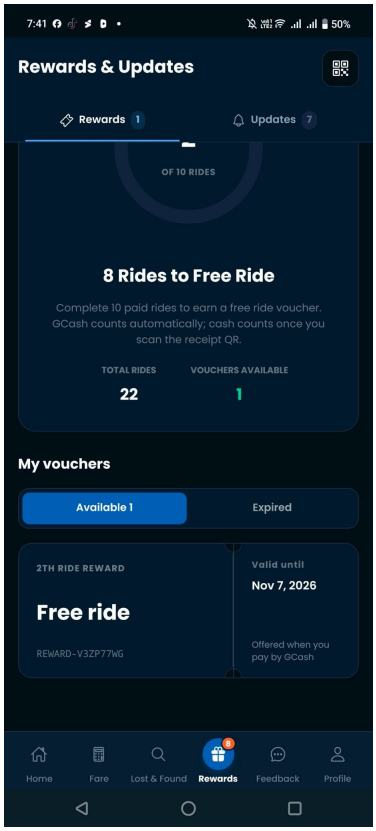


Figure 2.48: Loyalty Rewards

The voucher wallet provides commuters with a dedicated section for viewing and managing their available free-ride and discount vouchers. Users can check their available rewards and apply eligible vouchers when completing fare payments, including group transactions. This feature allows commuters to conveniently access and use the rewards they have earned or received.

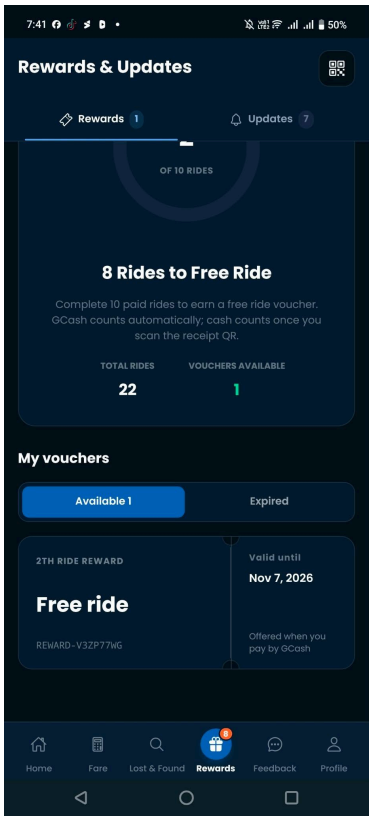


Figure 2.49: Voucher Wallet

Safety

The Share My Ride feature allows commuters to generate a public link containing their live trip location, which can be shared with family members or trusted contacts. Users can also stop sharing their location when they no longer need the service. This feature provides an additional safety measure by allowing selected individuals to monitor the commuter's ongoing journey.

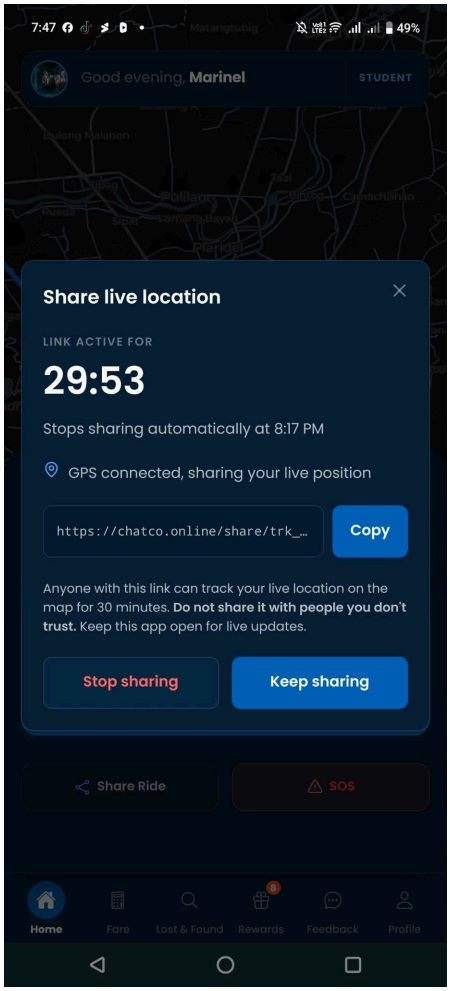


Figure 2.51: Share My Ride

The Lost and Found feature allows commuters to browse items reported as found within the transportation service. Users can monitor available found items, submit claims by providing supporting proof photos, and track the progress of their submitted requests. They may also cancel their claims when necessary, making it easier to coordinate the recovery of lost belongings.

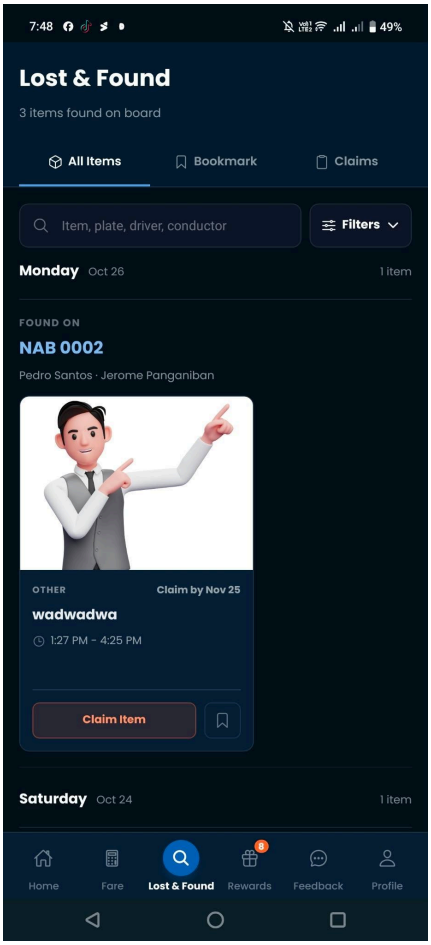


Figure 2.52: Lost and Found

Feedback

The ride feedback feature allows commuters to provide ratings and comments regarding their transportation experience and the crew assigned to the jeepney. Feedback can be submitted after completing a paid ride or by scanning the QR code associated with the unit. This feature allows commuters to share their experiences and provides administrators with information that can help evaluate transportation service quality.

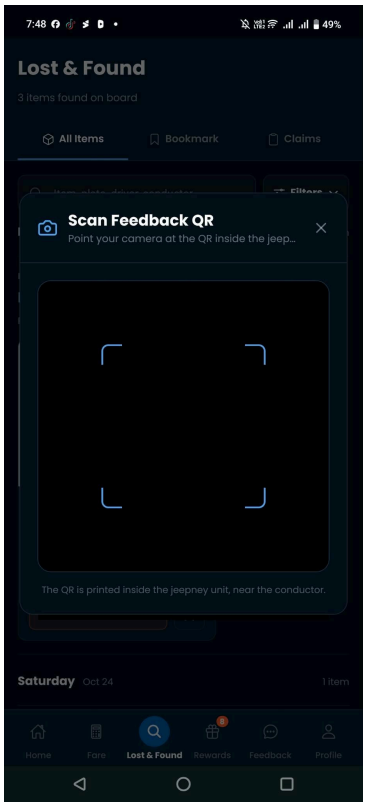


Figure 2.53: Ride Feedback

Profile and Discounts

The profile feature allows commuters to view and manage their personal account information within the system. It includes options for editing account details, updating their profile photo through an image cropper, and selecting between light and dark display themes. This feature provides commuters with control over their account information and application preferences.

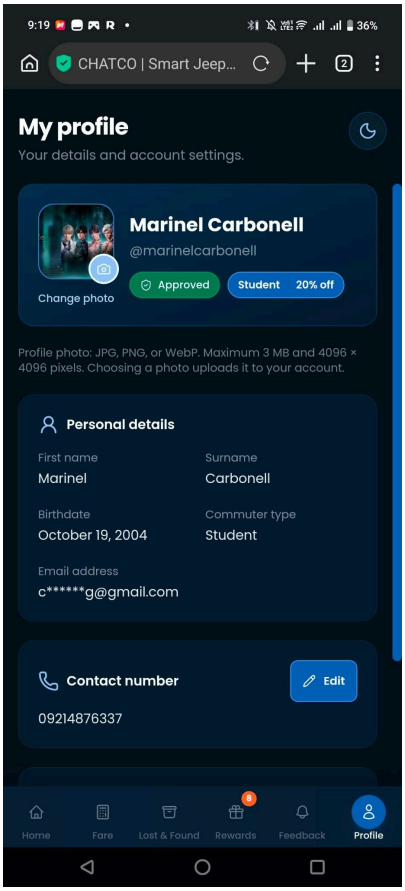


Figure 2.54: Profile

The change email and password feature allows commuters to update their registered email address or account password. The system requires verification through codes sent to the user's email to confirm the requested changes. This process helps maintain account security and ensures that important account updates are authorized by the account owner.

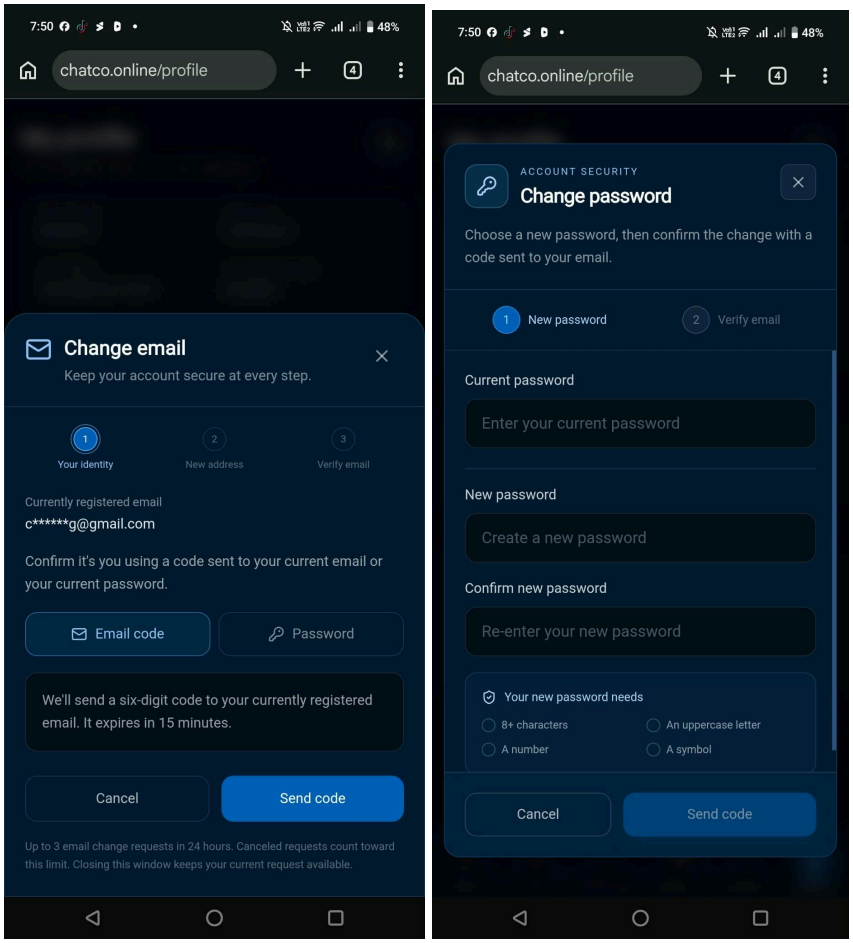


Figure 2.55: Change Email / Password

The discount application feature allows qualified commuters to apply for discounted transportation fares under the Student, Senior Citizen, or Persons with Disabilities (PWD) categories. Applicants must submit the required supporting identification or documents for administrator verification. Once submitted, the application is reviewed and either approved or rejected based on the provided information.

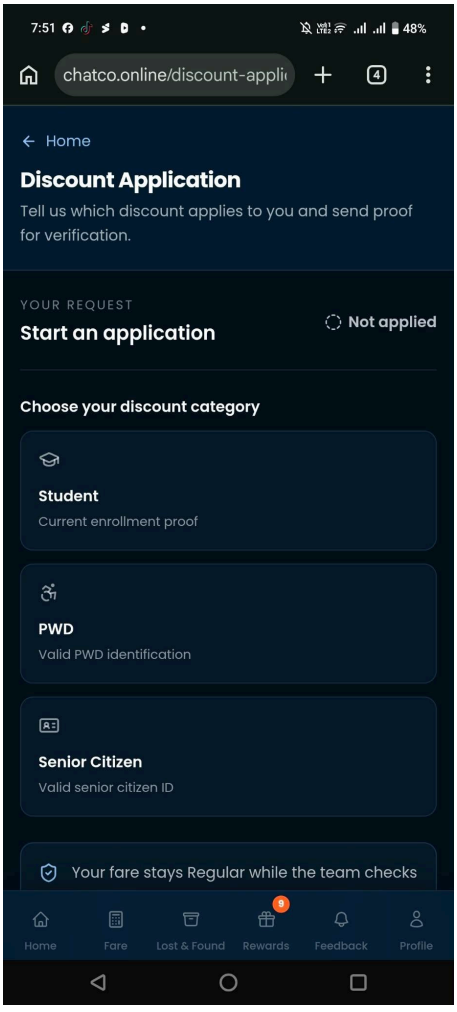


Figure 2.56: Discount Application

The student renewal feature allows commuters with student discount privileges to renew their eligibility at the beginning of each academic term. Users can submit updated information to maintain their student discount status or switch their account classification back to the regular fare category. The system also provides reminders to help students maintain their discount eligibility.

Figure 2.57: Student Renewal

The announcements feature allows commuters to receive important updates and notices published by the administrators. These announcements can be accessed through the notification bell, which also indicates the number of unread notifications. This feature helps keep commuters informed about transportation updates, service changes, and other relevant system announcements.

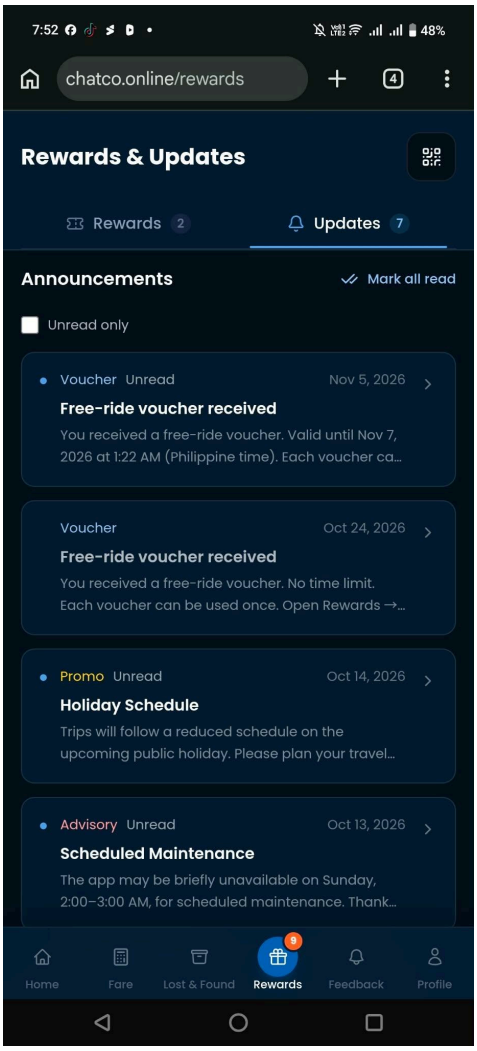


Figure 2.58: Announcements

Conductor Interface

The Conductor Interface is designed for the on-board personnel responsible for managing daily jeepney operations and assisting passengers during their journeys. It provides functions for vehicle verification, shift management, real-time location broadcasting, passenger hailing, fare collection, and transaction recording. It also includes end-of-day procedures, remittance management, and performance monitoring to support the conductor's operational responsibilities.

The unit verification feature allows conductors to select the jeepney and its assigned driver before beginning their daily shift. The selected vehicle and driver information are verified through the system to ensure that the correct personnel and unit are associated with the active operation. Once the required information is confirmed, the conductor can proceed with starting the shift.

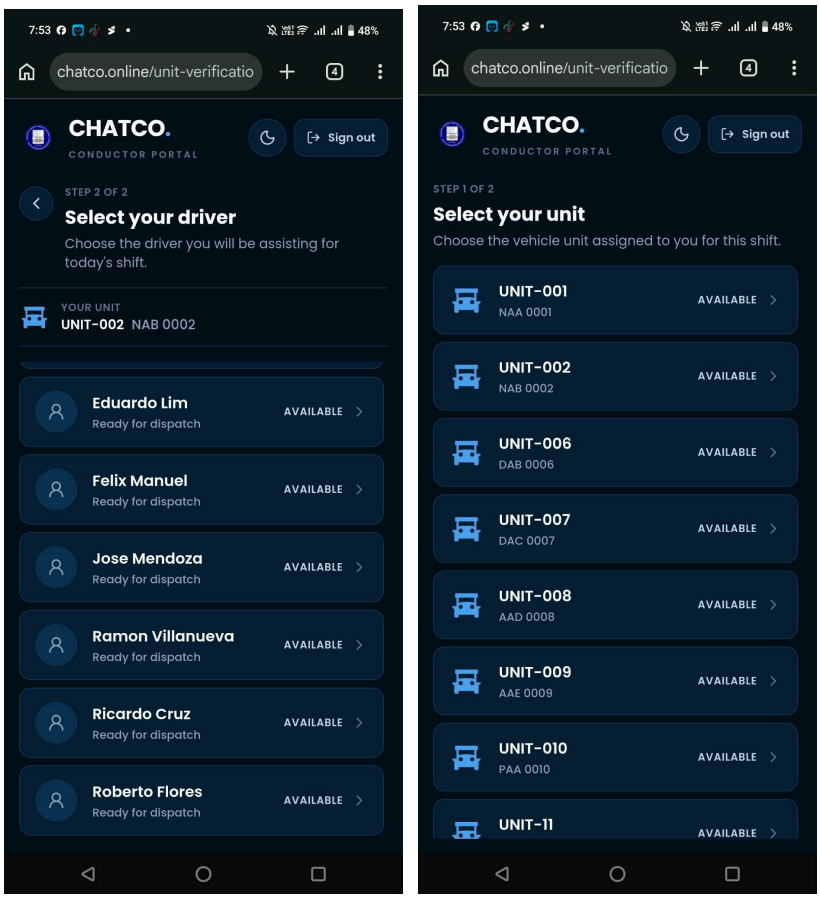


Figure 2.59: Unit Verification

The live location feature allows the conductor's mobile device to broadcast the current position of the assigned jeepney to the E-CHATCO system. The transmitted location is used to update the live map, allowing commuters and administrators to monitor the vehicle's movement along the designated route. This feature supports real-time transportation tracking and fleet monitoring.

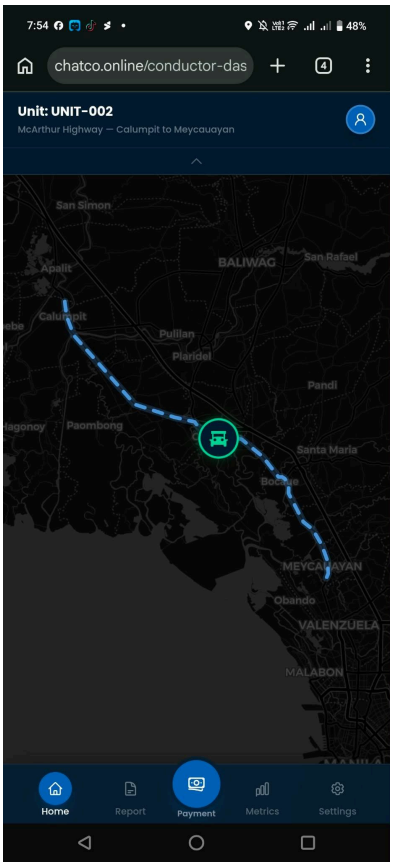


Figure 2.60: Live Location

The capacity and break status feature allows conductors to update the current availability of their assigned jeepney. Conductors can mark the vehicle as full or available depending on its passenger capacity and can also indicate whether they are currently on or off break. These updates help provide commuters and administrators with relevant information about the vehicle's operating status.

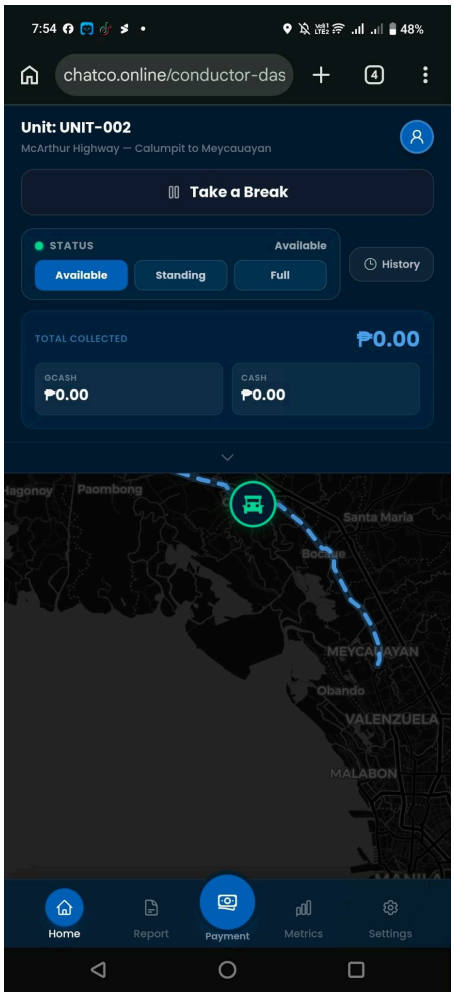


Figure 2.61: Capacity and Break Status

The hail requests feature allows conductors to view incoming Pick Me Up requests from commuters waiting for transportation. Requests are displayed with their corresponding locations on the map, allowing conductors to identify where passengers are waiting. The conductor can accept or reject each request depending on the vehicle's availability and current operating conditions.

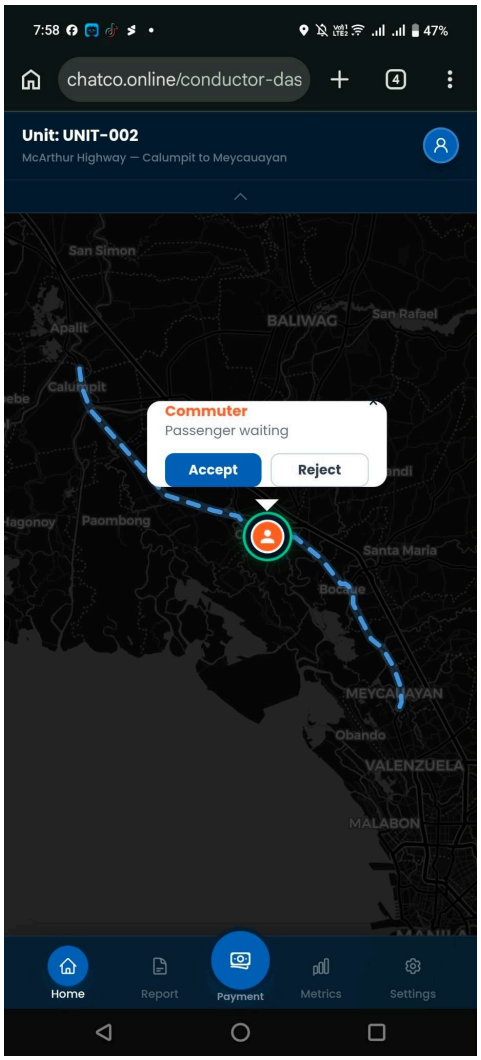


Figure 2.62: Hail Requests

The record fares feature allows conductors to process passenger payments during an active shift. It supports the recording of cash fares and the initiation of GCash QR payments, including

transactions involving group payments and applicable commuter discounts. This feature helps organize fare collection and ensures that passenger transactions are properly recorded within the system.

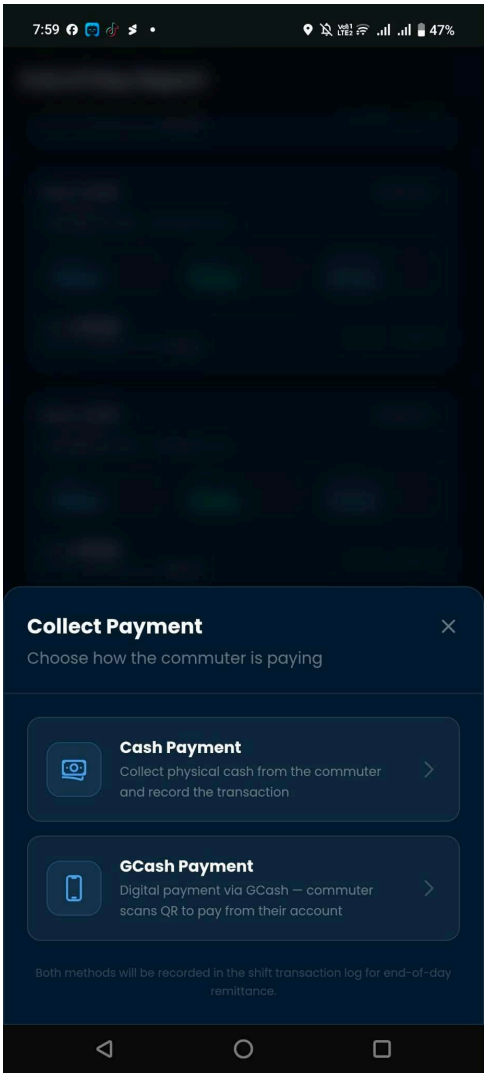


Figure 2.63: Record Fares

The fare calculator provides conductors with a quick way to determine the transportation fare between two selected stops. It displays the corresponding fare based on the route's fare information and applicable discount rates. This feature assists conductors in collecting the correct amount from passengers during their trips.

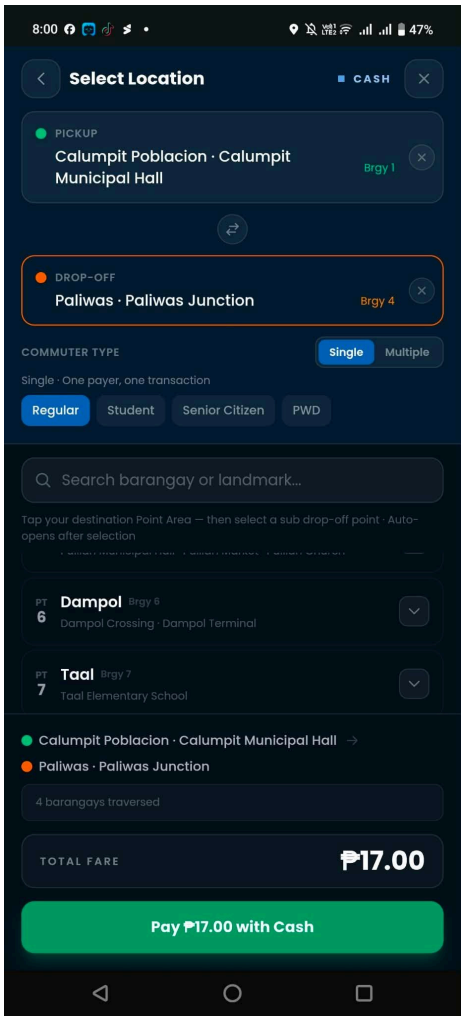


Figure 2.64: Fare Calculator

The printed receipts feature allows conductors to generate transaction receipts after completing passenger payments. Each receipt contains a unique reference number and a QR code that commuters can scan to claim and associate their cash-paid rides with their accounts. The system also supports optional automatic printing to make the receipt issuance process more convenient.

Figure 2.65: Printed Receipts

The transaction history feature displays the payment records generated during the conductor's active shift. It allows conductors to review recorded transactions and filter the information according to payment method and date. This feature helps conductors monitor their collected fares and verify transaction records throughout their operations.

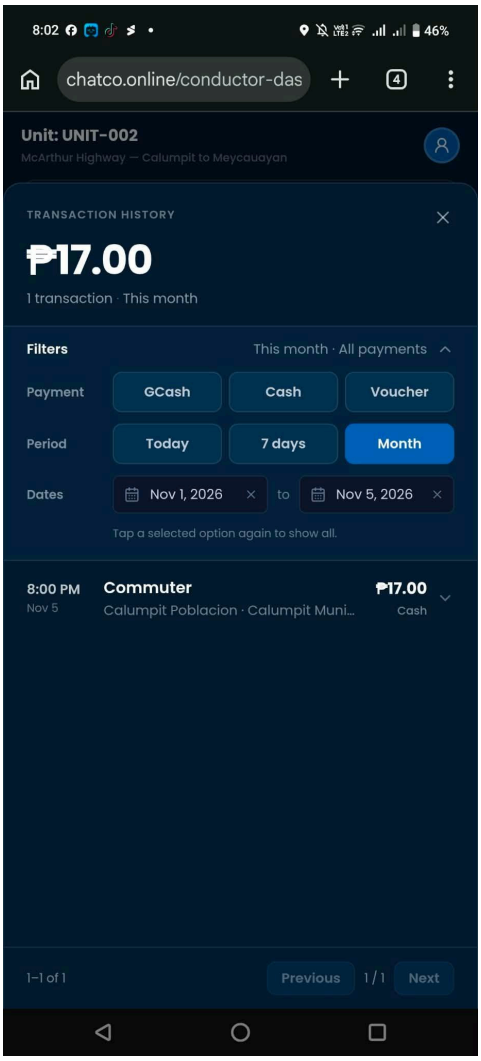


Figure 2.66: Transaction History

The offline support feature allows conductors to continue recording cash fare transactions when the device temporarily loses its internet connection. The system displays a connection status banner to inform the conductor about the current network condition. Once the connection is restored, the recorded offline cash transactions are synchronized with the system to update the corresponding records.

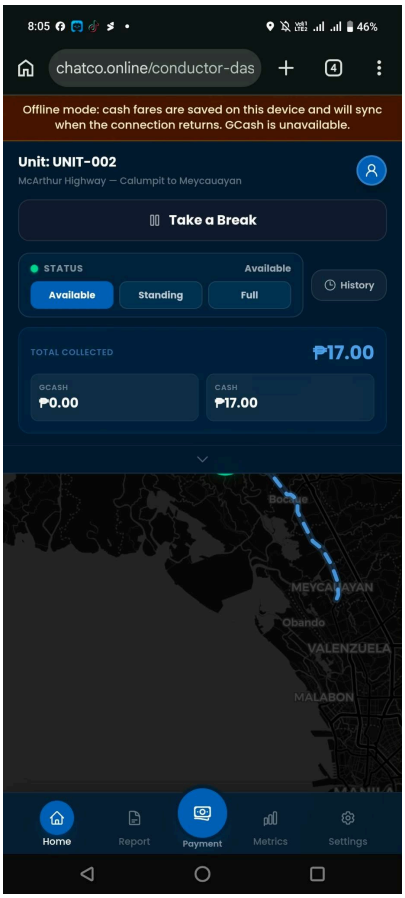


Figure 2.67: Offline Support

The end-of-day close-out feature allows conductors to formally end their active shift and review the summary of their daily operations. It presents the relevant shift totals, including the recorded transactions and collected fares, for checking before completing the closing process. This feature helps ensure that the conductor's daily activities are properly reviewed and finalized.

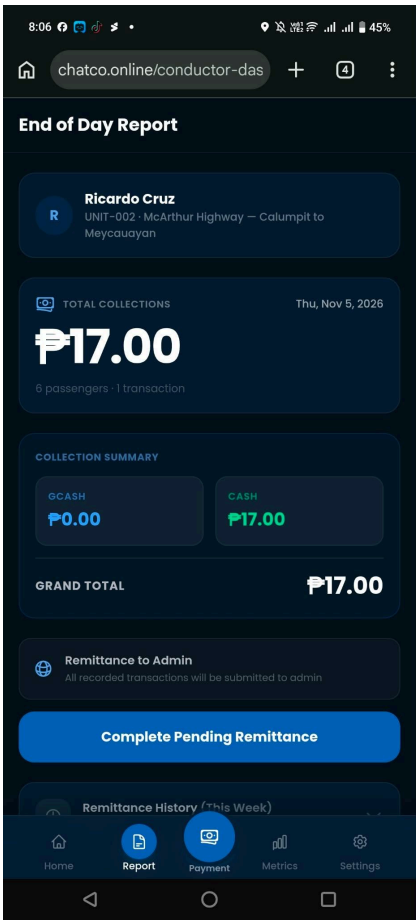


Figure 2.69: End-of-Day Close-out

The remittance feature allows conductors to submit the collected daily earnings to the designated management or operator. It provides access to the official remittance report and a history of previously submitted remittances. This feature supports the recording and monitoring of daily fare collections and the corresponding financial submissions.

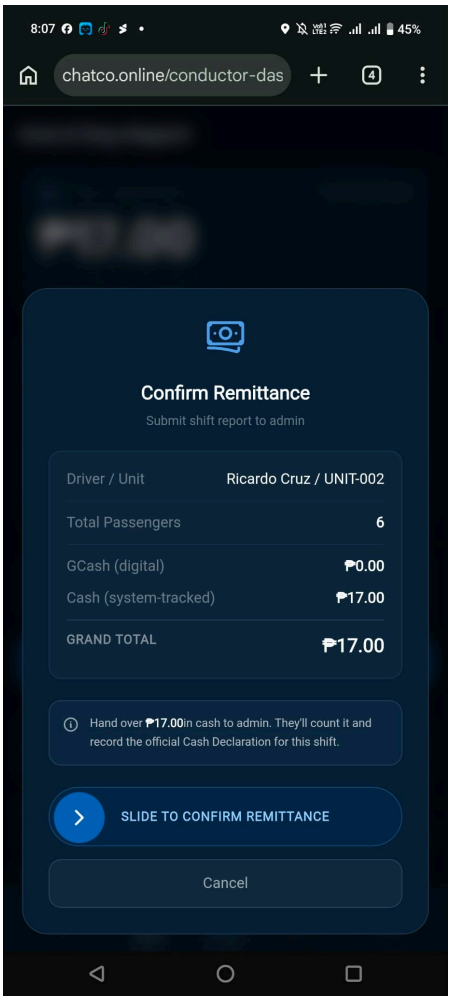


Figure 2.70: Remittance

The metrics and ratings feature provides conductors with a summary of their work performance and transportation activities. It displays information such as earnings, completed trips, and ratings submitted by commuters. This allows conductors to monitor their operational performance and review passenger feedback regarding their service.

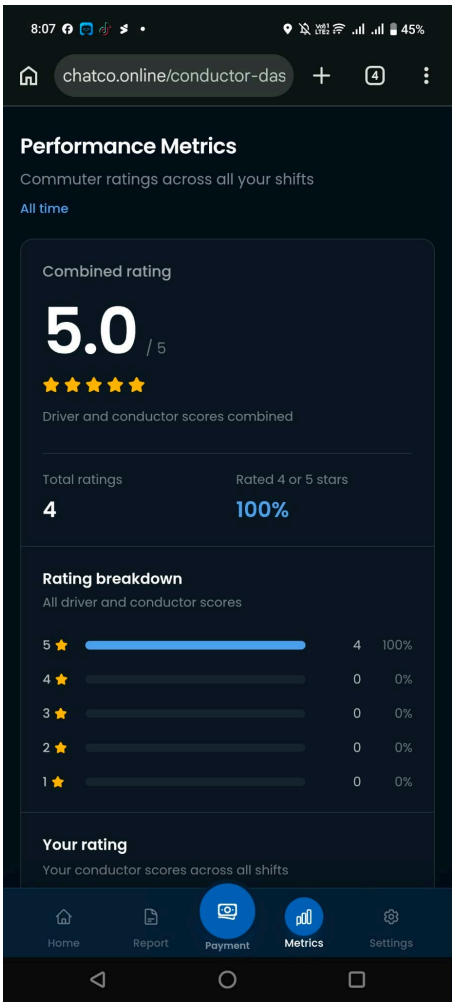


Figure 2.71: Metrics and Ratings

The settings feature allows conductors to manage selected application and account preferences. It includes options for updating their profile photo, accessing account settings, and clearing the application's cache. These functions help conductors maintain their account information and manage basic application settings.

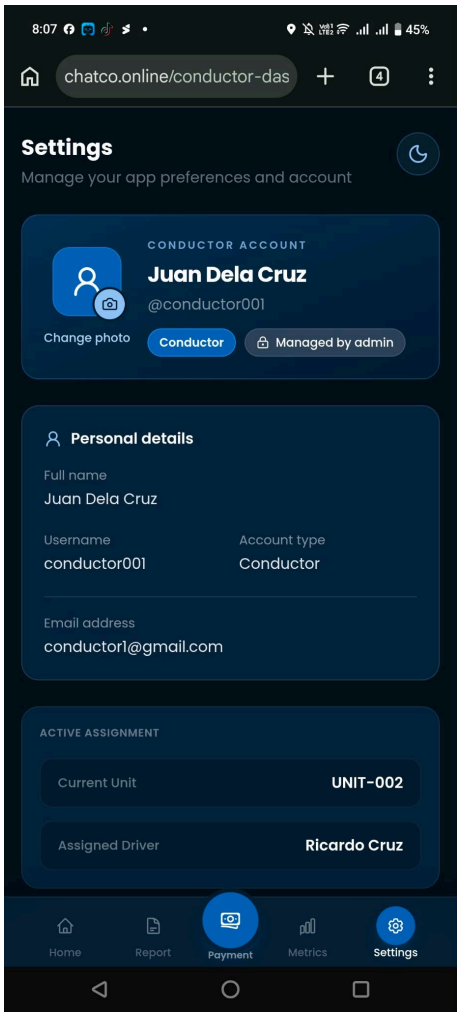


Figure 2.72: Settings

Administrator Interface

The Administrator Interface is designed for the personnel responsible for managing the E-CHATCO system, monitoring transportation operations, and overseeing the jeepney fleet. It provides access to user management, vehicle records, financial transactions, route configuration, safety monitoring, and system administration functions. Through this interface, administrators can supervise daily activities and manage the different operational components of the Calumpit-Meycauayan transportation service.

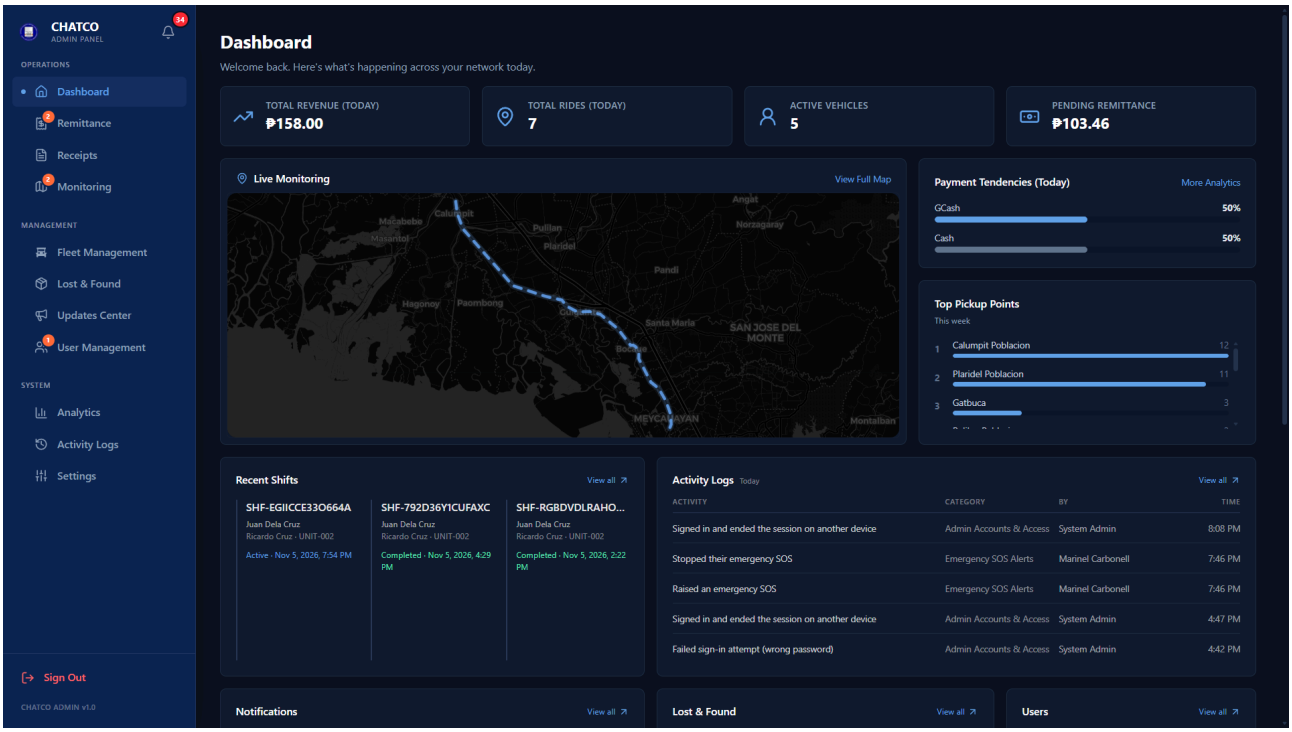


Figure 2.73: Dashboard

The analytics feature provides administrators with summarized information about the transportation service's operational and financial performance. It presents data related to ridership, revenue, and payment trends based on selected date ranges. This allows administrators to examine historical records, identify operational patterns, and evaluate the performance of the transportation service.

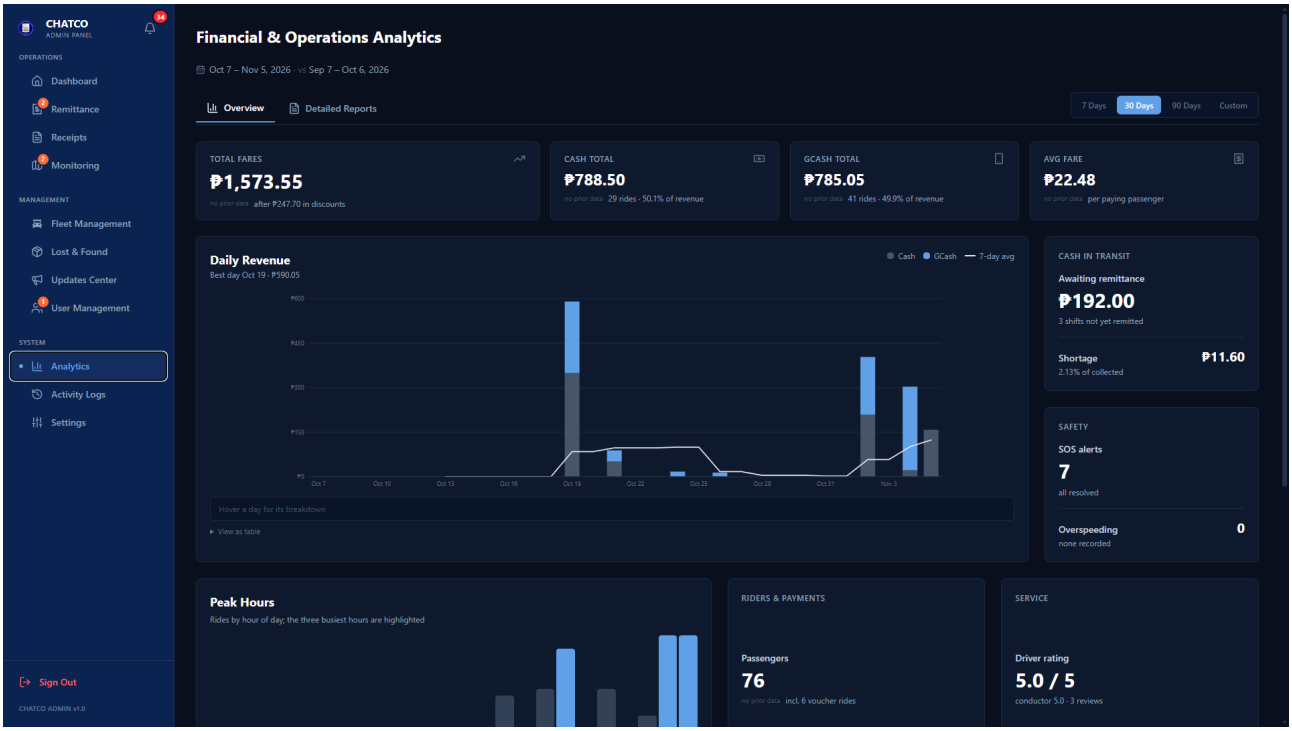


Figure 2.74: Analytics

The live monitoring feature allows administrators to observe the movement and operating conditions of jeepneys through a real-time fleet map. It also displays overspeeding events and commuter demand zones to help administrators identify relevant transportation activities. This feature supports fleet supervision and provides information that can assist in monitoring vehicle operations and passenger demand.

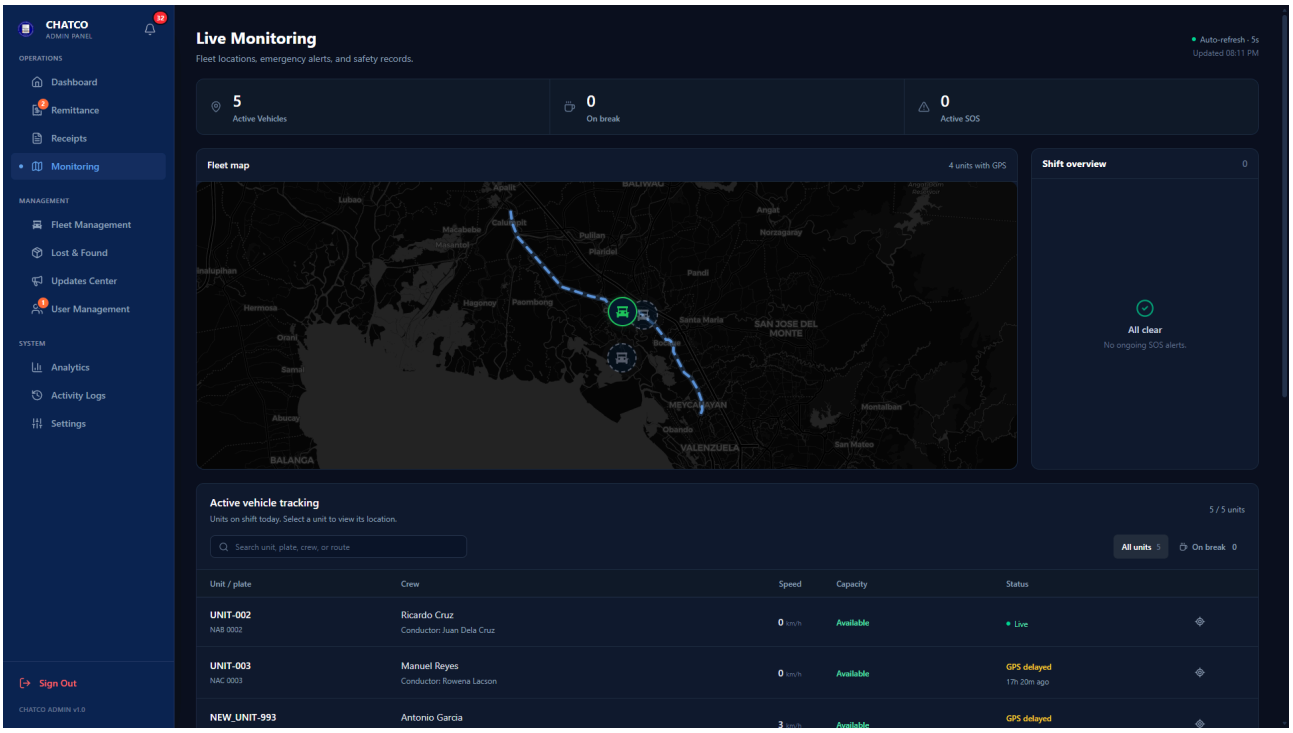


Figure 2.75: Live Monitoring

The user management feature allows administrators to access and manage registered user accounts within the E-CHATCO system. It provides options for viewing and editing user information, suspending or reactivating accounts, and reviewing user activity. This feature helps administrators maintain organized account records and manage access to the platform.

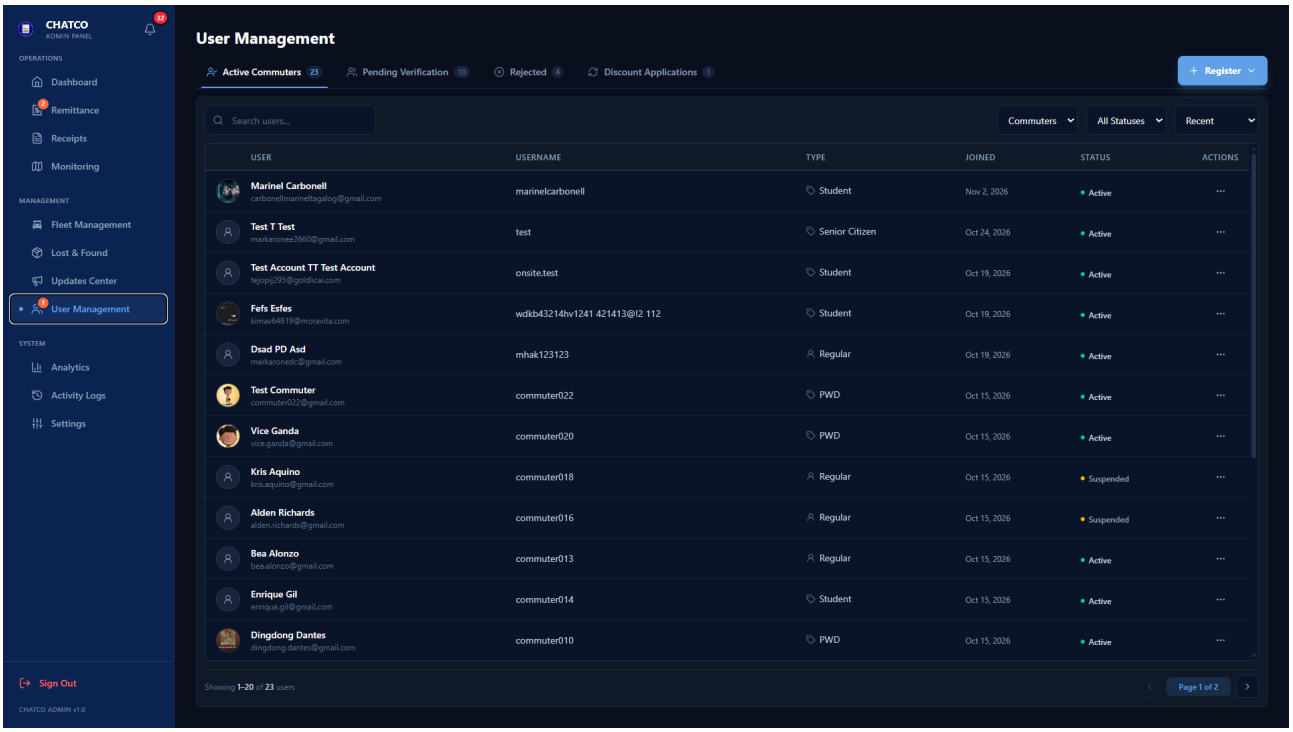


Figure 2.76: User Management

The registrations feature allows administrators to review commuter account applications submitted through the registration process. Administrators can examine the submitted information and supporting documents before approving or rejecting each application. The module also provides access to previously rejected registrations for reference and monitoring.

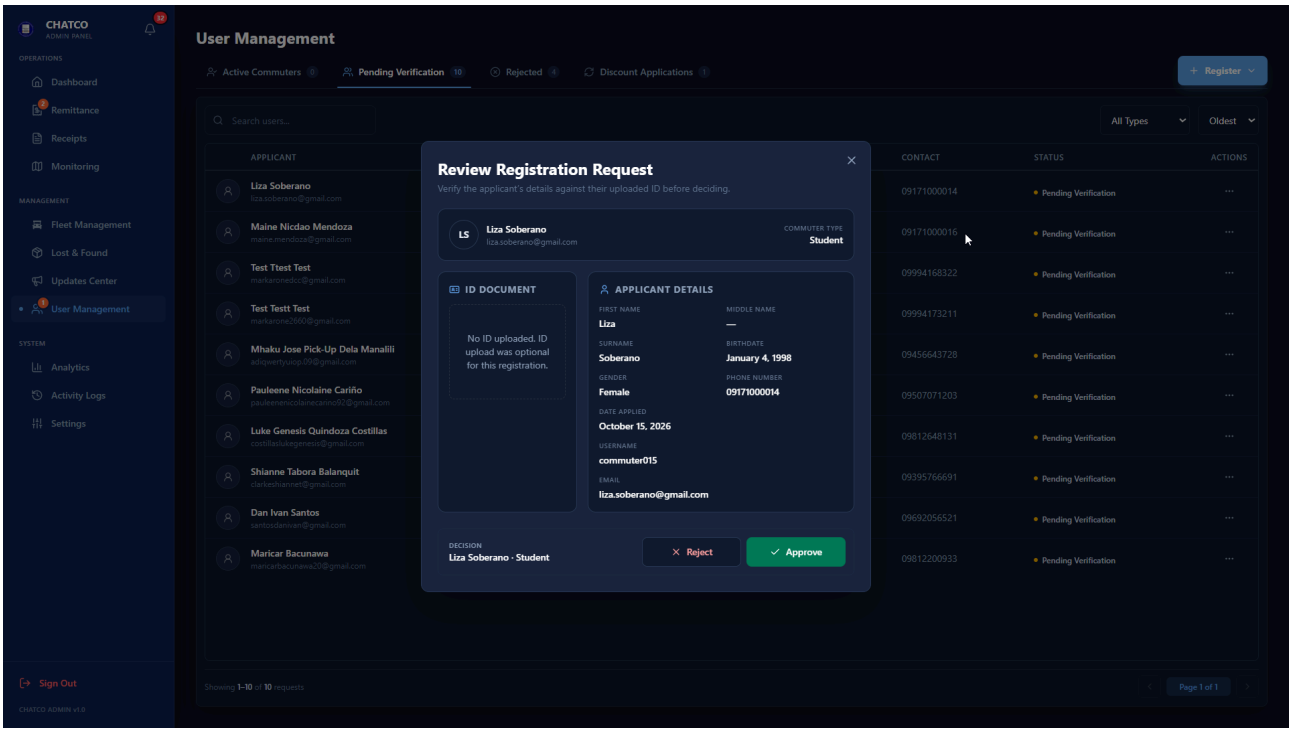


Figure 2.77: Registrations

The discount applications feature allows administrators to review requests submitted by commuters applying for Student, Senior Citizen, or PWD fare discounts. It provides access to the submitted identification and supporting documents needed to verify eligibility. Administrators can evaluate each application and determine whether the requested discount classification should be approved or rejected.

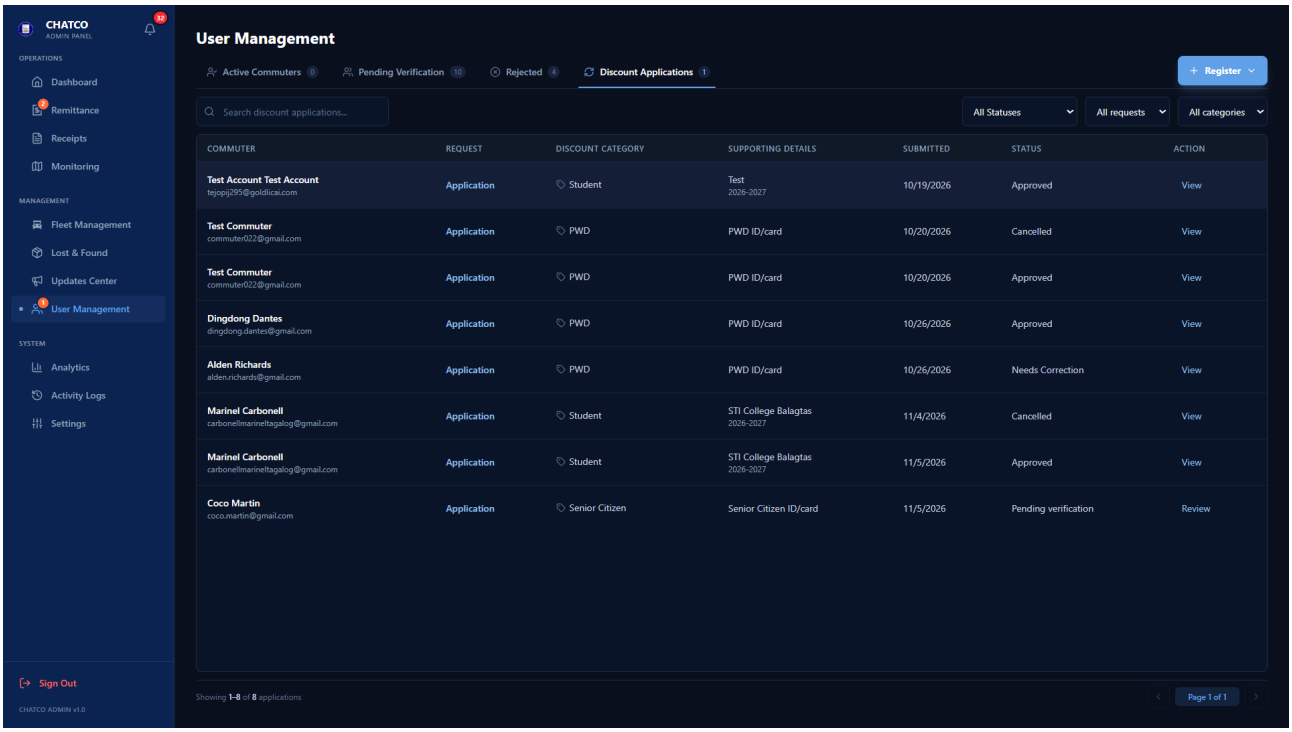


Figure 2.78: Discount Applications

The fleet management feature allows administrators to manage the transportation units and personnel associated with the E-CHATCO system. It includes functions for maintaining vehicle records, managing driver information with uploaded license images, and managing conductor records. The module also provides access to shift logs, allowing administrators to review the operational history of the assigned personnel and vehicles.

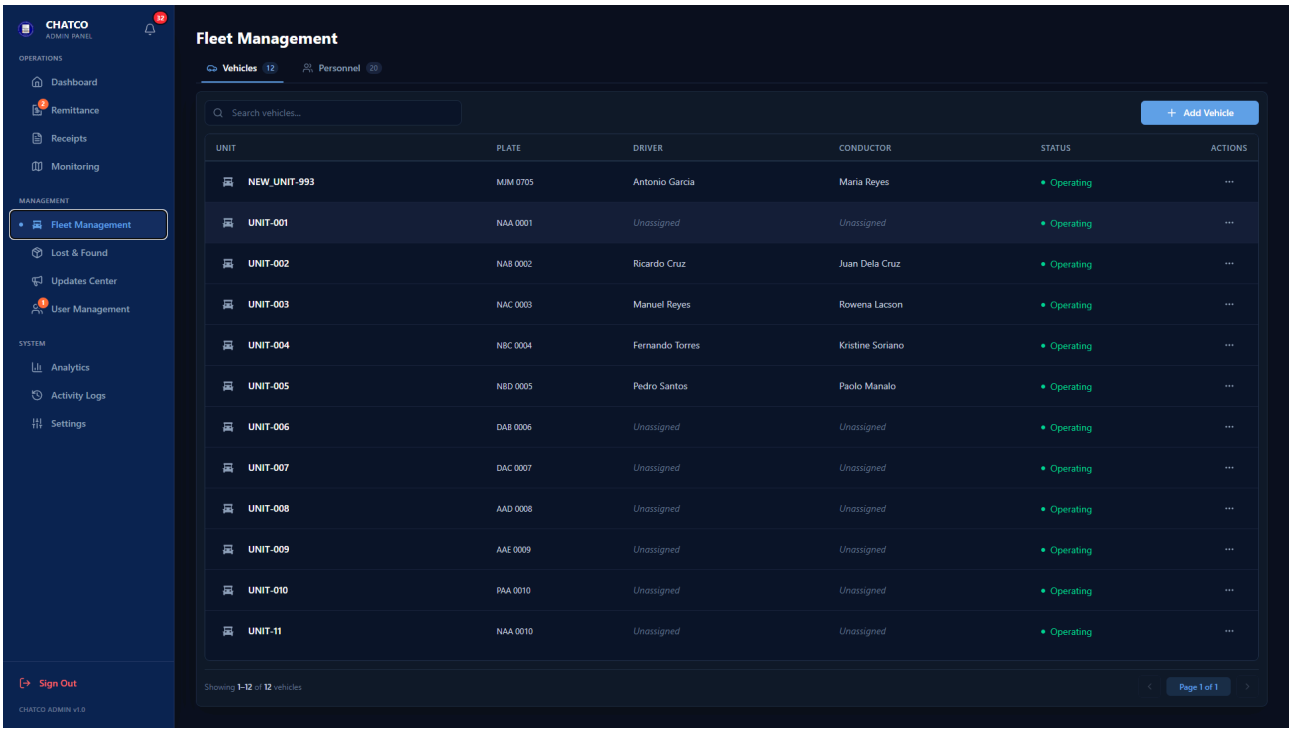


Figure 2.79: Fleet Management

The conductor accounts feature allows administrators to create and manage accounts assigned to conductors operating within the transportation service. It provides options for creating new accounts, disabling existing accounts, and resetting user credentials when necessary. This feature helps administrators control conductor access and maintain the accuracy of personnel account records.

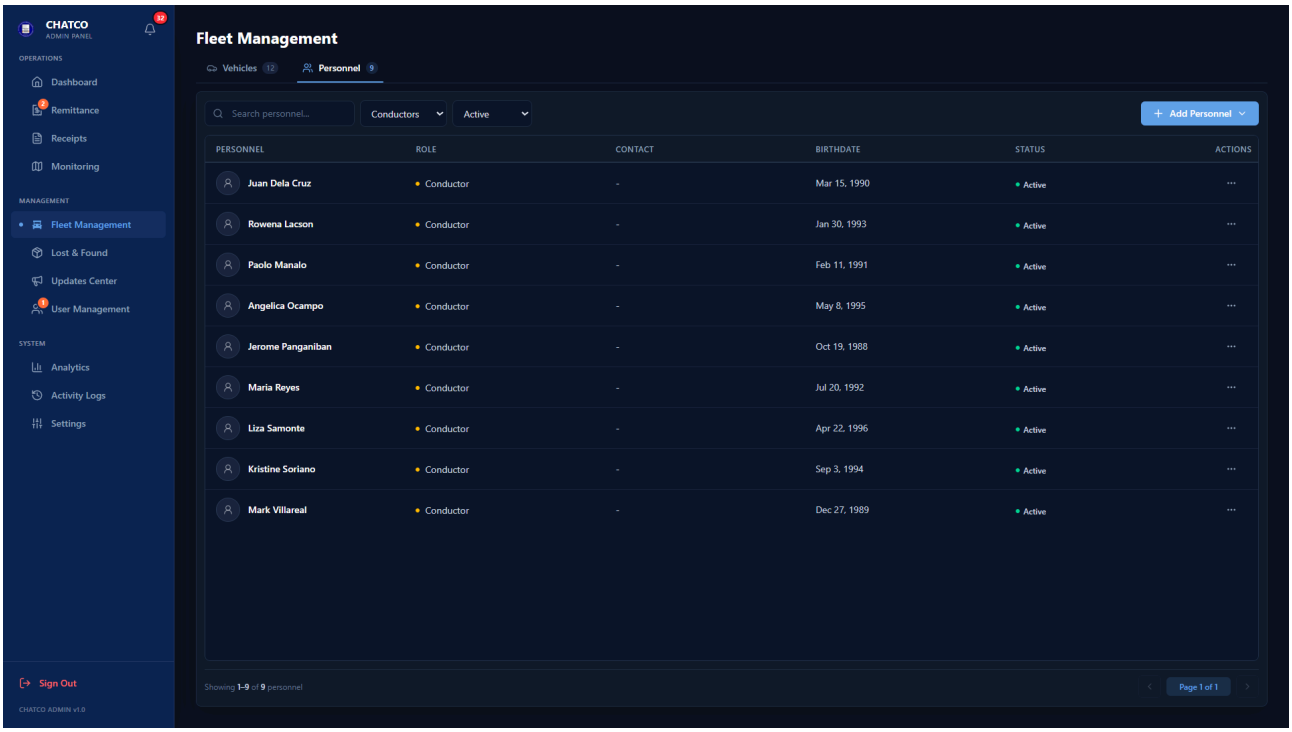


Figure 2.80: Conductor Accounts

The admin accounts feature allows authorized administrators to create additional administrator accounts for personnel responsible for managing the system. It supports the addition of other administrators who require access to the platform's management functions. This feature helps organize administrative access as the operational management requirements of the system increase.

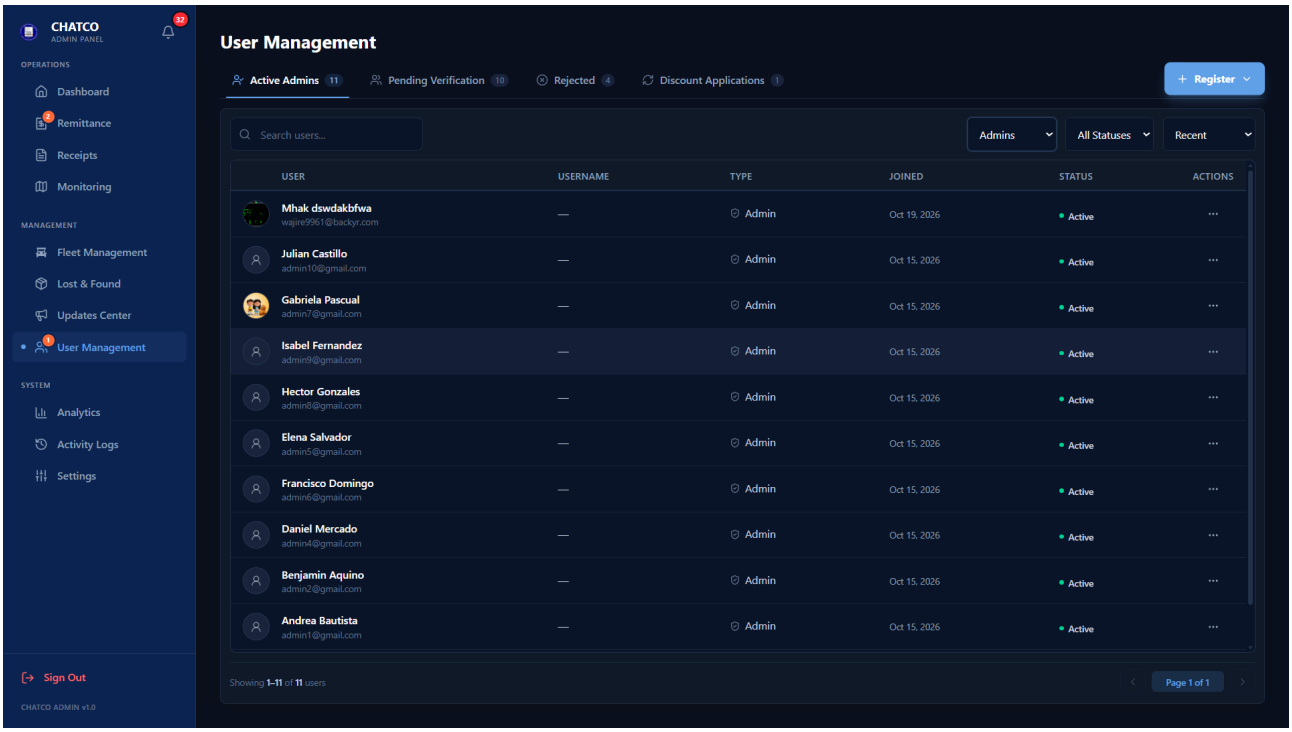


Figure 2.81: Admin Accounts

The receipts and transactions feature allows administrators to search and review payment records generated throughout the E-CHATCO system. It provides access to transaction details and their corresponding receipts, allowing administrators to examine recorded fare payments. This feature supports financial monitoring and helps verify the transactions made by commuters and conductors.

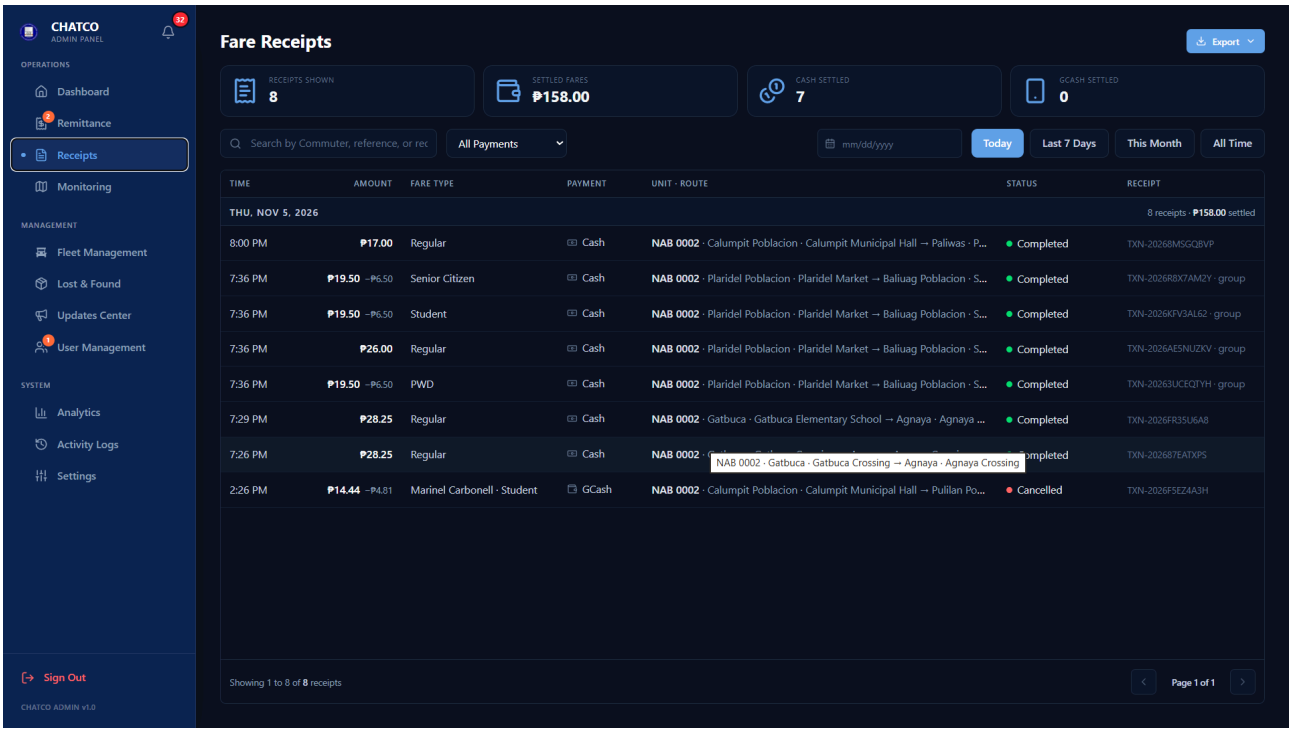


Figure 2.82: Receipts / Transactions

The remittance feature allows administrators to review the daily earnings submitted by conductors after completing their shifts. It provides access to remittance records and supports the recording of cash declarations to document the amounts submitted by conductors. This feature helps administrators monitor daily collections and maintain organized financial records.

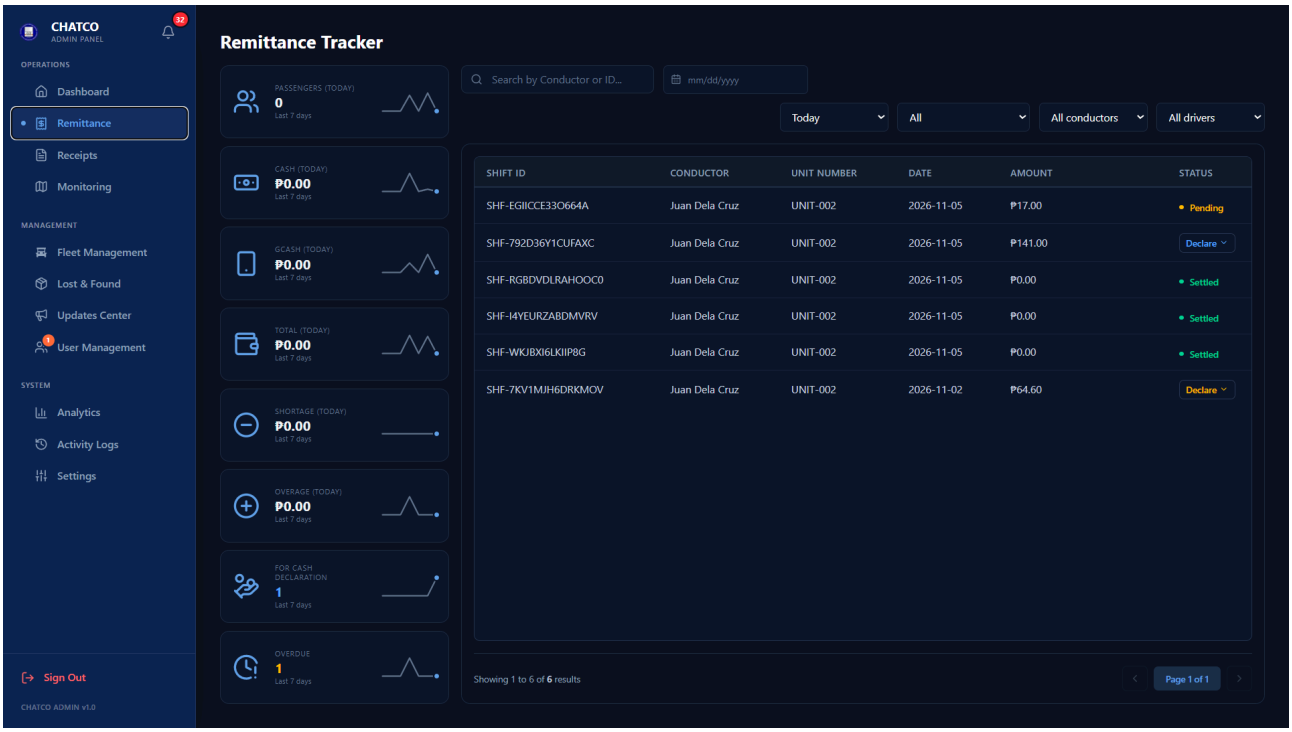


Figure 2.83: Remittance

The fare matrix feature allows administrators to manage the designated stops and corresponding fare amounts along the Calumpit-Meycauayan route. It provides functions for editing route stops and updating the fares assigned between different locations. This feature ensures that the transportation fare information used by commuters and conductors can be maintained according to the applicable fare structure.

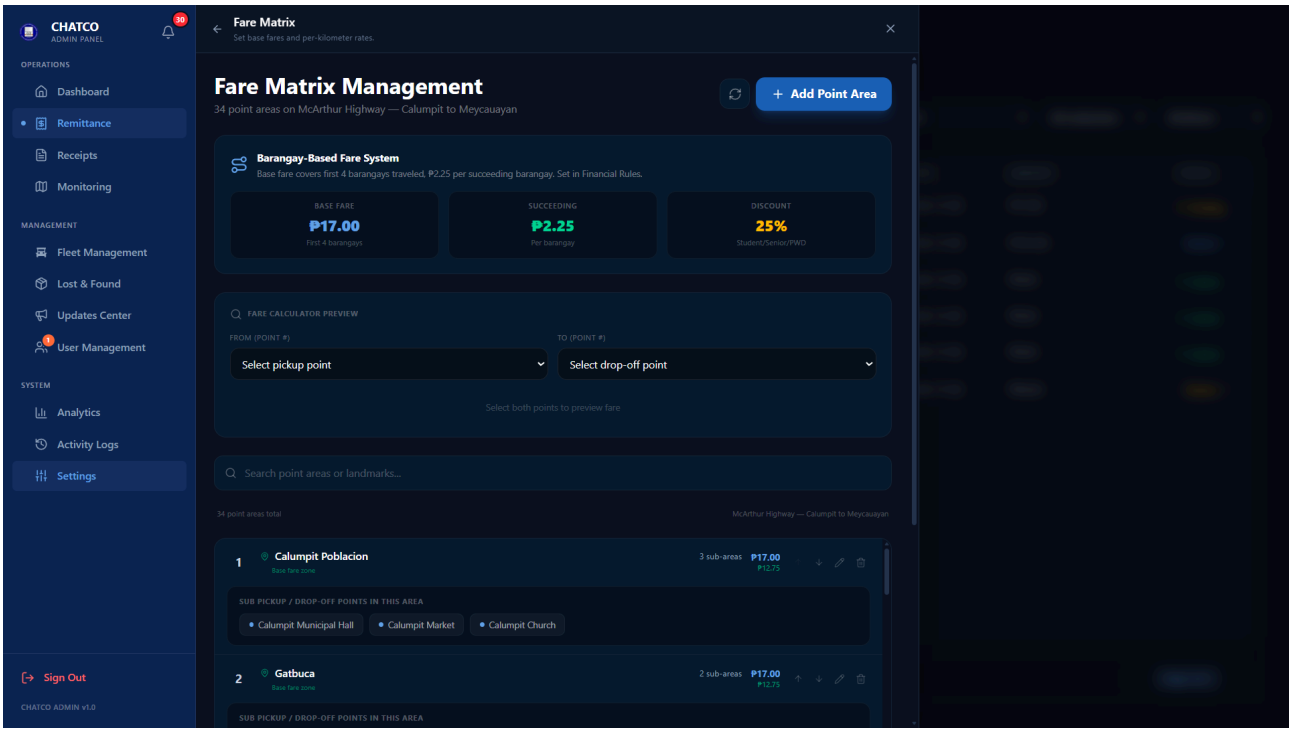


Figure 2.84: Fare Matrix

The financial rules feature allows administrators to configure important financial settings used by the E-CHATCO system. These include the base fare, commuter discount rates, and the required number of paid rides before a loyalty reward is granted. This feature helps maintain consistent fare calculations and reward conditions throughout the platform.

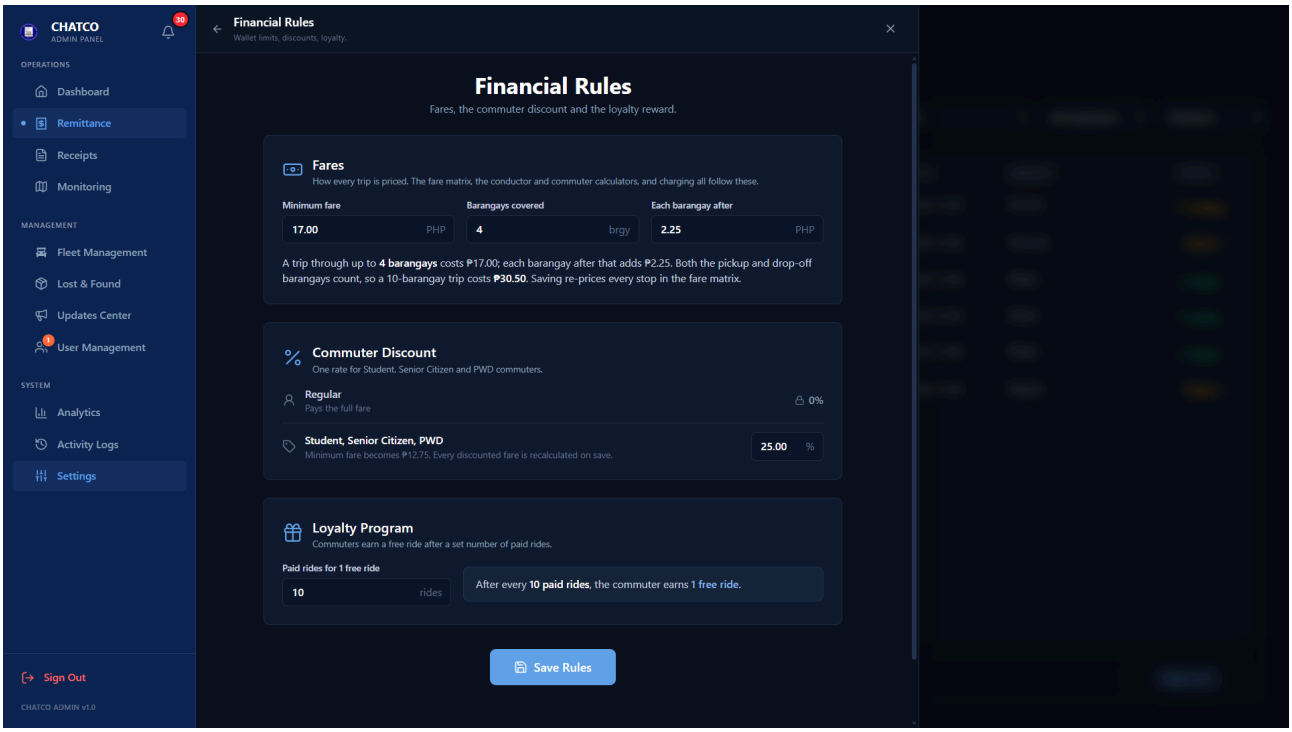


Figure 2.85: Financial Rules

The voucher generator feature allows administrators to create batches of voucher codes for distribution to eligible commuters. These vouchers may provide free rides or transportation discounts depending on their assigned purpose. This feature supports the management of promotional rewards and the loyalty program implemented within the system.

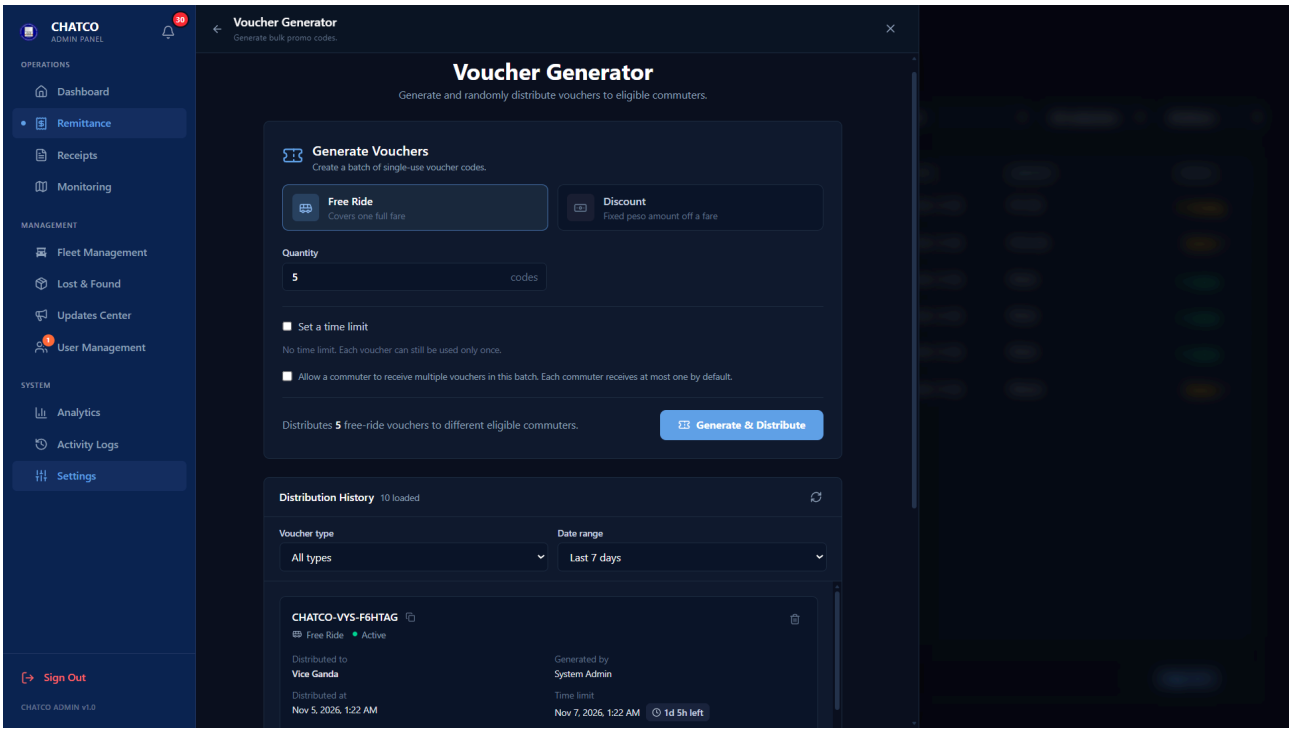


Figure 2.86: Voucher Generator

The receipt settings feature allows administrators to customize the information displayed on printed transaction receipts. It also provides an option to configure whether receipts should be printed automatically after transactions. This feature helps standardize the receipt format and manage the receipt printing behavior used by conductors.

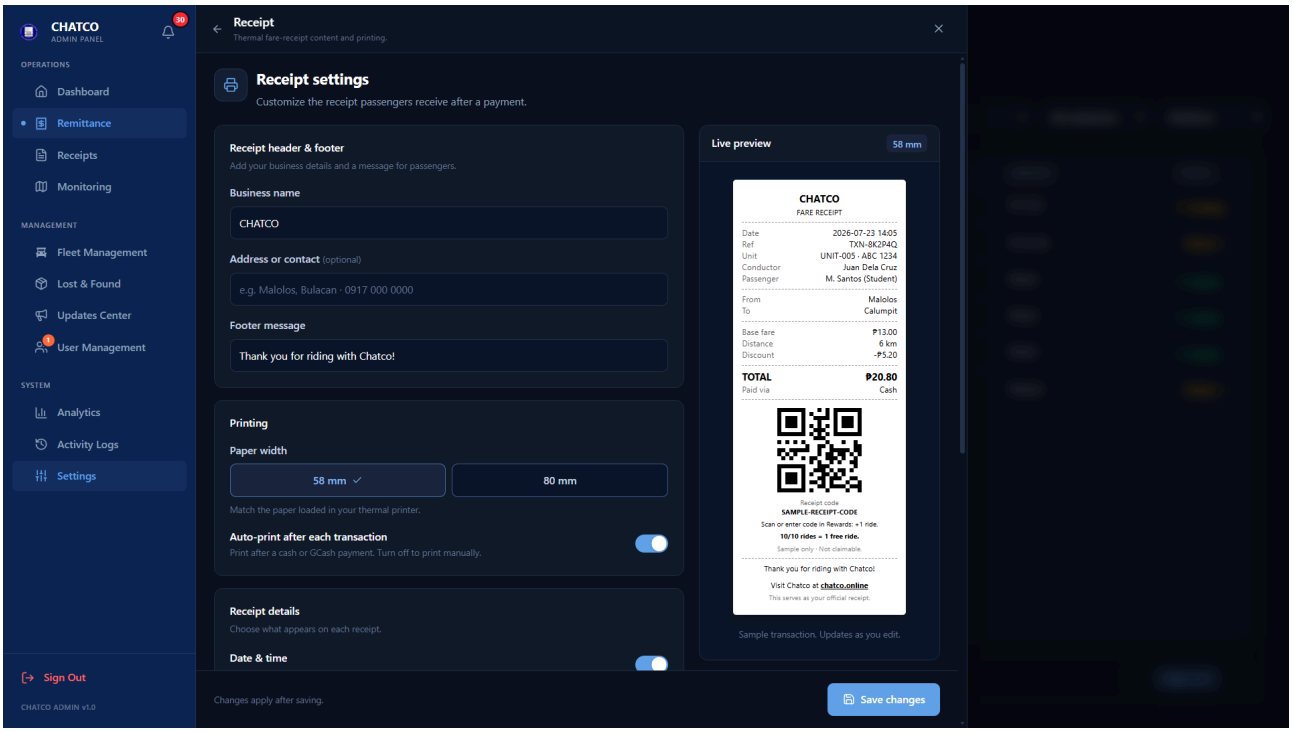


Figure 2.87: Receipt Settings

The routes feature allows administrators to create and manage the transportation route used by the E-CHATCO system. It provides tools for drawing and editing the route, saving changes as drafts, and publishing approved route updates. The system also maintains route versions to provide a record of previous configurations and changes.

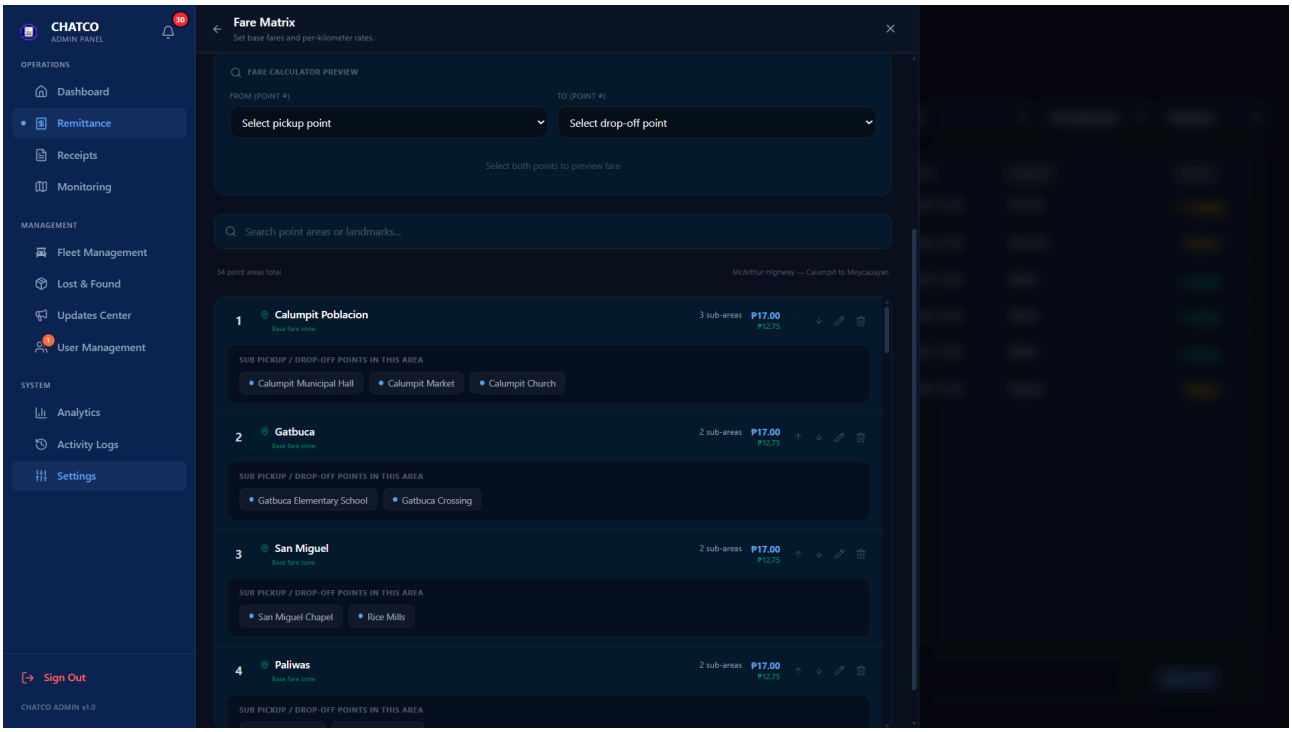


Figure 2.88: Routes

The operations rules feature allows administrators to configure system policies related to daily transportation activities. These settings include the overspeeding limit, maximum shift duration, and timing of remittance reminders. This feature helps establish consistent operational rules and allows the system to monitor activities according to the configured requirements.

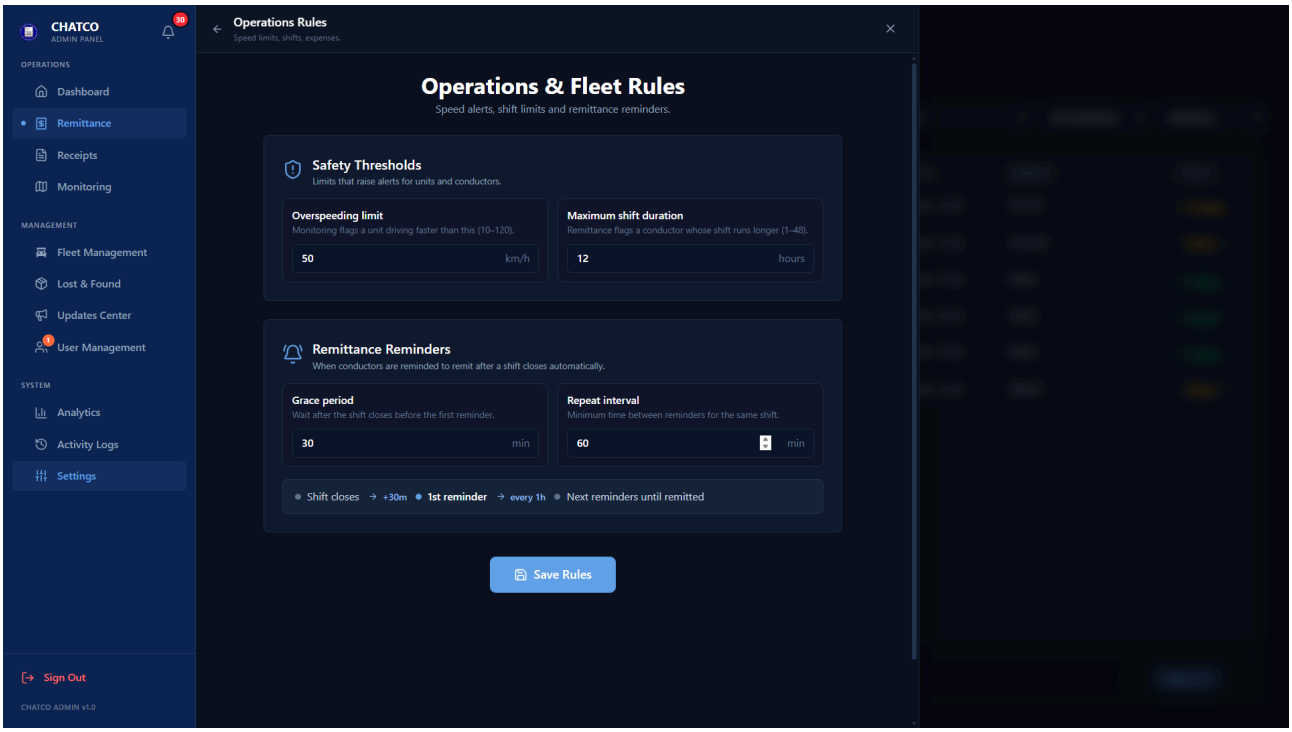


Figure 2.89: Operations Rules

The safety notifications feature allows administrators to configure the alerts and safety-related notices distributed through the E-CHATCO system. It provides control over which notifications should be sent to users based on the configured system requirements. This feature helps ensure that relevant safety information and operational alerts are communicated to the intended recipients.

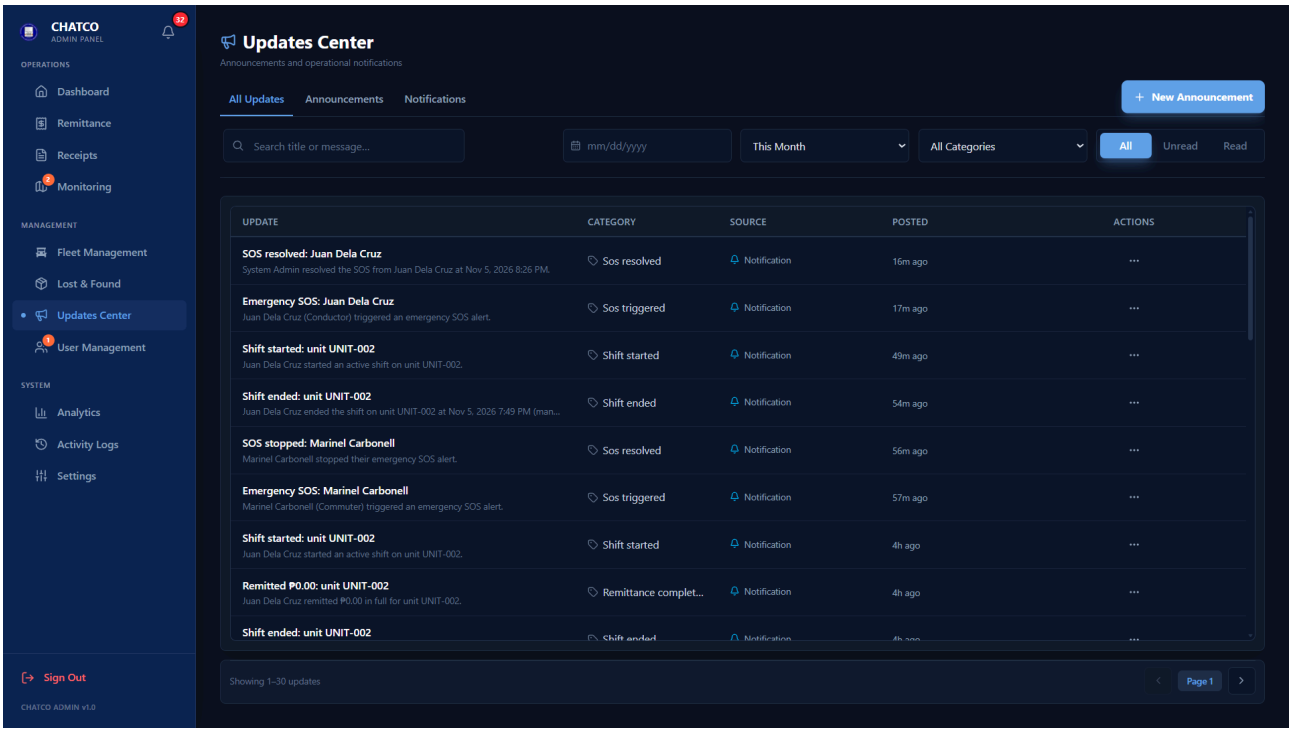


Figure 2.91: Safety Notifications

The Lost and Found feature allows administrators to manage reports of belongings found within the transportation service. It provides functions for recording found items with their corresponding photos and locations, reviewing submitted commuter claims, and approving or rejecting requests. Administrators can also coordinate the release of claimed belongings and close completed records to maintain an organized recovery process.

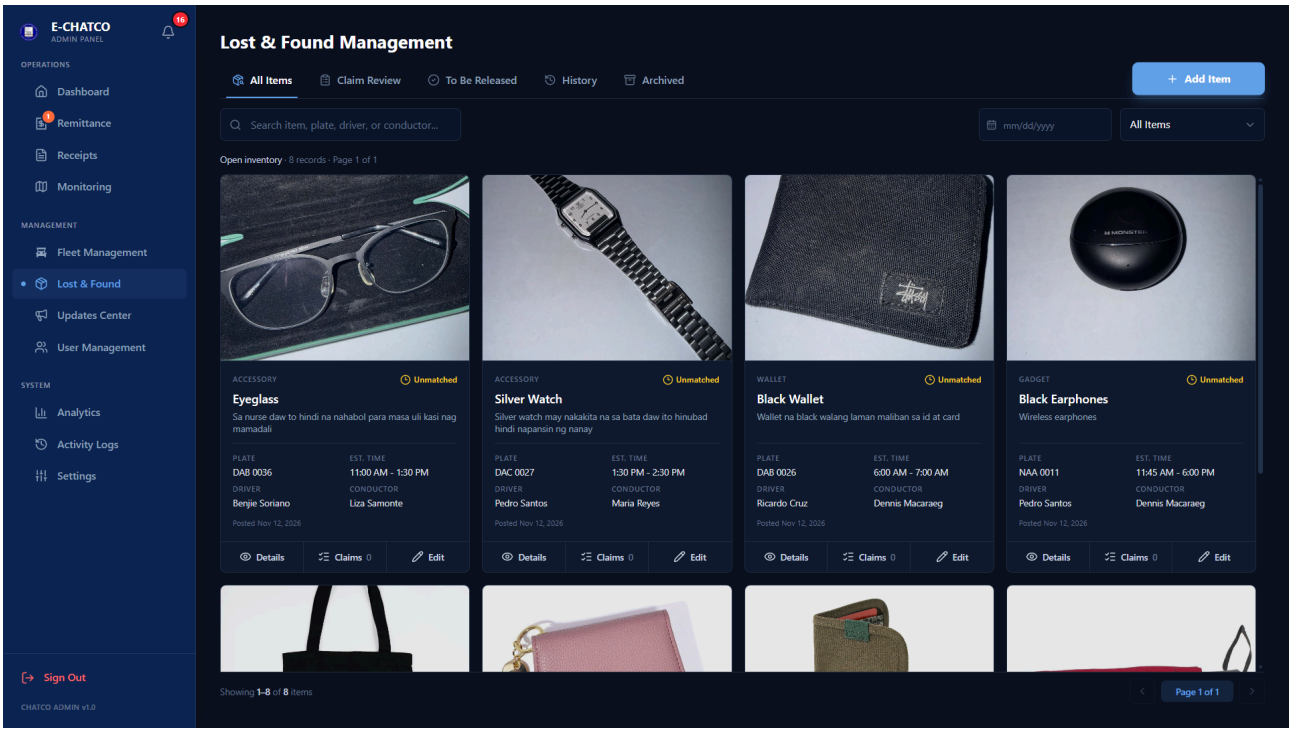


Figure 2.92: Lost and Found

The feedback feature allows administrators to access and review ratings and comments submitted by commuters regarding their rides and the assigned crew. It provides information about passenger experiences and the quality of transportation services. This feature helps administrators monitor commuter satisfaction and identify areas that may require operational improvement.

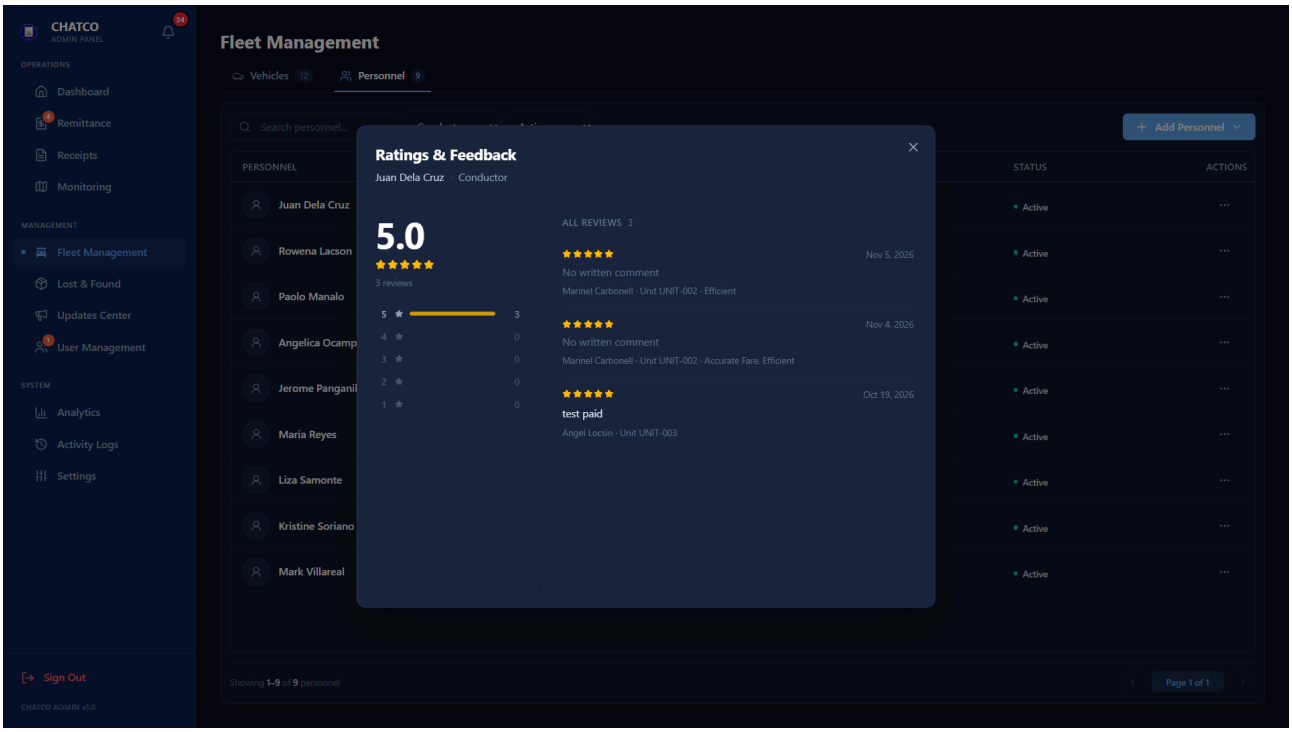


Figure 2.93: Admin Feedback

The announcements feature allows administrators to create, edit, and archive informational notices intended for commuters and conductors. These announcements can be used to communicate important transportation updates, service information, and other relevant notices. This feature helps administrators maintain communication with system users through a centralized announcement management module.

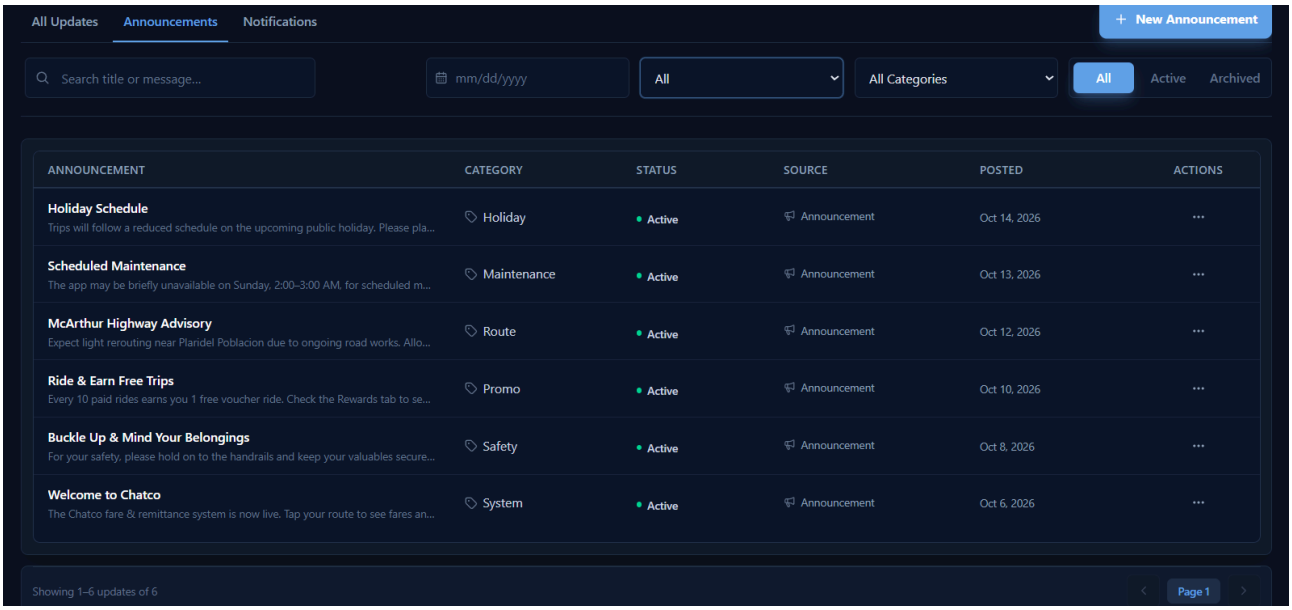


Figure 2.94: Announcements

The FAQ management feature allows administrators to maintain the questions and answers displayed in the public website's FAQ chat bubble. It provides functions for editing existing information and managing the content presented to visitors. This feature ensures that users can access updated answers to common questions and concerns regarding the E-CHATCO system.

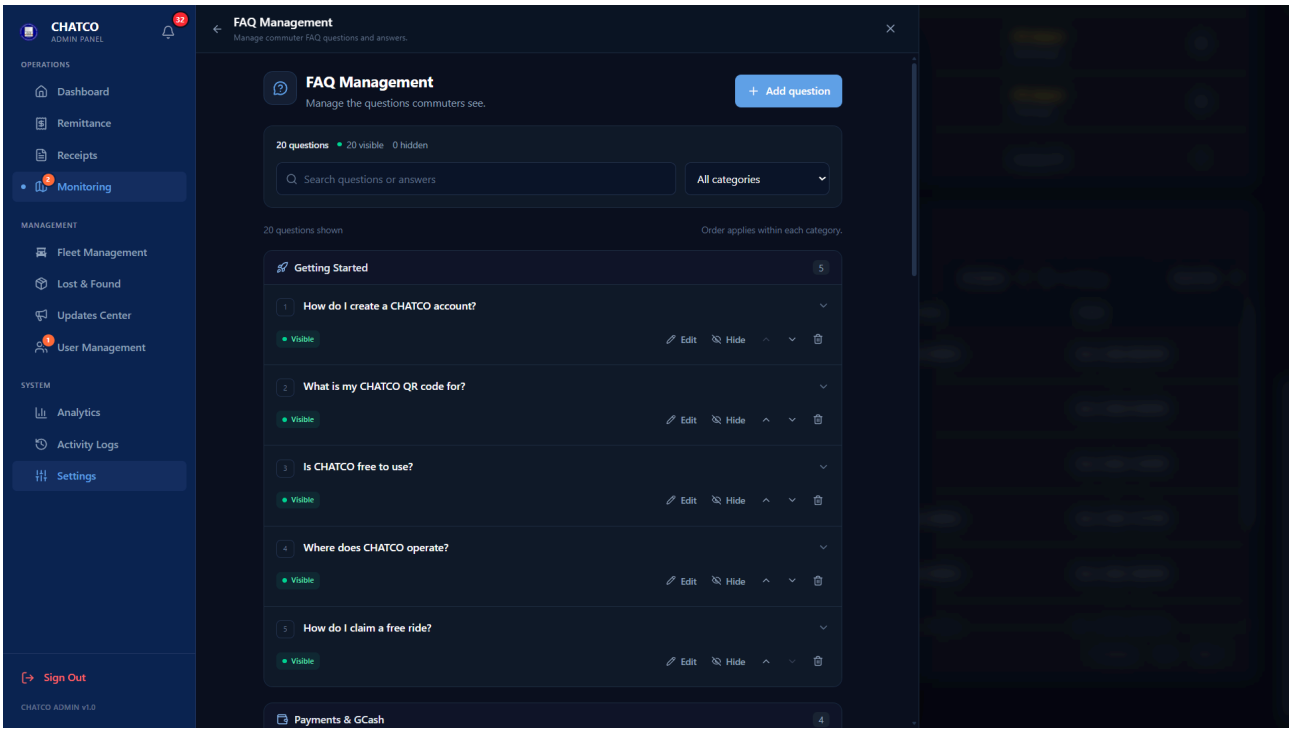


Figure 2.95: FAQ Management

The activity logs feature records important actions and activities performed within the E-CHATCO system. It provides administrators with an audit trail that can be reviewed to monitor system activities and account-related operations. The module also supports exporting recorded logs for documentation, review, and administrative reference.

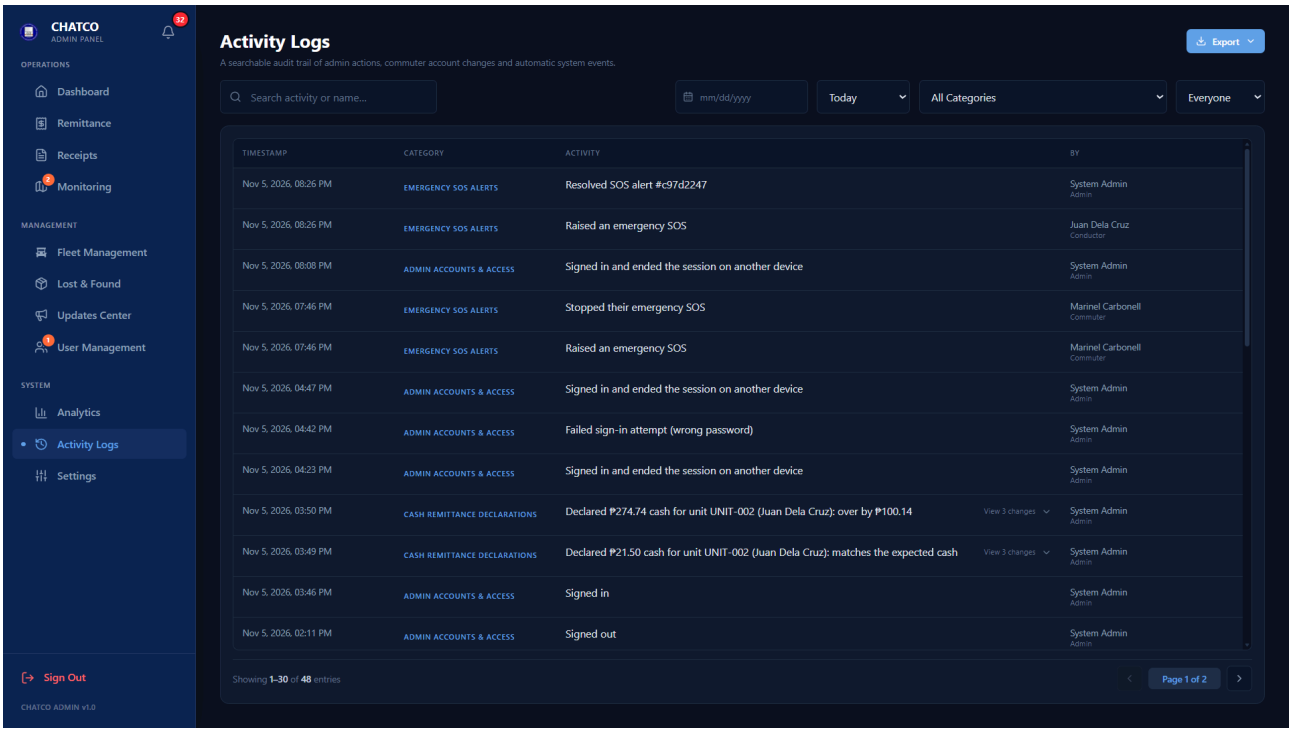


Figure 2.96: Activity Logs

The profile and security feature allows administrators to manage their own account information and security credentials. It provides options for updating the administrator's profile photo and changing their account password. This feature helps administrators maintain their personal account information while supporting secure access to the management interface.

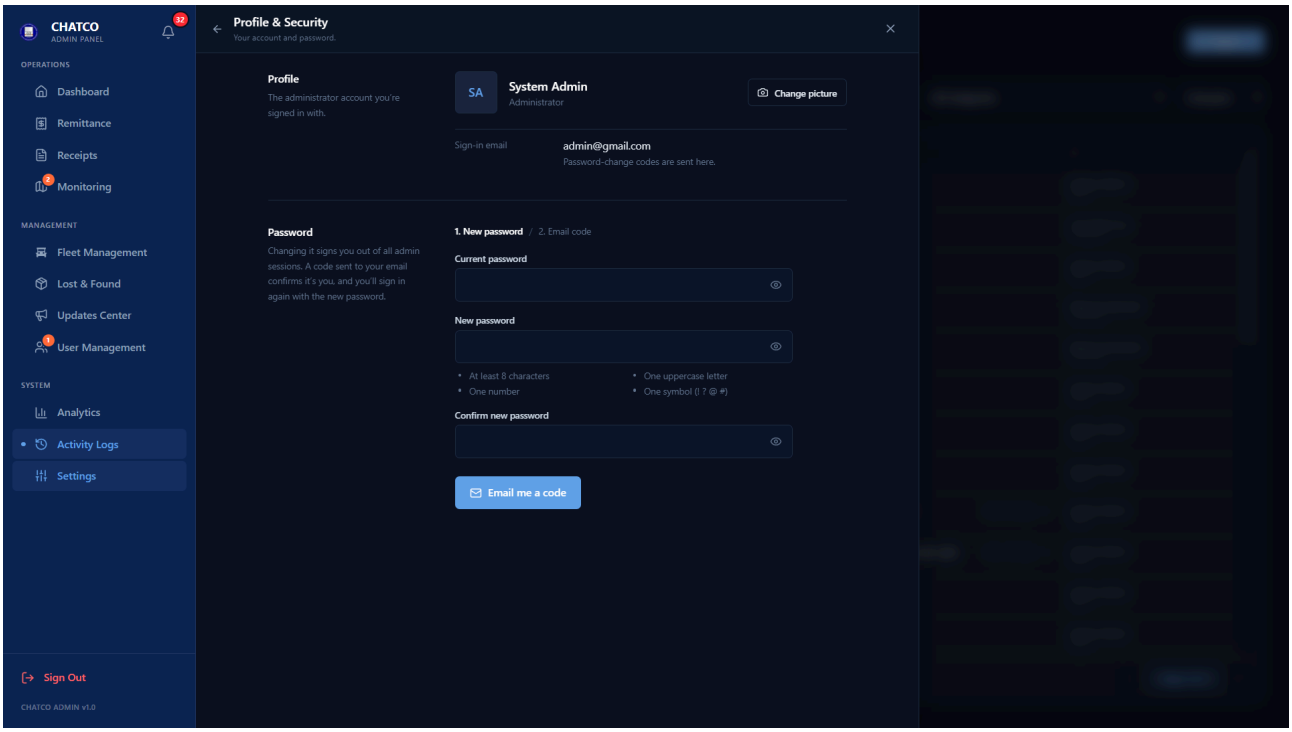


Figure 2.97: Profile and Security

Project Management and Collaboration Tools

The team carefully selected and utilized software tools to support the development of the E-CHATCO system using the Agile Scrum methodology. These tools were used to improve communication, organize project resources, manage assigned tasks, and ensure that the team could work efficiently throughout the development process. Google Docs, Google Drive, and ClickUp were utilized to support the team's documentation, file management, and project coordination activities.

Documentation and Collaboration Software

The team used Google Docs for preparing and managing the documentation of the E-CHATCO project. The software allowed team members to collaborate on the same document in real time, making it easier to write, edit, and review different sections of the capstone paper. Its commenting and suggestion features were also utilized to provide feedback, recommend revisions, and identify parts of the documentation that required improvements. Through these features, the team was able to maintain consistent communication and work together in completing the project documentation.

Figure 3.98 illustrates the team's utilization of Google Docs during the preparation of the E-CHATCO documentation. The document was shared among the members, allowing them to make simultaneous edits and provide comments and suggestions for necessary revisions. This helped simplify the documentation process and ensured that the members could contribute to improving the overall content of the paper.

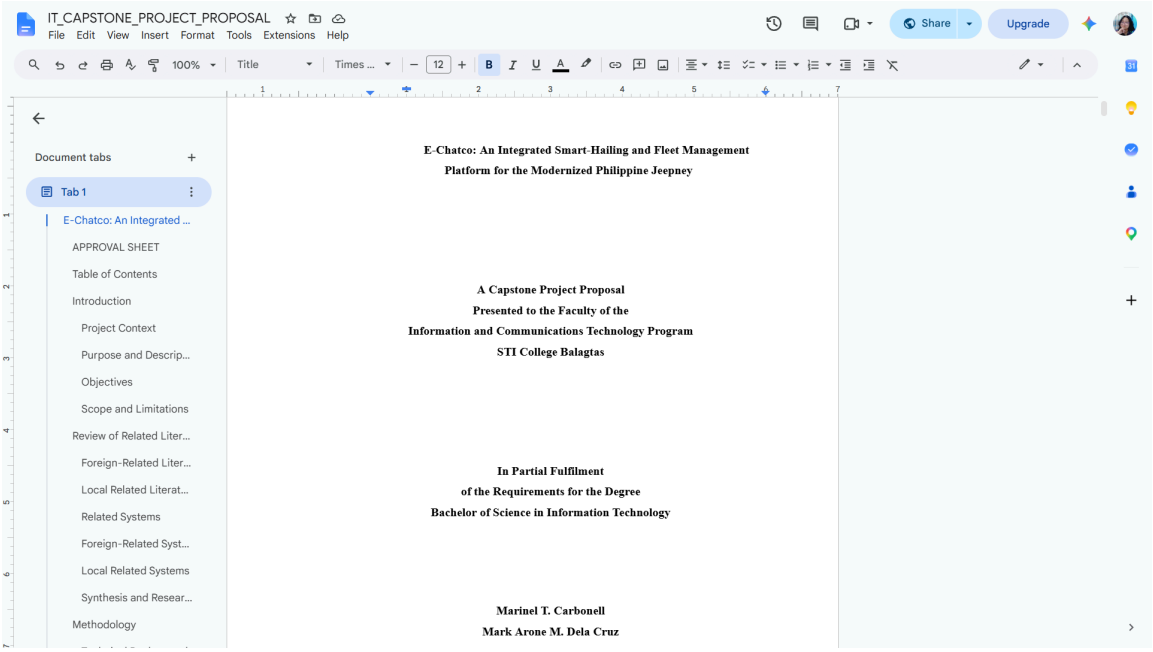


Figure 3.98

Project File Sharing and Cloud Storage

The team utilized Google Drive as the primary platform for storing and sharing files related to the e-CHATCO project. A shared folder was created specifically for the capstone project to provide team members with a centralized location for accessing important resources and documents. The files were organized into separate folders to maintain proper file management and make it easier to locate specific project materials. Through Google Drive, members were able to upload, access, and share files whenever needed, allowing the team to maintain an organized collection of project resources throughout the development process.

Figure 3.99 presents the team's GitHub shared folder, which served as the central storage location for the e-CHATCO project files. The folder contains organized subfolders used to manage different project materials and documentation. This allowed team members to access shared resources conveniently and maintain an organized workspace throughout the project.

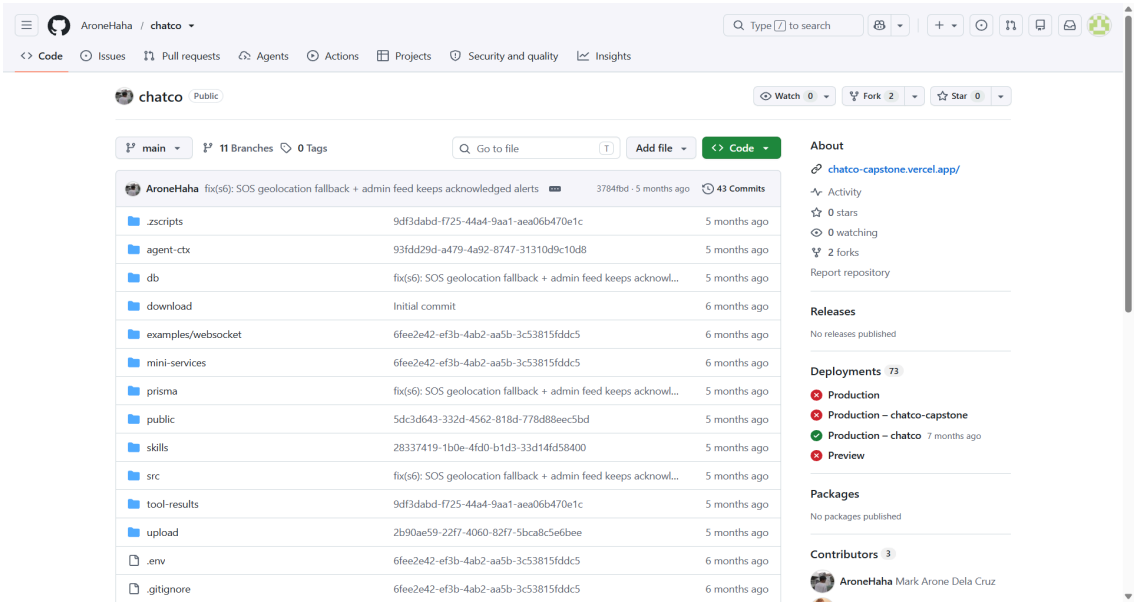


Figure 3.99

Project Management Software

The team decided to use ClickUp as its project management software to support the implementation of the Agile Scrum methodology. The platform provided an organized workspace for assigning responsibilities, managing tasks, and monitoring the progress of the project. The team leader assigned specific tasks to individual members, while each member was responsible for updating the status of their assigned activities. The team utilized the available task statuses to identify whether a task was pending, currently being worked on, or already completed. Through its task management and progress tracking features, ClickUp helped the team maintain accountability, organize development activities, and coordinate responsibilities throughout the project.

Figure 3.100 demonstrates the team's use of ClickUp through its project task board, which was utilized to manage and monitor the progress of e-CHATCO development activities. The board allowed the team leader to assign tasks to specific members and enabled each member to update the status of their assigned work. This provided the team with a clear overview of ongoing, pending, and completed activities, helping ensure that project tasks were properly monitored and managed.

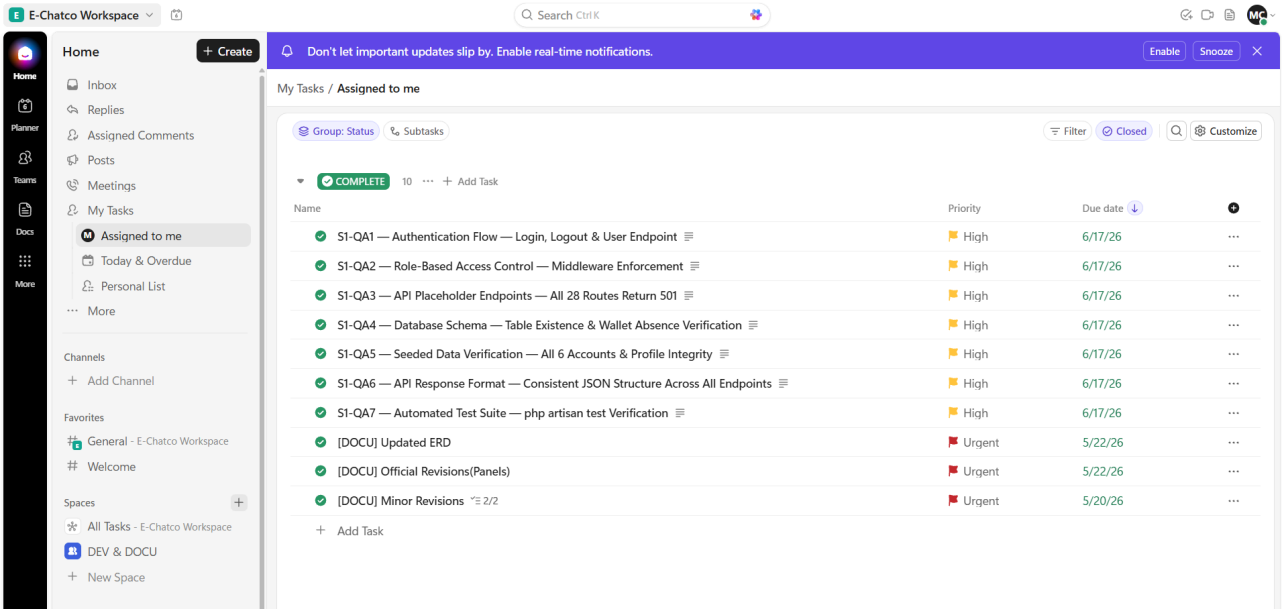

E-Chatco Workspace
Search Ctrl K
Home
Create
Inbox
Replies
Assigned Comments
Posts
Meetings
My Tasks
Assigned to me
Today & Overdue
Personal List
More
Channels
Add Channel
Favorites
General - E-Chatco Workspace
Welcome
Spaces
All Tasks - E-Chatco Workspace
DEV & DOCU
New Space
Planner
Teams
Docs
More
Don't let important updates slip by. Enable real-time notifications.
Enable
Snooze
My Tasks / Assigned to me
Group: Status
Subtasks
Filter
Closed
Customize
COMPLETE
10
Add Task
Name
Priority
Due date
S1-QA1 — Authentication Flow — Login, Logout & User Endpoint
High
6/17/26
S1-QA2 — Role-Based Access Control — Middleware Enforcement
High
6/17/26
S1-QA3 — API Placeholder Endpoints — All 28 Routes Return 501
High
6/17/26
S1-QA4 — Database Schema — Table Existence & Wallet Absence Verification
High
6/17/26
S1-QA5 — Seeded Data Verification — All 6 Accounts & Profile Integrity
High
6/17/26
S1-QA6 — API Response Format — Consistent JSON Structure Across All Endpoints
High
6/17/26
S1-QA7 — Automated Test Suite — php artisan test Verification
High
6/17/26
[DOCU] Updated ERD
Urgent
5/22/26
[DOCU] Official Revisions(Panels)
Urgent
5/22/26
[DOCU] Minor Revisions 2/2
Urgent
5/20/26
Add Task

Figure 3.100

RESULTS AND DISCUSSION

Testing

Testing Methodologies

- **System Testing**

  System Testing is conducted to ensure that the developed system functions as specified in the requirements and objectives of E-Chatco. This phase validates the integration of all modules, including user authentication, real-time GPS tracking and hailing, GCash e payment, fare calculation using predefined operational point areas, boundary remittance tracking with separate cash and GCash recording, fleet and traffic visualization, lost and found management, discount application, and safety and loyalty features. The process follows a requirement-based testing approach where every feature listed in the objectives is mapped to one or more test cases. This ensures that no critical function is overlooked. The black-box testing method is applied, meaning the testers do not need to check the internal program code, but instead verify that the system delivers accurate results when used according to the specified requirements.

- **Unit Testing**

  Unit Testing is the first level of software testing conducted to evaluate the individual components of the E-Chatco platform. At this stage, the focus is on verifying that the smallest testable parts of the system perform their intended functions correctly in isolation, without interference from other modules. The purpose of this phase is to identify and resolve errors early in development, ensuring that each component is stable and reliable before it is integrated into the complete system. This testing is performed by

the developers during the coding and implementation phase. Each unit is subjected to specific inputs to examine whether it produces the expected outputs. Examples of these units include the login validation, which ensures that only authorized users can access the system; the real-time GPS broadcast, which continuously transmits the location of active jeepney units to commuters and administrators; the fare calculator, which computes fares based on the predefined CHATCO operational point areas; the GCash payment QR generator, which produces unique QR codes for each transaction; and the Pick Me Up hailing function, which only activates when a CHATCO unit is within a one-kilometer radius of the commuter. The process follows a requirement-driven approach, where every feature is broken down into smaller components and tested separately. Normal scenarios are tested, such as submitting a fare transaction with complete origin and destination point areas, while abnormal scenarios are also considered, such as submitting a fare with an invalid GCash reference ID or attempting to hail a unit outside the one-kilometer radius. By addressing both valid and invalid cases, the testing ensures that the units behave correctly under different conditions. Through unit testing, errors can be detected and corrected at the earliest stage of development, reducing the possibility of major issues appearing during integration and system testing. This ensures that the foundation of the E-Chatco platform is robust and that each function is dependable before it becomes part of the overall system.

- **Acceptance Testing**

  Acceptance Testing is the final phase of evaluation conducted to confirm that the E Chatco platform meets the operational needs and expectations of CHATCO Cooperative as the client. This stage ensures that the system not only functions correctly but also supports the actual workflows and processes of the jeepney operations along the Calumpit Meycauayan route. The testing is carried out directly by the end users,

including the administrators, conductors, and commuters of CHATCO Cooperative, who validate the system by performing tasks that reflect their real business operations. During acceptance testing, the system is evaluated in terms of its ability to handle real time GPS tracking, proximity-based hailing, GCash payment verification, fare calculation using the CHATCO fare matrix, boundary remittance tracking with separate cash and GCash declarations, fleet and traffic visualization, lost and found matching, and safety features such Share My Ride. The assessment focuses on verifying whether the platform provides a reliable, user-friendly, and efficient solution for jeepney operations. Test scenarios are based on practical operational cases, such as commuters sending a Pick Me Up signal within the one-kilometer radius, conductors verifying GCash payments via the operational device, and administrators validating end-of-day remittances with separate cash and GCash breakdowns.

The evaluation criteria are aligned with the ISO/IEC 25010:2023 software quality standards, which include functional suitability, usability, reliability, performance efficiency, compatibility, security, maintainability, and portability. By measuring these attributes, the testing ensures that the system is acceptable for deployment and can be fully integrated into CHATCO Cooperative's operations.

The results of acceptance testing will serve as the basis for determining whether E Chatco can be officially adopted by CHATCO Cooperative for use in their smart-hailing and fleet management operations along the Calumpit–Meycauayan route.

Test Summary

The following test case summaries specify the number of tests executed for each module or backlog feature of the E-Chatco platform. Each module was tested against its corresponding set of test cases derived from the current implementation, ensuring that all routes, services, validation rules, and on-screen messages behave as expected.

User Authentication and Account Access			
Version	**Total Tests**	**Number of Passes**	**Number of Fails**
1.0	**65**	**63**	**2**
2.0	**65**	**65**	**0**

Live Map, Tracking, and Smart Hailing			
Version	**Total Tests**	**Number of Passes**	**Number of Fails**
1.0	**41**	**39**	**2**
2.0	**41**	**41**	**0**

Fare Calculator and Fare Guide			
Version	**Total Tests**	**Number of Passes**	**Number of Fails**
1.0	**6**	**6**	**0**
2.0	**6**	**6**	**0**

GCash E-Payment Integration			
Version	**Total Tests**	**Number of Passes**	**Number of Fails**
1.0	**17**	**16**	**1**
2.0	**17**	**17**	**0**

Vouchers, Rewards, and Loyalty			
Version	**Total Tests**	**Number of Passes**	**Number of Fails**
1.0	**17**	**14**	**3**
2.0	**17**	**17**	**0**

Cash Receipt and Payment History			
Version	**Total Tests**	**Number of Passes**	**Number of Fails**
1.0	**7**	**7**	**0**
2.0	**7**	**7**	**0**

Share My Ride			
Version	**Total Tests**	**Number of Passes**	**Number of Fails**
1.0	**10**	**10**	**0**
2.0	**10**	**10**	**0**

Lost and Found Management			
Version	**Total Tests**	**Number of Passes**	**Number of Fails**
1.0	**32**	**31**	**1**
2.0	**32**	**32**	**0**

Ride Feedback			
Version	**Total Tests**	**Number of Passes**	**Number of Fails**
1.0	**6**	**6**	**0**
2.0	**6**	**6**	**0**

Profile and Account Management			
Version	**Total Tests**	**Number of Passes**	**Number of Fails**
1.0	**14**	**13**	**1**
2.0	**14**	**14**	**0**

Discount Application and Student Renewal			
Version	**Total Tests**	**Number of Passes**	**Number of Fails**
1.0	**22**	**21**	**1**
2.0	**22**	**22**	**0**

Conductor Shift, Device, and Operations			
Version	**Total Tests**	**Number of Passes**	**Number of Fails**
1.0	**29**	**28**	**1**
2.0	**29**	**29**	**0**

Admin Dashboard, User Management, and Fleet			
Version	**Total Tests**	**Number of Passes**	**Number of Fails**
1.0	**44**	**43**	**1**
2.0	**44**	**44**	**0**

Remittance Tracking			
Version	**Total Tests**	**Number of Passes**	**Number of Fails**
1.0	**10**	**9**	**1**
2.0	**10**	**10**	**0**

Settings, Configuration, and System Activities			
Version	**Total Tests**	**Number of Passes**	**Number of Fails**
1.0	**17**	**17**	**0**
2.0	**17**	**17**	**0**

Announcements and Cross-Role Integration			
Version	**Total Tests**	**Number of Passes**	**Number of Fails**
1.0	**9**	**9**	**0**
2.0	**9**	**9**	**0**

Description of Prototype

This part includes the system requirements, the preliminary design, and how the system is being evaluated and tested.

System Requirements

Tables 4.2 and 4.3 list the hardware and software requirements for the system to run on the client's side.

Hardware	Requirements
CPU	Intel/AMD 2.0 GHz or higher
RAM	8 GB
Storage	100 GB Free Space
Graphics	Integrated GPU (any)
Monitor	HD 1080p
Internet	Stable 10 Mbps
Mobile Device (Conductor)	Operational device similar to a mobile phone with built-in thermal printer

Table 4.2: Hardware Requirements

Software	Requirements
Operating System	Windows 10 SL 64-bit or later
Web Browser	Any latest, stable browser version (Chrome, Edge)
Backend Framework	PHP 8.2+, Laravel 12.x (Laravel Sanctum for API Authentication)
Frontend Framework	Next.js 16 (React 19, Typescript 5), Tailwind CSS 4
Runtime Environment	Node.js v20.x LTS or later
Database	MySQL
Real-Time Engine	Pusher + Laravel Echo
Mapping Library	Leaflet.js 1.9 with React-Leaflet 5
QR Code Library	qrcode.react (generating QR codes), html5-qrcode (scanning)
Payment Gateway	PayMongo (GCash)
File Storage	Cloudfare R2 (S3-compatible)
Email Service	Brevo SMTP
Version Control	Git/GitHub

Table 4.3: Software Requirements

System Screenshots

The system prototype screenshots for the E-Chatco platform are presented in this part. It is organized around the core modules that are specific to each type of user. The prototype emphasizes key processes and interface components that represent the responsibilities and requirements of various users, including commuters, conductors, and administrators. The prototype reflects the output design of the system developed for CHATCO Cooperative.

Landing Interface

The landing interface serves as the public-facing section of the E-Chatco system. It provides visitors with information about the platform, its available features, safety services, and the Calumpit to Meycauayan route. Visitors can open the Sign in and Sign up dialogs directly from this page, browse the FAQ chat bubble for common questions, and access the Terms of Service and Privacy Policy pages without signing in. A signed-in user who opens the landing or login page is automatically redirected to their role-based dashboard to avoid redundant authentication steps.

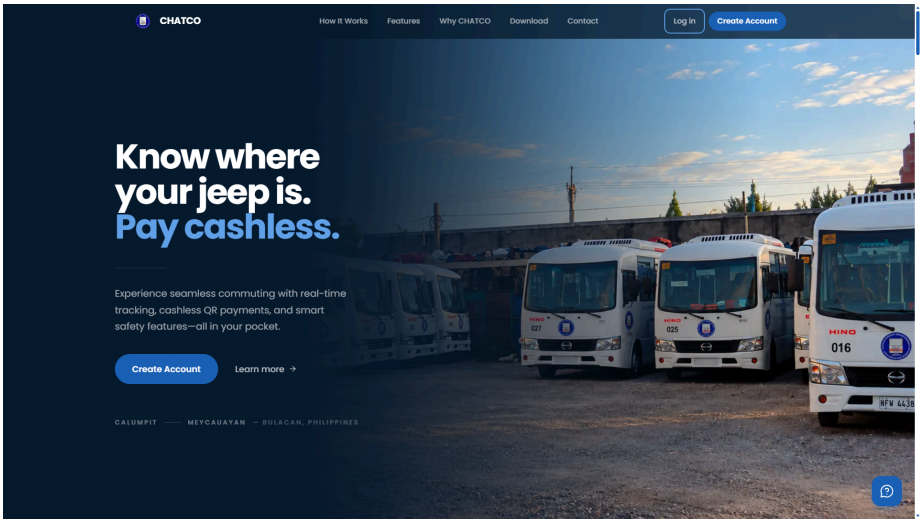


Figure 4.1: Landing Page

Commuter Interface

The commuter interface is designed for passengers who regularly use jeepneys operating along the Calumpit to Meycauayan route. It provides access to transportation services, including real-time vehicle tracking, smart hailing within a 1-kilometer radius, fare information, cashless GCash payments, voucher and rewards management, safety features, and account management. The interface modules are illustrated in Figures 4.2 to 4.10.

The Live Map displays the real-time locations of jeepneys operating along the designated route. Only on-duty units that are not on break appear on the map. Each unit is shown with its plate number, unit number, assigned crew, and current capacity status (Available, Standing, or Full). When a commuter is more than 1 kilometer away from every unit, the system displays an Outside service radius notice and disables the hail function while still showing the units on the map for visibility.

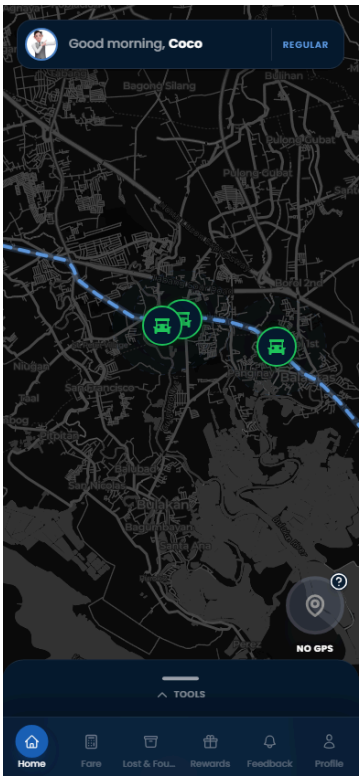


Figure 4.2: Live Map

The Pick Me Up (Hail) feature allows commuters to send a hailing request to a nearby jeepney when they are within 1 kilometer of an on-duty unit. The hail remains pending for three minutes, during which the commuter receives continuous vibration feedback and the conductor sees the waiting commuter on the dashboard map. The hail is automatically cancelled if the commuter walks out of range or if it expires without conductor response.

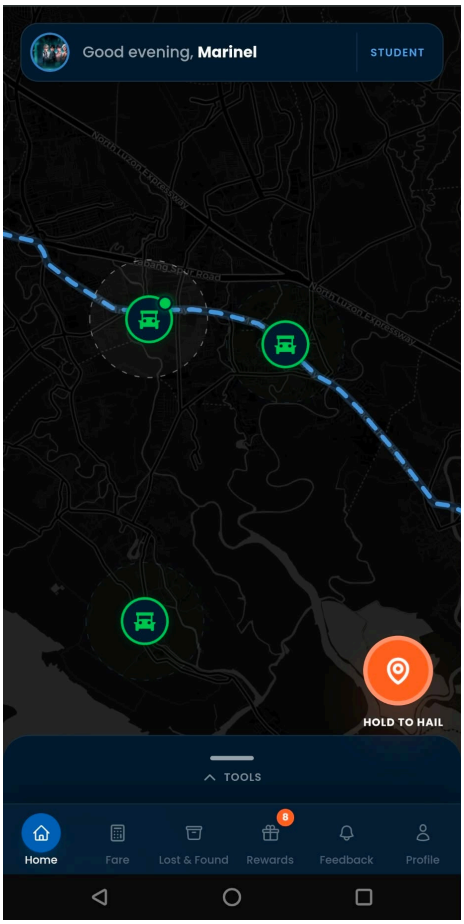


Figure 4.3: Pick Me Up (Hail)

The Fare Calculator provides commuters with fare information between two selected stops along the Calumpit to Meycauayan route. It displays both the regular fare and the discounted fare for

verified Student, Senior Citizen, and PWD commuters. The fare is computed based on the predefined CHATCO operational point areas, with the minimum fare of PHP 17.00 covering the first four barangays and PHP 2.25 added for each succeeding barangay, subject to the rates saved by the administrator in the Financial Rules settings

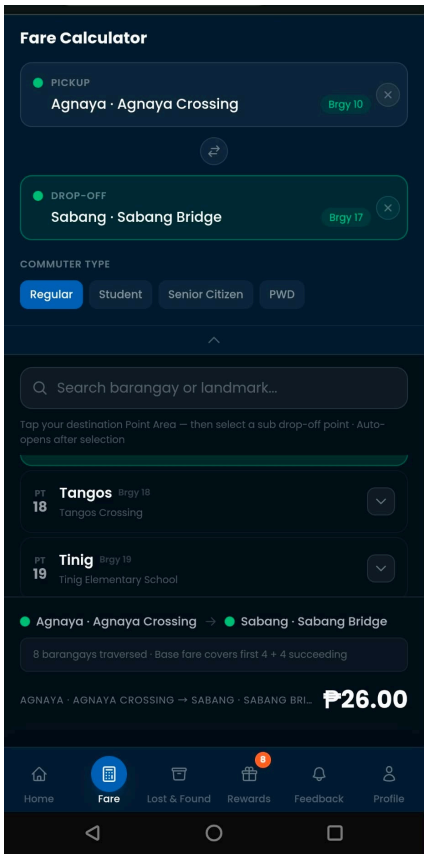


Figure 4.4: Fare Calculator

The GCash Payment feature allows commuters to settle their jeepney fares through GCash by scanning the QR code presented by the conductor. After scanning, the commuter is directed through the GCash checkout process and returns to the E-Chatco application with a confirmation.

The QR code expires after two minutes to prevent reuse, and a QR that has already been claimed by another commuter is refused with the appropriate message.

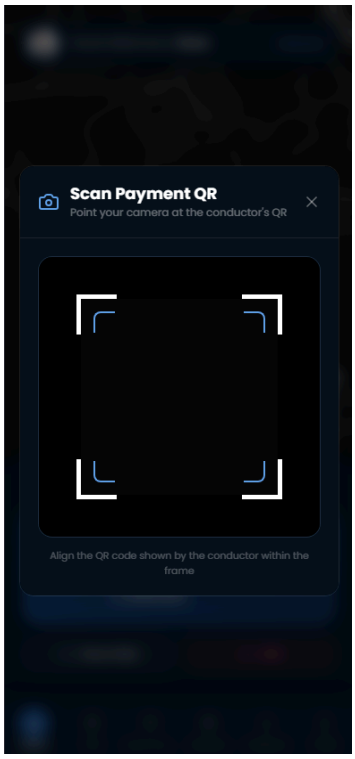


Figure 4.5: GCash / QR Fare Payment

The Cash Receipt Claim feature allows commuters to associate a cash-paid ride with their E-Chatco account by scanning the QR code printed on their thermal receipt. Each cash receipt can be claimed once and is valid for six hours from the time of printing. Claimed cash rides count toward the commuter's reward progress, allowing cash-paying passengers to participate in the loyalty program.

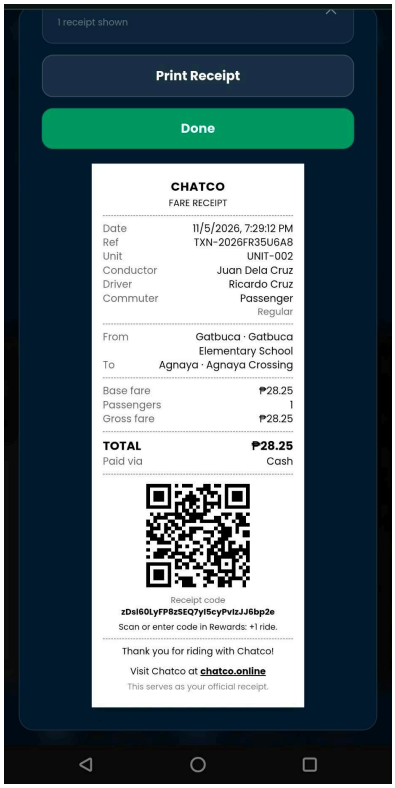


Figure 4.6: Cash Receipt Claim

The Loyalty Rewards feature grants commuters a free-ride voucher after completing a configurable number of qualifying paid rides (default: every 10 paid rides). The voucher appears in the Rewards page under the Available tab and is valid for three days. The Voucher Wallet provides commuters with a dedicated section for viewing and managing their available, used, and expired vouchers.

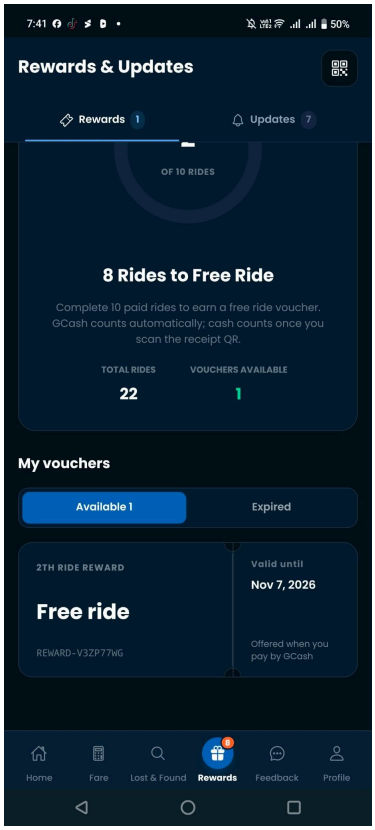


Figure 4.7: Loyalty Rewards and Voucher Wallet

The Share My Ride feature allows commuters to generate a public link containing their live trip location, which can be shared with family members and stops automatically after 30 minutes.

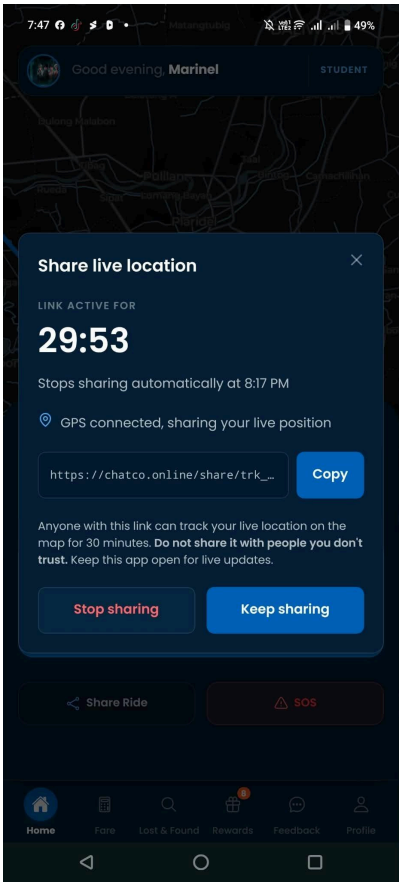


Figure 4.8: Share My Ride

The Lost and Found feature allows commuters to browse items reported as found within the transportation service. Commuters can filter items by category and date, watch items they are interested in, submit claims by providing proof of ownership and supporting photos, and track the progress of their submitted requests through My Claims.

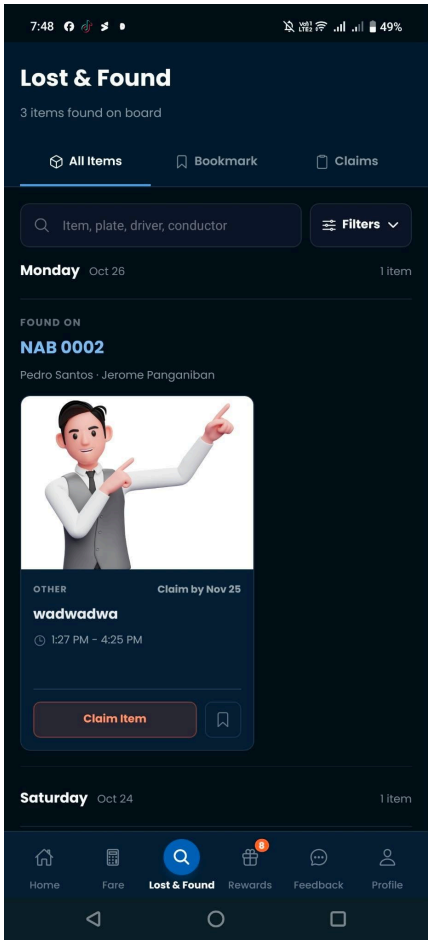


Figure 4.9: Lost and Found (Commuter)

The Ride Feedback feature allows commuters to provide ratings (1 to 5 stars) and comments regarding their transportation experience and the crew assigned to the jeepney. Feedback can be submitted after completing a paid ride or by scanning the QR code associated with the unit. Each commuter can submit only one feedback per shift.

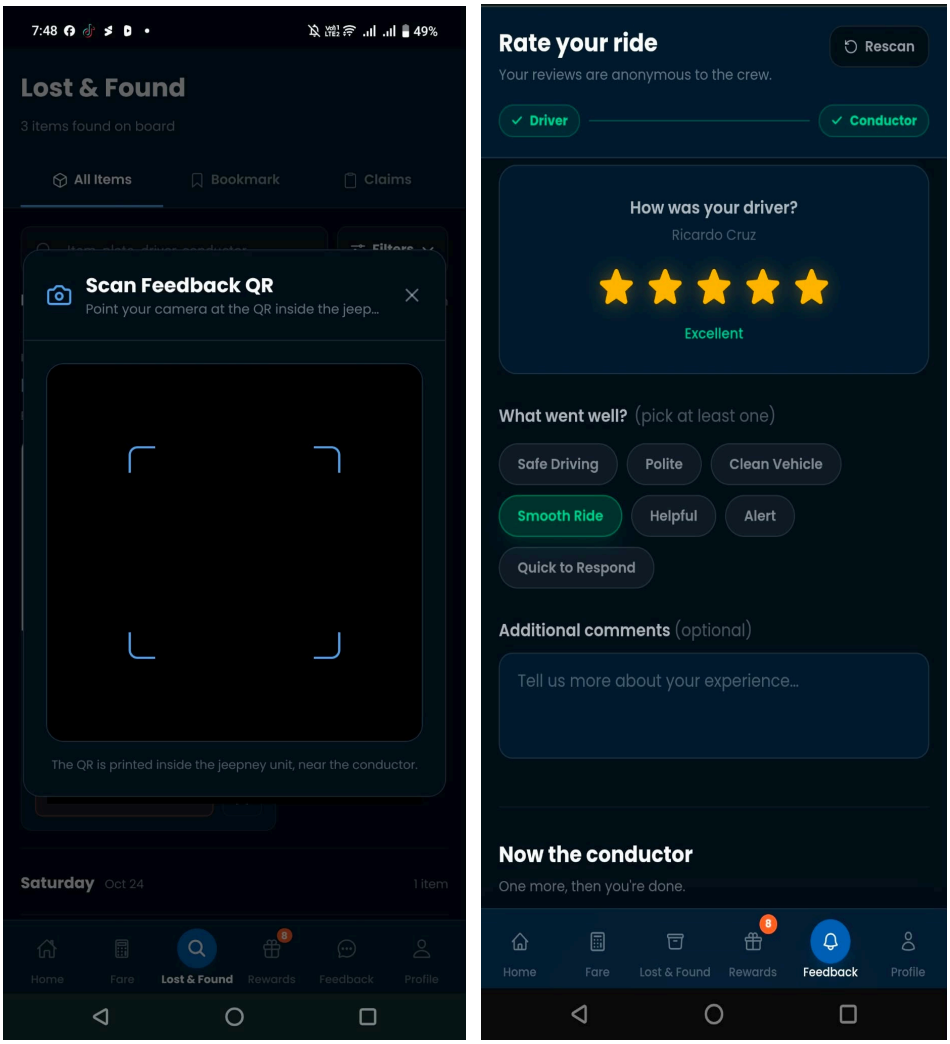


Figure 4.10: Ride Feedback

Profile, Discount Application, and Student Renewal (Commuter)

The profile feature allows commuters to view and manage their personal account information within the system. It includes options for editing account details, updating the profile photo through an image cropper, switching between light and dark display themes, changing the email

address (with verification), and changing the password (with verification). The change-email workflow is rate-limited to three changes within a 24-hour period, and only one email-change session can be active at a time across devices.

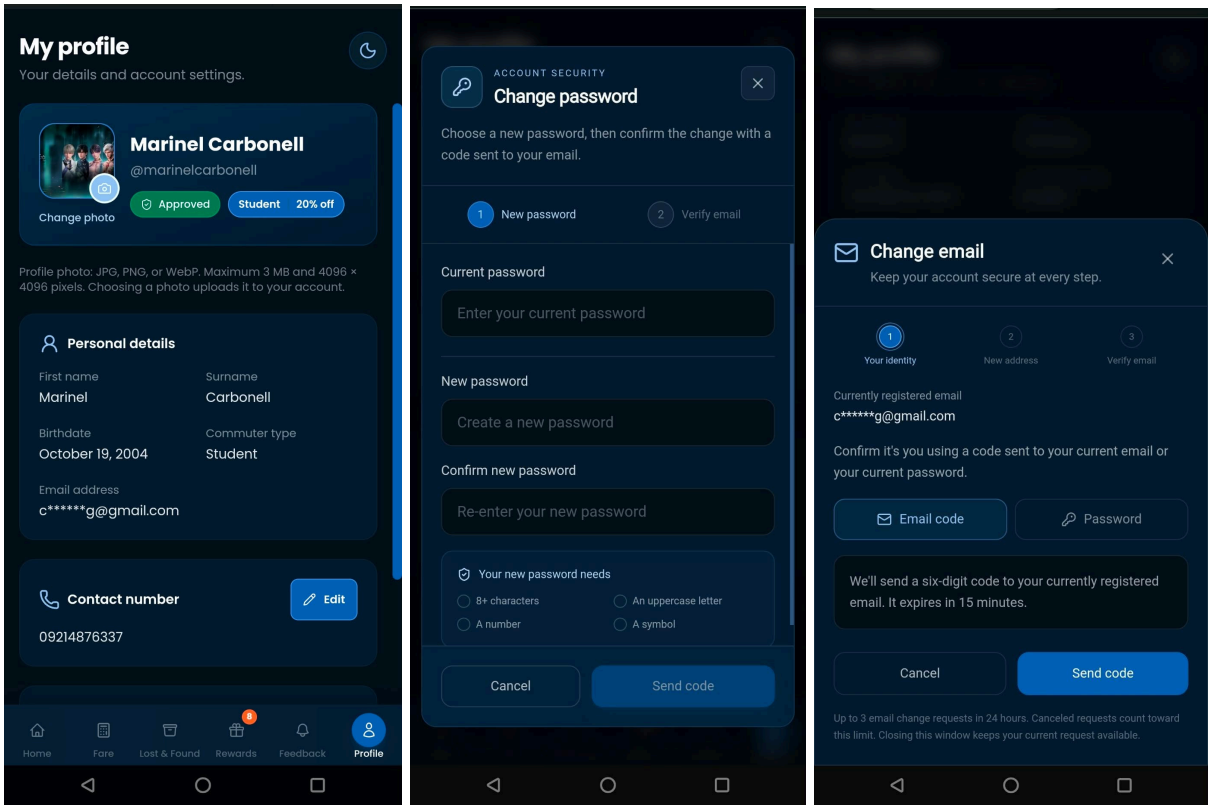


Figure 4.11: Profile and Account Management

Of particular relevance to the cooperative's compliance with Philippine fare discount regulations is the Application of Discounts feature, illustrated in Figure 4.12. This feature allows a Regular commuter to apply for a discounted fare classification under the Student, Senior Citizen, or PWD category by submitting a valid supporting document in JPG, PNG, WebP, or PDF format (up to 5 MB). Once submitted, the application enters the Pending state and the commuter continues to pay the regular fare until an administrator reviews and approves the request. The system enforces

the following rules: only Regular commuters may apply; only one active application is allowed at a time; pending applications may be cancelled by the commuter but reviewed applications may not; and if an administrator requests corrections, the commuter may resubmit with a corrected document but may not change the discount category during the correction cycle.

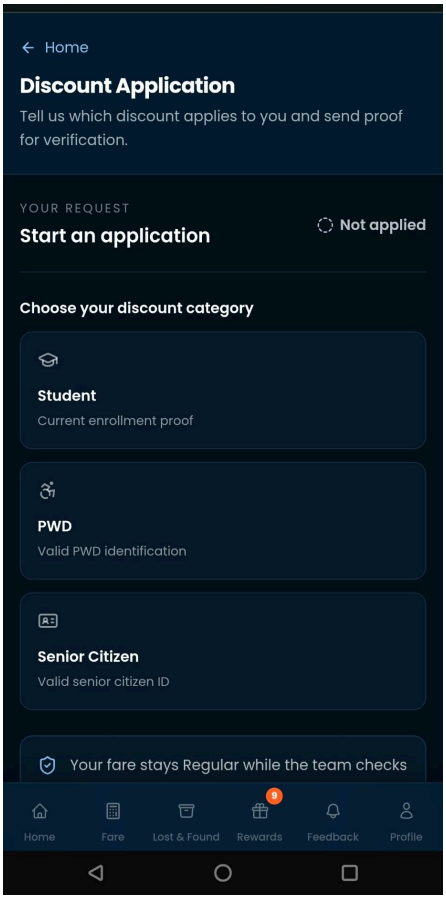


Figure 4.12: Discount Application (Commuter Side)

Complementing the Application of Discounts feature is the Renewal of Student Discount feature, illustrated in Figure 4.13. Because student discount eligibility is valid only for the duration of an academic term, the system requires Student commuters to renew their eligibility at the start of

each renewal cycle. The Student Renewal page is accessible only to active Student commuters and to former students whose discount has expired. Non-students who attempt to open the page receive a message indicating that renewal is unavailable to their account type.

Figure 4.13: Student Renewal (Commuter Side)

To renew, the student commuter uploads a current school ID or enrollment proof and submits the renewal request, which enters the Pending state for administrator review. Only one active application is allowed per renewal cycle. If the student is no longer enrolled, they may use the No Longer a Student option to switch their account classification back to Regular, after which their fares are no longer discounted. The system also displays renewal reminders to Student commuters approaching the renewal deadline; these reminders can be snoozed or dismissed as needed. If a student fails to renew before the deadline, the midnight scheduled job automatically converts the commuter type back to Regular and notifies the commuter that their student discount has expired. The next GCash fare is then charged at the regular rate. This automated enforcement ensures that the cooperative does not inadvertently extend discounted fares to commuters who are no longer eligible, in compliance with the applicable regulations governing student fare discounts.

The Announcements feature allows commuters to receive important updates and notices published by the administrators. These announcements can be accessed through the notification bell, which displays the count of unread notifications and allows the commuter to mark all notices as read.

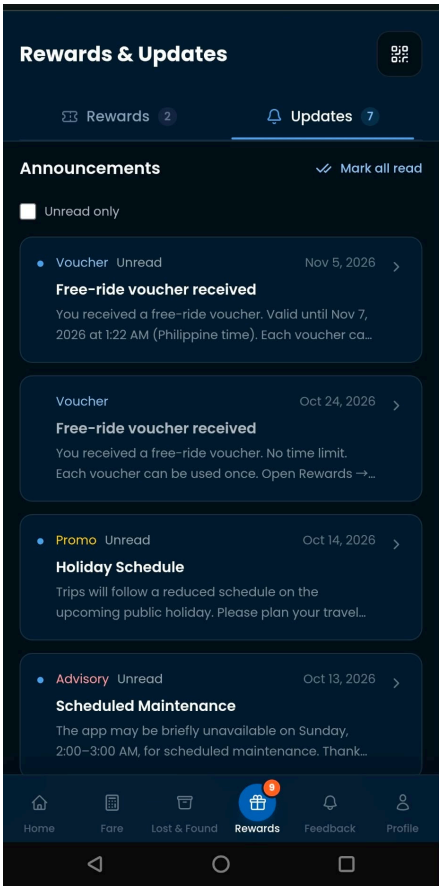


Figure 4.14: Announcements (Commuter)

Conductor Interface

The Conductor Interface is designed for the on-board personnel responsible for managing daily jeepney operations and assisting passengers during their journeys. It provides functions for vehicle verification, shift management, real-time location broadcasting, passenger hailing, fare collection, transaction recording, and end-of-day procedures. The interface modules are illustrated in Figures 4.15 to 4.25.

The Unit Verification feature allows conductors to select the jeepney and its assigned driver before beginning their daily shift. Units and drivers that are already on a shift, assigned to other personnel, or under maintenance are greyed out with their reason displayed. Once the unit and driver are confirmed, the conductor starts the shift and the dashboard opens.

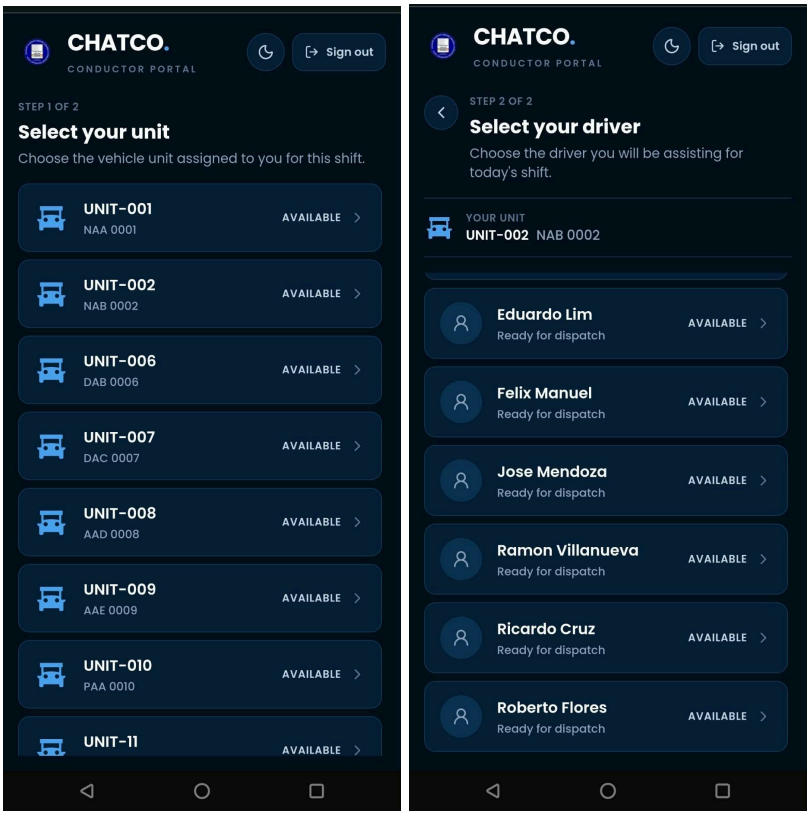


Figure 4.15: Unit Verification

The Live Location, Capacity, and Break Status feature allows conductors to broadcast the unit's current GPS position to the commuter and administrator maps, set the unit's capacity status (Available, Standing, or Full), and take a break. While on break, the unit disappears from the

commuter map and hails to it are refused. The conductor dashboard continues to display the On break status, and administrators monitoring the fleet see the break time recorded

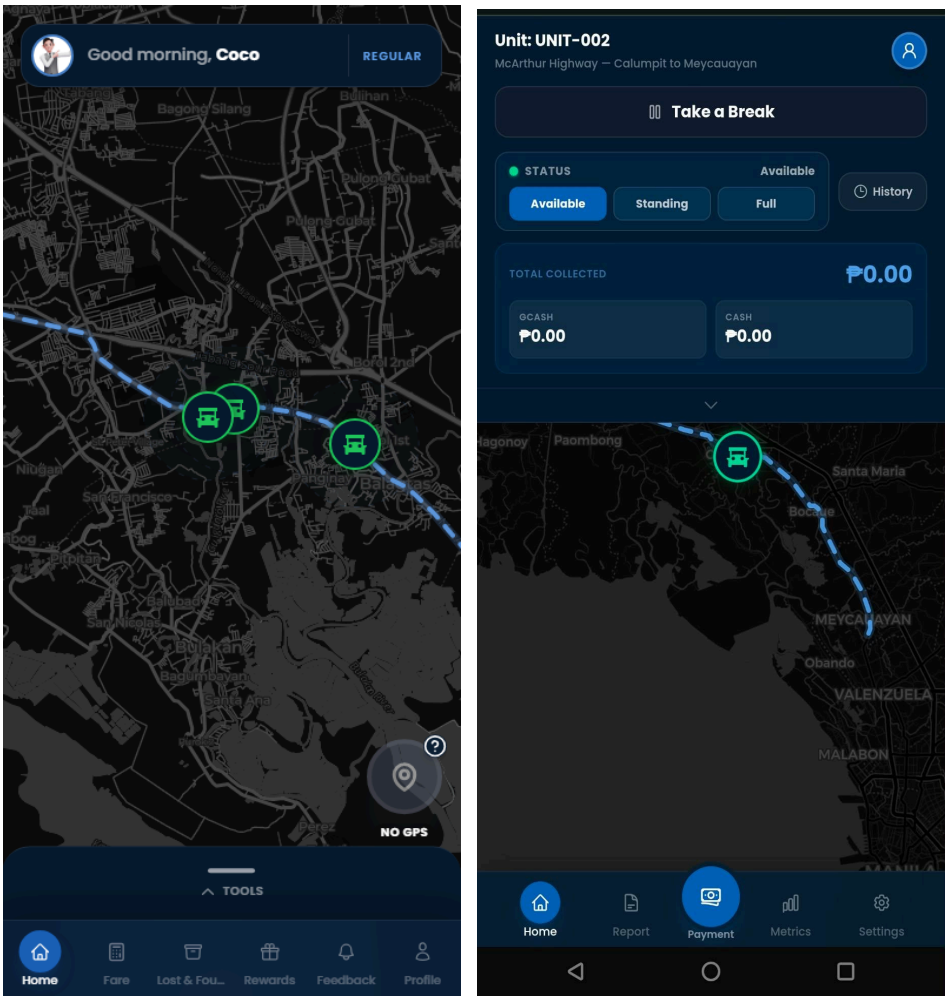


Figure 4.17: Live Location, Capacity, and Break Status

The Hail Requests feature allows conductors to view incoming Pick Me Up requests from commuters within the 1-kilometer radius. Requests are displayed with their corresponding

locations on the map. The conductor may accept or reject each request, and expired or already-handled hails are returned with the appropriate message.

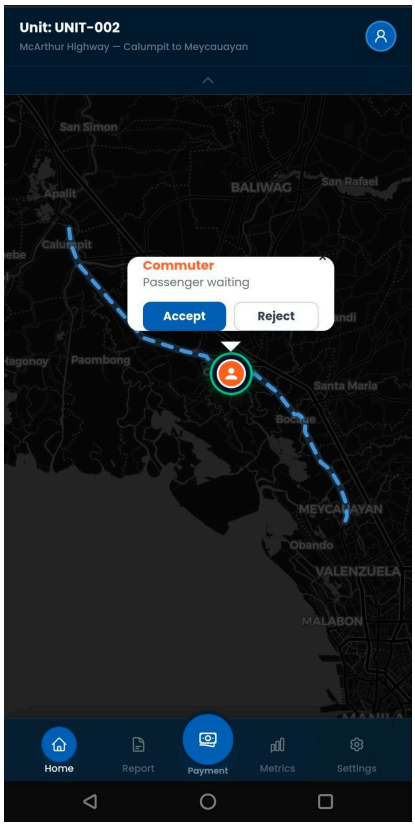


Figure 4.18: Hail Requests (Conductor)

The Record Fares feature allows conductors to process passenger payments during an active shift. It supports the recording of cash fares, the initiation of GCash QR payments (including group payments and voucher applications), and the application of commuter-type discounts. For discounted commuters, the verified commuter type stored on the account takes precedence over the conductor's manual selection, ensuring accurate fare application.

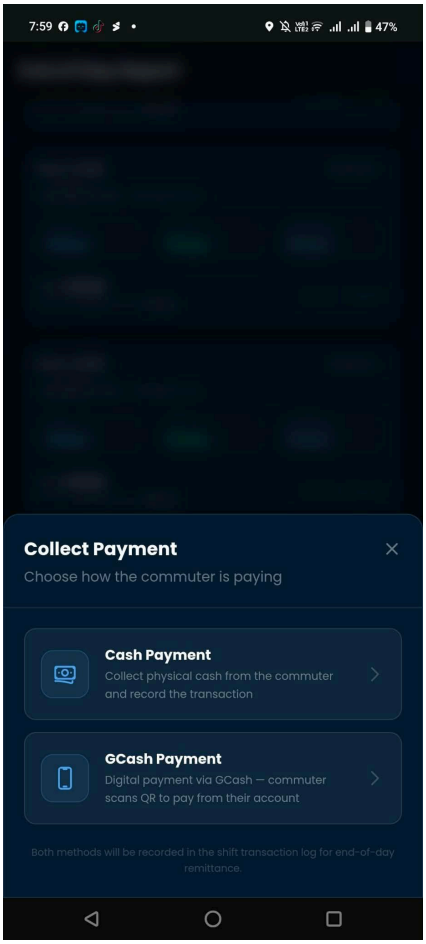


Figure 4.19: Record Fares

The Fare Calculator provides conductors with a quick way to determine the transportation fare between two selected stops, displaying the fare basis, barangays traveled, and the fare for each commuter type. The system uses the minimum fare and succeeding fare rates saved in the administrator's Financial Rules settings, with the discount rate applied automatically for Student, Senior Citizen, and PWD commuters.

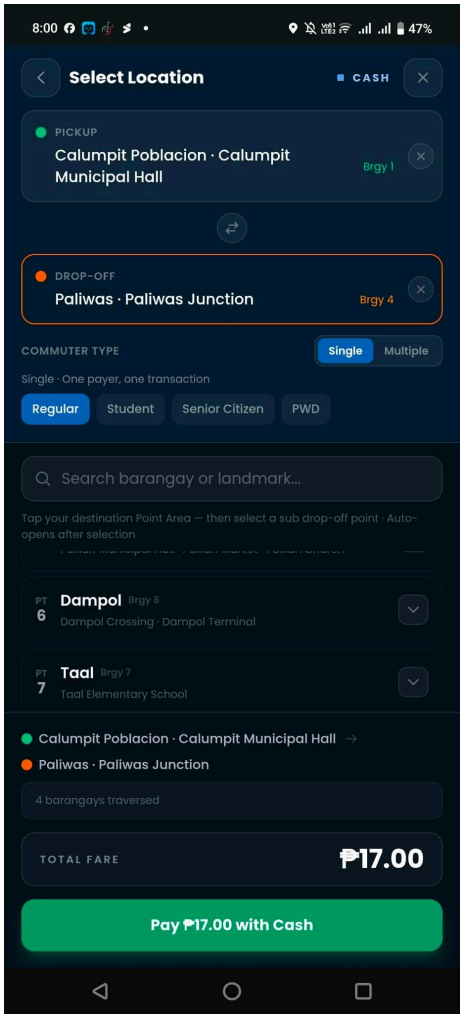


Figure 4.20: Fare Calculator

The Printed Receipts feature allows conductors to generate transaction receipts after completing passenger payments. Each receipt contains a unique reference number and a QR code that commuters can scan to claim and associate their cash-paid rides with their accounts. The administrator may enable Auto-print after each transaction, in which case the receipt prints without requiring an additional tap.

Figure 4.21: Printed Receipts

The Transaction History feature displays the payment records generated during the conductor's active shift, with filters by payment method (Cash, GCash, Voucher) and date. The Offline Support feature allows conductors to continue recording cash fare transactions when the device temporarily loses internet connection; recorded offline cash transactions are synchronized with the system once connectivity is restored. GCash and voucher validation require an internet connection and are unavailable while offline.

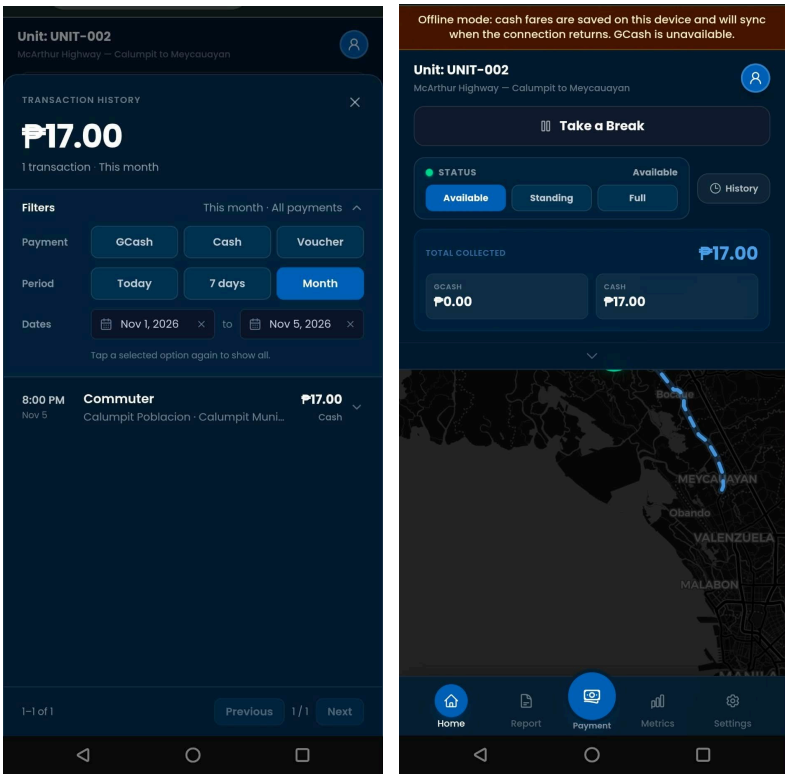


Figure 4.22: Transaction History and Offline Support

The End-of-Day Close-out feature allows conductors to formally end their active shift and review the summary of their daily operations before completing the closing process.

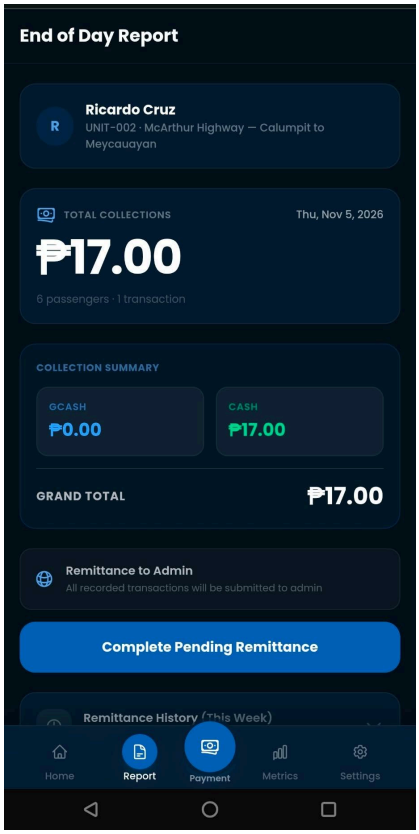


Figure 4.23: End-of-Day Close-out

The Remittance feature allows conductors to submit the collected daily earnings to the designated management. The system requires all offline cash to be synced before remittance can be submitted, and the end-of-day report presents the cash, GCash, and total collections separately for transparent declaration. The shift auto-ends if it exceeds the maximum duration or if it is still open at midnight, after which the next fare attempt is refused.

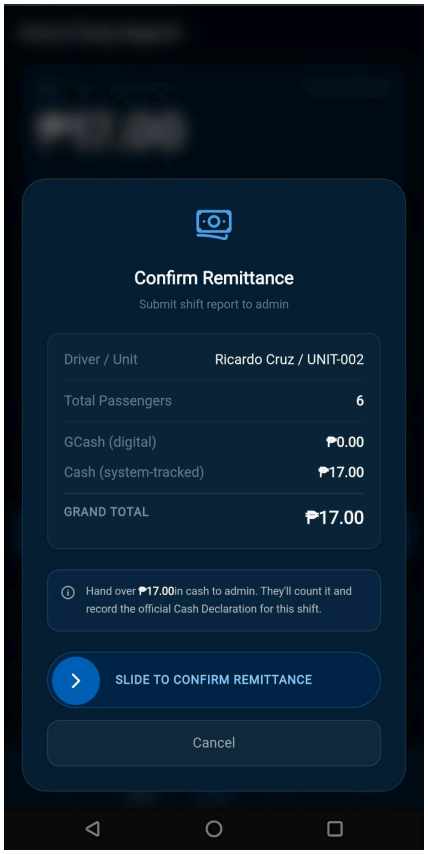


Figure 4.24: Remittance (Conductor)

The Metrics and Ratings feature provides conductors with a summary of their work performance, including earnings, completed trips, and ratings submitted by commuters. The Settings feature allows conductors to manage their profile photo, toggle the scan sound, clear the application cache, and access account settings.

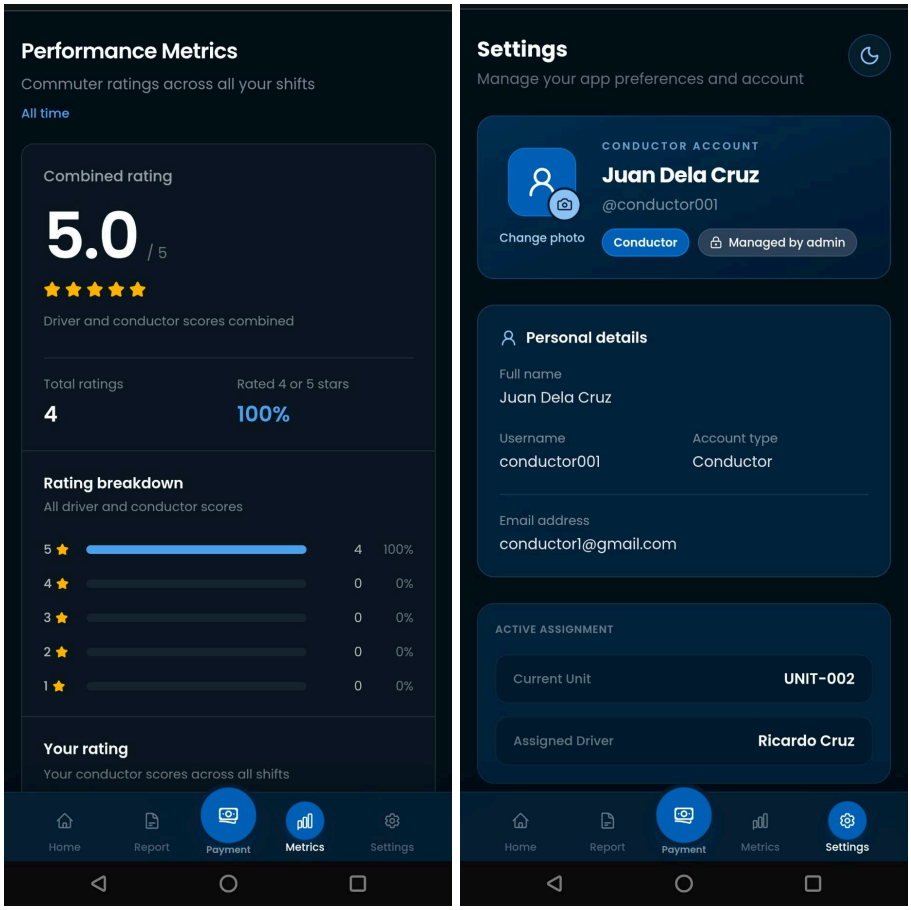
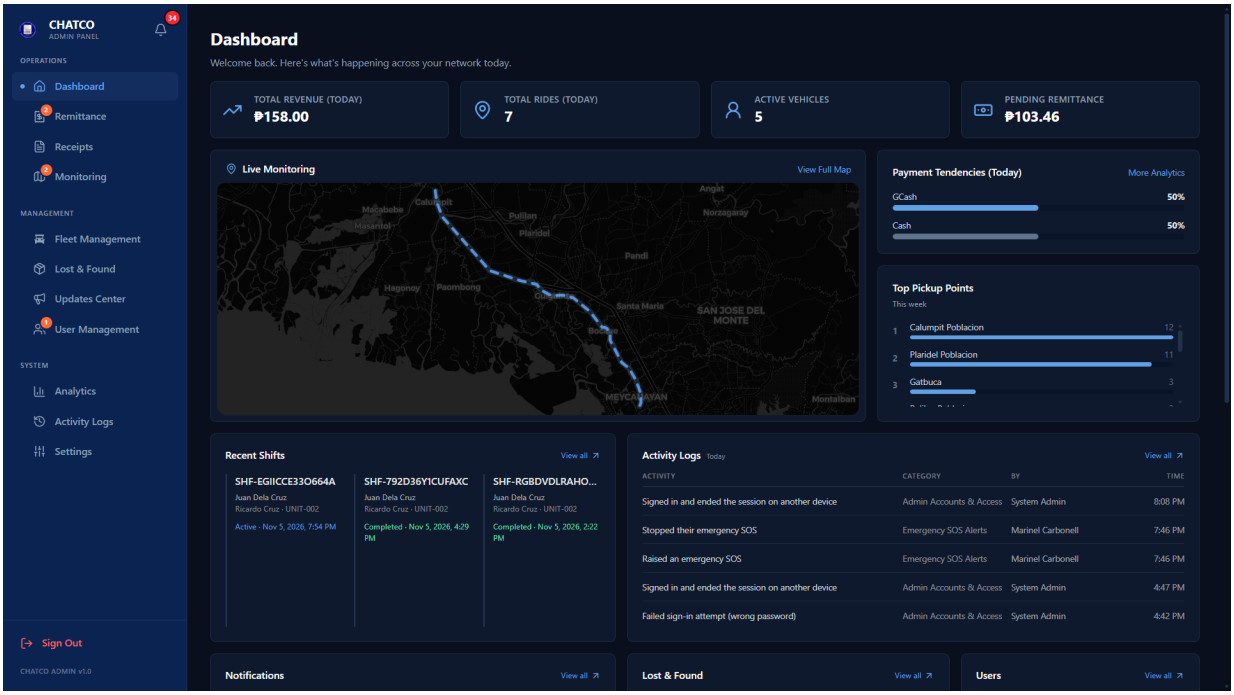


Figure 4.25: Metrics and Settings (Conductor)

Administrator Interface

The Administrator Interface is designed for the personnel responsible for managing the E-Chatco system, monitoring transportation operations, and overseeing the jeepney fleet. It provides access to user management, vehicle records, financial transactions, route configuration, safety monitoring, and system administration functions. The interface modules are illustrated in Figures 4.26 to 4.35.

The Dashboard displays summary cards for Total Rides (Today), Total Revenue (Today), Pending Remittance, and Active Vehicles, alongside a live map of currently on-duty units. The Analytics feature provides administrators with summarized information about ridership, revenue (cash, GCash, voucher rides), and shift remittance for selected date ranges.



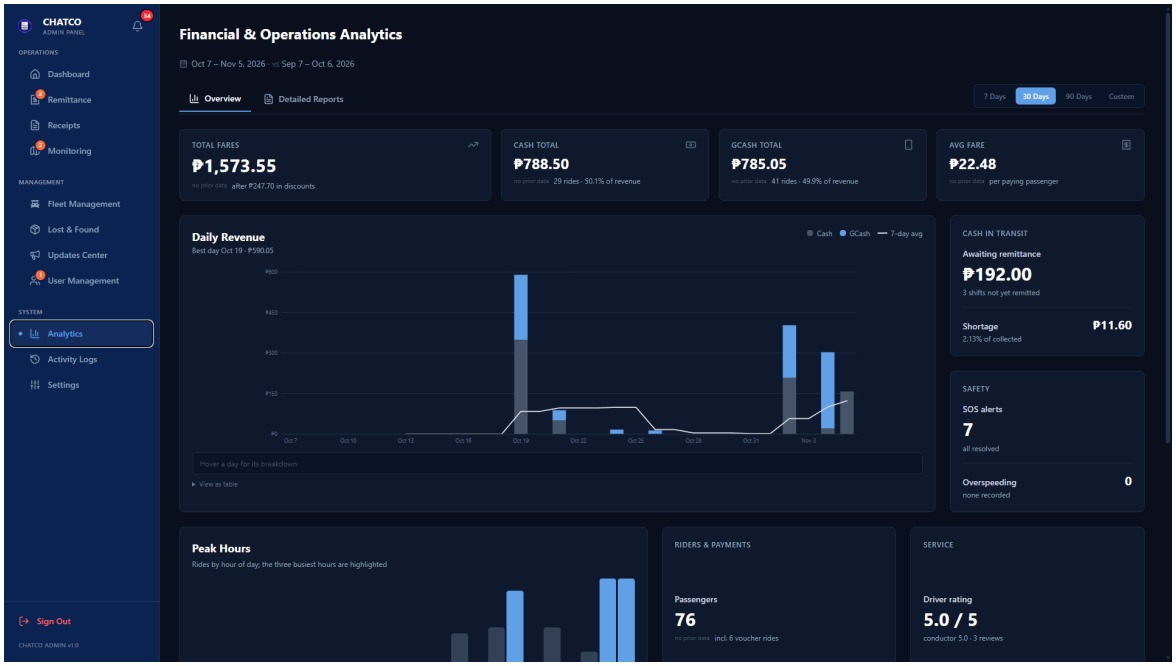


Figure 4.26: Admin Dashboard and Analytics

The Live Monitoring feature allows administrators to observe the movement and operating conditions of jeepneys through a real-time fleet map. The map also displays overspeeding events and commuter demand zones. Units on break are marked with their break time.

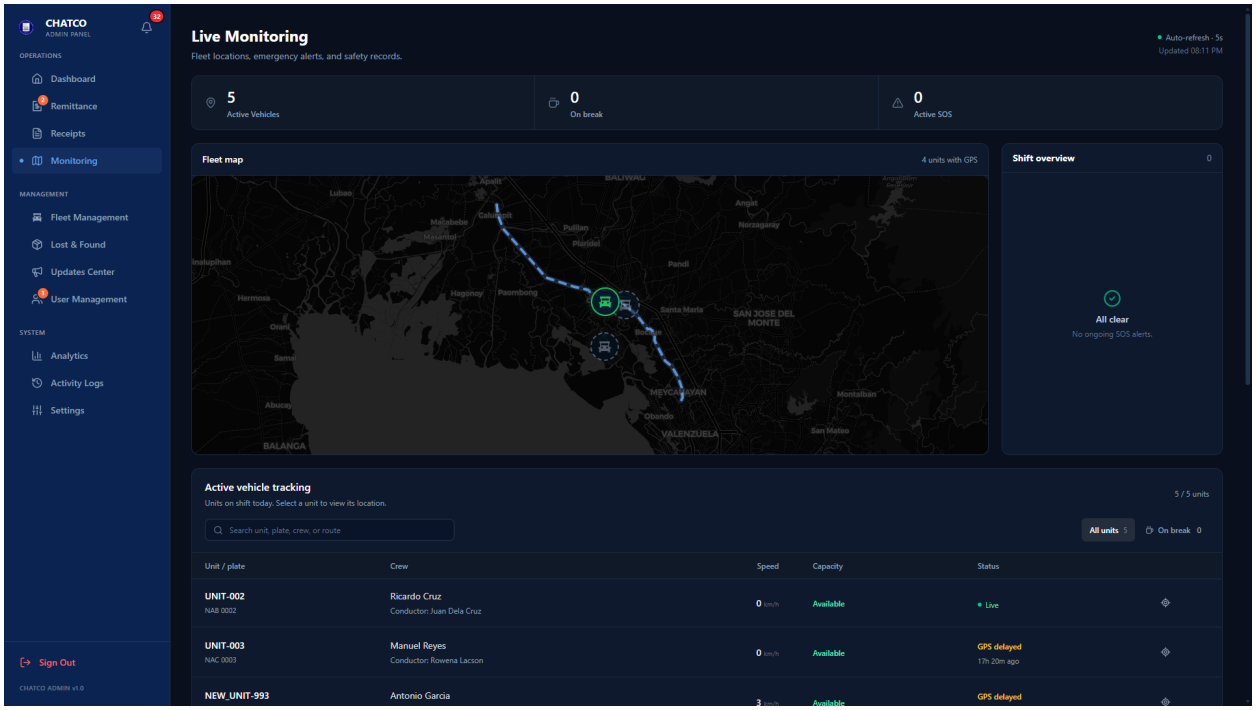
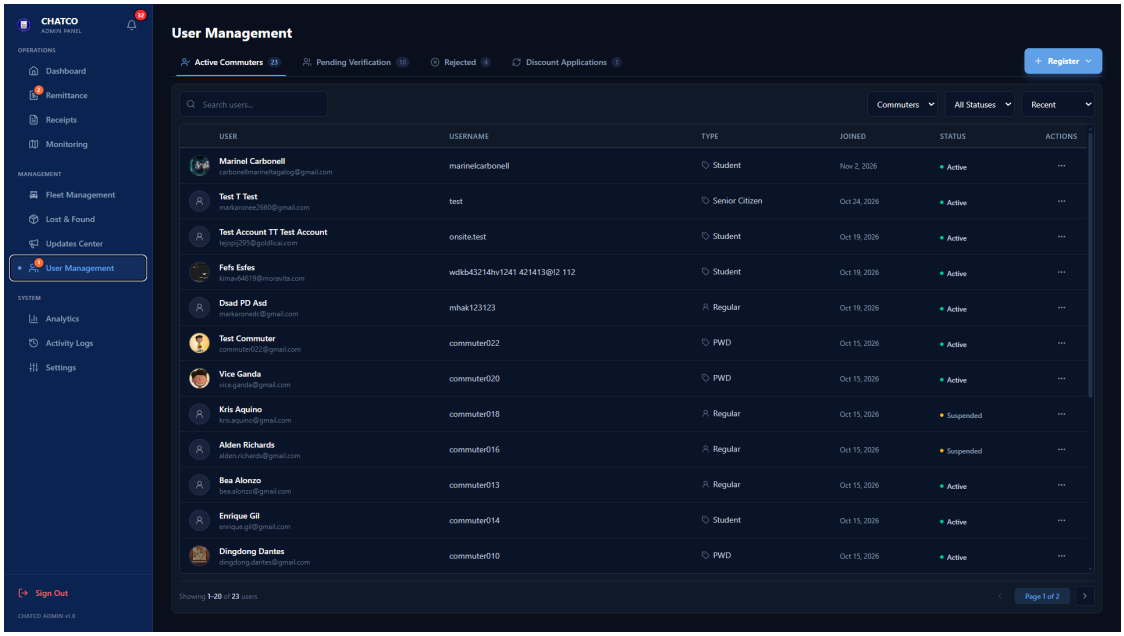


Figure 4.27: Live Monitoring

The User Management feature allows administrators to manage registered user accounts, including commuters, conductors, and other administrators. The Registrations sub-module lists pending commuter applications with the uploaded valid ID for review. Administrators may approve an application (the commuter receives an approval email and can sign in with the type they applied for), reject it with a reason of at least three characters (the commuter receives a rejection email and the email becomes available for re registration), or add a commuter on-site at the terminal with a default password that must be changed at first sign-in.



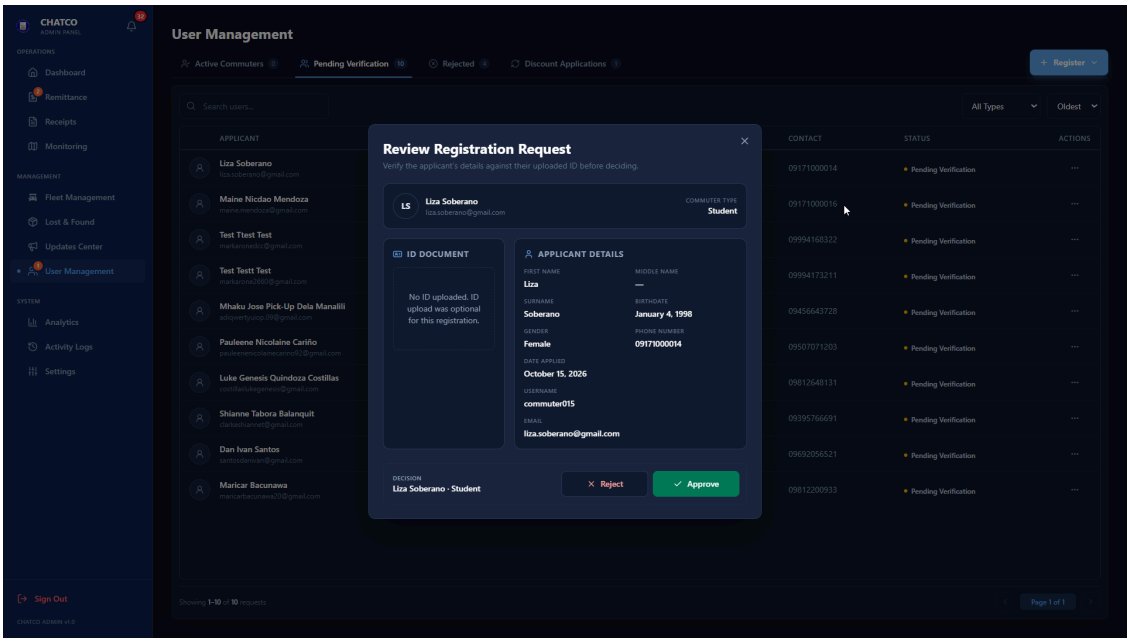


Figure 4.28: User Management and Registrations

The Accounts sub-module supports editing, suspending (with reason codes and durations of 1, 3, 7, 14, 30, 90 days, or permanent), unsuspending, disabling (for conductors), resetting credentials, and deleting accounts. The system prevents self-suspension, self deletion, deletion of the last administrator account, and suspension of conductor accounts (conductors use Disable Account instead).

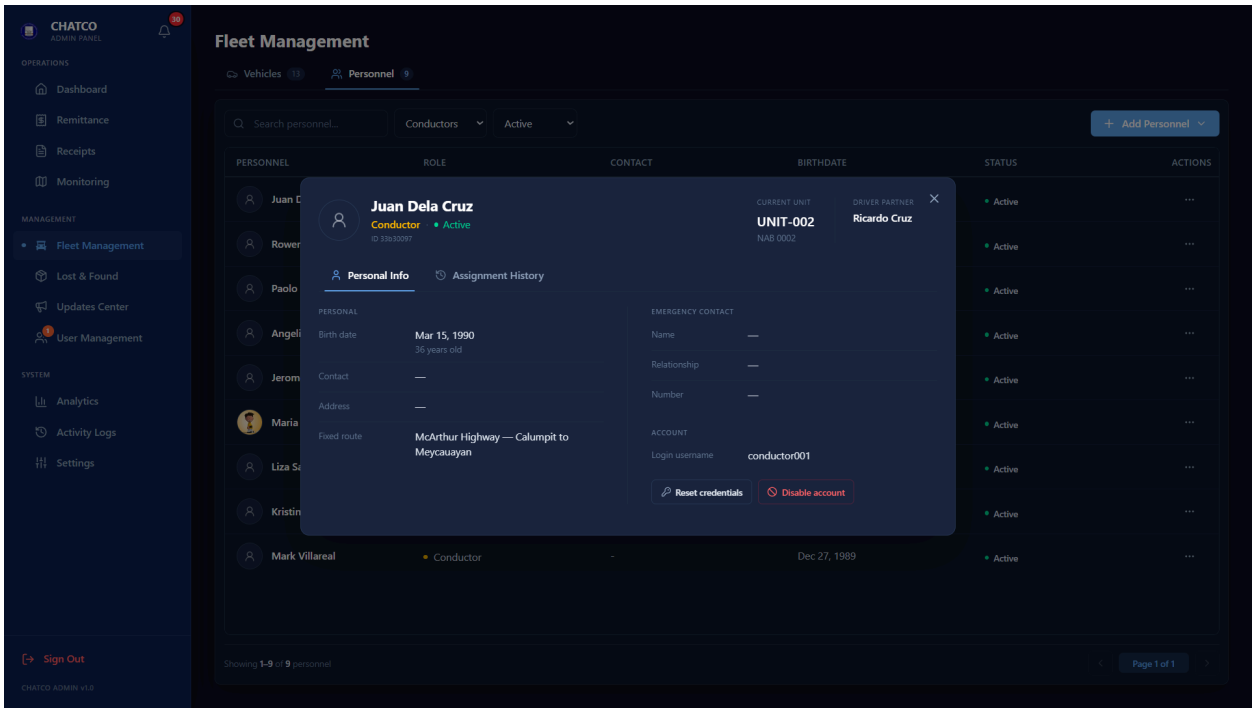


Figure 4.29: User Accounts Sub-module

The Discount Applications and Renewals sub-module, illustrated in Figure 4.30, is the administrator-side counterpart of the commuter's Application of Discounts and Student Renewal features. It lists all pending discount applications and student renewals submitted by commuters, with the uploaded supporting documents available for full-size review. For each pending item, the administrator may approve it (the commuter's type is updated and the configured discount

rate is applied to subsequent fares), reject it with a reason (the commuter stays Regular and is notified with the reason), or request corrections with a reason (the commuter is notified and the form reopens for resubmission with the same category). The system prevents reviewing an application whose supporting document has changed since the review was opened, prevents reviewing the same application twice, and supports filtering by status, type (Application or Renewal), and category.

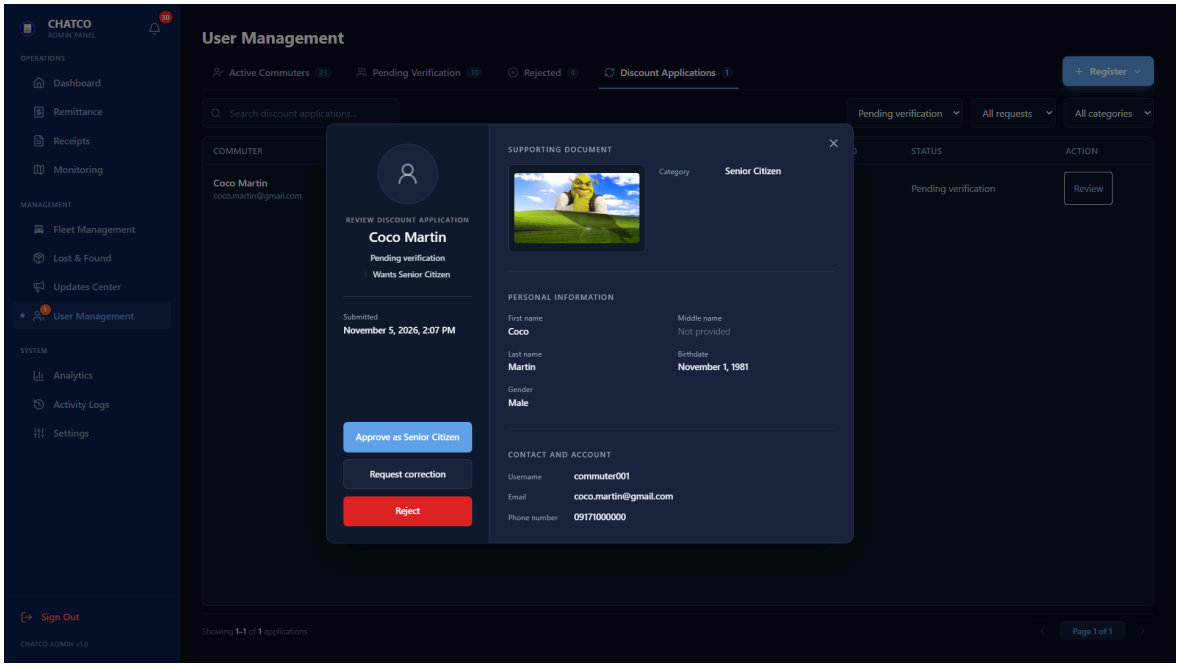
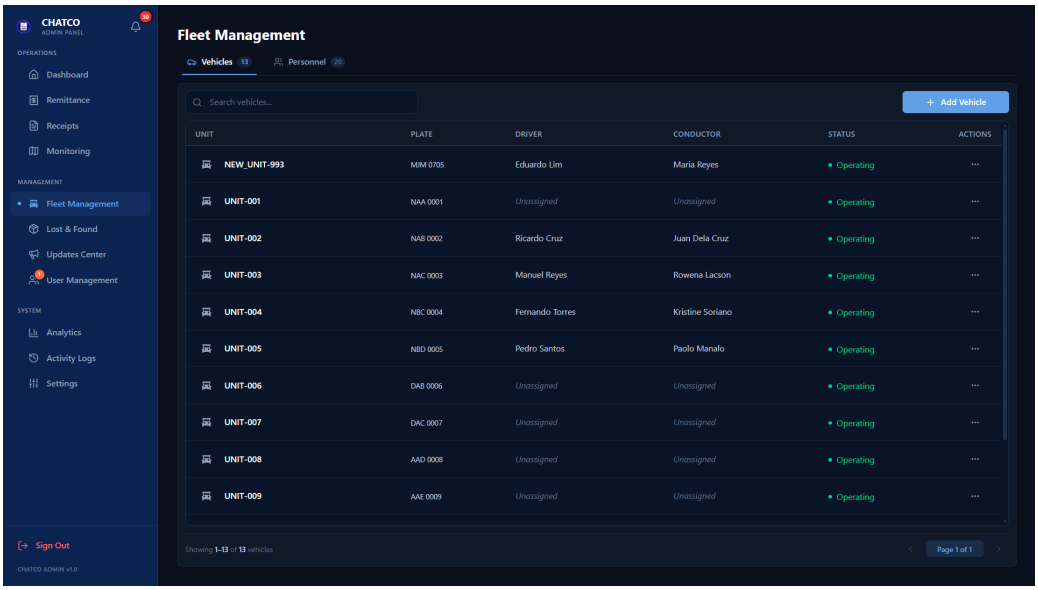
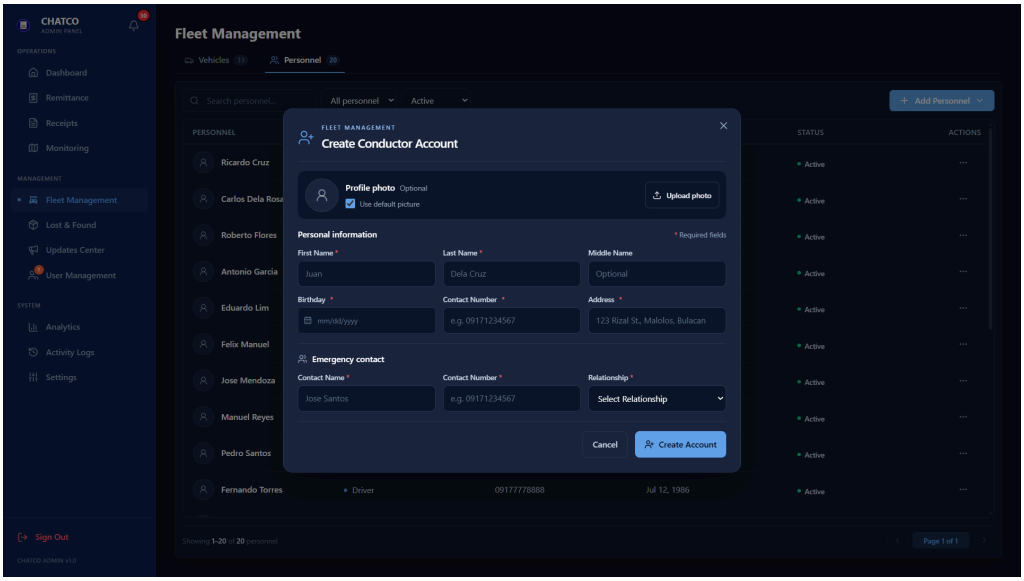


Figure 4.30: Discount Applications and Renewals (Admin Side)

The Fleet Management feature allows administrators to manage the transportation units and personnel associated with the E-Chatco system. The Vehicles sub-module supports adding vehicles with plate, brand, and model (the next unit number is generated automatically), assigning crew for the day (route, driver, and conductor), changing vehicle status (Operating or Under Maintenance), and deleting vehicles that are not on a shift. The Personnel sub-module

supports adding drivers with license details and images, creating conductor accounts with generated credentials, disabling conductors (with admin password verification), resetting conductor credentials, removing personnel (Terminated or Resigned), and recovering lost devices mid-shift. Shift logs and ratings are available for each driver and conductor.





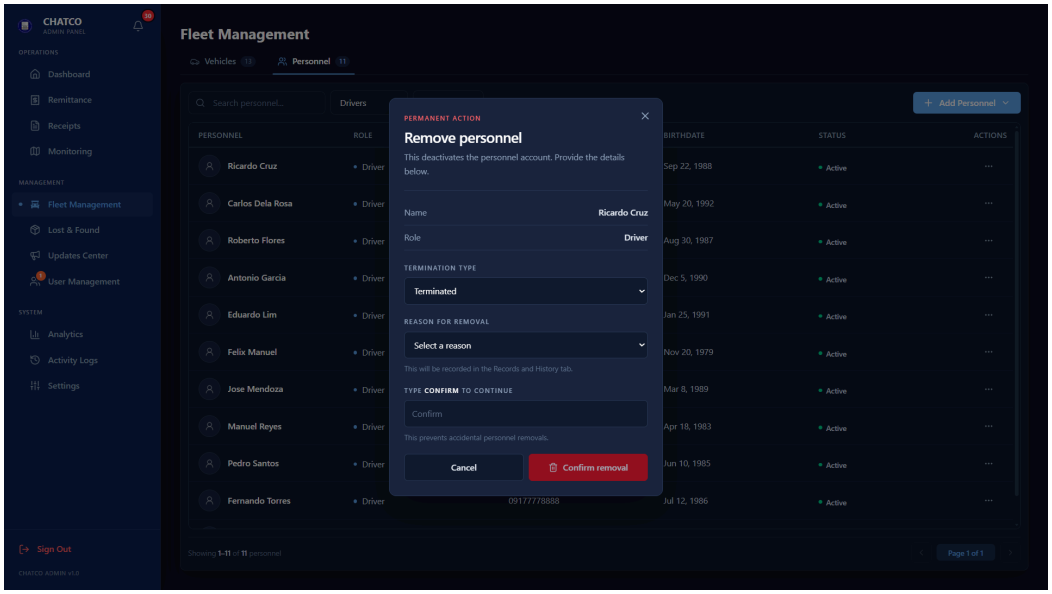
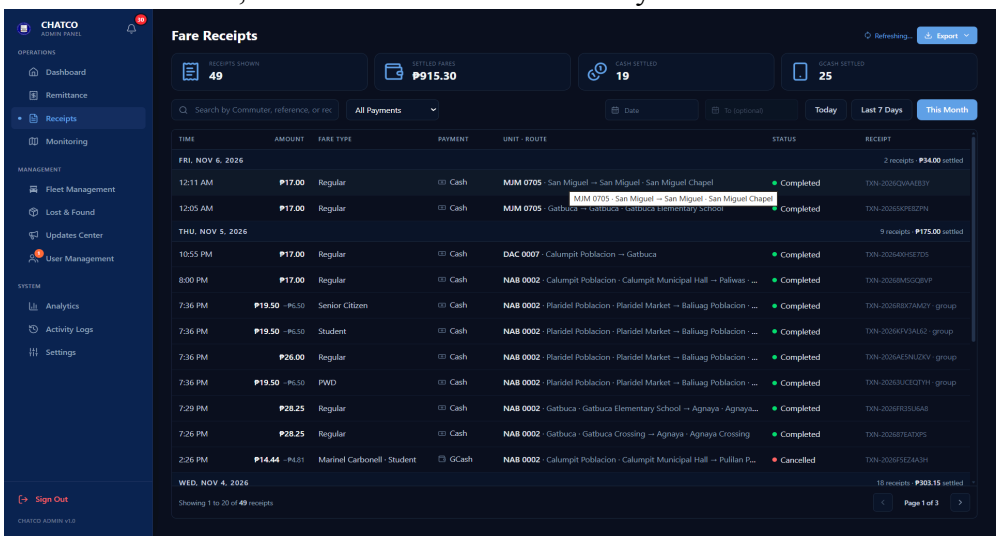


Figure 4.31: Fleet Management

The Receipts and Transactions feature allows administrators to search and review payment records by reference, payment method, fare type, and date, with export capability. The Remittance feature allows administrators to review the daily earnings submitted by conductors, filter by conductor and driver, and declare cash (with shortage or overage automatically computed against the expected amount). Remittance reminders repeat at the configured interval until the cash is declared, and overdue remittances notify administrators.



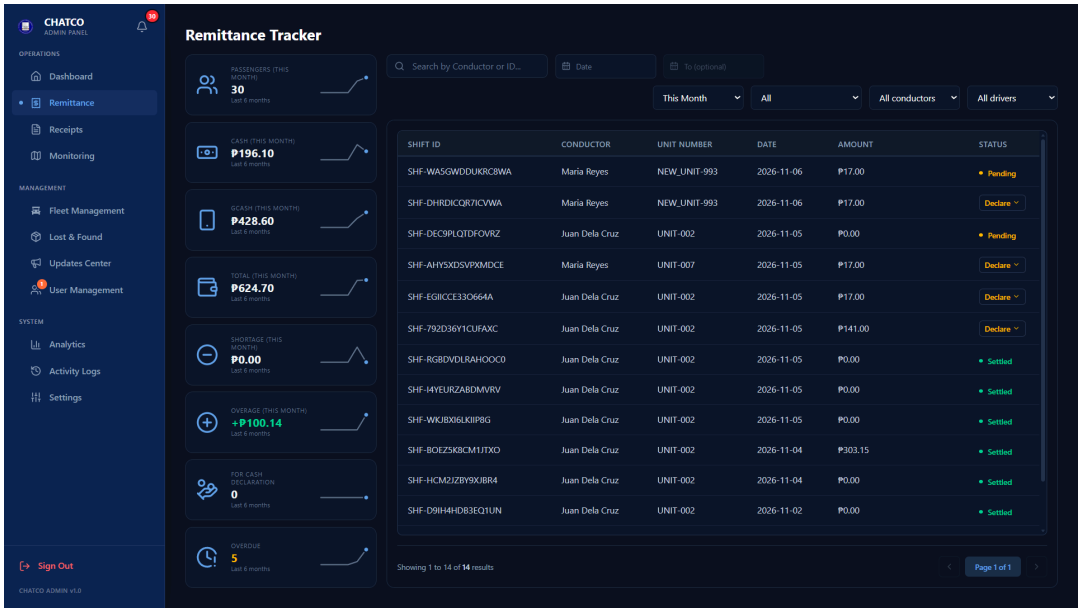
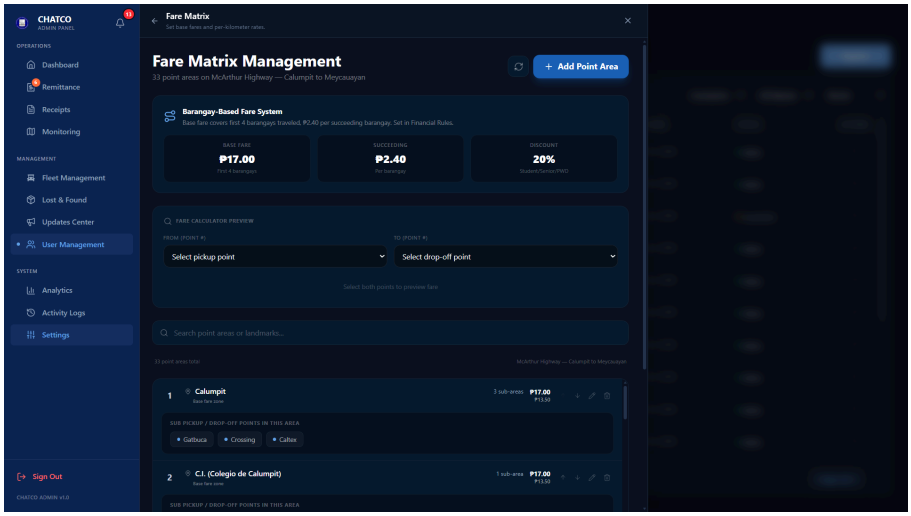
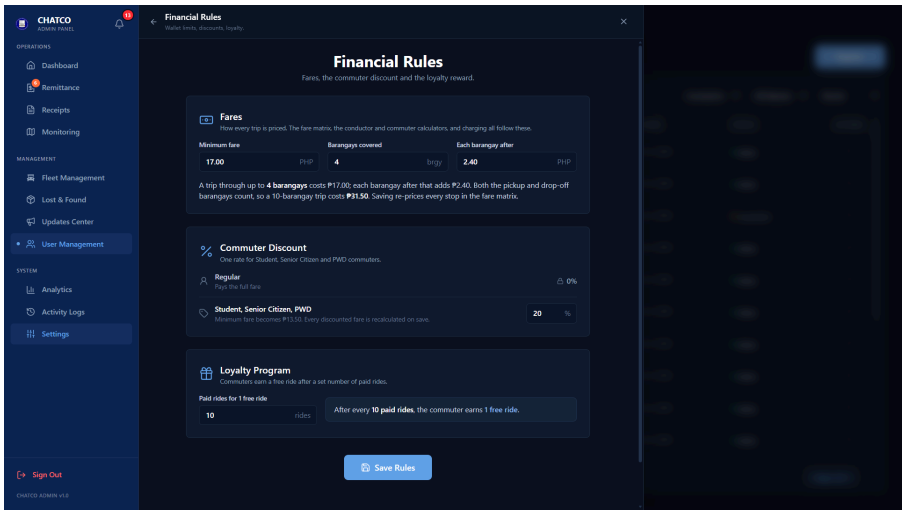


Figure 4.32: Receipts and Remittance (Admin)

The Settings module groups the configurable parameters of the E-Chatco system. The Fare Matrix sub-module allows administrators to manage the designated stops (fare points) and their landmarks along the route. The Route sub-module allows administrators to draw and publish the route line shown on the commuter, conductor, and administrator maps. The Financial Rules sub-module configures the minimum fare (1 to 1000), succeeding fare (0 to 1000), base barangays (0 to 100), discount rate (0% to 100%), and the number of paid rides required to earn a free ride (1 to 100). When the financial rules are changed, the conductor is signed out after a 20-second notice and the same shift reloads with the new fares. The Operations Rules sub-module configures the speed limit (10 to 120 km/h), maximum shift duration (1 to 48 hours), remittance grace period (1 to 1440 minutes), and reminder repeat interval (5 to 10080 minutes).


CHATCO
ADMIN PANEL
OPERATIONS
Dashboard
Remittance
Receipts
Monitoring
MANAGEMENT
Fleet Management
Lost & Found
Updates Center
User Management
SYSTEM
Analytics
Activity Logs
Settings
Sign Out
Fare Matrix
Fare Matrix Management
+ Add Point Area
Barangay-Based Fare System
BASE FARE
₱17.00
SUCCEEDING
₱2.40
DISCOUNT
20%
FARE CALCULATOR PREVIEW
Select pickup point
Select drop-off point
Search point areas or landmarks...
1
Calumpit
2
C.I. (Colegio de Calumpit)
Gatbuca
Crossing
Calfox


CHATCO
ADMIN PANEL
Financial Rules
Financial Rules
Fares, the commuter discount and the loyalty reward.
Fares
Minimum fare
17.00
PHP
Barangays covered
4
brgy
Each barangay after
2.40
PHP
A trip through up to 4 barangays costs ₱17.00; each barangay after that adds ₱2.40. Both the pickup and drop-off barangays count, so a 10-barangay trip costs ₱31.50. Saving re-prices every stop in the fare matrix.
Commuter Discount
One rate for Student, Senior Citizen and PWD commuters.
Regular
Pays the full fare
0%
Student, Senior Citizen, PWD
Minimum fare becomes ₱13.50. Every discounted fare is recalculated on save.
20
%
Loyalty Program
Commuters earn a free ride after a set number of paid rides.
Paid rides for 1 free ride
10
rides
After every 10 paid rides, the commuter earns 1 free ride.
Save Rules

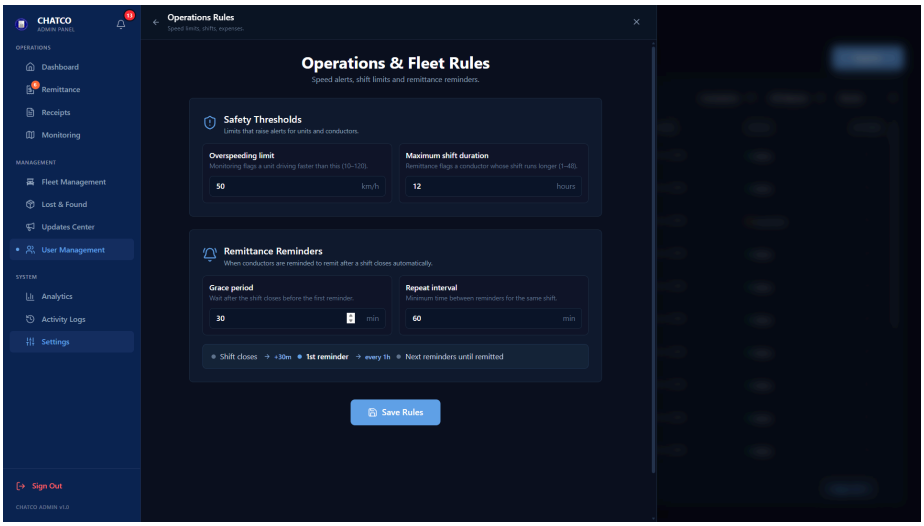


Figure 4.33: Fare Matrix, Route, Financial Rules, and Operations Rules

The Voucher Generator sub-module allows administrators to create batches of voucher codes for distribution to eligible commuters, with each voucher valid for a minimum of 24 hours. The Receipt Settings sub-module configures the business name, footer, and auto print behavior shown on the conductor's printed receipts. The Notification Templates sub module provides a template editor with sample-data preview, allowing administrators to customize the wording of approval, rejection, correction, and other notices. The Maintenance Mode sub-module allows administrators to enable maintenance for the Commuter, Conductor, or both applications with a custom message. The FAQ Management sub-module maintains the questions and answers shown in the landing page chat bubble. The Profile and Security sub-module allows administrators to update their own photo and password.

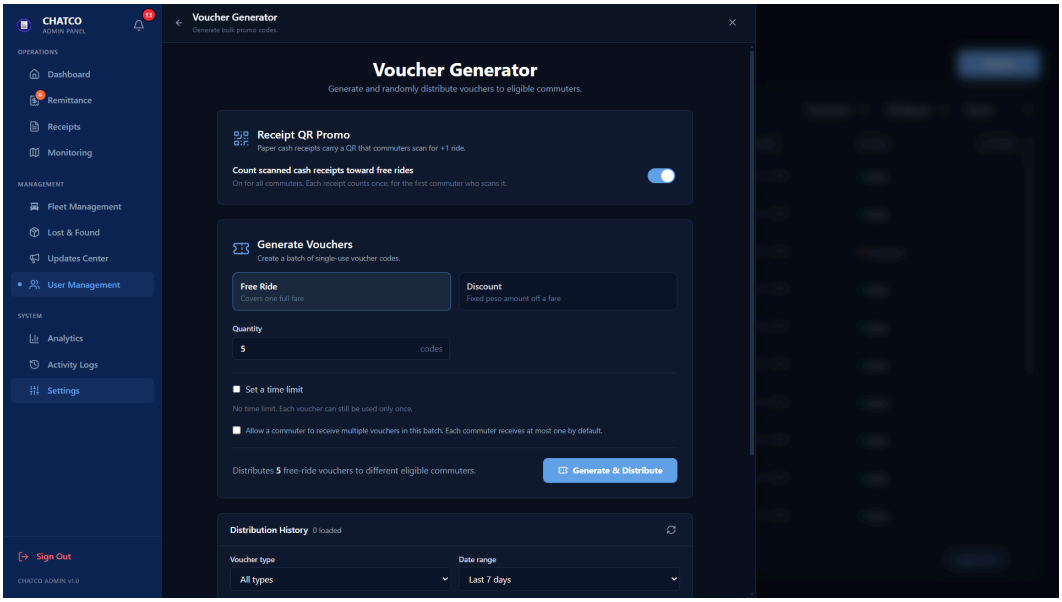
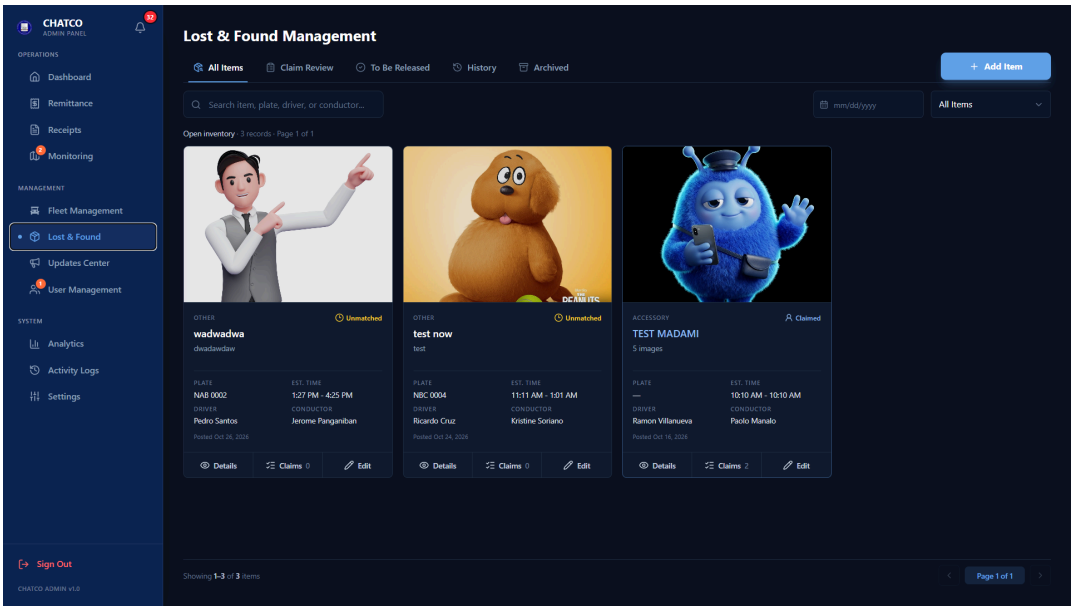
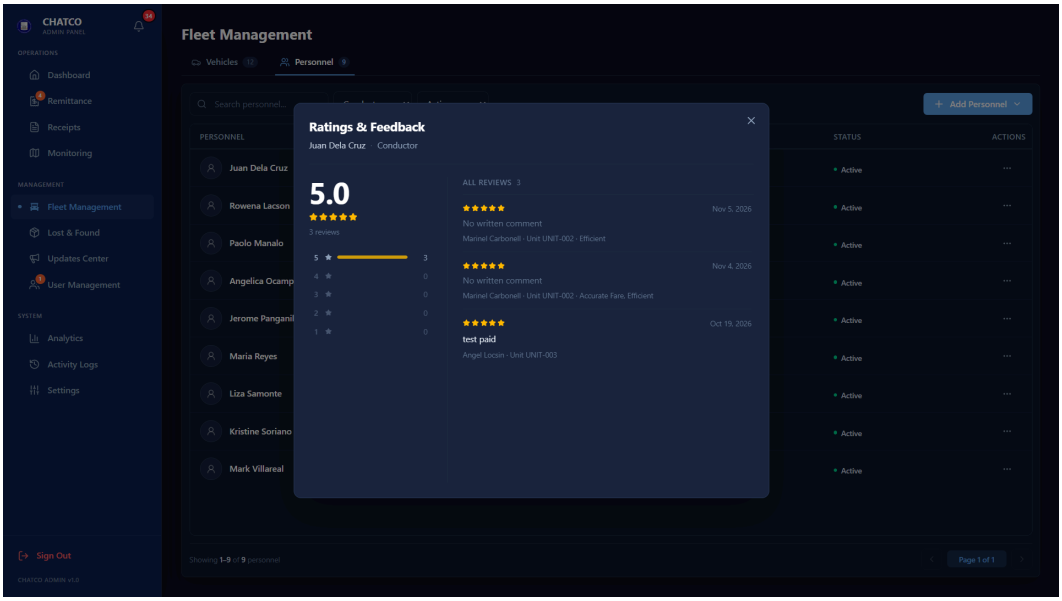


Figure 4.34: Voucher Generator and Other Settings

The Lost and Found feature allows administrators to record found items, review commuter claims, approve or reject claims with pickup schedules, release items, and close completed records. The Feedback feature provides access to commuter ratings and comments per conductor or driver. The Announcements feature allows administrators to create, edit, archive, and delete announcements intended for commuters (announcements do not appear in the conductor application, which has no announcement feed). The Activity Logs feature records all important actions performed within the system, with filtering by category and export capability.


CHATCO
ADMIN PANEL
OPERATIONS
Dashboard
Remittance
Receipts
Monitoring
MANAGEMENT
Fleet Management
Lost & Found
Updates Center
User Management
SYSTEM
Analytics
Activity Logs
Settings
Sign Out
Lost & Found Management
All Items
Claim Review
To Be Released
History
Archived
Add Item
Search item, plate, driver, or conductor...
mm/dd/yyyy
All Items
Open inventory · 3 records · Page 1 of 1
OTHER
Unmatched
wadwadwa
dwadawdaw
PLATE
NAB 0002
EST. TIME
1:27 PM - 4:25 PM
DRIVER
Pedro Santos
CONDUCTOR
Jerome Panganiban
Posted Oct 26, 2026
Details
Claims 0
Edit
OTHER
Unmatched
test now
test
PLATE
NBC 0004
EST. TIME
11:11 AM - 1:01 AM
DRIVER
Ricardo Cruz
CONDUCTOR
Kristine Soriano
Posted Oct 24, 2026
Details
Claims 0
Edit
ACCESSORY
Claimed
TEST MADAMI
5 images
PLATE
—
EST. TIME
10:10 AM - 10:10 AM
DRIVER
Ramon Villanueva
CONDUCTOR
Paolo Manalo
Posted Oct 16, 2026
Details
Claims 2
Edit
Showing 1-3 of 3 items
Page 1 of 1


CHATCO
ADMIN PANEL
OPERATIONS
Dashboard
Remittance
Receipts
Monitoring
MANAGEMENT
Fleet Management
Lost & Found
Updates Center
User Management
SYSTEM
Analytics
Activity Logs
Settings
Sign Out
Fleet Management
Vehicles
Personnel
Add Personnel
PERSONNEL
STATUS
ACTIONS
Juan Dela Cruz
Rowena Lacson
Paolo Manalo
Angelica Ocamp
Jerome Panganil
Maria Reyes
Liza Samonte
Kristine Soriano
Mark Villareal
Active
Ratings & Feedback
Juan Dela Cruz · Conductor
5.0
3 reviews
ALL REVIEWS 3
Nov 5, 2026
No written comment
Marinel Carbonell · Unit UNIT-002 · Efficient
Nov 4, 2026
No written comment
Marinel Carbonell · Unit UNIT-002 · Accurate Fare, Efficient
Oct 19, 2026
test paid
Angel Locsin · Unit UNIT-003
Showing 1-9 of 9 personnel
Page 1 of 1

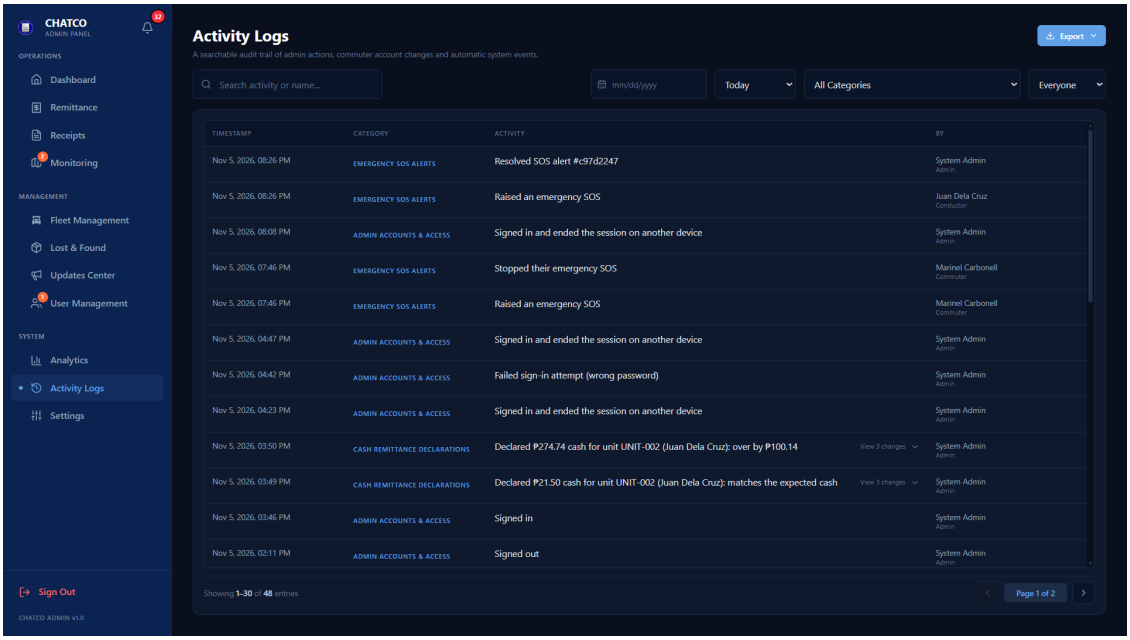


Figure 4.35: Lost and Found, Feedback, Announcements, and Activity Logs

System Evaluation and Testing

This section presents how the E-Chatco system was evaluated and tested by the client to verify that it meets the functional, usability, and operational requirements necessary for use in the business processes of CHATCO Cooperative.

User Acceptance Testing (UAT)

User Acceptance Testing was conducted with the client, who explored and tested the system's core functionalities based on the modules provided. The client tested each feature according to its intended purpose and verified that the system performed as expected across the commuter, conductor, and administrator applications.

The client primarily tested the following areas:

- **General Access and Authentication** - covering sign-in for commuters, conductors, and administrators; invalid login handling; role-based page access; and the password recovery and set-new-password flows.
- **Commuter Modules** - covering live map browsing, Pick Me Up hailing within the 1-kilometer radius, GCash QR payment scanning, cash receipt claiming, voucher application during payment, free-ride reward earning, Share My Ride, Lost and Found claim submission, Ride Feedback submission, profile editing, Application of Discounts for Student, Senior Citizen, and PWD categories, and Student Renewal submission.
- **Conductor Modules** - covering Unit Verification and shift start, cash and GCash fare recording, voucher application, receipt printing, capacity and break status, offline cash recording, End-of-Day close-out, and remittance submission with separate cash and GCash declarations.
- **Administrator Modules** - covering user registration review, account suspension and disabling, discount application and student renewal review, fleet and personnel management, live monitoring, remittance declaration, voucher generation, fare matrix and financial rules configuration, lost and found item logging, announcement creation, FAQ management, and activity log review.

The feedback gathered from this testing was used to refine minor system features, including the wording of certain error messages and the layout of the discount application review panel, and to confirm readiness for deployment.

Client Evaluation Form

A client evaluation form was distributed to individuals involved in the jeepney operations of CHATCO Cooperative, including the general manager, conductors, and selected commuters who

served as respondents. It was used to collect feedback on the system's compliance with the ISO/IEC 25010:2023 standard, specifically under the following characteristics: Functional Suitability, Reliability, Performance Efficiency, Usability, Compatibility, Security, Maintainability, and Portability.

The form also served as the primary instrument for User Acceptance Testing, as the client had to test the system comprehensively before providing their evaluation. The evaluation form provided criteria to be rated using a five-point Likert scale, where 5 represents Strongly Agree, 4 represents Agree, 3 represents Neutral, 2 represents Disagree, and 1 represents Strongly Disagree. This feedback produced both qualitative and quantitative insights into the system's readiness for actual deployment.

The actual form can be found in Appendix D of this documentation, while the summarized results are discussed in the Implementation Results section.

Implementation Plan

The implementation of E-Chatco: An Integrated Smart-Hailing and Fleet Management Platform for the Modernized Philippine Jeepney involves the deployment of the web-based system in a production environment where authorized staff and commuters can access it. The implementation ensures that all necessary hardware, software, and network configurations are in place for proper system operation along the Calumpit to Meycauayan route.

Implementation Tasks

The implementation of E-Chatco involved several major tasks to ensure the system's successful deployment and transition from the development phase to operational use. Each task was carefully planned to ensure that the system could function smoothly in the cooperative's business environment.

1. **System Deployment**

The E-Chatco platform is structured as a web-based application with a Laravel backend and a Next.js frontend. The backend provides the system's application logic, API endpoints, authentication, database operations, and scheduled processes, while the frontend provides the interfaces for commuters, conductors, and administrators. Deployment requires hosting environments that support the application's respective runtime requirements. The Laravel backend requires PHP 8.2 or later and the necessary PHP extensions, while the Next.js frontend requires a compatible Node.js runtime. The backend and frontend must be configured to communicate through the appropriate application URLs and environment variables. The deployment configuration must also provide the required database connection, broadcasting credentials, payment gateway credentials, file storage settings, and email service configuration when these external services are enabled. The application should be tested in the intended hosting environment before it is made available for operational use.

2. **Database Configuration**

E-Chatco uses MySQL as its primary relational database management system. The database stores information related to user accounts, commuter and conductor profiles, vehicles, routes, fare points, transactions, payments, remittances, discount applications, student renewals, lost and found records, announcements, and other system activities. Before deployment, the database must be prepared and configured using the application's migration files. These migrations establish the required tables, fields, indexes, and relationships used by the system. Initial configuration data, including fare points, financial rules, operational settings, and other required records, must also be prepared according to the cooperative's approved operational requirements. Database access credentials must be configured securely through the backend environment settings.

Appropriate backup and recovery procedures should be established to protect the system's stored information.

3. **User Account Setup**

E-Chatco supports three primary user roles: Administrator, Conductor, and Commuter. Each role is provided with access to the functions appropriate to its responsibilities. Administrator accounts are used to manage system settings, registrations, vehicles, personnel, financial records, remittances, announcements, and other administrative activities. Conductor accounts are used for unit verification, shift operations, fare recording, payment processing, and remittance submission. Commuter accounts provide access to vehicle monitoring, hailing, fare information, payment-related functions, discount applications, rewards, and safety features. Account creation and approval procedures must follow the system's registration and account-management workflows. Role-based access controls should be verified to ensure that users can access only the functions authorized for their assigned roles.

4. Testing and Verification

Before operational use, the system must undergo functional and integration testing in the intended deployment environment. Verification should cover authentication, role-based access, live vehicle location updates, commuter hailing, fare computation, GCash payment workflows, voucher redemption, receipt claiming, discount applications, student renewals, conductor shift operations, remittance submission, fleet monitoring, lost and found management, announcements, and safety-related functions. Testing should also verify the communication between the frontend, backend, database, and configured external services. Payment gateway callbacks, real-time broadcasting, email delivery, and file uploads should be checked when their corresponding services are configured. The

updated CHATCO system test case document provides a reference for the functional scenarios and expected results across the five major system components. Additional verification should be performed after deployment to confirm that the configured production environment behaves as expected.

5. **Training and Orientation**

   Before the system is adopted for regular operations, the cooperative's designated personnel should receive appropriate orientation on the functions relevant to their roles. The orientation may include commuter registration and discount applications, conductor unit verification and shift management, fare and payment recording, remittance submission, administrator fleet monitoring, user management, and safety-related functions. User guidance and technical documentation should be made available to help users understand the system's workflows and report issues that may occur during operation.

6. **System Launch**

   The system may be made available for operational use after the deployment configuration, database setup, functional verification, and user preparation have been completed and approved by the cooperative. Following deployment, the system should be monitored for errors, service availability, performance issues, and data consistency. Regular maintenance should include software updates, security reviews, database backups, configuration checks, and the resolution of reported issues. Any future changes to the system should be tested before being introduced into the operational environment.

Resources Needed

Table 4.4 lists the resources required for the implementation of E-Chatco within the cooperative's business environment. These include the software, hardware, and personnel necessary to support the system's deployment and operation**.**

Category	Resource	Description / Purpose
Software Requirements	**Laravel 12 / PHP 8.2**	**Backend framework for managing application logic, API endpoints, authentication, and database operations.**
	Next.js 16 / React 19	**Frontend framework for building the commuter, conductor, and administrator interfaces.**
	MySQL	**Relational database management system configured for storing system records and operational data.**
	Leaflet / React-Leaflet	**Mapping technologies used for displaying maps and location-related information.**
	Web Hosting Service	**Used for deploying the live system and database.**
	Web Browser	**Required for accessing and testing the deployed web application.**
Hardware Requirements	**Client Devices (Laptop, Tablet, Smartphone)**	**Used by administrators, commuters, and conductors to access the system online.**
	Operational Device with Thermal Printer	**Dedicated device used by conductors for fare processing, GCash payment verification, and receipt or ticket issuance.**
	Stable Internet Connection	**Ensures smooth communication with the hosted system, database, and GCash payment gateway.**
Personnel Requirements	**Project Developers**	**Responsible for system deployment, maintenance, and technical support.**
	Client-side Users (Manager, Conductors, Commuters)	**End-users of the system who will operate and utilize its functionalities in daily business operations.**

Table 4.4: Resources Needed

Site-Specific Requirements

The implementation of E-Chatco requires certain site-specific provisions to ensure smooth system operation and maintenance within the client's business environment. These requirements focus on the facilities, personnel, and operational setup at CHATCO Cooperative.

- **Administrative Workstation.** A computer or laptop with internet access should be available to authorized administrative personnel for account management, discount application reviews, fleet monitoring, remittance review, and system configuration.
- **Designated Personnel.** The cooperative should assign personnel responsible for user account administration, operational monitoring, and communication with system users.
- **Stable Internet Connection.** A reliable internet connection is necessary for accessing the web application and using features that depend on network communication, such as location broadcasting, hailing, payment processing, and safety notifications.
- **Compatible User Devices.** Commuters, conductors, and administrators should have access to supported devices and updated web browsers appropriate for their assigned system functions.
- **Thermal Printing Device.** Where printed fare receipts are required, a compatible printer should be available and properly configured with the device and browser used by the conductor. The system uses browser-based printing functionality.
- **External Service Configuration.** Payment gateway, real-time broadcasting, email delivery, and file storage services must have valid credentials and appropriate environment settings before their respective integrations can be used in a live environment.
- **Database Backup Procedures.** Regular database backups and recovery procedures should be established to protect important operational records and support business continuity.

Implementation Results

Following the development of the E-Chatco platform, the system underwent testing and evaluation to assess its functionality, usability, and suitability for the operational requirements of CHATCO Cooperative. The evaluation covered the system's major modules, including user authentication, fleet management, smart hailing, fare computation, payment processing, remittance monitoring, lost and found management, and safety-related functions. The results of the testing activities, User Acceptance Testing, and Client Evaluation Form were used to assess the system's performance and alignment with the identified user requirements.

Based on the evaluation, the client confirmed that the system's core functionalities, including real-time GPS tracking and hailing, GCash payment verification, fare calculation using predefined operational point areas, automated discount application for verified Student, Senior Citizen, and PWD commuters, renewal of student discount eligibility, boundary remittance tracking with separate cash and GCash recording, fleet monitoring, lost and found management, and safety features such as Share My Ride, performed according to their operational needs. Minor interface and usability adjustments were identified and addressed accordingly. The detailed evaluation results, including the computed average ratings per ISO/IEC 25010 quality characteristics, are presented in the following sections.

Functional Suitability

Functional suitability evaluates how effectively E-Chatco performs its intended functions and meets the operational requirements of CHATCO Cooperative. Among the criteria, functional appropriateness obtained the highest mean ($x = 5.00$; $SD = 0.00$), indicating that the system's modules and configurations appropriately support the workflow of jeepney operations, including real-time tracking, smart hailing, fare collection, discount application, boundary remittance tracking, and fleet monitoring. The other criteria, functional completeness and functional

correctness (x = 4.86, 4.71; SD = 0.38, 0.49), further affirm the system's capability to deliver accurate and comprehensive functionalities aligned with the users' needs. Overall, the system achieved an excellent rating with a mean score of 4.86 and a standard deviation of 0.15, reflecting the respondents' strong confidence in E-Chatco's performance and reliability in fulfilling its core operational objectives.

Table 4.5 presents the summarized results of the functional suitability evaluation of the system.

Criterion	5	4	3	2	1	Mean (x)	SD	Descriptive Interpretation
1. Functional Completeness - The system provides a complete set of functions that covers real-time tracking, GCash e-payment, fare calculation, discount application, remittance tracking, and lost and found management as intended.	**6**	**1**	**0**	**0**	**0**	**4.86**	**0.38**	**Excellent**
2. Functional Correctness - The system provides accurate results in fare calculation, GCash verification, discount application, and remittance tracking, aligned with jeepney operations.	**5**	**2**	**0**	**0**	**0**	**4.71**	**0.49**	**Excellent**
3. Functional Appropriateness - The system provides features that effectively support smart hailing, fare collection, fleet monitoring, and safety tasks.	**7**	**0**	**0**	**0**	**0**	**5.00**	**0.00**	**Excellent**
Overall Mean						**4.86**	**0.15**	**Excellent**

Table 4.5: Frequency Distribution and Descriptive Statistics of Respondents' Ratings on the System's Functional Suitability

Reliability

Reliability evaluates how consistently E-Chatco performs its processes and maintains stable operation during jeepney management tasks. Among the criteria, recoverability obtained the highest mean (x = 4.86; SD = 0.38), suggesting that the system performs reliably without unexpected failures or interruptions during usage. The other criteria, faultlessness, availability, and fault tolerance (x = 4.71, 4.57; SD = 0.49, 0.79), indicate that the system maintains consistent performance and continues functioning effectively even under varying operational conditions. Overall, the system garnered an excellent rating with an overall mean of 4.71 and a standard deviation of 0.12, reflecting the respondents' high confidence in E-Chatco's dependability, stability, and ability to sustain continuous service for jeepney operations.

Table 4.6: Frequency Distribution and Descriptive Statistics of Respondents' Ratings on the System's Reliability

Criterion	5	4	3	2	1	Mean (x)	SD	Descriptive Interpretation
1. Faultlessness - The system runs without errors in tracking, payment, discount application, and remittance tasks.	**5**	**2**	**0**	**0**	**0**	**4.71**	**0.49**	**Excellent**
2. Availability - The system is accessible during the required operational periods.	**5**	**2**	**0**	**0**	**0**	**4.71**	**0.49**	**Excellent**
3. Fault Tolerance - The system continues operating despite minor hardware or software issues (e.g., offline cash recording).	**5**	**1**	**1**	**0**	**0**	**4.57**	**0.79**	**Excellent**
4. Recoverability - The system can restore important records (transactions, remittances, discount applications) after crashes or failures.	**6**	**1**	**0**	**0**	**0**	**4.86**	**0.38**	**Excellent**
Overall Mean						**4.71**	**0.12**	**Excellent**

Performance Efficiency

Performance efficiency measures how effectively E-Chatco utilizes system resources and maintains responsiveness during jeepney operations. Among the criteria, time behavior obtained the highest mean (x = 5.00; SD = 0.00), indicating that the system performs tasks such as GPS broadcasting, hailing, GCash payment verification, and fare calculation with minimal delays. The other criteria, resource utilization and capacity (x = 4.57, 4.43; SD = 0.53, 0.53), demonstrate that the system efficiently manages server resources and sustains smooth performance even when handling multiple concurrent users or large transaction volumes. Overall, the system achieved an excellent rating with an overall mean of 4.67 and a standard deviation of 0.30, reflecting the respondents' confidence in E-Chatco's optimized performance and ability to deliver fast and efficient operations.

Table 4.7: Frequency Distribution and Descriptive Statistics of Respondents' Ratings on the System's Performance Efficiency

Criterion	5	4	3	2	1	Mean (x)	SD	Descriptive Interpretation
1. Time Behavior - The system responds quickly to user actions such as hailing, GCash payment scanning, and fare calculation.	**7**	**0**	**0**	**0**	**0**	**5.00**	**0.00**	**Excellent**
2. Resource Utilization - The system uses server and device resources efficiently without affecting performance.	**4**	**3**	**0**	**0**	**0**	**4.57**	**0.53**	**Excellent**
3. Capacity - The system supports multiple concurrent users (commuters, conductors, admins) and large transaction data without slowdown.	**3**	**4**	**0**	**0**	**0**	**4.43**	**0.53**	**Excellent**
Overall Mean						**4.67**	**0.30**	**Excellent**

Usability

Usability assesses how easily users can learn, navigate, and operate E-Chatco in performing jeepney-related tasks. Among the criteria, learnability obtained the highest mean (x = 4.86; SD = 0.38), indicating that users find the system intuitive and convenient to use when managing hailing, payments, discount applications, and remittance tracking. The other criteria suggest that the system's design supports quick familiarization and provides a visually organized interface that enhances user experience. Overall, the system received an excellent rating with an overall mean of 4.59 and a standard deviation of 0.15, reflecting the respondents' strong satisfaction with E-Chatco's ease of use, accessibility, and user-centered interface design.

Criterion	5	4	3	2	1	Mean (x)	SD	Descriptive Interpretation
1. Learnability - The system is easy to understand and operate for new users (commuters, conductors, administrators).	**6**	**1**	**0**	**0**	**0**	**4.86**	**0.38**	**Excellent**
2. Operability - The interface supports smooth use for commuters, conductors, and administrators.	**4**	**3**	**0**	**0**	**0**	**4.57**	**0.53**	**Excellent**
3. Error Prevention - The system prevents or alerts users of mistakes (e.g., duplicate hails, invalid GCash QR, ineligible discount application).	**4**	**3**	**0**	**0**	**0**	**4.57**	**0.53**	**Excellent**
4. User Satisfaction - Users are satisfied with the design, interaction, and experience of the system.	**4**	**2**	**1**	**0**	**0**	**4.43**	**0.79**	**Excellent**
5. Appropriateness Recognizability - Users can easily recognize whether the system is suitable for their tasks.	**3**	**4**	**0**	**0**	**0**	**4.43**	**0.53**	**Excellent**

6. User Interface Aesthetics - The interface is visually appealing and appropriate for a transport management platform.	**4**	**3**	**0**	**0**	**0**	**4.57**	**0.53**	**Excellent**
7. Accessibility - The system can be used by people with a wide range of abilities and disabilities.	**5**	**2**	**0**	**0**	**0**	**4.71**	**0.49**	**Excellent**
Overall Mean						**4.59**	**0.15**	**Excellent**

Table 4.8: Frequency Distribution and Descriptive Statistics of Respondents' Ratings on the System's Usability

Compatibility

Compatibility evaluates how well E-Chatco operates alongside other systems and environments used in jeepney management. Among the criteria, co-existence obtained the highest mean (x = 4.57; SD = 0.53), indicating that the system performs efficiently without conflicts when running simultaneously with other applications or tools on the same platform. The other criterion, interoperability (x = 4.29; SD = 0.49), shows that the system effectively exchanges and integrates data across modules such as tracking, payments, discount application, and remittance, as well as with the GCash payment gateway and the thermal printer used by conductors. Overall, the system achieved an excellent rating with an overall mean of 4.43 and a standard deviation of 0.20, signifying the respondents' confidence in E-Chatco's compatibility and integration capabilities within the cooperative's digital system.

Table 4.9 presents the summarized results of the compatibility evaluation of the system.

Criterion	5	4	3	2	1	Mean (x)	SD	Descriptive Interpretation
1. Co-existence - The system operates well on common devices and browsers without interfering with other applications.	**4**	**3**	**0**	**0**	**0**	**4.57**	**0.53**	**Excellent**
2. Interoperability - The system integrates with external devices (thermal printers) and services (GCash payment gateway) used in jeepney operations.	**2**	**5**	**0**	**0**	**0**	**4.29**	**0.49**	**Excellent**
Overall Mean						**4.43**	**0.20**	**Excellent**

Table 4.9: Frequency Distribution and Descriptive Statistics of Respondents' Ratings on the System's Compatibility'

Security

Security assesses how well E-Chatco protects data, user accounts, and system operations from unauthorized access or misuse. Among the criteria, non-repudiation obtained the highest mean (x = 5.00; SD = 0.00), indicating that the system ensures the accuracy and consistency of stored information, such as transaction records, remittances, discount applications, and activity logs, with all administrator actions traceable through the audit trail. Overall, the system garnered an excellent rating with an overall mean of 4.85 and a standard deviation of 0.12, reflecting the respondents' confidence in E-Chatco's ability to maintain data protection, secure GCash transactions, and uphold user privacy through role based access control and verified account management.

Table 4.10: Frequency Distribution and Descriptive Statistics of Respondents' Ratings on the System's Security

Criterion	5	4	3	2	1	Mean (x)	SD	Descriptive Interpretation
1. Confidentiality - The system ensures that sensitive transaction, remittance, and commuter information is accessible only to users with proper authorization, preventing unauthorized disclosure.	**6**	**1**	**0**	**0**	**0**	**4.86**	**0.38**	**Excellent**
2. Accountability - The system provides detailed records of user activities through activity logs, allowing administrators to trace actions back to specific users for responsibility and transparency.	**5**	**0**	**1**	**0**	**0**	**4.67**	**0.82**	**Excellent**
3. Authenticity - The system validates the identity of each user through secure login and role-based access to guarantee that only legitimate users can access system functions.	**6**	**1**	**0**	**0**	**0**	**4.86**	**0.38**	**Excellent**
4. Integrity - Data is accurate and not modified improperly, with all discount and remittance records preserved as submitted.	**6**	**1**	**0**	**0**	**0**	**4.86**	**0.38**	**Excellent**
5. Non-repudiation - Users cannot deny their actions in the system (actions are provable through activity logs and reference numbers).	**7**	**0**	**0**	**0**	**0**	**5.00**	**0.00**	**Excellent**
Overall Mean						**4.85**	**0.12**	**Excellent**

Maintainability

Maintainability evaluates how easily E-Chatco can be modified, updated, and improved to adapt to future business requirements. Among the criteria, testability obtained the highest mean (x =

5.00; SD = 0.00), indicating that the system's modular architecture allows efficient updates and adjustments to its features without disrupting existing functionalities. The other criteria, modularity, reusability, analyzability, and modifiability (x = 4.71, 4.71, 4.71, 4.86; SD = 0.76, 0.49, 0.49, 0.38), show that the system's structure facilitates effective troubleshooting, performance analysis, and testing of new components such as the configurable financial rules and the discount and renewal workflows. Overall, the system received an excellent rating with an overall mean of 4.80 and a standard deviation of 0.13, demonstrating the respondents' confidence in E-Chatco's flexibility, maintainable design, and potential for continuous enhancement..

Criterion	5	4	3	2	1	Mean (x)	SD	Descriptive Interpretation
1. Modularity - Each module (tracking, payments, discount application, remittance, fleet) can be updated independently.	**6**	**0**	**1**	**0**	**0**	**4.71**	**0.76**	**Excellent**
2. Reusability - Components can be reused for future features or other transport operations.	**5**	**2**	**0**	**0**	**0**	**4.71**	**0.49**	**Excellent**
3. Analyzability - Issues and impacts of changes can be easily assessed.	**5**	**2**	**0**	**0**	**0**	**4.71**	**0.49**	**Excellent**
4. Modifiability - The system can be updated (e.g., new fare point or discount category) without affecting other modules.	**6**	**1**	**0**	**0**	**0**	**4.86**	**0.38**	**Excellent**
5. Testability - Functions can be tested effectively to ensure quality.	**7**	**0**	**0**	**0**	**0**	**5.00**	**0.00**	**Excellent**
Overall Mean						**4.80**	**0.13**	**Excellent**

Table 4.11: Frequency Distribution and Descriptive Statistics of Respondents' Ratings on the System's Maintainability

Portability

Portability measures how easily E-Chatco can be installed, transferred, or adapted to different environments without compromising functionality. Among the criteria, adaptability obtained the highest mean (x = 5.00; SD = 0.00), indicating that the system performs efficiently across various devices and setups, such as different browsers, smartphones, tablets, and the conductor's operational device. The other criteria, installability and replaceability (x = 4.71; SD = 0.49), demonstrate that the system can be smoothly deployed and updated with minimal effort, ensuring continuity and ease of maintenance. Overall, the system earned an excellent rating with an overall mean of 4.81 and a standard deviation of 0.17, reflecting the respondents' confidence in E-Chatco's adaptability, flexibility, and capability to operate effectively in diverse environments.

Table 4.12 presents the summarized results of the portability evaluation of the system.

Criterion	5	4	3	2	1	Mean (x)	SD	Descriptive Interpretation
1. Adaptability - The system works on different devices (desktop, laptop, tablet, smartphone, conductor operational device).	**7**	**0**	**0**	**0**	**0**	**5.00**	**0.00**	**Excellent**
2. Installability - The system can be deployed easily in various environments.	**5**	**2**	**0**	**0**	**0**	**4.71**	**0.49**	**Excellent**
3. Replaceability - The system can replace manual jeepney workflows without disrupting operations.	**5**	**2**	**0**	**0**	**0**	**4.71**	**0.49**	**Excellent**
Overall Mean						**4.81**	**0.17**	**Excellent**

Table 4.12: Frequency Distribution and Descriptive Statistics of Respondents' Ratings on the System's Portability

Overall Summary of ISO/IEC 25010 Evaluation

Overall, the evaluation of E-Chatco across all eight ISO/IEC 25010 quality characteristics yielded an overall mean rating of 4.71, descriptively interpreted as Excellent. This result signifies the system's high level of effectiveness, reliability, and readiness for real-world implementation in jeepney operations. The findings demonstrate that E-Chatco successfully meets the core quality attributes of software products, particularly in functional suitability, security, portability, and maintainability, while maintaining strong levels of reliability, performance efficiency, usability, and compatibility. These results affirm the system's capability to operate efficiently and securely under the operational demands of CHATCO Cooperative, supporting its long-term viability and scalability as a comprehensive smart-hailing and fleet management platform.

Table 4.13 presents the summarized results of the overall ISO/IEC 25010 quality evaluation of the system, emphasizing its excellence and suitability for deployment within the client's business operations.

Property	Mean Rating	Descriptive Interpretation
Functional Suitability	**4.86**	**Excellent**
Reliability	**4.71**	**Excellent**
Performance Efficiency	**4.67**	**Excellent**
Usability	**4.59**	**Excellent**
Compatibility	**4.43**	**Excellent**
Security	**4.85**	**Excellent**
Maintainability	**4.80**	**Excellent**
Portability	**4.81**	**Excellent**
Overall Mean	**4.71**	**Excellent**

Table 4.13: Overview of Respondents' Ratings on the System's ISO/IEC 25010 Quality Characteristics

Among all the evaluated quality characteristics, Functional Suitability obtained the highest mean rating of 4.86, reflecting the system's ability to deliver accurate, complete, and appropriate

functionalities that align with the operational requirements of CHATCO Cooperative. This indicates that E-Chatco effectively supports the end-to-end workflow of jeepney operations, from real-time tracking and smart hailing to fare collection, discount application, student renewal, boundary remittance tracking, and fleet monitoring. Security (x = 4.85) and Portability (x = 4.81) followed closely, demonstrating the system's capability to protect sensitive commuter, transaction, and remittance data while maintaining adaptability across various platforms and devices. Maintainability (x = 4.80) and Reliability (x = 4.71) also achieved excellent ratings, highlighting the system's stability, flexibility for future updates, and consistent performance. Meanwhile, Performance Efficiency (x = 4.67), Usability (x = 4.59), and Compatibility (x = 4.43) were rated slightly lower but still within the excellent range, indicating that the system performs efficiently, remains user-friendly, and integrates effectively with external services such as the GCash payment gateway and the conductor's thermal printer. Collectively, these results emphasize E-Chatco's strong operational capability, adaptability to user needs, and reliable performance as a comprehensive smart-hailing and fleet management platform for the modernized Philippine jeepney.

CONCLUSION

The study developed E-Chatco: An Integrated Smart-Hailing and Fleet Management Platform for the Modernized Philippine Jeepney, which aimed to streamline the operations of CHATCO Cooperative along the Calumpit to Meycauayan route. The system was designed and implemented as a web-based platform integrating features such as real-time GPS tracking, proximity-based smart hailing within a 1-kilometer radius, GCash e-payment integration, fare calculation using predefined operational point areas, boundary remittance tracking with separate cash and GCash recording, fleet and traffic visualization, lost and found management, application of discounts for Student, Senior Citizen, and PWD commuters, renewal of student

discount eligibility, and safety features such as Share My Ride. The development of the system followed the Agile Scrum methodology, which allowed the team to design, build, and test the platform in iterative sprints while continuously improving its features based on feedback from the client.

Based on the system evaluation conducted using ISO/IEC 25010:2023 standards, the platform obtained an overall mean of 4.71, interpreted as Excellent. This indicates that E-Chatco met the users' expectations and successfully achieved its objectives. A total of 346 system test cases were executed across the commuter, conductor, and administrator applications as well as the cross-role interactions among them, with all cases passing in the second round of testing after the defects identified in the initial round were resolved.

Among the software quality characteristics, Functional Suitability obtained the highest mean rating, confirming that the system's functions accurately support the workflows of CHATCO Cooperative, from real-time tracking and hailing to fare collection, discount application, boundary remittance tracking, and fleet monitoring. Security, Portability, and Maintainability also scored high, signifying that the platform provides a secure environment for data management, can adapt to different operating conditions and devices, and allows easy modification for future enhancements.

Of particular note is the successful integration of the Application of Discounts and Renewal of Student Discount features, which directly address the cooperative's need for accurate fare discount administration in compliance with Philippine regulations on Student, Senior Citizen, and PWD discounted fares. The Application of Discounts feature allows Regular commuters to submit a valid supporting document for Student, Senior Citizen, or PWD classification through the discount-application page; the application is reviewed by an administrator through the Discount Applications and Renewals sub-module; and upon approval, the commuter's type is automatically updated and the configured discount rate (default 20%) is applied to all subsequent

GCash payment scans, without requiring the conductor to manually select the discounted type. The Renewal of Student Discount feature complements this workflow by requiring Student commuters to renew their eligibility at the start of each renewal cycle through the /student-renewal page, with the system automatically reverting the commuter type to Regular and removing the discounted fare if renewal is not completed before the deadline. Together, these features ensure that only verified commuters receive discounted fares, that student discounts are renewed each academic term to maintain eligibility, and that all discount-related transactions are properly recorded and reviewable by the administrator through the activity logs.

In conclusion, E-Chatco effectively addressed the operational challenges of CHATCO Cooperative by providing an efficient, user-friendly, and secure platform for managing jeepney operations along the Calumpit to Meycauayan route. The system met its primary goal of digitizing transit operations while promoting transparency, safety, and coordination among commuters, conductors, and cooperative management. The proximity-based hailing within the 1-kilometer radius addressed the cooperative's reported issue of commuters being missed due to overcrowding and poor visibility, while the separate cash and GCash recording in the boundary remittance module addressed the manual cash declaration discrepancies reported during the formal interview. The results affirm that the system is ready for full deployment and continuous use in the cooperative's daily operations.

Based on the findings and conclusions of the study, it is recommended that future development of E-Chatco focus on expanding its functionalities according to the future needs of CHATCO Cooperative. At present, the system requires conductors to use dedicated operational devices with built-in thermal printers for fare processing. Future iterations may explore the integration of the conductor application into a broader mobile application that can run on standard smartphones, provided that the cooperative's policy on phone usage during operations is updated.

The existing loyalty and rewards system, which distributes random digital vouchers and grants free-ride vouchers after a configurable number of paid rides (default 10), may be further enhanced by introducing account-specific loyalty tiers and personalized promotions based on commuter travel patterns. The voucher distribution defect identified during testing, where commuters approved through the normal sign-up flow may not receive generated vouchers due to a status mismatch between APPROVED and ACTIVE, has been resolved in the deployed version but should be monitored during ongoing operations to ensure consistent behavior. In addition, the system may be further improved to support external operations of CHATCO Cooperative, such as coordinating with fuel suppliers, vehicle maintenance providers, and local transport regulators. The announcement feed, which is currently visible to commuters but not to conductors in the conductor application, may be extended to the conductor interface to provide a unified communication channel across all roles. The route configuration, which currently supports a single fixed Calumpit to Meycauayan route, may be extended to support multiple routes should the cooperative expand its operations in the future. Continuous maintenance and regular system evaluation are encouraged to ensure that the platform remains stable, secure, and responsive to the cooperative's business requirements. As the Public Utility Vehicle Modernization Program (PUVMP) continues to evolve, E-Chatco can be updated to align with new regulatory requirements and to support potential expansion to other routes and cooperatives in the future.

Overall, the results make it reasonable to conclude that E-Chatco is an effective solution for modernized jeepney management. With sustained updates, system optimization, and design refinements, it can continue to improve efficiency, transparency, and commuter satisfaction within the Philippine public transportation sector.

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APPENDICES

APPENDIX A. RESOURCE PERSON(S)

Resource Person

This section documents the industry professionals and stakeholders consulted during the requirements gathering phase of the project. Their insights were critical in validating the operational workflows, boundary remittance challenges, and feature requirements of the system.

Resource Person 1

- **Name:** Mrs. Alicia Tiongco
- **Position:** General Manager
- **Company/Organization:** CHATCO Cooperative
- **Date of Interview:** April 6, 2026
- **Mode of Interview:** via Phone Call
- **Role in the Project:** Mrs. Tiongco provided the foundational requirements for the system, detailing the daily operational struggles of jeepney operators and conductors. She specifically highlighted the issues with manual boundary remittance tracking, cash declaration discrepancies, and the lack of visibility, which directly inspired the "Remittance Tracker" and "Pick Me Up" features of ChatCo.
- **Contribution Summary:** Validated the system's proposed modules; provided access to the cooperative's 39 modern jeepney units for the pilot testing phase; and supplied actual operational data (e.g., standard boundary rates and fuel expense averages) used to design the Admin Financial Dashboard.

Resource Person 2

- **Name:** Mr. Jose Crisanto S. Austira
- **Position:** Capstone Project Adviser
- **Company/Organization:** STI College Balagats
- **Role in the Project:** Mr. Austria serves as the capstone project adviser, providing academic guidance and oversight throughout the project development lifecycle. He reviewed the project proposal, endorsed it for proposal defense, and provided continuous feedback on the system's technical architecture, feature scope, and documentation standards.
- **Contribution Summary:** Mr. Austria serves as the capstone project adviser, providing academic guidance and oversight throughout the project development lifecycle. He reviewed the project proposal, endorsed it for proposal defense, and provided continuous feedback on the system's technical architecture, feature scope, and documentation standards.

Resource Person 3

- **Name:** Mr. Alejandro B. Burgos Jr.
- **Position:** Capstone Project Coordinator
- **Company/Organization:** STI College Balagtas
- **Role in the Project:** Mr. Burgos serves as the capstone project coordinator, overseeing the administrative and procedural aspects of the capstone project program. He approved the project for proposal defense and ensures compliance with institutional policies, deadlines, and quality standards.
- **Contribution Summary:** His coordination facilitates the scheduling of defense presentations, the assignment of panel members, and the communication of evaluation results. As coordinator, he also ensures that the project documentation, including this revision report, adheres to the formatting and content standards required by the Information and Communications Technology Program at STI College Balagtas.

APPENDIX B. INTERVIEW TRANSCRIPT (FORMAL)

Interview Details:

- **Date: April 6, 2026**
- **Time: 1:44 PM – 2:04 PM**
- **Location: Via Voice Call**
- **Interviewer: Marinel Carbonell**
- **Interviewee: Mrs. Alicia Tiongco (General Manager & HR, CHATCO Cooperative)**

Transcript:

Interviewer: Good afternoon and thank you for accepting our request for an interview.

Interviewee: It is an honor to be acquainted with you.

Interviewer: So before we start, may I ask for your permission to record this call for academic purposes and for proof that we have your permission?

Interviewee:: Yes, of course you may.

Interviewer: Now we can proceed with the interview. Could you please share your name and your role within the organization?

Interviewee:: Very well, my name is Alicia Tiongco, and I am the general manager and also the HR of our organization.

Interviewer: Next question, what are the most common and major concerns encountered on a daily basis? And what is the best approach to solve it?

Interviewee: There are many instances where some of our employees make mistakes in handling the fare of each commuter. We haven't had any solution as to how to solve that, unfortunately.

Interviewer: How many jeepneys/units does your company currently have?

Interviewee: Currently we have 39 units in total.

Interviewer: What is the end-point destination of your organization's modern jeepney?

Interviewee: Our jeepney drivers start their trip from here at the main office/terminal in Calumpit until they reach Meycauayan and from there back to the terminal. Each driver and conductor does 6 roundtrips every day with only one route to take.

Interviewer: What are the operating hours of your drivers and conductors?

Interviewee: We have no exact operating time, because each of our drivers has a fixed 10-15 min waiting time before their trip, and then either at 9:00 or 10:00 AM they start their trip.

Interviewer: Are the drivers and conductors assigned to a specific unit, or do they use different ones?

Interviewee: You see, each driver is assigned to only one unit, while the conductors can use or go on a trip with a different unit. Our drivers can also use another unit if the one they are assigned to is under maintenance.

Interviewer: Is there a target or quota that each driver must meet?

Interviewee: Ah, yes, there is. Back then they had to meet the quota of 8,000 to 9,000 per day. Now since the fuel expenses are getting higher, each of them must meet the 10,000+ quota a day.

Interviewer: Do they wait for passengers at designated stops like traditional jeepneys, or do they continue driving until they notice commuters signaling for a ride?

Interviewee: Currently they continue driving until they notice commuters and offer them a ride, but they also pick them up at designated stops sometimes.

Interviewer: How does your company calculate commuter fares?

Interviewee: We have a gadget that calculates the exact amount of the fare per kilometer. Each location has designated prices per kilometer, but it may spike depending on how fast the trip is or the spiking of fuel prices.

Interviewer: How many passengers can your modern jeepney accommodate?

Interviewee: Each unit can accommodate about 26 passengers, including the driver and conductor.

Interviewer: How do you currently handle fare collection?

Interviewee: We currently handle it via cash payment as of now.

Interviewer: How do conductors record cash fare payments?

Interviewee: The fare is recorded in a gadget, and that gadget produces a ticket for the conductor to give to the commuter.

Interviewer: How is an incorrect payment entry handled on the device for cash transactions?

Interviewee: Well, you see, as of now the incorrect payment entry is recorded, and we have not yet found a way to reverse or delete the entry.

Interviewer: Do you accept cashless payments?

Interviewee: As of now we are not accepting cashless payment. We do have the TAP card, but currently we have some trouble implementing it.

Interviewer: Are fuel expenses deducted from the earned fare, or is there a standard process followed?

Interviewee: Our cooperative is in charge of getting fuel for all of the units. We do not deduct the expenses from the earned fare, and we certainly do not deduct from their salary.

Interviewer: What difficulties do you face in tracking revenue and expenses?

Interviewee: The thing we are currently facing is the overpricing of the fuel that we buy for all of our units. We also face difficulty in tracking the revenue due to the oil spike and other expenses we are currently managing.

Interviewer: A question for your staff: Do all your conductors carry a phone during operating hours?

Interviewee: Yes, they do, but all of them are not allowed to use their phones during operating hours. If they do, they'll receive a violation.

Interviewer: Are there instances where commuters leave or forget items inside units?

Interviewee: Most of the time, yes, we do find some belongings inside our units, and it is then reported to the administrators.

Interviewer: How and where can passengers claim lost or forgotten items?

Interviewee: We will notify them immediately or via our page, and then the passenger can claim their belongings here in our office. But if ever another passenger took it, we may not be able to help them reclaim it.

Interviewer: Are there cases where a driver is required to refund the fares of passengers due to issues like an engine breakdown?

Interviewee: Yes, each driver is required to compensate the passengers and refund their fares.

Interviewer: Are there violations and corresponding penalties for drivers who cause problems?

Interviewee: Absolutely, some drivers are slacking off and on their phones during operating hours and will have warnings. If they repeat it again or do something they are not supposed to do, they are suspended and their salary is cut off.

Interviewer: Are there any instances where drivers or conductors miss picking up a commuter due to factors such as poor visibility, overcrowding, or safety concerns?

Interviewee: Yes, they do many times, and it's mostly because of overcrowding.

Interviewer: How do you ensure passenger safety during operations?

Interviewee: Every driver and conductor is tasked to train for challenges like this, and they also remind every passenger of the rules and regulations when they are riding our unit.

Interviewer: Are drivers, conductors, or administrators familiar with common pick-up points and peak times where commuters usually wait?

Interviewee: Every one of them undergoes training for these kinds of events, as it is our responsibility and job to do so.

Interviewer: Is it helpful for drivers to anticipate traffic conditions ahead?

Interviewee: Yes, it is. Knowing these conditions can help prepare them and also help them solve it when it is necessary.

Interviewer: That is it. Thank you so much for giving us your time today for this interview. We are very grateful for your insight. God bless all of you.

Interviewee: Thank you also for choosing our company and for the interview.

APPENDIX C. PERSONAL TECHNICAL VITAE

Curriculum Vitae of

MARINEL T. CARBONELL
654, Endless Love Compound, Poblacion, Guiguinto, Bulacan, 3015
lyraresflovilla@gmail.com
+639214876337

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	2023-present	STI College Balagtas
Vocational/Technical	2021-2022	Advance Montessori Education Center
High School	2017-2020	Tumauini National High School
Elementary	2015-2016	Tumauini North Central School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
month year	N/A	N/A
month year	N/A	N/A
month year	N/A	N/A
month year	N/A	N/A

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
2023-present	BITS Organization	Member
month year	N/A	N/A
month year	N/A	N/A
month year	N/A	N/A

SKILLS

SKILLS	Level of Competency	Date Acquired
Python	Beginner	August 2024
Database	Beginner	August 2024
Java	Intermediate	August 2023

TRAININGS, SEMINARS, OR WORKSHOPS ATTENDED

Inclusive Dates	Title of Training, Seminar, or Workshop
November 2025	8th ICT Research Colloquium
month year	N/A
month year	N/A
month year	N/A

Curriculum Vitae of
<Mark Arone M. Dela Cruz>
<2118 Purok 2, Bambang, Bulakan, Bulacans>
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+639994161584

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	2023 - Present	STI College Balagtas
Vocational/Technical	2021 - 2022	Assumpta Academy of Bulakan
High School	2017 - 2020	Assumpta Academy of Bulakan
Elementary	2010-2016	Bambang Elementary School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
month year		
month year		
month year		
month year		

Listed in reverse chronological order (most recent first).

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
month year		
month year		
month year		
month year		

Listed in reverse chronological order (most recent first).

SKILLS

SKILLS	Level of Competency	Date Acquired
		month year
		month year
		month year

TRAININGS, SEMINARS, OR WORKSHOPS ATTENDED

Inclusive Dates	Title of Training, Seminar, or Workshop
month year	
month year	
month year	
month year	

Listed in reverse chronological order (most recent first).

Curriculum Vitae of

Manalili, Mhaku Jose Marcelo
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EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	2023 - Present	STI College Balagtas
Vocational/Technical	2021 - 2022	Assumpta Academy of Bulakan
High School	2017 - 2020	Assumpta Academy of Bulakan
Elementary	2010-2016	Bambang Elementary School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
month year		
month year		
month year		
month year		

Listed in reverse chronological order (most recent first).

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
month year		
month year		
month year		
month year		

Listed in reverse chronological order (most recent first).

SKILLS

SKILLS	Level of Competency	Date Acquired
		month year
		month year
		month year

TRAININGS, SEMINARS, OR WORKSHOPS ATTENDED

Inclusive Dates	Title of Training, Seminar, or Workshop
month year	
month year	
month year	
month year	

Curriculum Vitae of
<GIVEN NAME MI. FAMILY NAME>
<complete address>
<email address>
contact number, either cellular phone or landline or both

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	month year	
Vocational/Technical	month year	
High School	month year	
Elementary	month year	

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
month year		
month year		
month year		
month year		

Listed in reverse chronological order (most recent first).

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
month year		
month year		
month year		
month year		

Listed in reverse chronological order (most recent first).

SKILLS

SKILLS	Level of Competency	Date Acquired
		month year
		month year
		month year

TRAININGS, SEMINARS, OR WORKSHOPS ATTENDED

Inclusive Dates	Title of Training, Seminar, or Workshop
month year	
month year	
month year	
month year	

Listed in reverse chronological order (most recent first).

APPENDIX D. CLIENT EVALUATION FORM

System Evaluation Form for E-Chatco (Based on ISO/IEC 25010 Standards)

Instruction: Please check (✓) the column that best represents your answer.

Criteria	5 - Strongly Agree	4 - Agree	3 - Neutral	2 - Disagree	1 - Strongly Disagree
A. Functional Suitability					
A1. The system provides all the functions needed for jeepney operations (smart hailing, fare calculation, payments, discounts, remittance, and fleet monitoring).					
A2. The system produces correct and reliable results (e.g., accurate fare calculations, payment records, discount applications, and remittance transactions).					
A3. The system provides features that are appropriate for the operational needs of commuters, conductors, and administrators.					
B. Reliability					
B1. The system runs without errors during use.					
B2. The system is available when I need it.					
B3. The system continues to work even if there are minor issues.					
B4. The system can recover important records after a crash or failure.					
C. Performance Efficiency					
C1. The system responds quickly to my actions.					
C2. The system uses device and server resources efficiently.					

C3. The system can handle many users or large event data without slowing down.					
D. Usability					
D1. The system is easy to learn and use.					
D2. The interface is user-friendly for commuters, conductors, and administrators					
D3. The system helps prevent or alert users about mistakes (e.g., invalid inputs, duplicate requests, and incorrect transaction details).					
D4. I am satisfied with the overall design and experience of the system.					
D5. The system's design is visually appealing and clearly tailored for jeepney transportation and fleet management operations.					
D6. The system can be used easily by people with different needs and abilities.					
D7. It is easy to interact with the system to get tasks done.					
E. Compatibility					
E1. The system works well on common devices and browsers.					
E2. The system can work with external devices (e.g., bluetooth thermal printers).					
F. Security					
F1. The system protects sensitive data (commuter information, payment records, and remittance details).					
F2. The system keeps records of user activities for accountability.					

F3. The system ensures only authorized users can access functions.					
F4. The system keeps data accurate and prevents unauthorized changes.					
F5. The system ensures that important user actions (such as transactions and remittance submissions) can be traced through system records and reference information.					
G. Maintainability					
G1. The system modules (tracking, payments, discount applications, and remittance) can be updated independently.					
G2. Parts of the system can be reused for future features.					
G3. The system allows issues and impacts of changes to be understood clearly.					
G4. The system can be updated (e.g., adding fare points or modifying financial rules) without affecting other functions.					
G5. The system's functions can be tested effectively.					
H. Portability					
H1. The system works on different devices (laptop, tablet, smartphone).					
H2. The system can be installed easily in various environments.					
H3. The system can replace manual jeepney operation workflows without unnecessary disruption.					

Open-Ended Questions

What is the most significant improvement we could make to the system?

_____

Please suggest any additional features that would be valuable for your catering operations.

_____

Acknowledgement

I hereby confirm that my responses are based on my actual experience using the system. I understand that the results will be used solely for academic purposes as part of the capstone project **"E-Chatco: An Integrated Smart-Hailing and Fleet Management Platform for the Modernized Philippine Jeepney."**

Evaluator's Name & Signature: _____

Date: _____